

Compact Registered Functional Encryption for Attribute-Weighted Sums with Access Control

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Abstract

In this work, we present Functional Encryption (FE) schemes for Attribute-Weighted Sums (AWS), introduced by Abdalla, Gong and Wee (Crypto 2020) in the registration-based setting (RFE). In such a setting, users sample their own public/private key pairs (pk_i, sk_i) ; a key curator registers user public keys along with their functions h_i ; encryption takes as input N attribute-value pairs $\{\mathbf{x}_j, \mathbf{z}_j\}_{j \in [N]}$ where $\mathbf{x}_j$ is public and $\mathbf{z}_j$ is private; and decryption recovers the weighted sum $\sum_{j \in [N]} h_i(\mathbf{x}_j)^\top \mathbf{z}_j$ while leaking no additional information about $\mathbf{z}_j$. Recently, Agrawal, Tomida and Yadav (Crypto 2023) studied the attribute-based case of AWS (AB-AWS) providing fine-grained access control, where the function is described by a tuple (g_i, h_i) , the input is extended to $(\mathbf{y}, \{\mathbf{x}_j, \mathbf{z}_j\}_{j \in [N]})$ and decryption recovers the weighted sum only if $g_i(\mathbf{y}) = 0$. Our main results are the following:

- We build the first RFE for (AB-)AWS functionality, where $N = 1$, that achieves adaptive indistinguishability-based security under the (bilateral) k -Lin assumption in prime-order pairing groups. Prior works achieve RFE for linear and quadratic functions without access control in the standard model, or for attribute-based linear functions in the generic group model.
- We develop the first RFE for AB-AWS functionality, where N is unbounded, that achieves very selective simulation-based security under the bilateral k -Lin assumption. Here, “very selective” means that the adversary declares challenge attribute values, all registered functions and corrupted users upfront. Previously, SIM-secure RFEs were only constructed for linear and quadratic functions without access control in the same security model.

We devise a novel nested encoding mechanism that facilitates achieving attribute-based access control and unbounded inputs in the registration-based setting for AWS functionalities, proven secure in the standard model. In terms of efficiency, our constructions feature short public parameters, secret keys independent of N , and compact ciphertexts unaffected by the length of public inputs. Moreover, as required by RFE properties, all objective sizes and algorithm costs scale poly-logarithmically with the total number of registered users in the system.

Keywords: registered functional encryption, attribute-weighted sums, access control

*Work done while the author was at ENS Paris.

Contents

1	Introduction	3
1.1	Our Results	4
2	Technical Overview	5
2.1	Constructing Slotted RFE for 1AWS	10
2.2	Adding Fine-Grained Access Control	13
2.3	Extension to Unbounded Inputs	16
3	Preliminaries	20
3.1	Notational Conventions	20
3.2	Pairing Groups and Hardness Assumptions	21
3.3	Arithmetic Branching Programs and Partial Garbling	22
3.4	Registered Functional Encryption	22
3.5	Slotted Registered Functional Encryption	25
3.6	Quasi-Adaptive Non-Interactive Zero-Knowledge Arguments	28
4	Adaptively IND-Secure sRFE for eAWS	29
4.1	Proof of Completeness, Correctness and Compactness	31
4.2	Proof of Security	32
5	Adaptively IND-Secure sRFE for AB-eAWS	38
5.1	Proof of Completeness, Correctness and Compactness	41
5.2	Proof of Security	43
6	Very Selectively SIM-Secure sRFE for AB-eAWS	49
6.1	Proof of Completeness, Correctness and Compactness	52
6.2	Simulator	54
6.3	Proof of Security	57
6.4	Special Case: Very Selectively SIM-Secure sRFE for eAWS	66
7	Very Selectively SIM-Secure sRFE for AB-AWS	70
7.1	Simulator	71
7.2	Proof of Security	72

1 Introduction

Registered Functional Encryption. Functional Encryption (FE) [SW05, BSW11, O’N10] is an elevated version of traditional public key encryption which provides fine-grained access control and computation abilities over encrypted data. A secret key sk_f of a FE is associated with a function f , a ciphertext ct is associated with a message $\mathbf{z}$ and the decryption of ct using sk_f yields $f(\mathbf{z})$ while concealing any other information about $\mathbf{z}$. Due to its extensive applicability, FE and its numerous subclasses have attracted substantial interest within the research community over the last decade [ABDP15, ALS16, BJK15, LV16, AS16, DDM16, BCFG17, BBL17, GKW17, Agr17, Wee17, CLT18, TT18, AV19, GW20, ACGU20, CDSG⁺20, JLS21, ALMT20, AY20, Wee20, AGW20, Gay20, AMVY21, KNT21, DP21, DPT22, JLS22, AKM⁺22, GGLW22, Tom23].

While FE possesses immense potential, offering a wide array of applications, it fundamentally reshapes the trust framework compared to conventional public key encryption. In particular, its dependence on a central trusted authority that manages a *master secret key* to generate user-specific keys poses a potential risk to the system’s long-term viability. If compromised, this key could expose all encrypted messages, making key exfiltration a significant threat. Registered FE (RFE) [FFM⁺23] addresses this risk by replacing the central authority with a *key curator*. In more detail, RFE uses a *one-time* trusted sampling of a common reference string which can be used across different systems and (possibly) generated via a multi-party computation (MPC) protocol. Like traditional public key encryption, users in RFE generate their own public/secret keys, and register the public keys along with specific functions. The key curator is a *public* algorithm which *deterministically* aggregates a list of public keys and functions to a *short* master public key and produces helper decryption keys for users. The master public key is changed whenever a new user joins the system. Thus, users must periodically update their helper decryption keys, which are bound to the current master public key, to decrypt messages. Notably, each user needs only $O(\log L)$ updates during the existence of the system, where L is the number of users, and the size of each update, the master public keys and helper decryption keys of RFE remain asymptotically $\text{poly}(\lambda, \log L)$.

The Attribute-Weighted Sums Function Family. Abdalla, Gong and Wee [AGW20] introduced the functionality of *Attribute-Weighted Sums* (AWS), enabling more fine-grained computation of aggregate statistics over encrypted data than simple linear/quadratic functions. Specifically, consider a database of N attribute-value pairs $\{(\mathbf{x}_j, \mathbf{z}_j)\}_{j \in [N]}$ where $\mathbf{x}_j$ is a public attribute of user i and $\mathbf{z}_j$ is private. Given a function f , the AWS functionality computes the weighted sum over $\mathbf{z}_j$ ’s as

$$\sum_{j \in [N]} f(\mathbf{x}_j)^\top \mathbf{z}_j.$$

FE for AWS functionality is a natural generalization of FE designed for linear functions [ABDP15, ALS16], opening the door to a range of interesting applications [AGW20]. This includes computation of (i) average salaries of minority groups holding a particular job title where $\mathbf{z}_j$ represents salary data, $f(\mathbf{x}_j)$ identifies membership in the minority group, (ii) approval ratings of an election candidate amongst targeted demographic groups in a particular state with $\mathbf{z}_j$ reflecting the ratings, $f(\mathbf{x}_j)$ determining group affiliation, (iii) total number of and average age of smokers with lung cancer in a state where $\mathbf{z}_j$ signifying lung cancer and $f(\mathbf{x}_j)$ testing the residency status in the state.

Building upon AWS functionality, Agrawal, Tomida, and Yadav [ATY23] introduced an enhanced Attribute-Based AWS (AB-AWS) framework by incorporating an additional layer of access control. Concretely, in AB-AWS, the public attribute consists of two parts: $\mathbf{y}$ is related to the access control layer and $\mathbf{x}_j$ is associated with the AWS layer as in the plain AWS functionality. The function f is a tuple (g, h) where g describes a policy and h computes the weights. On input a database $(\mathbf{y}, \{(\mathbf{x}_j, \mathbf{z}_j)\}_{j \in [N]})$, the AB-AWS functionality outputs

$$f(\mathbf{y}, \{(\mathbf{x}_j, \mathbf{z}_j)\}_{j \in [N]}) = \begin{cases} \sum_{j \in [N]} h(\mathbf{x}_j)^\top \mathbf{z}_j & \text{if } g(\mathbf{y}) = 0 \\ \perp & \text{otherwise.} \end{cases}$$

This extra layer of attribute-based access control adds a sophisticated level of security, safeguarding private input data while further increasing the practical utility of AWS in real-world applications. For example, the computation of average minority group salaries can now be limited to specific authorized users. Even with $N = 1$, the *single-input* AWS (1AWS) functionality [DPT22] computing $f(\mathbf{x})^\top \mathbf{z}$ and the single-input AB-AWS (AB-1AWS) functionality outputting $h(\mathbf{x})^\top \mathbf{z}$ if $g(\mathbf{y}) = 0$ capture IP [ABDP15, ALS16] and AB-IP [ACGU20], respectively. In this work, our objective is to design RFE for the AWS function family from standard assumptions.

All applications of (AB-)AWS become even more meaningful in the registration-based setting. Consider an analytics firm tasked with computing the total number of smokers in a state. Under traditional FE, the firm would rely on a central authority to generate a decryption key corresponding to the analytic function, introducing a single point of trust and potential vulnerability. In contrast, RFE allows the (specifically authorized) employees of the firm to independently register their functions, ensuring long-term security, decentralization, and system resilience. This approach not only enhances reliability but also eliminates dependence on a single trusted entity, making the system more scalable and sustainable. By shifting to a registration-based model, FE for AWS gains a stronger foundation for real-world deployment, fostering privacy-preserving analytics while maintaining decentralized trust.

Related Work. The concept of registered encryption originated with registration-based encryption (RBE) [GHMR18] initially built for identity-based encryption (IBE) [Sha84, BF01] and relies on non-black-box use of cryptography [GHM⁺19, GV20, CES21]. Recent research [GKMR23, DKL⁺23], however, has developed black-box methods to implement RBE based on pairings and lattices. Simultaneously, Hohenberger et al. [HLWW23] generalizes the notion of RBE to registered attribute-based encryption (RABE) where each user registers a predicate g , encryption occurs under an attribute $\mathbf{y}$, and decryption recovers the message if the attribute satisfies the predicate, i.e., $g(\mathbf{y}) = 0$. Subsequent works have focused on developing more general RABEs, supporting more expressive predicates [FWW23, ZZGQ23] or a large universe of attributes [FFM⁺23]. RFE aims to encompass prior registered encryption schemes and enhances the practical relevance of standard IBE, ABE, and FE schemes. However, existing RFEs for general functions [FFM⁺23, DPY24] either rely on indistinguishability obfuscation (iO) [JLS22] or offer only a weaker *bounded collusion* security [ZCHZ24]. Furthermore, even if (AB-)AWS is restricted to a bounded number of inputs such that circuits can handle the functionality, the RFE constructions in [FFM⁺23, DPY24, ZCHZ24] fail to support short ciphertexts (defined below), which is an essential efficiency feature of the FE scheme for AWS in [AGW20]. On the other hand, the list of RFEs for specific function classes only includes pairing-based schemes for inner products (IP) and quadratic functions (QF) [ZLZ⁺24, BLM⁺24], and attribute-based inner-products (AB-IP) [DPY24].

The decentralized digital ecosystem is rapidly evolving with the widespread adoption of Web3, decentralized identity, decentralized finance (DeFi), and decentralized storage. With the vision of advancing cryptographic systems for decentralized digital world by eliminating the central authority, several other variants of FE have been explored apart from RFE, particularly in the context of distributed data, leading to the study of decentralized multi-client FE (DMCFE) [CDG⁺18] and dynamic decentralized FE (DDFE) [CDSG⁺20]. Given its potential, DMCFE for inner product functionality has been extensively examined in subsequent research [ABKW19, ABG19, LT19, NPP23a, NPP23b, LWG⁺23, NPS24] with a focus on enhancing security or incorporating additional features such as verifiability and robustness. To facilitate the dynamic inclusion of new users, DMCFE has been extended to DDFE, which supports both distributed keys and ciphertexts. Existing constructions of DDFE have been instantiated for the inner product and AWS functionalities [CDSG⁺20, ATY23, Ngu25, NPS25]. To the best of our knowledge, AWS represents the most expressive function class known for these FE variants that does not rely on iO, and even AB-1AWS remains unexplored in the (D)MCFE and DDFE settings.

1.1 Our Results

We classify our results into two categories, addressing the 1AWS and AWS function families. The RFEs for the 1AWS function family meet standard adaptive indistinguishability-based (IND) security, while those for the

AWS function family achieve very selective simulation-based (SIM) security [ZLZ⁺24] which means that the adversary submits the challenge functions, public/private inputs and the corrupted users at the system setup. All our constructions are based on prime-order pairing groups and are proven secure in the standard model.

RFE for AB-1AWS. We design the first RFE for the AB-1AWS functionality. In this scheme, each user registers a function $f_i = (g_i, h_i)$ alongside a public key pk_i ; an encryptor produces a ciphertext ct associated with a public input $(\mathbf{y}, \mathbf{x})$ and a private input $\mathbf{z}$ using an aggregated master public key mpk ; a user with sk_i can decrypt ct using a helper decryption key hsk_i and obtains $h_i(\mathbf{x})^\top \mathbf{z}$ if $g_i(\mathbf{y}) = 0$ holds. The functions g_i, h_i are expressed as arithmetic branching programs (ABP), a robust model of computation capturing both boolean formula and branching programs. The ciphertext is *short* in the sense that it linearly grows with the length of $\mathbf{z}$, but not $(\mathbf{y}, \mathbf{x})$. Our RFE for AB-1AWS is secure under the bilateral k -Lin assumption and includes the following results:

- As an *intermediate result*, we obtain the first RFE for 1AWS functionality relying only on the k -Lin assumption. It generalizes the RFE framework realizing linear functions constructed by Zhu et al. [ZLZ⁺24] based on the same assumption.
- As a *special case*, we get the first IND secure RFE for AB-IP. Previously, Datta et al. [DPY24] built such an RFE in the ciphertext-policy setting achieving a ciphertext size scaling with the total number of attributes in the system, and the same security in the generic group model (GGM).
- As an *implication*, we achieve a concrete RABE for ABPs in the key-policy setting. Prior work [ZZGQ23] built such an RABE in the ciphertext-policy setting and relies on a “ciphertext-policy to key-policy” transformation due to [ABS17].

RFE for AB-AWS. We build the first RFE for the AB-AWS functionality. Akin to the plain FE for AWS [AGW20], our RFE for AB-AWS supports (1) functions g_i, h_i expressed as ABPs, (2) encrypting an arbitrary number of inputs $\{(\mathbf{x}_j, \mathbf{z}_j)\}_{j \in [N]}$ under an attribute $\mathbf{y}$, (3) compact public parameters, key sizes that are independent of N , (4) short ciphertexts that grow linearly with N and the size of private inputs $\mathbf{z}_j$, but not $\mathbf{y}$ and $\mathbf{x}_j$. Our RFE for AB-AWS is secure under the bilateral k -Lin assumption and implies the following results:

- As an *intermediate result*, we obtain the first RFE for AWS functionality. In the non-registration-based setting, prior work [AGW20] achieved FE for AWS with similar efficiency satisfying selective SIM security under the k -Lin assumption.
- As a *special case*, we get the first SIM secure RFE for AB-IP. Previously, Zhu et al. [ZLZ⁺24] constructed RFE for IP (without any access control) satisfying the same very selective SIM security, and Abdalla et al. [ACGU20] built AB-IPFE in the non-registration-based setting with selective SIM security under the SXDH assumption.

Similar to prior pairing-based schemes [HLWW23, FFM⁺23, ZZGQ23, ZLZ⁺24, DPY24], all our RFEs support *bounded* number of slots, and each user requires to update their decryption keys only $O(\log L)$ times during the life-time of the system where L is the total number of users. We achieve compact master public key and helper decryption keys of size $O((n' + mn) \log L)$, secret key size of $O(m)$ and ciphertext size of $O(N(n' + m) \log L)$ where n', n denote the lengths of *one* private, public input, N is the number of inputs and m is proportional to the size of functions. Additionally, the secret key size of our RFE for AWS, AB-(1)AWS depends on n' . Please see Table 1 for a detailed comparison.

2 Technical Overview

We start by reviewing the two main ingredients used by Abdalla, Gong and Wee [AGW20], henceforth AGW, in their construction of (plain) FE for AWS. For a group $\mathbb{G}$ of prime order p and $x \in \mathbb{Z}_p$, we write $[x] = g^x \in \mathbb{G}$ and

Work	Functionality	Security	Assumption	sk	mpk	hsk	ct
[FFM ⁺ 23, DPY24]	General	Adp-IND	iO + SSB	1	$\log L$	$\log L$	$n' \log L$
[DPY24]	IP	Adp-IND	GGM	1	$n' \log L$	$\log L$	$n' \log L$
	AB-IP	Adp-IND	GGM	1	$(n + n') \log L$	$n \log L$	$(n + n') \log L$
[ZLZ ⁺ 24]	IP	Adp-IND	k -Lin	1	$n' \log L$	$n' \log L$	$n' \log L$
	IP	Sel-SIM	k -Lin	1	$n' + \log L$	$\log L$	$n' + \log L$
	QF	Sel*-SIM	bi- k -Lin	1	$n' + \log L$	$\log L$	$n' + \log L$
[BLM ⁺ 24]	IP	SelSta-IND	q -type	n'	$n' \log L$	$n' \log L$	$n' \log L$
	weak QF	Adp-IND	GGM	1	$n' \log L$	$n' \log L$	$n' \log L$
Sec. 4	1AWS	Adp-IND	k -Lin	m	$(n' + mn) \log L$	$(n' + mn) \log L$	$(n' + m) \log L$
Sec. 5	AB-1AWS	Adp-IND	bi- k -Lin	$m + n'$	$(n' + mn) \log L$	$(n' + mn) \log L$	$(n' + m) \log L$
Sec. 6.4 + 7	AWS	Sel*-SIM	bi- k -Lin	$m + n'$	$(n' + mn) \log L$	$(n' + mn) \log L$	$N(n' + m) \log L$
Sec. 6 + 7	AB-AWS	Sel*-SIM	bi- k -Lin	$m + n'$	$(n' + mn) \log L$	$(n' + mn) \log L$	$N(n' + m) \log L$

Table 1: Comparison with prior fully collusion-secure RFEs beyond RABEs. Here, “weak QF” indicates a standard QF with fixed functions; “Adp”, “Sel”, “SelSta”, “Sel*” stand for adaptive, selective, selective with static corruptions, very selective; n, n' indicate the lengths of public, private inputs; and m is proportional to the size of functions. If the functionality supports an unbounded number N of inputs, then n, n' refer to the length of *one* input. L denotes the number of registered users in the system.

use additive notations for the group operations. Similarly, if $\mathbb{G} = (\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_t, g_1, g_2, g_t, e, p)$ is a pairing group and $\tau \in \{1, 2, t\}$, we write $\llbracket x \rrbracket_\tau = g_\tau^x \in \mathbb{G}_\tau$. The notations extend component-wise to vectors and matrices.

Recap 1: The ALS IPFE Scheme [ABDP15, ALS16, Wee17]. The Inner-Product Functional Encryption (IPFE) scheme of Agrawal, Libert and Stehlé [ALS16], referred to as ALS, is secure under the MDDH $_k$ assumption and works as follows: initially, sample $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_p^{k \times (k+1)}$, $\mathbf{W} \xleftarrow{\$} \mathbb{Z}_p^{(k+1) \times \ell}$ and define $\text{mpk} = ([\mathbf{A}], [\mathbf{AW}])$ and $\text{msk} = \mathbf{W}$. To encrypt a vector $\mathbf{z} \in \mathbb{Z}_p^\ell$ or to generate a secret key for a matrix $\mathbf{Y} \in \mathbb{Z}_p^{\ell \times m}$ ¹, compute

$$\text{ct} = \left(\underbrace{[\mathbf{s}^\top \mathbf{A}]}_{[\mathbf{c}_1^\top]}, \underbrace{[\mathbf{z}^\top - \mathbf{s}^\top \mathbf{AW}]}_{[\mathbf{c}_2^\top]} \right) \quad \text{and} \quad \text{sk}_Y = \underbrace{\mathbf{WY}}_{\mathbf{D}},$$

where $\mathbf{s} \xleftarrow{\$} \mathbb{Z}_p^k$. Decryption of ct with sk_Y computes $[\mathbf{c}_1^\top] \mathbf{D} + [\mathbf{c}_2^\top] \mathbf{Y} = [\mathbf{z}^\top \mathbf{Y}]$ and recovers the discrete logarithm via brute-force if the result lies in a poly-size interval.

Recap 2: Partial Garbling [IW14, Wee17]. Our second ingredient is the partial garbling scheme of [IW14, Wee17] for functionalities $f: \mathbb{Z}_p^n \times \mathbb{Z}_p^{n'} \rightarrow \mathbb{Z}_p$ of the form $f(\mathbf{x}, \mathbf{z}) = h(\mathbf{x})^\top \mathbf{z}$, where $h: \mathbb{Z}_p^n \rightarrow \mathbb{Z}_p^{n'}$ can be any ABP. Given h , there exist efficiently computable matrices $(\mathbf{L}_0, \mathbf{L}_1) \in \mathbb{Z}_p^{(m+n'-1) \times m} \times \mathbb{Z}_p^{(m+n'-1) \times mn}$ and given $(h, \mathbf{x})$, there exists an efficiently computable vector $\mathbf{d}_{h, \mathbf{x}} \in \mathbb{Z}_p^{n'+m}$ with the following properties:

Correctness. For each $\mathbf{w} \in \mathbb{Z}_p^{m+n'-1}$ and

$$\text{pgb}(h, \mathbf{x}, \mathbf{z}; \mathbf{w}) = (\mathbf{p}_1^\top, \mathbf{p}_2^\top) := (\mathbf{z}^\top - \mathbf{w}^\top, \mathbf{w}^\top \mathbf{L}_\mathbf{x}),$$

¹ Usually, IPFE considers key vectors instead of matrices, i.e. $m = 1$, but the extension to matrices is immediate.

we have $(\mathbf{p}_1^\top \parallel \mathbf{p}_2^\top) \cdot \mathbf{d}_{h,\mathbf{x}} = h(\mathbf{x})^\top \mathbf{z}$, where $\underline{\mathbf{w}}$ refers to the last n' entries of $\mathbf{w}$ and $\mathbf{L}_\mathbf{x} := \mathbf{L}_0 + \mathbf{L}_1(\mathbf{x} \otimes \mathbf{I}_m)$.

Privacy. For $\mathbf{w} \xleftarrow{\$} \mathbb{Z}_p^{m+n'-1}$, we have the statistical indistinguishability

$$\{(\mathbf{z}^\top - \underline{\mathbf{w}}^\top, \mathbf{w}^\top \mathbf{L}_\mathbf{x})\} \approx_s \{(-\underline{\mathbf{w}}^\top, \mathbf{w}^\top \mathbf{L}_\mathbf{x} + h(\mathbf{x})^\top \mathbf{z} \cdot \mathbf{e}_1^\top)\}.$$

The Classical FE Setting. We can view the vectors of the partial garbling scheme as a one-time secure, secret-key FE scheme for 1AWS² with master secret key $\text{msk} = \mathbf{w}$, ciphertext $\text{ct} = \mathbf{p}_1$ and secret key $\text{sk}_{h,\mathbf{x}} = \mathbf{p}_2$. To lift this one-time secure construction to full-fledged FE security, we can plug it into the ALS IPFE scheme. For this, we observe that the vectors $(\mathbf{p}_1, \mathbf{p}_2)$ of the partial garbling scheme have the same structure as a pair of an ALS ciphertext encrypting $\mathbf{z}$ and an ALS secret key for $\mathbf{Y} = \mathbf{L}_\mathbf{x}$, using fixed randomness $\mathbf{w} = \mathbf{s}^\top \mathbf{A}\mathbf{W}$. By combining the encryption and key-generation algorithm of ALS with the reconstruction procedure of the partial garbling scheme, we readily obtain a (public-key) FE scheme. In a bit more detail, for a key pair $\text{FE.mpk} = ([\mathbf{A}]_1, [\mathbf{A}\mathbf{W}]_1)$ and $\text{FE.msk} = \mathbf{W}$, where $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_p^{k \times (k+1)}$, $\mathbf{W} \xleftarrow{\$} \mathbb{Z}_p^{(k+1) \times (m+n'-1)}$ and $\underline{\mathbf{W}}$ denotes the last n' columns of $\mathbf{W}$, the FE scheme works as follows:

$$\left. \begin{array}{l} \text{FE.ct:} \quad [\mathbf{s}^\top \mathbf{A}], [\mathbf{z}^\top - \mathbf{s}^\top \mathbf{A}\mathbf{W}] \\ \text{FE.sk}_{h,\mathbf{x}}: \quad \mathbf{W}\mathbf{L}_\mathbf{x} \end{array} \right\} \Rightarrow \underbrace{[\mathbf{z}^\top - \mathbf{s}^\top \mathbf{A}\mathbf{W}]}_{=[\mathbf{p}_1^\top]}, \underbrace{[\mathbf{s}^\top \mathbf{A}\mathbf{W}\mathbf{L}_\mathbf{x}]}_{=[\mathbf{p}_2^\top]}. \quad (1)$$

The linear reconstruction of pgb can be carried out in the exponents to obtain $[(\mathbf{p}_1^\top \parallel \mathbf{p}_2^\top)] \cdot \mathbf{d}_{h,\mathbf{x}} = [h(\mathbf{x})^\top \mathbf{z}]$. We refer to such a pair of vectors $([\mathbf{p}_1], [\mathbf{p}_2])$ as an ALS *encoding* (of pgb). We also call $\mathbf{s}^\top \mathbf{A}\mathbf{W}$ the *random pad* of the encoding. Since ALS encodings are far simpler than RFE, we make some preliminary observations about this primitive before diving into the actual construction of RFE.

Challenges in the Registration-Based Setting. As illustrated in the above example, constructions of plain FE usually provide the vector $[\mathbf{p}_1]$ as part of the ciphertext and a term related to $\mathbf{p}_2$ in the secret keys. In the registration-based setting, this approach does not seem to work since secret keys are generated function-independently and are only linked to a function during registration. The registration process, however, is deterministic which hinders us from running a randomized algorithm to compute the garbling. It is therefore unclear how to generate and embed the pgb vectors $(\mathbf{p}_1, \mathbf{p}_2)$ into a registration-based FE construction. In particular, this observation rules out a straightforward adaptation of generic approaches for other FE primitives that combine pgb with inner-product functional encryption, as done e.g. in [ATY23] for the case of MIFE.

If we cannot embed some parts of the garbling in the secret keys, there seems to be no choice but embedding the entire ALS encoding $([\mathbf{p}_1], [\mathbf{p}_2])$ in the ciphertexts. More specifically, we note that in plain FE a single $[\mathbf{p}_1]$ can be decrypted by multiple secret keys, and each decryption process produces a different $\mathbf{p}_2$ corresponding to $\mathbf{p}_1$. This means that our RFE ciphertexts include not only a single $\mathbf{p}_2$ vector but one for each function registered by a user. Loosely speaking, this leads to the problem that all $\mathbf{p}_2$ vectors share the same random pad $\mathbf{s}^\top \mathbf{A}\mathbf{W}$, and a user could use their own secret key to evaluate the encoding with respect to another user's $\mathbf{p}_2$. A simple solution to avoid such attacks could be to provide independent ALS encodings with fresh random pads for each user. But this would lead to ciphertexts of size $O(L)$ whereas compactness permits only polylogarithmic dependency on L . Therefore, we can only afford to embed a single $\mathbf{p}_1$ vector into the ciphertext, and we need a mechanism to randomize the pad later during RFE decryption.

Randomizing the Pad of ALS Encodings. For our randomization mechanism, we switch to a pairing group $\mathbb{G} = (\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_t, g_1, g_2, g_t, e, p)$, place the ALS encoding over $\mathbb{G}_1$ and hope to provide additional “hints” over $\mathbb{G}_2$ that can be used for the randomization of the pad. Jumping ahead, these hints will be included in the helper secret keys of the RFE scheme which means that they can be user-specific but cannot depend on the input $\mathbf{z}$.

²Strictly speaking, this scheme does not provide the actual 1AWS functionality but a slight modification, where the vector $\mathbf{x}$ is given as part of the function instead of the input. We ignore this fact for now because FE schemes for the actual 1AWS functionality are more complex as e.g. in [AGW20].

First of all, we note that a simple multiplication of the encoding with a matrix $\mathbf{R} \xleftarrow{\$} \mathbb{Z}_p^{(m+n'-1) \times (m+n'-1)}$ chosen uniformly at random does not do the job:

$$\begin{aligned} \llbracket (\mathbf{p}'_1)^\top \rrbracket_t &:= \llbracket \mathbf{p}_1^\top \rrbracket_1 \cdot \llbracket \mathbf{R} \rrbracket_2 = \llbracket \mathbf{z}^\top - \mathbf{s}^\top \mathbf{A} \mathbf{W} \rrbracket_1 \cdot \llbracket \mathbf{R} \rrbracket_2 = \llbracket \mathbf{z}^\top \mathbf{R} - \mathbf{s}^\top \mathbf{A} \mathbf{W} \cdot \mathbf{R} \rrbracket_t \\ \llbracket (\mathbf{p}'_2)^\top \rrbracket_t &:= \llbracket \mathbf{p}_2^\top \rrbracket_1 \cdot \llbracket \mathbf{R} \rrbracket_2 = \llbracket \mathbf{s}^\top \mathbf{A} \mathbf{W} \mathbf{L}_x \rrbracket_1 \cdot \llbracket \mathbf{R} \rrbracket_2 = \llbracket \mathbf{s}^\top \mathbf{A} \mathbf{W} \mathbf{L}_x \cdot \mathbf{R} \rrbracket_t. \end{aligned}$$

This approach has at least three immediate problems.

1. The multiplication of $\mathbf{p}_1^\top$ by $\mathbf{R}$ does not only randomize the pad but also changes the input from $\mathbf{z}^\top$ to $\mathbf{z}^\top \mathbf{R}$ which of course should remain invariant.
2. If we view $\mathbf{s}^\top \mathbf{A} \mathbf{W} \mathbf{R}$ as the desired new pad, then the terms in $\mathbf{p}'_2$ are in the wrong order. The correctly randomized encoding should be of the form $\mathbf{p}'_2 = \mathbf{s}^\top \mathbf{A} \mathbf{W} \mathbf{R} \cdot \mathbf{L}_x$.
3. To maintain a valid ALS encoding, the randomized pad $\mathbf{s}^\top \mathbf{A} \mathbf{W} \cdot \mathbf{R}$ in $\mathbf{p}'_1$ has to be equal to the last n' columns of the pad $\mathbf{s}^\top \mathbf{A} \mathbf{W} \mathbf{R}$. Even if we had solved the second issue, it would still be unclear how to define $\mathbf{R}$ in $\mathbf{p}'_1$ because the last n' columns of $\mathbf{s}^\top \mathbf{A} \mathbf{W} \mathbf{R}$ depend on the entire matrix $\mathbf{W}$, not just the sub-matrix $\mathbf{W}$ contained in $\mathbf{p}_1$.

We discuss two different ways to resolve these issues.

Solution 1: Tensorized ALS Encodings. As we rely on a DDH-style assumption, the randomization matrix $\mathbf{R}$ can have a certain structure and still produce a pad that appears random to the adversary. It turns out that a particular choice of $\mathbf{R}$ can solve the second and third issue at once. Specifically, we let $\mathbf{R} = (\mathbf{I}_{m+n'-1} \otimes \mathbf{r})$ for $\mathbf{r} \xleftarrow{\$} \mathbb{Z}_p^k$ and slightly change the form of the original encoding to $\mathbf{p}_2 = \mathbf{s}^\top \mathbf{A} \mathbf{W} (\mathbf{L}_x \otimes \mathbf{I}_k)$. Then we can compute $\mathbf{p}'_2$ as

$$\begin{aligned} \llbracket (\mathbf{p}'_2)^\top \rrbracket_t &:= \llbracket \mathbf{p}_2^\top \rrbracket_1 \cdot \llbracket \mathbf{I}_m \otimes \mathbf{r} \rrbracket_2 = \llbracket \mathbf{s}^\top \mathbf{A} \mathbf{W} (\mathbf{L}_x \otimes \mathbf{I}_k) (\mathbf{I}_m \otimes \mathbf{r}) \rrbracket_t = \llbracket \mathbf{s}^\top \mathbf{A} \mathbf{W} (\mathbf{L}_x \otimes \mathbf{r}) \rrbracket_t \\ &= \llbracket \mathbf{s}^\top \mathbf{A} \mathbf{W} (\mathbf{I}_{m+n'-1} \otimes \mathbf{r}) \mathbf{L}_x \rrbracket_t = \llbracket \mathbf{s}^\top \mathbf{A} \mathbf{W} \mathbf{R} \mathbf{L}_x \rrbracket_t, \end{aligned}$$

where we solve the second issue by using the so-called *mixed-product* equality $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = \mathbf{AC} \otimes \mathbf{BD}$ for matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$ and $\mathbf{D}$ of correct format. Note that we implicitly increase the number of columns of $\mathbf{W}$ and $\mathbf{W}$ by a factor of k to maintain valid matrix operations. For the third issue, we observe that the last kn' columns of the pad $\mathbf{s}^\top \mathbf{A} \mathbf{W} \mathbf{R} \mathbf{L}_x$ can now be computed as $\mathbf{s}^\top \mathbf{A} \mathbf{W} (\mathbf{I}_{n'} \otimes \mathbf{r})$, so we can define $\mathbf{R} = \mathbf{I}_{n'} \otimes \mathbf{r}$.

It remains the first issue that randomization of $\mathbf{p}_1$ also changes the input $\mathbf{z}$. To fix this problem, we consider a variable-base version where randomization leads to an encoding in a new base. For this we slightly change the form of the original encoding to $\mathbf{p}_1 = \mathbf{z}^\top \otimes \mathbf{s}^\top \mathbf{A} - \mathbf{s}^\top \mathbf{A} \mathbf{W}$. Then the randomization yields

$$\llbracket (\mathbf{p}'_1)^\top \rrbracket_t := \llbracket \mathbf{p}_1^\top \rrbracket_1 \cdot \llbracket \mathbf{I}_{n'} \otimes \mathbf{r} \rrbracket_2 = \llbracket (\mathbf{z}^\top \otimes \mathbf{s}^\top \mathbf{A} - \mathbf{s}^\top \mathbf{A} \mathbf{W}) (\mathbf{I}_{n'} \otimes \mathbf{r}) \rrbracket_t = \llbracket \mathbf{z}^\top \cdot \underbrace{\mathbf{s}^\top \mathbf{A} \mathbf{r}}_{\llbracket b \rrbracket_t} - \mathbf{s}^\top \mathbf{A} \mathbf{W} \cdot \mathbf{R} \rrbracket_t.$$

If the encryptor additionally provides $\llbracket \mathbf{s}^\top \mathbf{A} \rrbracket_1$ (as it is the case in the original ALS scheme), then the decryptor can compute the new base $\llbracket b \rrbracket_t$ of the randomized encoding as $\llbracket b \rrbracket_t = \llbracket \mathbf{s}^\top \mathbf{A} \rrbracket_1 \llbracket \mathbf{r} \rrbracket_2$. In summary, if we embed the vectors

$$(\llbracket \mathbf{p}_1^\top \rrbracket_1, \llbracket \mathbf{p}_2^\top \rrbracket_1) := (\llbracket \mathbf{z}^\top \otimes \mathbf{s}^\top \mathbf{A} - \mathbf{s}^\top \mathbf{A} \mathbf{W} \rrbracket_1, \llbracket \mathbf{s}^\top \mathbf{A} \mathbf{W} (\mathbf{L}_x \otimes \mathbf{I}_k) \rrbracket_1)$$

in the original ciphertext and multiply them with $(\llbracket \mathbf{I}_{n'} \otimes \mathbf{r} \rrbracket_2, \llbracket \mathbf{I}_m \otimes \mathbf{r} \rrbracket_2)$, we obtain a valid ALS encoding of the original message $\mathbf{z}$ and a randomized pad in the new base $\llbracket \mathbf{s}^\top \mathbf{A} \mathbf{r} \rrbracket_t$.

Looking ahead, the usage of tensorized ALS encodings enables a proof technique known as nested dual-system method [LW11] which allows proving adaptive security. On the negative side, our compiler that lifts RFE for (AB-)1AWS to (AB-)AWS cannot handle variable-base schemes. Therefore, tensorized ALS encodings only help to obtain RFE for the single-input versions of the AWS function family, but we need a different encoding technique to achieve unbounded inputs.

Solution 2: Nested ALS Encodings. Our second method does not start from a single encoding but uses an interleaving of two ALS encodings to produce a new one with randomized pad. Intuitively, the $\mathbf{p}_1$ vector of the first (“inner”) encoding has a fixed random pad. This allows us to include the encoding in the ciphertext of an RFE scheme without violating compactness. But as explained above, we cannot provide the corresponding $\mathbf{p}_2$ vector for such a fixed-pad encoding as it would render the RFE scheme insecure. Therefore, we use a second (“outer”) encoding whose message is the pad of the inner encoding. The motivation for this nested approach is that now the outer encoding is independent of the encrypted message. So we can provide it as a hint rather than as part of the ciphertext which allows us to choose the outer pad user-specific without contradicting the compactness condition. We summarize the idea as follows. As before, elements in $\mathbb{G}_1$ can be thought of as given in the ciphertext and elements in $\mathbb{G}_2$ are given as (public) hints.

$$\begin{aligned}\llbracket \mathbf{p}_{1,\text{in}}^\top \rrbracket_t &= \llbracket \mathbf{z}^\top - \mathbf{s}^\top \mathbf{A} \mathbf{W}_{\text{in}} \rrbracket_1 \cdot \llbracket \mathbf{I}_{n'} \rrbracket_2 = \llbracket \mathbf{z}^\top - \mathbf{s}^\top \mathbf{A} \mathbf{W}_{\text{in}} \rrbracket_t \\ \llbracket \mathbf{p}_{1,\text{out}}^\top \rrbracket_t &= \llbracket \mathbf{s}^\top \mathbf{A} \rrbracket_1 \cdot \llbracket \mathbf{W}_{\text{in}} - \mathbf{W}_{\text{out}} \rrbracket_2 = \llbracket \mathbf{s}^\top \mathbf{A} \mathbf{W}_{\text{in}} - \mathbf{s}^\top \mathbf{A} \mathbf{W}_{\text{out}} \rrbracket_t \\ \llbracket \mathbf{p}_2^\top \rrbracket_t &= \llbracket \mathbf{s}^\top \mathbf{A} \rrbracket_1 \cdot \llbracket \mathbf{W}_{\text{out}} \mathbf{L}_x \rrbracket_2 = \llbracket \mathbf{s}^\top \mathbf{A} \mathbf{W}_{\text{out}} \mathbf{L}_x \rrbracket_t.\end{aligned}$$

Taking the sum of $\mathbf{p}_{1,\text{in}}$ and $\mathbf{p}_{1,\text{out}}$ gives $\mathbf{p}_1^\top = \mathbf{z}^\top - \mathbf{s}^\top \mathbf{A} \mathbf{W}_{\text{out}}$ which forms a valid ALS encoding together with $\mathbf{p}_2$.

It turns out that our nesting technique can *not only* be used to randomize pads *but also* to apply linear functions to the input vector $\mathbf{z}$. Consider any linear function represented by a matrix $\mathbf{M}^\top \in \mathbb{Z}_p^{n' \times n'}$. Then we can extend the above idea and define

$$\begin{aligned}\llbracket \mathbf{p}_{1,\text{in}}^\top \rrbracket_t &= \llbracket \mathbf{z}^\top - \mathbf{s}^\top \mathbf{A} \mathbf{W}_{\text{in}} \rrbracket_1 \cdot \llbracket \mathbf{M} \rrbracket_2 = \llbracket \mathbf{z}^\top \mathbf{M} - \mathbf{s}^\top \mathbf{A} \mathbf{W}_{\text{in}} \mathbf{M} \rrbracket_t \\ \llbracket \mathbf{p}_{1,\text{out}}^\top \rrbracket_t &= \llbracket \mathbf{s}^\top \mathbf{A} \rrbracket_1 \cdot \llbracket \mathbf{W}_{\text{in}} \mathbf{M} - \mathbf{W}_{\text{out}} \rrbracket_2 = \llbracket \mathbf{s}^\top \mathbf{A} \mathbf{W}_{\text{in}} \mathbf{M} - \mathbf{s}^\top \mathbf{A} \mathbf{W}_{\text{out}} \rrbracket_t.\end{aligned}$$

Taking again the sum of $\mathbf{p}_{1,\text{in}}$ and $\mathbf{p}_{1,\text{out}}$ yields $\mathbf{p}_1^\top = \mathbf{z}^\top \mathbf{M} - \mathbf{s}^\top \mathbf{A} \mathbf{W}_{\text{out}}$, and we can observe that $(\mathbf{p}_1, \mathbf{p}_2)$ forms a valid ALS encoding of $\mathbf{z}^\top \mathbf{M}$. This observation will prove useful when equipping our scheme with access control.

Finally, we note that both encoding techniques are compatible, giving us the ability to evaluate linear functions on $\mathbf{z}$ even in the case of tensored ALS encodings. Specifically, we can compute

$$\begin{aligned}\llbracket \mathbf{p}_{1,\text{in}}^\top \rrbracket_t &= \llbracket \mathbf{z}^\top \otimes \mathbf{s}^\top \mathbf{A} - \mathbf{s}^\top \mathbf{A} \mathbf{W}_{\text{in}} \rrbracket_1 \cdot \llbracket \mathbf{M} \otimes \mathbf{r} \rrbracket_2 = \llbracket \mathbf{z}^\top \mathbf{M} \otimes \mathbf{s}^\top \mathbf{A} \mathbf{r} - \mathbf{s}^\top \mathbf{A} \mathbf{W}_{\text{in}} (\mathbf{M} \otimes \mathbf{r}) \rrbracket_t \\ \llbracket \mathbf{p}_{1,\text{out}}^\top \rrbracket_t &= \llbracket \mathbf{s}^\top \mathbf{A} \rrbracket_1 \cdot \llbracket \mathbf{W}_{\text{in}} (\mathbf{M} \otimes \mathbf{r}) - \mathbf{W}_{\text{out}} (\mathbf{I}_{n'} \otimes \mathbf{r}) \rrbracket_2 = \llbracket \mathbf{s}^\top \mathbf{A} \mathbf{W}_{\text{in}} (\mathbf{M} \otimes \mathbf{r}) - \mathbf{s}^\top \mathbf{A} \mathbf{W}_{\text{out}} (\mathbf{I}_{n'} \otimes \mathbf{r}) \rrbracket_t,\end{aligned}$$

and their sum $\mathbf{p}_1 = \mathbf{z}^\top \mathbf{M} \otimes \mathbf{s}^\top \mathbf{A} \mathbf{r} - \mathbf{s}^\top \mathbf{A} \mathbf{W}_{\text{out}} (\mathbf{I}_{n'} \otimes \mathbf{r})$ forms a valid ALS encoding of $\mathbf{z}^\top \mathbf{M}$ together with the corresponding $\mathbf{p}_2 = \mathbf{s}^\top \mathbf{A} \mathbf{W}_{\text{out}} (\mathbf{I}_{m+n'-1} \otimes \mathbf{r}) \mathbf{L}_x$.

Roadmap Towards AB-AWS. In the remainder of the technical overview, we describe our construction in three steps. In Section 2.1, we present our basic RFE scheme for the 1AWS functionality. In Section 2.2, we discuss how the basic construction can be equipped with attribute-based access control. In Section 2.3, we first explain how to extend the construction for 1AWS to an intermediate notion that we call eAWS, which can eventually be lifted to full-fledged AWS for unbounded inputs. Both extensions of 1AWS, i.e. AB-1AWS and eAWS, are designed in such a way that they are compatible and yield a sRFE for AB-AWS when being combined. This combination works both in the adaptive IND setting and the very selective SIM setting. Furthermore, we note that an existing transformation [AGW20] in the case of (classical) SIM-secure FE from one input to unbounded inputs can be adapted to the registration-based setting w.r.t. our new notion of eAWS. Further inspection shows that it can also be enhanced with access control, yielding our generic “extended AGW” compiler. Plugging everything together leads to our final construction of sRFE for AB-AWS with very selective SIM-security. Figure 1 depicts a short outline of the construction and provides links to the technical details of each step.

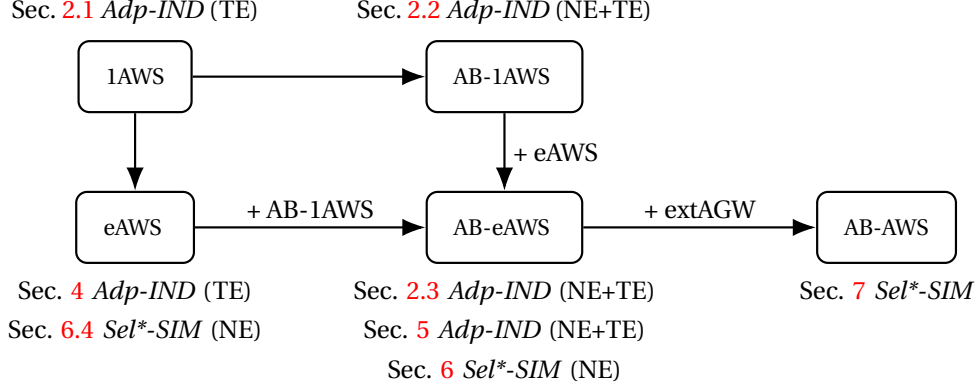


Figure 1: Roadmap for the technical part; (NE), (TE) and (NE+TE) indicate that the respective construction uses “Nested ALS Encodings” or “Tensorized ALS Encodings” or an interleaving of both

2.1 Constructing Slotted RFE for 1AWS

In this section, we describe our construction for the basic 1AWS functionality,

$$f(\mathbf{x}, \mathbf{z}) = h(\mathbf{x})^\top \mathbf{z},$$

where h can be computed by an ABP. Thanks to the powers-of-two compiler developed in [GHMR18, HLWW23, FFM⁺23], it suffices to design constructions for the simpler *slotted* RFE (sRFE) primitive. In a sRFE scheme, the number L of slots³ is fixed, hence it does not incorporate the complex update mechanism of RFE. In a bit more detail, each user $i \in [L]$ can generate their own key pair (pk_i, sk_i) and register pk_i along with a function f_i of their choice. After receiving all tuples $\{(pk_i, f_i)\}_{i \in [L]}$, the transparent *key aggregator* generates a short master public key mpk and a set of helper secret keys $\{hsk_i\}_{i \in [L]}$. Encryption of a public input $\mathbf{x}$ and a private input $\mathbf{z}$ uses mpk , and each user i can decrypt a ciphertext ct using (sk_i, hsk_i) to learn $f_i(\mathbf{x}, \mathbf{z})$. IND-security intuitively requires that the adversary cannot distinguish a ciphertext ct encrypting either $(\mathbf{x}, \mathbf{z}^0)$ or $(\mathbf{x}, \mathbf{z}^1)$ given mpk and $\{hsk_i\}_{i \in [L]}$ when satisfying the additional condition that $f_i(\mathbf{x}, \mathbf{z}^0) = f_i(\mathbf{x}, \mathbf{z}^1)$ for all slots i where sk_i is chosen by or known to the adversary⁴.

Our construction proceeds in three steps. First, we design a 1-slot sRFE for a variant of 1AWS denoted $1AWS_{\mathbf{x}}$, where the vector $\mathbf{x}$ is given as part of the function instead of the input. While this functionality itself seems not very useful, the construction can serve as an instructive intermediate step. In the second step, we show how to upgrade the scheme an L -slot sRFE for $L > 1$. In the third step, we discuss how to modify this construction to obtain a sRFE for the actual 1AWS functionality.

Step 1: 1-slot sRFE for $1AWS_{\mathbf{x}}$. Our first step consists in turning the FE scheme in Equation (1) into a 1-slot sRFE scheme for $1AWS_{\mathbf{x}}$ following an approach inspired by [ZLZ⁺24]. First of all, $FE.mpk$ is globally provided as crs . The user generates an ElGamal-style key pair $(pk, sk) = ([\mathbf{A}\mathbf{U}], \mathbf{U})$ for $\mathbf{U} \xleftarrow{\$} \mathbb{Z}_p^{(k+1) \times m}$, and ct consists of $FE.ct$ along with an encryption of $FE.sk_{h,\mathbf{x}}$ under pk . Then the user is the only party in the system that can recover $FE.sk_{h,\mathbf{x}}$ and run the FE decryption procedure. To make sure that the user cannot decrypt any other functionality, we hardcode pk together with $\mathbf{L}_{\mathbf{x}}$ into mpk during registration. This leads us to the following

³In this work, we use the terms “slots” and “users” interchangeably.

⁴More formally, the adversary is given a crs which allows deriving mpk , $\{hsk_i\}_{i \in [L]}$ by performing the same computation as the (deterministic) aggregator.

1-slot sRFE scheme.

$$\begin{aligned}
\text{crs; pk; sk: } & \underbrace{[\mathbf{A}], [\mathbf{AW}]}_{\text{FE.mpk}}; [\mathbf{AU}]; \mathbf{U} \\
\text{mpk: } & [\mathbf{A}], [\mathbf{AW}], [\mathbf{AU} + \mathbf{AWL}_x] \\
\text{ct: } & \underbrace{[\mathbf{s}^\top \mathbf{A}], [\mathbf{z}^\top - \mathbf{s}^\top \mathbf{AW}]}_{\text{FE.ct}}, \underbrace{[\mathbf{s}^\top \mathbf{AU} + \mathbf{s}^\top \mathbf{AWL}_x]}_{\text{Encode(pk, FE.sk}_{h,x})}
\end{aligned} \tag{2}$$

Using $\text{sk} = \mathbf{U}$, the user can recover

$$\underbrace{[\mathbf{z}^\top - \mathbf{s}^\top \mathbf{AW}]}_{[\mathbf{p}_1^\top]}, \underbrace{[\mathbf{s}^\top \mathbf{AU} + \mathbf{s}^\top \mathbf{AWL}_x] - [\mathbf{s}^\top \mathbf{A}]\mathbf{U} = [\mathbf{s}^\top \mathbf{AWL}_x]}_{[\mathbf{p}_2^\top]}$$

and run the pgb reconstruction procedure as in the case of FE decryption. Security basically follows from the following two observations.

1. If $\text{sk} = \mathbf{U}$ is secret, i.e., the user is honest, then it follows from the MDDH_k assumption that $[\mathbf{s}^\top \mathbf{AU}]$ can serve as a random mask for $[\mathbf{s}^\top \mathbf{AWL}_x]$ which in turn implies that $[\mathbf{s}^\top \mathbf{AW}]$ can entirely hide the secret input $\mathbf{z}^\top$.
2. If $\text{sk} = \mathbf{U}$ is leaked, i.e., the user is corrupted or malicious, then $[\mathbf{s}^\top \mathbf{AU}]$ can no longer serve as a random mask; but in this case we do not expect that ct completely hides $\mathbf{z}^\top$. Instead, we can rely on the security of pgb to argue that

$$\{[\mathbf{z}^0]^\top - \mathbf{s}^\top \mathbf{AW}], [\mathbf{s}^\top \mathbf{AU} + \mathbf{s}^\top \mathbf{AWL}_x]_1\} \approx_s \{[(\mathbf{z}^1)^\top - \mathbf{s}^\top \mathbf{AW}], [\mathbf{s}^\top \mathbf{AU} + \mathbf{s}^\top \mathbf{AWL}_x]_1\},$$

since $h(\mathbf{x})^\top \mathbf{z}^0 = h(\mathbf{x})^\top \mathbf{z}^1$ and $\mathbf{s}^\top \mathbf{AW}$ looks random assuming MDDH_k .

Step 2: Extension to $L > 1$ Slots. Our next step consists in the extension of the 1-slot scheme from the previous step to arbitrary $L > 1$ slots. We follow a similar path as previous works [ZZGQ23, HLWW23, ZLZ⁺24] by designing our L -slot scheme as the “sum” of L parallel instances of the 1-slot scheme. The $\text{crs} = ([\mathbf{A}], \{[\mathbf{AW}_i]\}_{i \in [L]})$ contains one matrix $\mathbf{A}$ that is shared across all slots, but provides independent matrices $\mathbf{W}_i$ for each slot $i \in [L]$. User key pairs remain the same (for now), i.e., each user $i \in [L]$ samples a key pair $(\text{pk}_i, \text{sk}_i) = ([\mathbf{AU}_i], \mathbf{U}_i)$ for $\mathbf{U}_i \xleftarrow{s} \mathbb{Z}_p^{(k+1) \times m}$. To register $\{(\text{pk}_i, h_i)\}_{i \in [L]}$, the aggregator generates the master public key

$$\text{mpk} = \left([\mathbf{A}], \sum_{i \in [L]} [\mathbf{AW}_i], \sum_{i \in [L]} [\mathbf{AU}_i + \mathbf{AW}_i \mathbf{L}_{i,x}] \right),$$

where $\mathbf{L}_{i,x}$ is computed with respect to the function h_i and the vector $\mathbf{x}$. Applying the same encryption procedure as in the 1-slot scheme gives ciphertexts of the form

$$\text{ct} = \left(\underbrace{[\mathbf{s}^\top \mathbf{A}]}_{[\mathbf{c}_0^\top]}, \underbrace{[\mathbf{z}^\top - \mathbf{s}^\top \mathbf{A} \sum_{i \in [L]} \mathbf{W}_i]}_{[\mathbf{c}_1^\top]}, \underbrace{[\mathbf{s}^\top \mathbf{A} \sum_{i \in [L]} \mathbf{U}_i + \mathbf{W}_i \mathbf{L}_{i,x}]}_{[\mathbf{c}_2^\top]} \right). \tag{3}$$

Unfortunately, the mere addition of L 1-slot master public keys breaks correctness because now decryption would require knowledge of all L secret keys. To overcome this issue, the sRFE model allows the use of additional helper secret keys hsk_j for each $j \in [L]$, which may be used for decryption along with $\text{sk}_j = \mathbf{U}_j$. Note that these helper secret keys are public since they are provided by the transparent key aggregator. Therefore, the challenge is to design the helper secret keys in such a way that hsk_j is useful in combination with sk_j (in the sense that it enables decryption), but useless otherwise. To achieve this, we can employ either of our two ALS encodings. For simplicity, we discuss the case of tensored ALS encodings. To use them, we switch to a pairing group and place the previously described parallel instances of the 1-slot scheme over $\mathbb{G}_1$. Furthermore,

we let $\mathbf{B} \xleftarrow{s} \mathbb{Z}_p^{(k+1) \times k}$ and $\mathbf{r}_1, \dots, \mathbf{r}_L \xleftarrow{s} \mathbb{Z}_p^k$ and additionally provide the following helper secret keys over $\mathbb{G}_2$:

$$\text{hsk}_j = \left(\underbrace{\llbracket \mathbf{Br}_j \rrbracket_2}_{\llbracket \mathbf{d}_0 \rrbracket_2}, \underbrace{- \sum_{i \in [L] \setminus \{j\}} \llbracket \mathbf{W}_i(\mathbf{I}_{n'} \otimes \mathbf{Br}_j) \rrbracket_2}_{\llbracket \mathbf{D}_1 \rrbracket_2}, \underbrace{\sum_{i \in [L] \setminus \{j\}} \llbracket \mathbf{U}_i(\mathbf{I}_m \otimes \mathbf{Br}_j) + \mathbf{W}_i(\mathbf{L}_{i,\mathbf{x}} \otimes \mathbf{Br}_j) \rrbracket_2}_{\llbracket \mathbf{D}_2 \rrbracket_2} \right).$$

Intuitively, hsk_j can be used to turn the ciphertext in Equation (3) back into a ciphertext of the 1-slot scheme (2) with respect to (pk_j, f_j) . During decryption, user j can use the pairing operation to multiply the components $\llbracket \mathbf{D}_1 \rrbracket_2, \llbracket \mathbf{D}_2 \rrbracket_2$ of the helper secret key with the ciphertext randomness $\llbracket \mathbf{c}_0^\top \rrbracket_1 = \llbracket \mathbf{s}^\top \mathbf{A} \rrbracket_1$ from left and the ciphertext components $\llbracket \mathbf{c}_1^\top \rrbracket_1, \llbracket \mathbf{c}_2^\top \rrbracket_1$ with the user-specific randomness $\llbracket \mathbf{d}_0 \rrbracket_2 = \llbracket \mathbf{Br}_j \rrbracket_2$ from right. After this mutual randomization, ciphertexts and helper secret keys have the same form and subtraction of the two will cancel all terms related to slots $i \in [L] \setminus \{j\}$. More specifically, we obtain:

$$\begin{aligned} \llbracket \mathbf{p}_1^\top \rrbracket_t &= \llbracket \mathbf{c}_1^\top \rrbracket_1 \llbracket \mathbf{I}_{n_{\text{pri}}} \otimes \mathbf{d}_0 \rrbracket_2 - \llbracket \mathbf{c}_0^\top \rrbracket_1 \llbracket \mathbf{D}_1 \rrbracket_2 = \left\llbracket \mathbf{z}^\top \left(\mathbf{I}_{n_{\text{pri}}} \otimes \mathbf{Br}_j \right) - \mathbf{s}^\top \mathbf{A} \mathbf{W}_j(\mathbf{I}_{n_{\text{pri}}} \otimes \mathbf{Br}_j) \right\rrbracket_t \\ \llbracket \mathbf{p}_2^\top \rrbracket_t &= \llbracket \mathbf{c}_2^\top \rrbracket_1 \llbracket \mathbf{I}_m \otimes \mathbf{d}_0 \rrbracket_2 - \llbracket \mathbf{c}_0^\top \rrbracket_1 \llbracket \mathbf{D}_2 \rrbracket_2 = \llbracket \mathbf{s}^\top \mathbf{A} \mathbf{U}_j(\mathbf{I}_m \otimes \mathbf{Br}_j) + \mathbf{s}^\top \mathbf{A} \mathbf{W}_j(\mathbf{L}_{i,\mathbf{x}} \otimes \mathbf{Br}_j) \rrbracket_t. \end{aligned}$$

This is basically a valid tensored ALS encoding masked by the term $\mathbf{s}^\top \mathbf{A} \mathbf{U}_j$ which can only be removed by user j using $\text{sk}_j = \mathbf{U}_j$. The only problem is that $\mathbf{z}$ is not preserved correctly as indicated by the gray box. As explained above, we can resolve this issue by switching to a variable-base scheme. That is, we let the encryptor provide

$$\llbracket \mathbf{c}_1^\top \rrbracket_1 = \left\llbracket \mathbf{z}^\top \otimes \mathbf{s}^\top \mathbf{A} \mathbf{V} - \mathbf{s}^\top \mathbf{A} \sum_{i \in [L]} \mathbf{W}_i \right\rrbracket_1 \quad \text{and} \quad \llbracket \mathbf{c}_3^\top \rrbracket_1 = \llbracket \mathbf{s}^\top \mathbf{A} \mathbf{V} \rrbracket_1.$$

Then the result can be recovered by solving the discrete log in the new basis $\llbracket \mathbf{c}_3^\top \rrbracket_1 \llbracket \mathbf{d}_0 \rrbracket_2 = \llbracket \mathbf{s}^\top \mathbf{A} \mathbf{V} \mathbf{Br}_j \rrbracket_t$.

Step 3: From 1AWS_x to 1AWS. In the registration-based setting, the 1AWS_x functionality considers a fixed attribute vector $\mathbf{x}$ hardcoded in mpk . This is conceptually similar to the inner-product functionality considered in [ZLZ⁺24]. On the other hand, the real 1AWS functionality takes the vector $\mathbf{x}$ only provided at the time of encryption and it can change for each ciphertext. Therefore, we need to modify the structure of mpk and hsk_j , for $j \in [L]$, such that they still fix the functions $h_1, \dots, h_L$ but allow plugging in different values of $\mathbf{x}$. In the spirit of AGW, we split all components of mpk and hsk_j that depend on $\mathbf{L}_{i,\mathbf{x}} := \mathbf{L}_{i,0} + \mathbf{L}_{i,1}(\mathbf{x} \otimes \mathbf{I}_m)$ into two components each, one for $\mathbf{L}_{i,0}$ and one for $\mathbf{L}_{i,1}$. For example, we replace the hsk_j component $\llbracket \mathbf{D}_2 \rrbracket_2 = \sum_{i \in [L] \setminus \{j\}} \llbracket \mathbf{U}_i(\mathbf{I}_m \otimes \mathbf{Br}_j) + \mathbf{W}_i(\mathbf{L}_{i,\mathbf{x}} \otimes \mathbf{Br}_j) \rrbracket_2$ with

$$\begin{aligned} \llbracket \mathbf{D}_{2,0} \rrbracket_2 &= \sum_{i \in [L] \setminus \{j\}} \llbracket \mathbf{U}_i(\mathbf{I}_m \otimes \mathbf{Br}_j) + \mathbf{W}_i(\mathbf{I}_{m+n'-1} \otimes \mathbf{Br}_j) \mathbf{L}_{i,0} \rrbracket_2 \\ \llbracket \mathbf{D}_{2,1} \rrbracket_2 &= \sum_{i \in [L] \setminus \{j\}} \llbracket \mathbf{W}_i(\mathbf{I}_{m+n'-1} \otimes \mathbf{Br}_j) \mathbf{L}_{i,1} \rrbracket_2. \end{aligned}$$

Then decryption can recompute $\llbracket \mathbf{D}_2 \rrbracket_2 = \llbracket \mathbf{D}_{2,0} \rrbracket_2 + \llbracket \mathbf{D}_{2,1} \rrbracket_2(\mathbf{x} \otimes \mathbf{I}_m)$ for any choice of $\mathbf{x}$, where we use the equality

$$\mathbf{W}_i(\mathbf{I}_{m+n'-1} \otimes \mathbf{Br}_j) \mathbf{L}_{i,\mathbf{x}} = \mathbf{W}_i(\mathbf{L}_{i,\mathbf{x}} \otimes \mathbf{Br}_j).$$

The modifications of mpk can be carried out in a similar fashion. We summarize our sRFE scheme for 1AWS which we obtain by following the above steps.

Construction 1 (Slotted RFE for 1AWS – Simplified).

$$\begin{aligned}
& \text{pk}_i; \text{sk}_i : ([\mathbf{A}\mathbf{U}_i]_1, \{[\mathbf{U}_i(\mathbf{I}_m \otimes \mathbf{B}\mathbf{r}_j)]_2\}_{j \in [L] \setminus \{i\}}; \mathbf{U}_i \\
& \text{mpk} : \left\| \begin{aligned} & \mathbf{A}, \sum_{i \in [L]} \mathbf{A}\mathbf{W}_i, \sum_{i \in [L]} \mathbf{A}\mathbf{W}_i(\mathbf{L}_{i,1} \otimes \mathbf{I}_{k+1}), \\ & \sum_{i \in [L]} \mathbf{A}\mathbf{U}_i + \mathbf{A}\mathbf{W}_i(\mathbf{L}_{i,0} \otimes \mathbf{I}_{k+1}), \mathbf{A}\mathbf{V} \end{aligned} \right\|_1 \\
& \text{ct} : \left\| \begin{aligned} & \mathbf{s}^\top \mathbf{A}, \mathbf{z}^\top \otimes \mathbf{s}^\top \mathbf{A}\mathbf{V} - \mathbf{s}^\top \mathbf{A} \sum_{i \in [L]} \mathbf{W}_i \\ & \mathbf{s}^\top \mathbf{A} \sum_{i \in [L]} \mathbf{U}_i + \mathbf{W}_i(\mathbf{L}_{i,x} \otimes \mathbf{I}_{2k+1}), \mathbf{s}^\top \mathbf{A}\mathbf{V} \end{aligned} \right\|_1 \\
& \text{hsk}_j : \left\| \begin{aligned} & \mathbf{B}\mathbf{r}_j, - \sum_{i \in [L] \setminus \{j\}} \mathbf{W}_i(\mathbf{I}_{n'} \otimes \mathbf{B}\mathbf{r}_j), \sum_{i \in [L] \setminus \{j\}} \mathbf{W}_i(\mathbf{L}_{i,1} \otimes \mathbf{B}\mathbf{r}_j) \\ & \sum_{i \in [L] \setminus \{j\}} \mathbf{U}_i(\mathbf{I}_m \otimes \mathbf{B}\mathbf{r}_j) + \mathbf{W}_i(\mathbf{L}_{i,0} \otimes \mathbf{B}\mathbf{r}_j), \end{aligned} \right\|_2,
\end{aligned}$$

where $\mathbf{s}, \mathbf{r}_1, \dots, \mathbf{r}_L \xleftarrow{\$} \mathbb{Z}_p^k$, $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_p^{k \times (2k+1)}$, $\mathbf{B} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times k}$, $\mathbf{V} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)}$, $\mathbf{W}_1, \dots, \mathbf{W}_L \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)(m+n'-1)}$ and $\mathbf{U}_1, \dots, \mathbf{U}_L \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)m}$.

This is almost our final scheme except for some extra components to deal with malicious keys. To prove security against these keys, we can use quasi-adaptive NIZK arguments in the same way as previous works [ZZGQ23, ZLZ⁺24]; we therefore omit the details here.

When incorporating these modifications, we obtain a scheme that is adaptively IND-secure in the standard model. The proof performs a hybrid argument over all slots $i \in [L]$. By using the nested dual system technique similar to [ZLZ⁺24], we can isolate the terms related to each slot and consider the ALS encodings one by one. Specifically, the relevant part of the adversary's view in hybrid i is

$$([\mathbf{z}^b]^\top \otimes \mathbf{s}^\top \mathbf{A}\mathbf{V} - \mathbf{s}^\top \mathbf{A}\mathbf{W}_i]_1, [\mathbf{s}^\top \mathbf{A}\mathbf{U}_i + \mathbf{s}^\top \mathbf{A}\mathbf{W}_i(\mathbf{L}_{i,x} \otimes \mathbf{I}_{2k+1})]_1).$$

Then the argument is very similar to the proof of the 1-slot scheme in Step 1.

1. If slot i is honest, then the ElGamal encryption under the public key $[\mathbf{A}\mathbf{U}_i]_1$ completely hides the ALS secret key $\mathbf{W}_i(\mathbf{L}_{i,x} \otimes \mathbf{I}_{2k+1})$, so the ALS encryption under the public key $[\mathbf{A}\mathbf{W}_i]_1$ can completely hide $\mathbf{z}$.
2. If slot i is corrupted, then we first rely on the security of ALS to conclude that the random pad $\mathbf{s}^\top \mathbf{A}\mathbf{W}_i$ of the pgb encoding is pseudorandom. Then it follows from the security of pgb that we only leak $h_i(\mathbf{x})^\top \mathbf{z}^b$ which is independent of the challenge bit b according to the adversary's admissibility condition.

2.2 Adding Fine-Grained Access Control

We now explain how our construction for 1AWS from the previous section can be equipped with attribute-based access control, i.e., we aim for the following AB-1AWS functionality

$$f(\mathbf{y}, \mathbf{x}, \mathbf{z}) = \begin{cases} h(\mathbf{x})^\top \mathbf{z} & \text{if } g(\mathbf{y}) = 0 \\ \perp & \text{otherwise,} \end{cases}$$

where g and h can be computed by ABPs. We observe that sRFE for AB-IP [DPY24] can be viewed as a special case of this AB-1AWS functionality where we omit the input $\mathbf{x}$ and consider $h_i(\mathbf{x})$ as a fixed vector $\mathbf{v}_i$, for all slots $i \in [L]$, registered with only g_i in place of (g_i, h_i) . Our approach is inspired by a recent work of Agrawal, Tomida and Yadav [ATY23] which we refer to as ATY in what follows. The main goal of ATY is the construction of multi-party FEs for (AB-)AWS. As a stepping stone, they build single-input FE for a functionality called AWSw/IP which is a generalization of the 1AWS functionality considered in the last section.

The relevant fact for us is that they show how to enhance AWSw/IP with access control. Their basic idea is to define $\mathbf{x}' = (\mathbf{y}, \mathbf{x})$, $\mathbf{z}' = (\mathbf{f}, \mathbf{z})$ and $\phi(\mathbf{x}') = (g(\mathbf{y}) \cdot \mathbf{g}^\top \parallel h(\mathbf{x})^\top)$ where $\mathbf{f}, \mathbf{g} \xleftarrow{\$} \mathbb{Z}_p^k$, and to use $(\mathbf{x}', \mathbf{z}')$ and ϕ for

encryption and key generation, respectively. Then decryption outputs $\llbracket g(\mathbf{y}) \cdot \mathbf{g}^\top \mathbf{f} + h(\mathbf{x})^\top \mathbf{z} \rrbracket_t$. Since $\llbracket g(\mathbf{y}) \cdot \mathbf{g}^\top \mathbf{f} \rrbracket_t$ looks random if $g(\mathbf{y}) \neq 0$ under the MDDH_k assumption, the decryptor can learn $\llbracket h(\mathbf{x})^\top \mathbf{z} \rrbracket_t$ only when $g(\mathbf{y}) = 0$. This simple approach does not work out completely for technical reasons, though it is a good starting point for us when we design sRFE for AB-1AWS.

First Attempt. To understand the core idea, we strip of all components from our 1AWS scheme related to the registration of users and return to the original ALS encoding of pgb for only one slot $i \in [L]$:

$$\begin{aligned} \mathbf{p}_1^\top &= \underbrace{(\mathbf{z}^\top \otimes \mathbf{s}^\top \mathbf{A} \mathbf{V} - \mathbf{s}^\top \mathbf{A} \mathbf{W}_i)}_{\text{ct}} \underbrace{(\mathbf{I}_{n'} \otimes \mathbf{B} \mathbf{r}_i)}_{\text{hsk}_i} = \mathbf{z}^\top \cdot \mathbf{s}^\top \mathbf{A} \mathbf{V} \mathbf{B} \mathbf{r}_i - \mathbf{s}^\top \mathbf{A} \mathbf{W}_i (\mathbf{I}_{n'} \otimes \mathbf{B} \mathbf{r}_i) \\ \mathbf{p}_2^\top &= \underbrace{\mathbf{s}^\top \mathbf{A} \mathbf{W}_i (\mathbf{I}_{m+n'-1} \otimes \mathbf{B} \mathbf{r}_i)}_{\text{ct}} \underbrace{\mathbf{L}_{i,\mathbf{x}}}_{\text{hsk}_i} = \mathbf{s}^\top \mathbf{A} \mathbf{W}_i (\mathbf{I}_{m+n'-1} \otimes \mathbf{B} \mathbf{r}_i) \mathbf{L}_{i,\mathbf{x}}. \end{aligned}$$

Following the approach of ATY, we replace the matrix $\mathbf{L}_{i,\mathbf{x}}$ generated for the ABP h_i and public input $\mathbf{x}$ by a matrix $\mathbf{L}_{i,\mathbf{y},\mathbf{x}}$ generated for the extended ABP ϕ_i and the tuple $(\mathbf{y}, \mathbf{x})$. Furthermore, we include random vectors $\mathbf{f}, \mathbf{g}_i \xleftarrow{\$} \mathbb{Z}_p^k$ into ct and hsk_i, respectively, such that multiplication yields their inner product:

$$\begin{aligned} \mathbf{p}_1^\top &= \underbrace{((\mathbf{f}^\top \parallel \mathbf{z}^\top) \otimes \mathbf{s}^\top \mathbf{A} \mathbf{V} - \mathbf{s}^\top \mathbf{A} \mathbf{W}_i)}_{\text{ct}} \underbrace{(\mathbf{I}_{0,\mathbf{g}_i} \otimes \mathbf{B} \mathbf{r}_i)}_{\text{hsk}_i} \\ &= (\mathbf{f}^\top \parallel \mathbf{g}_i^\top \parallel \mathbf{z}^\top) \cdot \mathbf{s}^\top \mathbf{A} \mathbf{V} \mathbf{B} \mathbf{r}_i - \mathbf{s}^\top \mathbf{A} \mathbf{W}_i (\mathbf{I}_{0,\mathbf{g}_i} \otimes \mathbf{B} \mathbf{r}_i) \\ \mathbf{p}_2^\top &= \underbrace{\mathbf{s}^\top \mathbf{A} \mathbf{W}_i (\mathbf{I}_{m-1,\mathbf{g}_i} \otimes \mathbf{B} \mathbf{r}_i)}_{\text{ct}} \underbrace{\mathbf{L}_{i,\mathbf{y},\mathbf{x}}}_{\text{hsk}_i} = \mathbf{s}^\top \mathbf{A} \mathbf{W}_i (\mathbf{I}_{m-1,\mathbf{g}_i} \otimes \mathbf{B} \mathbf{r}_i) \mathbf{L}_{i,\mathbf{y},\mathbf{x}} \end{aligned} \tag{4}$$

where the matrix $\mathbf{I}_{d,\mathbf{g}} \in \mathbb{Z}_p^{(d+k+n') \times (d+1+n')}$ is defined as

$$\mathbf{I}_{d,\mathbf{g}} := \begin{bmatrix} \mathbf{I}_d & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{g} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_{n'} \end{bmatrix},$$

and we implicitly increase the number of columns of $\mathbf{W}_i$ to maintain valid matrix multiplications. For correctness, we observe that

$$(\mathbf{p}_1^\top, \mathbf{p}_2^\top) = \text{pgb}(\phi_i, (\mathbf{y}^\top \parallel \mathbf{x}^\top), (\mathbf{f}^\top \parallel \mathbf{g}_i^\top \parallel \mathbf{z}^\top), \mathbf{s}^\top \mathbf{A} \mathbf{W}_i (\mathbf{I}_{m-1,\mathbf{g}_i} \otimes \mathbf{B} \mathbf{r}_i)),$$

so pgb reconstruction yields $g_i(\mathbf{y}) \cdot \mathbf{f}^\top \mathbf{g}_i + h_i(\mathbf{x})^\top \mathbf{z} = h_i(\mathbf{x})^\top \mathbf{z}$ if $g_i(\mathbf{y}) = 0$.

The security proof, however, runs into troubles because the vectors $(\mathbf{p}_1^\top, \mathbf{p}_2^\top)$ do not contain a valid ALS encoding anymore. The problem is that the random pad $\mathbf{s}^\top \mathbf{A} \mathbf{W}_i (\mathbf{I}_{m-1,\mathbf{g}_i} \otimes \mathbf{B} \mathbf{r}_i)$ depends on the vector $\mathbf{g}_i$ which is also a part of the message $(\mathbf{f}^\top \parallel \mathbf{g}_i^\top \parallel \mathbf{z}^\top)$ that we want to hide! Therefore any randomization attempt of the pad would fail. Note that the solution of ATY does not help here either as it uses *function-hiding* IPFE to decouple the vector $\mathbf{g}_i$ from the random pad. But function-hiding IPFE only exists in the secret-key setting⁵, whereas we aim for a public encryption algorithm.

Second Attempt. Taking a step back, we may view our problem from a slightly different perspective: what we really try to do here is applying a linear function $\mathbf{M} = \mathbf{I}_{0,\mathbf{g}_i}$ to the input $(\mathbf{f}^\top \parallel \mathbf{z}^\top)$ *without* changing the pad. But we already know how to do this using nested ALS encodings. Applying this technique, we obtain an “inner” layer that encodes the input $(\mathbf{f}^\top \parallel \mathbf{z}^\top)$ and its pad may depend on $\mathbf{g}_i$, hence it cannot be used as randomness

⁵In the public-key setting, the adversary can create encryptions of all unit vectors and recover the key vector component-wise.

for pgb. The “outer” layer encodes the pad of the inner layer and has itself a pad independent of $\mathbf{g}_i$ which may serve as randomness for pgb:

$$\begin{aligned}\mathbf{p}_{1,\text{in}}^\top &= \underbrace{(\mathbf{f}^\top \parallel \mathbf{z}^\top) \otimes \mathbf{s}^\top \mathbf{A} \mathbf{V} - \mathbf{s}^\top \mathbf{A} \mathbf{H}_i}_{\text{ct}} \underbrace{(\mathbf{I}_{0,\mathbf{g}_i} \otimes \mathbf{B} \mathbf{r}_i)}_{\text{hsk}_i} = (\mathbf{f}^\top \mathbf{g}_i \parallel \mathbf{z}^\top) \cdot \mathbf{s}^\top \mathbf{A} \mathbf{V} \mathbf{B} \mathbf{r}_i - \mathbf{s}^\top \mathbf{A} \mathbf{H}_i (\mathbf{I}_{0,\mathbf{g}_i} \otimes \mathbf{B} \mathbf{r}_i) \\ \mathbf{p}_{1,\text{out}}^\top &= \underbrace{(\mathbf{s}^\top \mathbf{A} \mathbf{H}_i (\mathbf{I}_{0,\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}) - \mathbf{s}^\top \mathbf{A} \mathbf{W}_i)}_{\text{ct}} \underbrace{(\mathbf{I}_{n'+1} \otimes \mathbf{B} \mathbf{r}_i)}_{\text{hsk}_i} = \mathbf{s}^\top \mathbf{A} \mathbf{H}_i (\mathbf{I}_{0,\mathbf{g}_i} \otimes \mathbf{B} \mathbf{r}_i) - \mathbf{s}^\top \mathbf{A} \mathbf{W}_i (\mathbf{I}_{n'+1} \otimes \mathbf{B} \mathbf{r}_i).\end{aligned}$$

Here, $\mathbf{H}_i$ is a fresh master secret key for the inner ALS encoding and $\mathbf{W}_i$ represents the master secret key of the outer encoding as before. To turn this into a complete ALS encoding, we can compute $\mathbf{p}_1$ as the sum of $\mathbf{p}_{1,\text{in}}$ and $\mathbf{p}_{1,\text{out}}$ and the corresponding vector $\mathbf{p}_2$ as follows:

$$\begin{aligned}\mathbf{p}_1^\top &= \mathbf{p}_{1,\text{in}} + \mathbf{p}_{1,\text{out}} = (\mathbf{f}^\top \mathbf{g}_i \parallel \mathbf{z}^\top) \cdot \mathbf{s}^\top \mathbf{A} \mathbf{V} \mathbf{B} \mathbf{r}_i - \mathbf{s}^\top \mathbf{A} \mathbf{W}_i (\mathbf{I}_{n'+1} \otimes \mathbf{B} \mathbf{r}_i) \\ \mathbf{p}_2^\top &= \underbrace{\mathbf{s}^\top \mathbf{A} \mathbf{W}_i (\mathbf{I}_{m+n'-1} \otimes \mathbf{B} \mathbf{r}_i)}_{\text{ct}} \underbrace{\mathbf{L}_{i,\mathbf{y},\mathbf{x}}}_{\text{hsk}_i} = \mathbf{s}^\top \mathbf{A} \mathbf{W}_i (\mathbf{I}_{m+n'-1} \otimes \mathbf{B} \mathbf{r}_i) \mathbf{L}_{i,\mathbf{y},\mathbf{x}}\end{aligned}$$

We can observe that $(\mathbf{p}_1, \mathbf{p}_2)$ are almost the same as in first attempt (Equation (4)) except that the pad is now independent of $\mathbf{g}_i$ which allows relying on the security of pgb during the security proof. Being equipped with our new encoding, we can tackle the construction of sRFE for AB-1AWS. We basically follow the same steps as in the case of plain 1AWS with the only difference that we now have two independent ALS encodings to deal with. To see why this is relevant, we recall from Step 1 of the 1AWS construction that user registration is enabled by encrypting the ALS secret key $\mathbf{s}^\top \mathbf{A} \mathbf{W}_i \mathbf{L}_{i,\mathbf{x}}$ under an ElGamal key $\mathbf{U}_i$ (see Section 2.1). As our new pgb encoding in the attribute-based setting consists of two independent ALS encodings, it naturally happens that we also need two ElGamal keys $\mathbf{U}_{1,i}$ and $\mathbf{U}_{2,i}$ to hide the inner and outer ALS secret key, respectively.

We summarize our final AB-1AWS scheme below. Modifications from the plain 1AWS scheme in Construction 1 are highlighted using **boxes**. Even though the scheme might seem quite complex, most of the new components are only introduced as the result of using two independent ALS encodings.

Construction 2 (Slotted RFE for AB-1AWS – Simplified).

$$\begin{aligned}\text{pk}_i &: \left(\begin{array}{c} [\mathbf{A} \mathbf{U}_{1,i}]_1, \{[\mathbf{U}_{1,i} (\mathbf{I}_{n'+1} \otimes \mathbf{B} \mathbf{r}_j)]_2\}_{j \in [L] \setminus \{i\}}, \\ [\mathbf{A} \mathbf{U}_{2,i}]_1, \{[\mathbf{U}_{2,i} (\mathbf{I}_m \otimes \mathbf{B} \mathbf{r}_j)]_2\}_{j \in [L] \setminus \{i\}} \end{array} \right) \quad \text{for} \quad \text{sk}_i = \begin{pmatrix} \mathbf{U}_{1,i} \\ \mathbf{U}_{2,i} \end{pmatrix} \\ \text{mpk} &: \left\| \begin{array}{c} \sum_{i \in [L]} \mathbf{A} \mathbf{H}_i, \sum_{i \in [L]} \mathbf{A} \mathbf{U}_{1,i} - \mathbf{A} \mathbf{H}_i (\mathbf{I}_{0,\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}) - \mathbf{A} \mathbf{W}_i, \\ \sum_{i \in [L]} \mathbf{A} \mathbf{W}_i (\mathbf{L}_{i,1} \otimes \mathbf{I}_{k+1}), \sum_{i \in [L]} \mathbf{A} \mathbf{U}_{2,i} + \mathbf{A} \mathbf{W}_i (\mathbf{L}_{i,0} \otimes \mathbf{I}_{k+1}), \mathbf{A}, \mathbf{A} \mathbf{V} \end{array} \right\|_1 \\ \text{ct} &: \left\| \begin{array}{c} \mathbf{z}^\top \otimes \mathbf{s}^\top \mathbf{A} \mathbf{V} - \mathbf{s}^\top \mathbf{A} \sum_{i \in [L]} \mathbf{H}_i, \mathbf{s}^\top \mathbf{A} \sum_{i \in [L]} \mathbf{U}_{1,i} - \mathbf{H}_i (\mathbf{I}_{0,\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}) - \mathbf{W}_i, \\ \mathbf{s}^\top \mathbf{A} \sum_{i \in [L]} \mathbf{U}_{2,i} + \mathbf{W}_i (\mathbf{L}_{i,\mathbf{x}} \otimes \mathbf{I}_{2k+1}), \mathbf{s}^\top \mathbf{A}, \mathbf{s}^\top \mathbf{A} \mathbf{V} \end{array} \right\|_1 \\ \text{hsk}_j &: \left\| \begin{array}{c} \mathbf{B} \mathbf{r}_j, \mathbf{I}_{0,\mathbf{g}_j} \otimes \mathbf{B} \mathbf{r}_j, \sum_{i \in [L] \setminus \{j\}} \mathbf{H}_i (\mathbf{I}_{0,\mathbf{g}_i} \otimes \mathbf{B} \mathbf{r}_j), \\ \sum_{i \in [L] \setminus \{j\}} \mathbf{U}_{1,i} (\mathbf{I}_{n'+1} \otimes \mathbf{B} \mathbf{r}_j) - \mathbf{H}_i (\mathbf{I}_{0,\mathbf{g}_i} \otimes \mathbf{B} \mathbf{r}_j) - \mathbf{W}_i (\mathbf{I}_{n'+1} \otimes \mathbf{B} \mathbf{r}_j), \\ \sum_{i \in [L] \setminus \{j\}} \mathbf{W}_i (\mathbf{L}_{i,1} \otimes \mathbf{B} \mathbf{r}_j), \sum_{i \in [L] \setminus \{j\}} \mathbf{U}_{2,i} (\mathbf{I}_m \otimes \mathbf{B} \mathbf{r}_j) + \mathbf{W}_i (\mathbf{L}_{i,0} \otimes \mathbf{B} \mathbf{r}_j), \end{array} \right\|_2\end{aligned}$$

where $\mathbf{U}_{1,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)(n'+1)}$, $\mathbf{H}_i \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)(k+n')}$ and all other components have the same sizes as in Construction 1.

The proof proceeds in a similar fashion as in the case of 1AWS. The relevant difference lies in the analysis of a corrupted slot $i \in [L]$. In this case, the security of pgb implies that the adversary only learns $[\mathbf{g}_i(\mathbf{x}) \cdot \mathbf{f}^\top \mathbf{g}_i +$

$h_i(\mathbf{x})^\top \mathbf{z}^b \mathbb{I}_1$. If $g_i(\mathbf{y}) = 0$ then we can proceed as in the case of 1AWS since the decryption result $h_i(\mathbf{x})^\top \mathbf{z}^b$ may be revealed. The new part of the analysis happens if $g_i(\mathbf{y}) \neq 0$ because in this case the adversary must not learn anything about $\mathbf{z}^b$ even when knowing $\text{sk}_i = (\mathbf{U}_{1,i}, \mathbf{U}_{2,1})$. To argue this, we rely on the MDDH_k assumption which implies that $\llbracket \mathbf{f}^\top \mathbf{g}_i \rrbracket_1 \approx_c \llbracket r_i \rrbracket_1$ for a random $r_i \xleftash \mathbb{Z}_p$. More precisely, we note that the simulation of mpk and hsk_i needs $\mathbf{g}_i$ in both $\mathbb{G}_1$ and $\mathbb{G}_2$. Therefore, we use bilateral MDDH_k [Wee20, GLW23].

2.3 Extension to Unbounded Inputs

In this section, we explain our sRFE construction for the AWS functionality with unbounded inputs. To reduce the complexity of the scheme, let us disregard the access control feature discussed in the previous section and only concentrate on the plain AWS functionality

$$f(\{\mathbf{x}_\ell, \mathbf{z}_\ell\}_{\ell \in [N]}) = \sum_{\ell \in [N]} h(\mathbf{x}_\ell)^\top \mathbf{z}_\ell,$$

where h can be expressed as an ABP. For the construction, we attempt to follow the blueprint of AGW, who present a “single-input-to-unbounded-inputs” transformation in the non-registration-based FE setting.

Recap: From One Input to Unbounded Inputs [AGW20]. In the construction of [AGW20], a ciphertext for N inputs $\{\mathbf{x}_\ell, \mathbf{z}_\ell\}_{\ell \in [N]}$ basically consists of N independent encryptions ct_ℓ of $(\mathbf{x}_\ell, \mathbf{z}_\ell)$. Obviously, this would leak the intermediate summands which is why they must be computationally masked using MDDH_k tuples, by using the single-input scheme to compute

$$\llbracket h(\mathbf{x}_\ell)^\top \mathbf{z}_\ell + \mathbf{p}_\ell^\top \mathbf{q} \rrbracket_t \quad \text{for all } \ell \in [N],$$

where $\mathbf{q} \in \mathbb{Z}_p^k$ is fresh per secret key and the $\mathbf{p}_\ell \in \mathbb{Z}_p^k$ are sampled during encryption subject to the constraint $\sum_{\ell \in [N]} \mathbf{p}_\ell = 0$. It is not hard to see that the sum of all partial decryptions still yields $\llbracket \sum_{\ell \in [N]} h(\mathbf{x}_\ell)^\top \mathbf{z}_\ell \rrbracket_t$. Conversely, the term $\llbracket \mathbf{p}_\ell^\top \mathbf{q} \rrbracket_t$ serves as a mask that computationally hides the ℓ -th summand $\llbracket h(\mathbf{x}_\ell)^\top \mathbf{z}_\ell \rrbracket_t$. To make the security proof work for an unbounded number N of inputs, AGW uses a sophisticated hybrid argument over the inputs $\ell \in [N]$. In the ℓ -th step, one relies on the MDDH_k assumption to obtain a random looking masking term $\llbracket r_\ell \rrbracket_2 \approx_c \llbracket \mathbf{p}_\ell^\top \mathbf{q} \rrbracket_2$ for $r_\ell \xleftash \mathbb{Z}_p$. For this argument to work out, we must make sure not to disclose any information about $\mathbf{p}_\ell$ except for the inner product $\llbracket \mathbf{p}_\ell^\top \mathbf{q} \rrbracket_2$, i.e., our underlying single-input scheme must achieve *SIM-based* security. We omit the details of the transformation here and instead focus on how to integrate the masking terms into our 1AWS scheme from the previous section.

Extended 1AWS with Inner Product. We first observe that the 1AWS functionality itself is insufficient to accomplish the task of hiding the intermediate summands, even though one might view the inner products between $\mathbf{p}_\ell$ and $\mathbf{q}$ as a special case of 1AWS. This is because the 1AWS functionality takes *scalar* inputs $(\mathbf{x}, \mathbf{z}) \in \mathbb{Z}_p^n \times \mathbb{Z}_p^{n'}$. However, we want to rely on MDDH_k for the randomization of the masking terms $\mathbf{p}_\ell^\top \mathbf{q}$, so we must simulate ciphertexts and secret keys given only the encoding $(\llbracket \mathbf{q} \rrbracket_2, \llbracket \mathbf{p}_\ell^\top \mathbf{q} \rrbracket_2)$ in the exponent of a group. In the non-registration-based setting, the definition of this so-called *extended* 1AWS functionality (eAWS) is straightforward: ciphertexts encrypt a tuple $(\mathbf{x}, \mathbf{z}, \mathbf{p})$ and secret keys are generated with respect to a pair $(h, \llbracket \mathbf{q} \rrbracket_2)$. Decryption returns $\llbracket h(\mathbf{x})^\top \mathbf{z} + \mathbf{p}_\ell^\top \mathbf{q} \rrbracket_t$ but hides any other information about $\mathbf{z}$ and $\mathbf{p}_\ell$.

It seems natural to aim for a similar definition in the registration-based setting. In particular, users may register their public key pk_i along with tuples $(h_i, \llbracket \mathbf{q}_i \rrbracket_2)$. However, it is unclear to us how aggregation could be performed in this case. The problem becomes apparent when looking at our 1AWS scheme in Construction 1: since the crs contains the matrices $\mathbf{W}_i$ only as encodings in $\mathbb{G}_1$ and $\mathbb{G}_2$, it is crucial for the derivation of mpk and hsk_i that the function-specific components $\mathbf{L}_{i,0}$ and $\mathbf{L}_{i,1}$ are known as scalars. Otherwise, the multiplications cannot be performed efficiently.

As we do not know how to design a sRFE scheme that directly allows users to register vectors of group elements, we need an alternative method to integrate the $\mathbf{q}_i$'s into the scheme. A simple solution could be to

delay the sampling until the time of aggregation. But this is incompatible with the sRFE model which requires a transparent and deterministic aggregator, so in particular it cannot sample $\{\llbracket \mathbf{q}_i \rrbracket_2\}_{i \in [L]}$. Therefore, the only plausible option seems to be that we provide the $\llbracket \mathbf{q}_i \rrbracket_2$'s already at the time of setup which means we need to embed them into the crs. This idea can be made work, though the embedding of the $\mathbf{q}_i$ vectors requires some care as it should enable the derivation of both mpk and the hsk_i 's, but prevent a trivial decryption without using a secret key. Looking ahead, since mpk contains elements over $\mathbb{G}_1$ and hsk_i contains elements over $\mathbb{G}_2$, we will provide the setup algorithm with the tuple $\llbracket \mathbf{q}_i \rrbracket_{1,2} := (\llbracket \mathbf{q}_i \rrbracket_1, \llbracket \mathbf{q}_i \rrbracket_2)$ and rely on the bilateral MDDH $_k$ assumption during the conversion to unbounded inputs as a compensation.

Construction of sRFE for eAWS – First Attempt. We follow a bottom-up approach to build our sRFE for eAWS. Given that $\llbracket \mathbf{q}_i \rrbracket_{1,2} := (\llbracket \mathbf{q}_i \rrbracket_1, \llbracket \mathbf{q}_i \rrbracket_2)$ is known at the time of setup for all $i \in [L]$, we can precompute and include the terms related to these vectors in the crs. Since the $\mathbf{q}_i$'s can be considered part of the functions, a naive attempt to construct eAWS would be to identify the terms related to the functions h_i in Construction 1. Looking at hsk_j , the terms related to h_i are $\mathbf{W}_i(\mathbf{L}_{i,1} \otimes \mathbf{B}\mathbf{r}_j)$ and $\mathbf{W}_i(\mathbf{L}_{i,0} \otimes \mathbf{B}\mathbf{r}_j)$. This suggests we precompute and include the terms $\llbracket \mathbf{H}_i(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j) \rrbracket_2$ to crs for all $j \neq i$ where $\{\mathbf{H}_i\}_{i \in [L]}$ is a set of new matrices which will play a similar role as $\mathbf{W}_i$ did for h_i . Now, to encrypt $\mathbf{p} \in \mathbb{Z}_p^k$, we introduce a new ciphertext component

$$\left\llbracket \mathbf{p}^\top \otimes \mathbf{s}^\top \mathbf{A} \mathbf{V} - \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A} \mathbf{H}_i \right\rrbracket_1$$

following the encoding of $\mathbf{z}$ in Construction 1. This indicates that mpk must also include the term $\llbracket \sum_{i \in [L]} \mathbf{A} \mathbf{H}_i \rrbracket_1$, and consequently, the crs must contain $\{\llbracket \mathbf{A} \mathbf{H}_i \rrbracket_1\}_{i \in [L]}$. Then, as before, decryption computes

$$\left\llbracket \mathbf{p}^\top \otimes \mathbf{s}^\top \mathbf{A} \mathbf{V} - \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A} \mathbf{H}_i \right\rrbracket_1 \cdot \llbracket \mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j \rrbracket_2 = \left\llbracket (\mathbf{p}^\top \mathbf{q}_j) \cdot (\mathbf{s}^\top \mathbf{A} \mathbf{V} \mathbf{B}\mathbf{r}_j) - \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A} \mathbf{H}_i (\mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j) \right\rrbracket_t. \quad (5)$$

As one might observe, we run into two problems. Firstly, the decryptor cannot obtain $\llbracket (\mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j) \rrbracket_2$ from $\llbracket \mathbf{q}_j \rrbracket_2$ and $\llbracket \mathbf{B}\mathbf{r}_j \rrbracket_2$ which are parts of crs. Secondly, given $\llbracket \mathbf{s}^\top \mathbf{A} \rrbracket_1$, the components $\{\llbracket \mathbf{H}_i(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j) \rrbracket_2\}_{i \neq j}$ cannot eliminate the masking term $\llbracket \mathbf{H}_j(\mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j) \rrbracket_t$ of Equation (5). The first issue can be easily resolved by adding $\llbracket \mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j \rrbracket_2$ to crs for all $j \in [L]$. To solve the second problem, one might hope to add $\llbracket \mathbf{H}_j(\mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j) \rrbracket_2$ to hsk_j . Although it helps to eliminate the masking term from Equation (5), this would reveal $\llbracket \mathbf{p}^\top \mathbf{q}_j \rrbracket_t$ which is information not desired to be disclosed by the eAWS functionality. Indeed, the sRFE for eAWS should not reveal anything but $h_j(\mathbf{x})^\top \mathbf{z} + \mathbf{q}_j^\top \mathbf{p}$.

Second Attempt. We will fix the second issue. To avoid the partial leakage of the eAWS functionality, we have to carefully embed the *actual* masking term $\mathbf{H}_j(\mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j)$ into some other component, but, which one? In the non-registration-based setting, AGW embed the masking term into the secret key. However, the same strategy would not work for us because the secret keys of sRFE are independent of functions. Therefore, the only possible option left is to encode the term into a ciphertext component. Moreover, since this actual masking term hides partial information of $\mathbf{q}_j^\top \mathbf{p}$ about the functional value, the computation of such a term must require the knowledge of the secret key sk_j during decryption. This narrows down the choice to the ciphertext component

$$\mathbf{c}_2^\top = \mathbf{s}^\top \mathbf{A} \sum_{i \in [L]} \mathbf{U}_i + \mathbf{W}_i(\mathbf{L}_{i,\mathbf{x}} \otimes \mathbf{I}_{2k+1})$$

into which one should encode $\mathbf{H}_j(\mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j)$. Now, we decompose the masking term as $\mathbf{H}_j(\mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j) = \mathbf{H}_j(\mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) \cdot (\mathbf{I}_m \otimes \mathbf{B}\mathbf{r}_j)$ and embed $\mathbf{H}_j(\mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1})$, for all $j \in [L]$, into $\mathbf{c}_2$ as follows

$$\mathbf{c}_2^\top = \mathbf{s}^\top \mathbf{A} \sum_{i \in [L]} \mathbf{U}_i + \mathbf{H}_i(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_i(\mathbf{L}_{i,\mathbf{x}} \otimes \mathbf{I}_{2k+1})$$

To allow decryption and cancel the cross terms $\{\mathbf{H}_i(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j)\}_{i \neq j}$, we again seek help of hsk_j . In particular, we add these cross terms to those of plain 1AWS in Construction 1 by modifying the very last component of

the helper secret key hsk_j as

$$\sum_{i \in [L] \setminus \{j\}} \mathbf{U}_i(\mathbf{I}_m \otimes \mathbf{B}\mathbf{r}_j) + \mathbf{H}_i(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j) + \mathbf{W}_i(\mathbf{L}_{i,0} \otimes \mathbf{B}\mathbf{r}_j).$$

Our final sRFE for eAWS is now ready and can be described as follows. We indicate new components using boxes.

Construction 3 (Slotted RFE for eAWS – Simplified).

$$\begin{aligned} \text{pk}_i &: (\|\mathbf{A}\mathbf{U}_i\|_1, \{\|\mathbf{U}_i(\mathbf{I}_m \otimes \mathbf{B}\mathbf{r}_j)\|_2\}_{j \in [L] \setminus \{i\}}) \text{ for } \text{sk}_i = \mathbf{U}_i \\ \text{mpk} &: \left\| \begin{aligned} &\mathbf{A}, \sum_{i \in [L]} \mathbf{A}\mathbf{H}_i, \sum_{i \in [L]} \mathbf{A}\mathbf{W}_i, \sum_{i \in [L]} \mathbf{A}\mathbf{W}_i(\mathbf{L}_{i,1} \otimes \mathbf{I}_{k+1}), \\ &\sum_{i \in [L]} \mathbf{A}\mathbf{U}_i + \mathbf{A}\mathbf{H}_i(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{A}\mathbf{W}_i(\mathbf{L}_{i,0} \otimes \mathbf{I}_{k+1}), \mathbf{A}\mathbf{V} \end{aligned} \right\|_1 \\ \text{ct} &: \left\| \begin{aligned} &\mathbf{s}^\top \mathbf{A}, \mathbf{z}^\top \otimes \mathbf{s}^\top \mathbf{A}\mathbf{V} - \mathbf{s}^\top \mathbf{A} \sum_{i \in [L]} \mathbf{W}_i, \mathbf{p}^\top \otimes \mathbf{s}^\top \mathbf{A}\mathbf{V} - \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A}\mathbf{H}_i, \\ &\mathbf{s}^\top \mathbf{A} \sum_{i \in [L]} \mathbf{U}_i + \mathbf{H}_i(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_i(\mathbf{L}_{i,x} \otimes \mathbf{I}_{2k+1}), \mathbf{s}^\top \mathbf{A}\mathbf{V} \end{aligned} \right\|_1 \\ \text{hsk}_j &: \left\| \begin{aligned} &\mathbf{B}\mathbf{r}_j, \mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j, - \sum_{i \in [L] \setminus \{j\}} \mathbf{H}_i(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j), \\ &- \sum_{i \in [L] \setminus \{j\}} \mathbf{W}_i(\mathbf{I}_{n'} \otimes \mathbf{B}\mathbf{r}_j), \sum_{i \in [L] \setminus \{j\}} \mathbf{W}_i(\mathbf{L}_{i,1} \otimes \mathbf{B}\mathbf{r}_j), \\ &\sum_{i \in [L] \setminus \{j\}} \mathbf{U}_i(\mathbf{I}_m \otimes \mathbf{B}\mathbf{r}_j) + \mathbf{H}_i(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j) + \mathbf{W}_i(\mathbf{L}_{i,0} \otimes \mathbf{B}\mathbf{r}_j) \end{aligned} \right\|_2 \end{aligned}$$

At the time of decryption, we obtain

$$\|\mathbf{p}_1^\top\|_t = \|\mathbf{z}^\top \cdot \mathbf{s}^\top \mathbf{A}\mathbf{V}\mathbf{B}\mathbf{r}_j - \mathbf{s}^\top \mathbf{A}\mathbf{W}_j(\mathbf{I}_{n'} \otimes \mathbf{B}\mathbf{r}_j)\|_t$$

as before, but most importantly the knowledge of $\text{sk}_j = \mathbf{U}_j$ yields

$$\|\mathbf{p}_2^\top\|_t = \|\mathbf{s}^\top \mathbf{A}\mathbf{W}_j(\mathbf{I}_{m+n'-1} \otimes \mathbf{B}\mathbf{r}_j) \mathbf{L}_{j,x} + \mathbf{q}_j^\top \mathbf{p} \cdot \mathbf{s}^\top \mathbf{A}\mathbf{V}\mathbf{B}\mathbf{r}_j \cdot \mathbf{e}_1^\top\|_t$$

and finally the reconstruction algorithm of pgb outputs $\|(\mathbf{h}_j(\mathbf{x})^\top \mathbf{z} + \mathbf{q}_j^\top \mathbf{p}) \cdot \mathbf{s}^\top \mathbf{A}\mathbf{V}\mathbf{B}\mathbf{r}_j\|_t$. Our final scheme achieves adaptive IND-based security which can be argued similar to our 1AWS functionality.

Conversion to Fixed-Base Decryption. So far, we considered all our constructions in the variable-base setting. On the other hand, our “single-input-to-unbounded-inputs” compiler (which is an adaptation of the AGW compiler in the plain FE setting) requires a *fixed base*, i.e., the underlying sRFE for AB-eAWS must output $\|(\mathbf{h}_j(\mathbf{x})^\top \mathbf{z} + \mathbf{q}_j^\top \mathbf{p})\|_t$ without the slot-dependent varying base $\mathbf{s}^\top \mathbf{A}\mathbf{V}\mathbf{B}\mathbf{r}_j$. Before we discuss our fixed-base scheme, let us observe that the previous encoding of $\mathbf{z}$ with pad $\mathbf{s}^\top \mathbf{A} \sum_{i \in [L]} \mathbf{W}_i$ does not work anymore because the step in decryption to remove the masks $\mathbf{s}^\top \mathbf{A}\mathbf{W}_i$ of the other users $i \neq j$ inherently relies on the tensored ALS encoding, thus the variable basis. We therefore need another approach. For simplicity, let us only focus on obtaining a fixed-base sRFE for 1AWS rather than eAWS.

Fortunately for us, [ZLZ⁺24] already present a fixed-base sRFE for inner products, and we may hope to generalize their techniques to the case of 1AWS. Transferring their ideas on a conceptual level, we place the input $\mathbf{z}$ in an encoding $\mathbf{z}^\top - \mathbf{s}^\top \mathbf{A}\mathbf{\hat{H}}$, where $\mathbf{\hat{H}}$ is a fresh but fixed ALS master secret key. During decryption, [ZLZ⁺24] convert this encoding into another ALS encoding $\mathbf{p}_1 = \mathbf{z} - \mathbf{s}^\top \mathbf{A}\mathbf{\hat{W}}$ with respect to the pad $\mathbf{\hat{W}}$, and the security proof uses $\mathbf{\hat{W}}$ to hide the “actual” masking terms $\mathbf{W}_j(\mathbf{I}_{m+n'-1} \otimes \mathbf{B}\mathbf{r}_j)$ (as in the variable-base scheme) in crs. However, this argument does not work anymore in the case of 1AWS. The main difference between the proofs for inner products and 1AWS happens when dealing with corrupted or malicious identities. In the former case, it suffices to argue that $\mathbf{s}^\top \mathbf{A}\mathbf{\hat{W}}$ hides the inner product, whereas in the latter case we want to rely on the security of pgb with $\mathbf{s}^\top \mathbf{A}\mathbf{\hat{W}}$ being the pad of the encoding. Therefore, we care about the distribution

of $\mathbf{s}^\top \mathbf{A} \widehat{\mathbf{W}}$ itself which was irrelevant in the case of inner products. In a bit more detail, we must replace $\mathbf{s}^\top \mathbf{A} \widehat{\mathbf{W}}$ with a uniformly random element for each malicious or corrupted slot using a statistical argument. However, this is not possible since the information of $\widehat{\mathbf{W}}$ is still available as pgb randomness of the other slots. Thus, to make the security proof work, we must replace $\widehat{\mathbf{W}}$ with $\widehat{\mathbf{W}}_j$ to produce slot-specific pgb randomness $\mathbf{s}^\top \mathbf{A} \widehat{\mathbf{W}}_j$.

While this resolves the issue with security, we now lose compactness. This is because the master public key must contain $\mathbf{A} \widehat{\mathbf{W}}_j$ for all $j \in [L]$ and the ciphertext cannot produce a succinct encoding of the input vector $\mathbf{z}$ from $\{\mathbf{z} - \mathbf{s}^\top \mathbf{A} \widehat{\mathbf{W}}_j\}_{j \in [L]}$. In order to surpass this obstacle, we need to produce a single encoding of $\mathbf{z}$ at the time of encryption which can later be transformed into a slot-specific ALS encoding of the form $\mathbf{z} - \mathbf{s}^\top \mathbf{A} \widehat{\mathbf{W}}_j$. This sounds exactly like a task that can be accomplished by our nested ALS encoding. Specifically, we can use a fixed (inner) encoding generated by the encryptor and a slot-specific (outer) encoding which is placed in hsk_j . In more detail, we sample a fresh master secret key $\widehat{\mathbf{H}}$ during setup and get

$$\begin{aligned} \mathbf{p}_{1,\text{in}}^\top &= \underbrace{(\mathbf{z}^\top - \mathbf{s}^\top \mathbf{A} \widehat{\mathbf{H}})}_{\text{ct}} \cdot \mathbf{I}_{n'} = \mathbf{z}^\top - \mathbf{s}^\top \mathbf{A} \widehat{\mathbf{H}} \\ \mathbf{p}_{1,\text{out}}^\top &= \underbrace{\mathbf{s}^\top \mathbf{A} (\widehat{\mathbf{H}} - \widehat{\mathbf{W}}_j)}_{\text{ct}} + \underbrace{\mathbf{W}_j (\mathbf{I}_{n'} \otimes \mathbf{Br}_j)}_{\text{hsk}_j} = \mathbf{s}^\top \mathbf{A} \widehat{\mathbf{H}} - \mathbf{s}^\top \mathbf{A} \widehat{\mathbf{W}}_j + \mathbf{s}^\top \mathbf{A} \widehat{\mathbf{W}}_j (\mathbf{I}_{n'} \otimes \mathbf{Br}_j), \end{aligned}$$

where $\mathbf{s}^\top \mathbf{A} \widehat{\mathbf{W}}_j (\mathbf{I}_{n'} \otimes \mathbf{Br}_j)$ is the “actual” masking term that was already used in the IND-secure scheme. Finally, by adding $\mathbf{p}_{1,\text{in}}^\top$ and $\mathbf{p}_{1,\text{out}}^\top$ we obtain the desired resulting encoding

$$\mathbf{p}_{1,\text{in}}^\top + \mathbf{p}_{1,\text{out}}^\top = \underbrace{\mathbf{z}^\top - \mathbf{s}^\top \mathbf{A} \widehat{\mathbf{W}}_j}_{\mathbf{p}_1^\top} + \mathbf{s}^\top \mathbf{A} \widehat{\mathbf{W}}_j (\mathbf{I}_{n'} \otimes \mathbf{Br}_j).$$

Achieving SIM-Security. Another requirement of the “single-input-to-unbounded-inputs” transformation is that the underlying (AB-)eAWS must provide SIM security. To our knowledge, there is no IPFE scheme at all in the literature (even in the non-registration-based setting) that enables computing on encrypted data with group-based functional keys and achieves adaptive SIM security. On the other hand, the only known registration-based FE scheme that achieves SIM security [ZLZ⁺24] is proven secure in the very selective model where all the challenge function vectors along with the challenge input and corrupted slots are known in advance during system setup. We therefore focus on the same security notion as [ZLZ⁺24]. An in-depth analysis reveals that by converting our scheme from tensored to nested ALS encodings, we can not only enable decryption with respect to a fixed basis but also achieve SIM security. Due to the complexity of the scheme, we omit the details here and refer the reader to Section 6.4 for the formal construction of our SIM-secure sRFE for eAWS.

Comparison With [DCC⁺24]. The concurrent work [DCC⁺24] also presents an RFE scheme for the AWS functionality (not AB-AWS). Their construction is significantly simpler than ours and the authors claim it to be even adaptively IND-secure (whereas our work achieves very selective SIM security). However, we believe their security proof is flawed. Specifically, when they reduce the security in the malicious and corrupted case to the security of the partial garbling, the simulator seems to use the *same* random coins for the pgb encodings of *all* users (see page 36 of [DCC⁺24], application of Lemma 5). This is not captured by the one-time security of pgb. As we have repeatedly emphasized throughout our technical overview (e.g., in the paragraph **Conversion to Fixed-Base Decryption**), providing independent coins to all users while keeping the scheme compact was one of the key challenges we had to solve. To do so, we use either tensored or nested ALS encodings requiring an additional secret key for each user which significantly contributes to the complexity of our constructions. In terms of efficiency, the size of the ciphertexts in [DCC⁺24] depends on both the public and the private inputs. We believe it to be an important feature of the construction in AGW (and ours) that ciphertext sizes depend only on the private input and are independent of the public input.

3 Preliminaries

In this section, we present the preliminaries necessary to understand this work.

3.1 Notational Conventions

Let $\lambda \in \mathbb{N}$ be the security parameter. Except in the definitions, we will suppress λ in subscripts for brevity. A nonnegative function $\varepsilon: \mathbb{N} \rightarrow \mathbb{R}$ is negligible if $\varepsilon(\lambda) = O(\lambda^{-n})$ for all $n \in \mathbb{N}$. An algorithm is said to be *efficient* if it runs in probabilistic polynomial time (PPT) in the security parameter.

To avoid confusion, we always write vectors $\mathbf{v}$ and matrices $\mathbf{A}$ in boldface and use uppercase letters for the latter. Scalars s are written in italics. Unless otherwise stated, all vectors $\mathbf{v}$ are viewed as column vectors. The corresponding row vector is denoted by $\mathbf{v}^\top$. We denote the i -th unit vector in $\mathbb{Z}_p^n$ by $\mathbf{e}_i^{(n)}$ and sometimes omit the superscript if n is clear from the context.

For conciseness, we may write $\sum_i a_i + b_i$ as a shorthand for $\sum_i (a_i + b_i)$.

We write $\mathbf{A} \otimes \mathbf{B}$ to denote the tensor product between matrices $\mathbf{A}$ and $\mathbf{B}$. In the absence of parentheses, multiplication takes precedence over tensor products and tensor products take precedence over addition. For matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$ and $\mathbf{D}$ of such size that the matrix products $\mathbf{AC}$ and $\mathbf{BD}$ are defined, we have the *mixed-product* equality: $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = \mathbf{AC} \otimes \mathbf{BD}$.

Security Experiments and Distributions. Let $\mathbf{Exp}$ be an interactive *experiment* that interacts with an algorithm $\mathcal{A}$ (called the *adversary*), depends on the security parameter λ and has binary outcome. We also refer to such objects as *games* or *hybrids*. We let “ $\mathbf{Exp}_{\mathcal{A}}(1^\lambda) \rightarrow 1$ ” denote the event that the outcome of running $\mathbf{Exp}$ with $\mathcal{A}$ on security parameter λ is 1. For two experiments $\mathbf{Exp}^0$ and $\mathbf{Exp}^1$, we define the distinguishing advantage of $\mathcal{A}$ against the tuple $(\mathbf{Exp}^0, \mathbf{Exp}^1)$ as

$$\text{Adv}_{\mathbf{Exp}^0, \mathbf{Exp}^1, \mathcal{A}}(\lambda) := \left| \Pr \left[\mathbf{Exp}_{\mathcal{A}}^1(1^\lambda) \rightarrow 1 \right] - \Pr \left[\mathbf{Exp}_{\mathcal{A}}^0(1^\lambda) \rightarrow 1 \right] \right|.$$

We write $\mathbf{Exp}^0 \approx_c \mathbf{Exp}^1$ if the experiments are *computationally indistinguishable*, i.e. their distinguishing advantage is negligible for all efficient adversaries $\mathcal{A}$. We write $\mathbf{Exp}^0 \approx_s \mathbf{Exp}^1$ if the experiments are *statistically indistinguishable*, i.e. their distinguishing advantage is negligible for all (even unbounded) adversaries. We write $\mathbf{Exp}^0 \equiv \mathbf{Exp}^1$ if the experiments are *identically distributed*, i.e. their distinguishing advantage is 0 for all (even unbounded) adversaries. By default, the term *indistinguishable* refers to computational indistinguishability.

More general, the same notations can be used for sequences of distributions. Let $D^0 = \{D_\lambda^0\}_{\lambda \in \mathbb{N}}$ and $D^1 = \{D_\lambda^1\}_{\lambda \in \mathbb{N}}$ be two sequences of distributions. For $b \in \{0, 1\}$, we define $\mathbf{Exp}_{\mathcal{A}}^b(1^\lambda)$ as follows: sample $x \xleftarrow{s} D_\lambda^b$, run $\mathcal{A}(1^\lambda, x)$ and use the output of $\mathcal{A}$ as the outcome of the experiment. Then we write $D^0 \approx_c D^1$ (resp. $D^0 \approx_s D^1$, $D^0 \equiv D^1$) if $\mathbf{Exp}_{\mathcal{A}}^0 \approx_c \mathbf{Exp}_{\mathcal{A}}^1$ (resp. $\mathbf{Exp}_{\mathcal{A}}^0 \approx_s \mathbf{Exp}_{\mathcal{A}}^1$, $\mathbf{Exp}_{\mathcal{A}}^0 \equiv \mathbf{Exp}_{\mathcal{A}}^1$).

Sets and Indexing. We denote by $\mathbb{Z}$ and $\mathbb{N}$ the sets of integers and natural numbers (positive integers). For integers m and n , we write $[m; n]$ to denote the set $\{z \in \mathbb{Z} : m \leq z \leq n\}$ and let $[n] := [1; n]$. For a prime number p , $\mathbb{Z}_p$ denotes the finite field of integers modulo p . For a finite set S , we let 2^S denote the power set of S .

To index a vector or the columns of a matrix, we write $\mathbf{v}[i]$ and $\mathbf{A}[j]$. In contrast, objects of some collection that is not regarded as a vector or matrix are indexed using subscripts (or superscripts in some cases). For instance, $\mathbf{v}_i$ represents a vector, not a component of some vector. If i runs through some index set $[n]$, it means that there are n vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$. If the n objects are scalars (or not explicitly vectors), we will write $v_1, \dots, v_n$ instead.

3.2 Pairing Groups and Hardness Assumptions

Pairing Groups. Our constructions use a sequence of pairing groups

$$\mathbb{G} = \{\mathbb{G}_\lambda = (\mathbb{G}_{\lambda,1}, \mathbb{G}_{\lambda,2}, \mathbb{G}_{\lambda,t}, g_{\lambda,1}, g_{\lambda,2}, g_{\lambda,t}, e_\lambda, p_\lambda)\}_{\lambda \in \mathbb{N}},$$

where $\mathbb{G}_{\lambda,1}$ (resp. $\mathbb{G}_{\lambda,2}, \mathbb{G}_{\lambda,t}$) is a cyclic group of order p_λ generated by $g_{\lambda,1}$ (resp. $g_{\lambda,2}, g_{\lambda,t}$), and $e_\lambda: \mathbb{G}_{\lambda,1} \times \mathbb{G}_{\lambda,2} \rightarrow \mathbb{G}_{\lambda,t}$ is the pairing operation satisfying $e_\lambda(g_{\lambda,1}^a, g_{\lambda,2}^b) = g_{\lambda,t}^{ab}$ for all integers a, b . The group operations and the pairing map are required to be efficiently computable.

Following the implicit notation in [EHK⁺13], we write $\llbracket a \rrbracket_i$ to denote $g_{\lambda,i}^a$ for $i \in \{1, 2, t\}$. This notation extends component-wise to matrices and vectors having entries in $\mathbb{Z}_p$. Equipped with these notations, group operations are written additively and the pairing operation multiplicatively, e.g. $\llbracket \mathbf{A} \rrbracket_1 - \llbracket \mathbf{B} \rrbracket_1 \llbracket \mathbf{C} \rrbracket_1 \mathbf{D} = \llbracket \mathbf{A} - \mathbf{BCD} \rrbracket_1$ and $\llbracket \mathbf{A} \rrbracket_1 \llbracket \mathbf{B} \rrbracket_2 = \llbracket \mathbf{AB} \rrbracket_t$. For convenience, we may also write $\llbracket a \rrbracket_{1,2}$ to denote the tuple $(\llbracket a \rrbracket_1, \llbracket a \rrbracket_2)$.

Computational Assumptions. We recall the MDDH assumption defined in [EHK⁺13].

Definition 4 (MDDH $_{k,\ell}^q$ Assumption). Let $\alpha \in \{1, 2, t\}$, $k \geq 1$ an integer constant and $\ell = \ell(\lambda)$, $q = q(\lambda)$ some polynomials. For a sequence of pairing groups $\mathbb{G}$, the MDDH $_{k,\ell}^q$ assumption holds in $\mathbb{G}_\alpha$ if $\{\llbracket \mathbf{A} \rrbracket_\alpha, \llbracket \mathbf{AS} \rrbracket_\alpha\}_{\lambda \in \mathbb{N}} \approx_c \{\llbracket \mathbf{A} \rrbracket_\alpha, \llbracket \mathbf{U} \rrbracket_\alpha\}_{\lambda \in \mathbb{N}}$ for $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_p^{\ell \times k}$, $\mathbf{S} \xleftarrow{\$} \mathbb{Z}_p^{k \times q}$ and $\mathbf{U} \xleftarrow{\$} \mathbb{Z}_p^{\ell \times q}$.

In [EHK⁺13], it was shown that k -Lin implies MDDH $_{k,k+1}^1$ which in turn implies MDDH $_{k,\ell}^q$ for all $\ell > k$ and $q \geq 1$ with a tight security reduction. For convenience, we refer to the MDDH $_{k,k+1}^1$ assumption by MDDH $_k$. Similarly, we define the bilateral MDDH assumption bi-MDDH $_{k,\ell}^q$ with the special case bi-MDDH $_k = \text{bi-MDDH}_{k,k+1}^1$.

Definition 5 (bi-MDDH $_{k,\ell}^q$ Assumption). Let $k \geq 1$ an integer constant and $\ell = \ell(\lambda)$, $q = q(\lambda)$ some polynomials. The bi-MDDH $_{k,\ell}^q$ assumption holds in $\mathbb{G}$ if $\{\llbracket \mathbf{A} \rrbracket_{1,2}, \llbracket \mathbf{AS} \rrbracket_{1,2}\}_{\lambda \in \mathbb{N}} \approx_c \{\llbracket \mathbf{A} \rrbracket_{1,2}, \llbracket \mathbf{U} \rrbracket_{1,2}\}_{\lambda \in \mathbb{N}}$ for $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_p^{\ell \times k}$, $\mathbf{S} \xleftarrow{\$} \mathbb{Z}_p^{k \times q}$ and $\mathbf{U} \xleftarrow{\$} \mathbb{Z}_p^{\ell \times q}$.

Extension to Vector Spaces. For a matrix $\mathbf{A}$ over $\mathbb{Z}_p$, we use $\text{span}(\mathbf{A})$ to denote the column span of $\mathbf{A}$ and $\text{basis}(\mathbf{A})$ to denote a basis of $\text{span}(\mathbf{A})$. Let $\ell_1, \ell_2, \ell_3 \geq 1$ and $\ell := \ell_1 + \ell_2 + \ell_3$. Consider random matrices

$$\mathbf{B}_1 \xleftarrow{\$} \mathbb{Z}_p^{\ell \times \ell_1}, \quad \mathbf{B}_2 \xleftarrow{\$} \mathbb{Z}_p^{\ell \times \ell_2}, \quad \mathbf{B}_3 \xleftarrow{\$} \mathbb{Z}_p^{\ell \times \ell_3},$$

and let $[\mathbf{B}_1^\parallel \parallel \mathbf{B}_2^\parallel \parallel \mathbf{B}_3^\parallel]^\top$ denote its inverse such that for all $i, j \in \{1, 2, 3\}$, we have that

$$\mathbf{B}_i^\top \mathbf{B}_j^\parallel = \begin{cases} \mathbf{I}_{\ell_i} & \text{if } i = j \text{ (non-degeneracy)} \\ \mathbf{0} & \text{if } i \neq j \text{ (orthogonality)} \end{cases}.$$

We note that for all $\mathbf{v} \in \mathbb{Z}_p^\ell$, we can uniquely decompose $\mathbf{v} = \sum_{i \in [3]} \mathbf{v}^{(i)}$, where $\mathbf{v}^{(i)} \in \text{span}(\mathbf{B}_i^\parallel)$. In the same way, a vector $\mathbf{w} \in \mathbb{Z}_p^{n\ell}$ uniquely decomposes into $\mathbf{w} = \sum_{i \in [3]} \mathbf{w}^{(i)}$ such that $\mathbf{w}^{(i)} \in \text{span}(\mathbf{I}_n \otimes \mathbf{B}_i^\parallel)$. Note that if $\mathbf{w}$ is of the form $\mathbf{u} \otimes \mathbf{v}$ for $\mathbf{u} \in \mathbb{Z}_p^n$ and $\mathbf{v} \in \mathbb{Z}_p^\ell$, then we have $(\mathbf{u} \otimes \mathbf{v})^{(i)} = \mathbf{u} \otimes (\mathbf{v}^{(i)})$.

We recall the definition of the subspace decision assumption in prime-order groups introduced in [CGW18]. Let $i, j \in [3]$, $i \neq j$ and $\alpha \in \{1, 2, t\}$.

Definition 6 (SD $_{\mathbf{B}_i \rightarrow \mathbf{B}_j}$ Assumption). In the above notation, the subspace decision assumption SD $_{\mathbf{B}_i \rightarrow \mathbf{B}_j}$ holds in $\mathbb{G}_\alpha$ if there exists an efficient sampler outputting random $[\mathbf{B}_1]_\alpha \in \mathbb{G}_\alpha^{\ell \times \ell_1}$, $[\mathbf{B}_2]_\alpha \in \mathbb{G}_\alpha^{\ell \times \ell_2}$ and $[\mathbf{B}_3]_\alpha \in \mathbb{G}_\alpha^{\ell \times \ell_3}$ along with its dual basis $(\mathbf{B}_1^\parallel, \mathbf{B}_2^\parallel, \mathbf{B}_3^\parallel)$ such that $\{\llbracket \mathbf{t}_0 \rrbracket_\alpha, D\}_{\lambda \in \mathbb{N}} \approx_c \{\llbracket \mathbf{t}_1 \rrbracket_\alpha, D\}_{\lambda \in \mathbb{N}}$, where $\mathbf{t}_0 \xleftarrow{\$} \text{span}(\mathbf{B}_i)$, $\mathbf{t}_1 \xleftarrow{\$} \text{span}(\mathbf{B}_j)$ and $D = ([\mathbf{B}_1 \mathbf{B}_2 \mathbf{B}_3]_\alpha, \text{basis}(\mathbf{B}_i^\parallel), \text{basis}(\mathbf{B}_j^\parallel))$ for $k \in [3] \setminus \{i, j\}$.

In [CGW18], it was shown that MDDH $_{\ell_i}$ implies SD $_{\mathbf{B}_i \rightarrow \mathbf{B}_j}$ in $\mathbb{G}_\alpha$.

3.3 Arithmetic Branching Programs and Partial Garbling

Arithmetic Branching Programs. We recall the definition of arithmetic branching programs.

Definition 7 (Arithmetic Branching Program (ABP)). An arithmetic branching program $f: \mathbb{Z}_p^n \rightarrow \mathbb{Z}_p$ is defined by a tuple $(V, E, s, t, q, n, \sigma)$ consisting of a directed acyclic graph (V, E) with two distinguished vertices $s, t \in V$, a prime q , an arity n and a labelling function $\sigma: E \rightarrow \mathcal{F}^{\text{aff}}$, where $\mathcal{F}^{\text{aff}}$ contains all affine functions $g: \mathbb{Z}_p^n \rightarrow \mathbb{Z}_p$. Let P be the set of all paths from s to t . The output of the ABP on input $\mathbf{x} \in \mathbb{Z}_p^n$ is defined as $f(\mathbf{x}) = \sum_{p \in P} \prod_{e \in p} \sigma(e)(\mathbf{x})$. We refer to $|V| + |E|$ as the size of f .

More general, we denote by $\mathcal{F}_{n,n'}^{\text{abp}}$ the class of functions $f: \mathbb{Z}_p^n \rightarrow \mathbb{Z}_p^{n'}$ that evaluate an ABP in each coordinate.

Partial Garbling Scheme. We recall the partial garbling scheme of [IW14, Wee17] for $f(\mathbf{x})^\top \mathbf{z}$ with $f \in \mathcal{F}_{n,n'}^{\text{abp}}$. We follow the presentation in [AGW20].

Lemma 8 (Partial Garbling Scheme With Shifts). *There exist four efficient algorithms ($\text{lgen}, \text{pgb}, \text{rec}, \text{pgb}^*$) with the following properties:*

- Syntax. On input $f \in \mathcal{F}_{n,n'}^{\text{abp}}$, $\text{lgen}(f)$ outputs two matrices $(\mathbf{L}_0 \in \mathbb{Z}_p^{(m+n'-1) \times m}, \mathbf{L}_1 \in \mathbb{Z}_p^{(m+n'-1) \times mn})$, and

$$\begin{aligned} \text{pgb}(f, \mathbf{x}, \mathbf{z}; \mathbf{w}) &= (\mathbf{z}^\top - \underline{\mathbf{w}}^\top, \mathbf{w}^\top (\mathbf{L}_0 + \mathbf{L}_1 (\mathbf{x} \otimes \mathbf{I}_m))) \\ \text{pgb}^*(f, \mathbf{x}, \mu; \mathbf{w}) &= (-\underline{\mathbf{w}}^\top, \mathbf{w}^\top (\mathbf{L}_0 + \mathbf{L}_1 (\mathbf{x} \otimes \mathbf{I}_m)) + \mu \cdot \mathbf{e}_1^\top), \end{aligned}$$

where $\mathbf{w} \in \mathbb{Z}_p^{m+n'-1}$, $\underline{\mathbf{w}}$ consists of the last n' entries of $\mathbf{w}$, and m is linear in the size of f .

- Reconstruction. On input $f \in \mathcal{F}_{n,n'}^{\text{abp}}$ and $\mathbf{x} \in \mathbb{Z}_p^n$, $\text{rec}(f, \mathbf{x})$ outputs a vector $\mathbf{d}_{f,\mathbf{x}} \in \mathbb{Z}_p^{n'+m}$ such that for all $\mathbf{z} \in \mathbb{Z}_p^{n'}$, $\mathbf{w} \in \mathbb{Z}_p^{m+n'-1}$ and $\delta \in \mathbb{Z}_p$, it holds that

$$((\mathbf{p}_1^\top \parallel \mathbf{p}_2^\top) + (\mathbf{0}^\top, \delta \cdot \mathbf{e}_1^\top)) \cdot \mathbf{d}_{f,\mathbf{x}} = f(\mathbf{x})^\top \cdot \mathbf{z} + \delta,$$

where $(\mathbf{p}_1^\top, \mathbf{p}_2^\top) = \text{pgb}(f, \mathbf{x}, \mathbf{z}; \mathbf{w})$.

- Privacy. For all $f \in \mathcal{F}_{n,n'}^{\text{abp}}$, $\mathbf{x} \in \mathbb{Z}_p^n$, $\mathbf{z} \in \mathbb{Z}_p^{n'}$ and $\delta \in \mathbb{Z}_p$, it holds that

$$\text{pgb}(f, \mathbf{x}, \mathbf{z}; \mathbf{w}) + (\mathbf{0}^\top, \delta \cdot \mathbf{e}_1^\top) \approx_s \text{pgb}^*(f, \mathbf{x}, f(\mathbf{x})^\top \cdot \mathbf{z} + \delta; \mathbf{w}),$$

where the randomness is over $\mathbf{w} \xleftarrow{s} \mathbb{Z}_p^{m+n'-1}$.

Overloading notation, we may also run the algorithms pgb and pgb^* given certain inputs as elements of an MDDH-hard group $\mathbb{G}_1$ instead of $\mathbb{Z}_p$:

$$\begin{aligned} \text{pgb}(f, \mathbf{x}, \llbracket \mathbf{z} \rrbracket_1; \llbracket \mathbf{w} \rrbracket_1) &= (\llbracket \mathbf{z} \rrbracket_1^\top - \llbracket \underline{\mathbf{w}} \rrbracket_1^\top, \llbracket \mathbf{w} \rrbracket_1^\top \cdot (\mathbf{L}_0 + \mathbf{L}_1 (\mathbf{x} \otimes \mathbf{I}_m))) \\ \text{pgb}^*(f, \mathbf{x}, \llbracket \mu \rrbracket_1; \llbracket \mathbf{w} \rrbracket_1) &= (-\llbracket \underline{\mathbf{w}} \rrbracket_1^\top, \llbracket \mathbf{w} \rrbracket_1^\top \cdot (\mathbf{L}_0 + \mathbf{L}_1 (\mathbf{x} \otimes \mathbf{I}_m)) + \llbracket \mu \rrbracket_1 \cdot \mathbf{e}_1^\top). \end{aligned}$$

3.4 Registered Functional Encryption

In this section, we provide our notion of Registered Functional Encryption (RFE). The definitions mostly follow [ZLZ⁺24]. Let $\{\mathcal{X}_\lambda\}_{\lambda \in \mathbb{N}}$ and $\{\mathcal{Y}_\lambda\}_{\lambda \in \mathbb{N}}$ be sequences of domain and range spaces, where $\mathcal{X}_\lambda = \mathcal{X}_{\lambda, \text{pub}} \times \mathcal{X}_{\lambda, \text{pri}}$ consists of a public and a private component. We consider a function class $\mathcal{F} = \{\mathcal{F}_\lambda\}_{\lambda \in \mathbb{N}}$ where each $\mathcal{F}_\lambda$ contains functions $f: \mathcal{X}_\lambda \rightarrow \mathcal{Y}_\lambda$.

Definition 9 (Syntax of (Bounded) RFE). A RFE scheme for the functionality $\mathcal{F}$ consists of six efficient algorithms:

Setup($1^\lambda, 1^L$) $\rightarrow$ **crs**: On input the security parameter 1^λ and the maximum number of users 1^L , this algorithm outputs a common reference string **crs**.

$\text{Gen}(\text{crs}, \text{aux}) \rightarrow (\text{pk}_i, \text{sk}_i)$: On input the crs and a state aux , this algorithm outputs a pair of a public and a secret key $(\text{pk}_i, \text{sk}_i)$.

$\text{Reg}(\text{crs}, \text{aux}, \text{pk}, f) \rightarrow (\text{mpk}, \text{aux}')$: On input the crs , a state aux , a public key pk and a function $f \in \mathcal{F}_\lambda$, this algorithm outputs a master public key mpk and an updated state aux' . We require Reg to be deterministic.

$\text{Enc}(\text{mpk}, x_{\text{pub}}, x_{\text{pri}}) \rightarrow \text{ct}$: On input the master public key mpk , a public input $x_{\text{pub}} \in \mathcal{X}_{\lambda, \text{pub}}$ and a private input $x_{\text{pri}} \in \mathcal{X}_{\lambda, \text{pri}}$, this algorithm outputs a ciphertext ct .

$\text{Update}(\text{crs}, \text{aux}, \text{pk}) \rightarrow \text{hsk}$: On input the crs , a state aux and a public key pk , this algorithm outputs a helper secret key hsk . We require Update to be deterministic.

$\text{Dec}(\text{sk}, \text{hsk}, \text{ct}) \rightarrow y \vee \perp \vee \text{GetUpdate}$: On input a secret key sk , a helper secret key hsk and a ciphertext ct , this algorithm either outputs a value $y \in \mathcal{Y}_\lambda$, a special symbol $\perp$ indicating decryption failure, or a special message GetUpdate indicating an updated helper secret key is needed to decrypt the ciphertext. We require Dec to be deterministic.

We recall the definitions of correctness, compactness and update efficiency.

Definition 10 (Correctness, Compactness and Update Efficiency of (Bounded) RFE). Given a RFE scheme FE and a PPT adversary $\mathcal{A}$, we define the experiment $\text{Exp}_{\text{FE}, \mathcal{A}}^{\text{rfe}}$ as follows:

- *Setup*. Launch $\mathcal{A}(1^\lambda)$ and receive from it $L \in \mathbb{N}$. Run $\text{crs} \leftarrow \text{Setup}(1^\lambda, 1^L)$ and send crs to $\mathcal{A}$. Initialize the auxiliary input $\text{aux} := \perp$, two empty dictionaries $\mathcal{E}, \mathcal{R}$ and counters $i_{\text{reg}}, i_{\text{reg}}^*, i_{\text{enc}} := 1$ to keep track of QRegNT, QRegT and QEnc queries. Let $b = 0$ and $f^*, \text{pk}^*, \text{sk}^*, \text{hsk}^* := \perp$.
- *Query*. Repeat the following for arbitrarily many rounds determined by $\mathcal{A}$. The oracle QRegNT can be queried at most $L - 1$ times and QRegT can be queried exactly once. In each round, $\mathcal{A}$ has four options.
 - QRegNT(pk, f): upon $\mathcal{A}$ submitting a public key pk and a function $f \in \mathcal{F}_\lambda$, run $(\text{mpk}, \text{aux}') \leftarrow \text{Reg}(\text{crs}, \text{aux}, \text{pk}, f)$ and return $(i_{\text{reg}}, \text{mpk}, \text{aux}')$ to $\mathcal{A}$. Update $\mathcal{R}[i_{\text{reg}}] := (\text{mpk}, \text{aux}')$, $\text{aux} := \text{aux}'$ and $i_{\text{reg}} := i_{\text{reg}} + 1$.
 - QRegT(f): upon $\mathcal{A}$ submitting a function $f \in \mathcal{F}_\lambda$, run $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(\text{crs}, \text{aux})$, $(\text{mpk}, \text{aux}') \leftarrow \text{Reg}(\text{crs}, \text{aux}, \text{pk}, f)$, $\text{hsk} := \text{Update}(\text{crs}, \text{aux}', \text{pk})$ and return $(i_{\text{reg}}, \text{mpk}, \text{aux}', \text{pk}, \text{sk}, \text{hsk})$ to $\mathcal{A}$. Update $\mathcal{R}[i_{\text{reg}}] := (\text{mpk}, \text{aux}')$, $\text{aux} := \text{aux}'$, $i_{\text{reg}} := i_{\text{reg}} + 1$ and $i_{\text{reg}}^* := i_{\text{reg}}$. Furthermore, set $f^* := f$, $\text{pk} := \text{pk}^*$, $\text{sk} := \text{sk}^*$ and $\text{hsk} := \text{hsk}^*$,
 - QEnc($j, x_{\text{pub}}, x_{\text{pri}}$): upon $\mathcal{A}$ submitting an index $j \in [i_{\text{reg}}^*; i_{\text{reg}}]$ and a message $(x_{\text{pub}}, x_{\text{pri}}) \in \mathcal{X}_{\lambda, \text{pub}}$, retrieve $(\text{mpk}, \star) := \mathcal{R}[j]$, run $\text{ct} \leftarrow \text{Enc}(\text{mpk}, x_{\text{pub}}, x_{\text{pri}})$ and return $(i_{\text{enc}}, \text{ct})$ to $\mathcal{A}$. Set $\mathcal{E}[i_{\text{enc}}] := (x_{\text{pub}}, x_{\text{pri}}, \text{ct})$ and $i_{\text{enc}} := i_{\text{enc}} + 1$.
 - QDec(j): upon $\mathcal{A}$ submitting an index $j \in [i_{\text{enc}}]$, if $\text{sk}^* = \perp$ return $\perp$. Otherwise, retrieve the tuple $(x_{\text{pub}}, x_{\text{pri}}, \text{ct}) := \mathcal{E}[j]$ and run $y \leftarrow \text{Dec}(\text{sk}^*, \text{hsk}^*, \text{ct})$. If $y = \text{GetUpdate}$, compute $\text{hsk}^* \leftarrow \text{Update}(\text{crs}, \text{aux}, \text{pk}^*)$ and recompute $y \leftarrow \text{Dec}(\text{sk}^*, \text{hsk}^*, \text{ct})$. Set $b = 1$ if $y \neq f^*(x_{\text{pub}}, x_{\text{pri}})$.

We say that FE is

- correct if $\Pr[\text{Exp}_{\text{FE}, \mathcal{A}}^{\text{rfe}}(1^\lambda) \rightarrow 0] = 1$ for all PPT adversaries $\mathcal{A}$,
- compact if $|\text{mpk}| = \text{poly}(\lambda, \log L)$ and $|\text{hsk}| = \text{poly}(\lambda, \log L)$ at any stage during the execution of the experiment $\text{Exp}_{\text{FE}, \mathcal{A}}^{\text{rfe}}(1^\lambda)$, and
- update efficient if the oracle QDec invokes Update at most $O(\log |\mathcal{R}|)$ times and each invocation runs in time $\text{poly}(\log |\mathcal{R}|)$.

Security. We define adaptive IND-CPA and very-selective SIM security.

Definition 11 (Adaptive IND-Security of (Bounded) RFE). A RFE scheme FE for a functionality $\mathcal{F} = \{\mathcal{F}_\lambda\}_{\lambda \in \mathbb{N}}$ is (adaptively) IND-secure if $\text{Exp}_{\text{FE}, \mathcal{A}}^{\text{rfe-0}}(1^\lambda) \approx_c \text{Exp}_{\text{FE}, \mathcal{A}}^{\text{rfe-1}}(1^\lambda)$ for all PPT adversaries $\mathcal{A}$, where $\text{Exp}_{\text{FE}, \mathcal{A}}^{\text{rfe-}b}$ for $b \in \{0, 1\}$ proceeds as follows.

- *Setup.* Launch $\mathcal{A}(1^\lambda)$ and receive from it $L \in \mathbb{N}$. Run $\text{crs} \leftarrow \text{Setup}(1^\lambda, 1^L)$ and send crs to $\mathcal{A}$. Initialize the auxiliary input $\text{aux} := \perp$, the master public key $\text{mpk} := \perp$, a counter $i_{\text{reg}} := 0$ to keep track of QRegHK queries, an empty set $\mathcal{C} := \emptyset$ and empty dictionaries $\mathcal{D}, \mathcal{R}$.
- *Query.* Repeat the following for arbitrarily many rounds determined by $\mathcal{A}$. In each round, $\mathcal{A}$ has three options.
 - QRegCK(pk, f): upon $\mathcal{A}$ submitting a public key pk and a function $f \in \mathcal{F}_\lambda$, run $(\text{mpk}', \text{aux}') \leftarrow \text{Reg}(\text{crs}, \text{aux}, \text{pk}, f)$ and return $(\text{mpk}', \text{aux}')$ to $\mathcal{A}$. Set $\text{mpk} := \text{mpk}'$, $\text{aux} := \text{aux}'$, $\mathcal{D}[\text{pk}] := \mathcal{D}[\text{pk}] \cup \{f\}$ and add pk to $\mathcal{C}$.
 - QRegHK(f): upon $\mathcal{A}$ submitting a function $f \in \mathcal{F}_\lambda$, run $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(\text{crs}, \text{aux})$, $(\text{mpk}', \text{aux}') \leftarrow \text{Reg}(\text{crs}, \text{aux}, \text{pk}, f)$ and return $(i_{\text{reg}}, \text{mpk}', \text{aux}', \text{pk})$ to $\mathcal{A}$. Set $\text{mpk} := \text{mpk}'$, $\text{aux} := \text{aux}'$, $\mathcal{D}[\text{pk}] := \mathcal{D}[\text{pk}] \cup \{f\}$, $\mathcal{R}[i_{\text{reg}}] := (\text{pk}, \text{sk})$ and $i_{\text{reg}} := i_{\text{reg}} + 1$.
 - QCorHK(j): upon $\mathcal{A}$ submitting an index $j \in [i_{\text{reg}}]$, retrieve $(\text{pk}, \text{sk}) := \mathcal{R}[j]$ and return sk to $\mathcal{A}$. Add pk to $\mathcal{C}$.
- *Challenge.* The adversary submits a pair of messages $(x_{\text{pub}}, x_{\text{pri}}^0), (x_{\text{pub}}, x_{\text{pri}}^1) \in \mathcal{X}_{\lambda, \text{pub}} \times \mathcal{X}_{\lambda, \text{pri}}$. Run $\text{ct}^* \leftarrow \text{Enc}(\text{mpk}, x_{\text{pub}}, x_{\text{pri}}^b)$ and return ct^* to $\mathcal{A}$.
- *Guess.* The adversary outputs a bit $b' \in \{0, 1\}$. The outcome of the experiment is b' if $f(x_{\text{pub}}, x_{\text{pri}}^0) = f(x_{\text{pub}}, x_{\text{pri}}^1)$ for all $f \in \bigcup_{\text{pk} \in \mathcal{C}} \mathcal{D}[\text{pk}]$. Otherwise, the outcome is set to 0.

Definition 12 (Very Selective SIM-Security of (Bounded) RFE). A RFE scheme FE for a functionality $\mathcal{F} = \{\mathcal{F}_\lambda\}_{\lambda \in \mathbb{N}}$ is said to be very selectively SIM-secure if there exist three PPT algorithms $\widetilde{\text{Setup}}$, $\widetilde{\text{Gen}}$ and $\widetilde{\text{Enc}}$ satisfying $\text{Exp}_{\text{FE}, \mathcal{A}}^{\text{rfe-real}}(1^\lambda) \approx_c \text{Exp}_{\text{FE}, \mathcal{A}}^{\text{rfe-sim}}(1^\lambda)$ for all PPT adversaries $\mathcal{A}$, where $\text{Exp}_{\text{FE}, \mathcal{A}}^{\text{rfe-real}}$ and $\text{Exp}_{\text{FE}, \mathcal{A}}^{\text{rfe-sim}}$ proceed as follows.

- *Setup.* Launch $\mathcal{A}(1^\lambda)$ and receive from it the maximum number of users $L \in \mathbb{N}$, the actual number of users $L' \in [L]$, the set of malicious users $\mathcal{I}_{\text{mal}} \subseteq [L']$, the set of corrupted users $\mathcal{I}_{\text{cor}} \subseteq [L'] \setminus \mathcal{I}_{\text{mal}}$, the challenge message $(x_{\text{pub}}^*, x_{\text{pri}}^*) \in \mathcal{X}_{\lambda, \text{pub}} \times \mathcal{X}_{\lambda, \text{pri}}$ and the challenge functions $f_1^*, \dots, f_{L'}^* \in \mathcal{F}_\lambda$. Run

$$\begin{aligned} \text{in } \text{Exp}_{\text{FE}, \mathcal{A}}^{\text{rfe-real}}: & \quad \text{crs} \leftarrow \text{Setup}(1^\lambda, 1^L), \\ \text{in } \text{Exp}_{\text{FE}, \mathcal{A}}^{\text{rfe-sim}}: & \quad (\text{crs}, \text{td}) \leftarrow \widetilde{\text{Setup}}(1^\lambda, 1^L, x_{\text{pub}}^*, \{f_i^*\}_{i \in [L']}, \{f_i^*(x_{\text{pub}}^*, x_{\text{pri}}^*)\}_{i \in \mathcal{I}}), \end{aligned}$$

where $\mathcal{I} = \mathcal{I}_{\text{mal}} \cup \mathcal{I}_{\text{cor}}$, and send crs to $\mathcal{A}$. Initialize a counter $\ell := 1$ and $\text{pk}_1^*, \dots, \text{pk}_{L'}^* := \perp$.

- *Query.* The adversary has access to the oracle QReg(i, pk_i) which returns $\perp$ if $i \neq \ell$, and behaves according to the following case distinction otherwise.
 - If $\ell \in \mathcal{I}_{\text{mal}}$, set $\text{pk}_i^* := \text{pk}_i$, run $(\text{mpk}', \text{aux}') \leftarrow \text{Reg}(\text{crs}, \text{aux}, \text{pk}_i^*, f_i^*)$ and return $(\text{mpk}', \text{aux}')$ to $\mathcal{A}$. Set $\text{mpk} := \text{mpk}'$, $\text{aux} := \text{aux}'$ and $\ell := \ell + 1$.
 - If $\ell \in [L'] \setminus \mathcal{I}_{\text{mal}}$, run

$$\begin{aligned} \text{in } \text{Exp}_{\text{FE}, \mathcal{A}}^{\text{rfe-real}}: & \quad (\text{pk}_i^*, \text{sk}_i^*) \leftarrow \text{Gen}(\text{crs}, \text{aux}), \\ \text{in } \text{Exp}_{\text{FE}, \mathcal{A}}^{\text{rfe-sim}}: & \quad (\text{pk}_i^*, \text{sk}_i^*) \leftarrow \widetilde{\text{Gen}}(\text{aux}, \text{td}), \end{aligned}$$

and $(\text{mpk}', \text{aux}') \leftarrow \text{Reg}(\text{crs}, \text{aux}, \text{pk}_i^*, f_i^*)$. If $i \in \mathcal{I}_{\text{cor}}$, return $(\text{mpk}', \text{aux}', \text{pk}_i^*, \text{sk}_i^*)$ to $\mathcal{A}$, otherwise return $(\text{mpk}', \text{aux}', \text{pk}_i^*)$. Set $\text{mpk} := \text{mpk}'$, $\text{aux} := \text{aux}'$ and $\ell := \ell + 1$.

- *Challenge.* Run

$$\begin{aligned} &\text{in } \mathbf{Exp}_{\mathcal{FE}, \mathcal{A}}^{\text{rfe-real}}: & \text{ct}^* &\leftarrow \text{Enc}(\text{mpk}, x_{\text{pub}}^*, x_{\text{pri}}^*) , \\ &\text{in } \mathbf{Exp}_{\mathcal{FE}, \mathcal{A}}^{\text{rfe-sim}}: & \text{ct}^* &\leftarrow \widetilde{\text{Enc}}(\{\text{pk}_i^*\}_{i \in [L]}, \text{td}) , \end{aligned}$$

and return ct^* to $\mathcal{A}$.

- *Guess.* The adversary outputs a guess $\alpha \in \{\text{real}, \text{sim}\}$ which is also the output of the experiment.

3.5 Slotted Registered Functional Encryption

Our definition of slotted RFE is almost the same as [ZLZ⁺24] except that we consider functions that may depend on an additional parameter specified at setup.

Let $\{\mathcal{X}_\lambda\}_{\lambda \in \mathbb{N}}$, $\{\mathcal{Y}_\lambda\}_{\lambda \in \mathbb{N}}$ and $\{\text{Param}_\lambda\}_{\lambda \in \mathbb{N}}$ be sequences of domain, range and parameter spaces, respectively, and $\mathcal{X}_\lambda = \mathcal{X}_{\lambda, \text{pub}} \times \mathcal{X}_{\lambda, \text{pri}}$ consists of a public and a private component. We consider a function class $\mathcal{F} = \{\mathcal{F}_\lambda\}_{\lambda \in \mathbb{N}}$ where each $\mathcal{F}_\lambda$ contains functions $f_\lambda: \text{Param}_\lambda \times \mathcal{X}_\lambda \rightarrow \mathcal{Y}_\lambda$.

Definition 13 (Syntax of Slotted RFE (sRFE)). A slotted RFE scheme for the functionality $\mathcal{F}$ consists of six efficient algorithms:

Setup($1^\lambda, 1^L, \{\text{param}_i\}_{i \in [L]}\rangle \rightarrow \text{crs}$: On input the security parameter 1^λ , the maximum number of slots 1^L and a parameter $\text{param}_i \in \text{Param}_\lambda$ for each slot $i \in [L]$, this algorithm outputs a common reference string crs . If $\text{Param}_\lambda = \{\top\}$ is a singleton for all $\lambda \in \mathbb{N}$, we may omit the algorithm's third input for convenience.

Gen($\text{crs}, i\rangle \rightarrow (\text{pk}_i, \text{sk}_i)$: On input the crs and an index $i \in [L]$, this algorithm outputs a pair of a public and a secret key $(\text{pk}_i, \text{sk}_i)$.

Ver($\text{crs}, i, \text{pk}_i\rangle \rightarrow b$: On input the crs , an index $i \in [L]$ and a public key pk_i , this algorithm outputs $b \in \{0, 1\}$. We require **Ver** to be deterministic.

Agg($\text{crs}, \{(\text{pk}_i, f_i)\}_{i \in [L]}\rangle \rightarrow (\text{mpk}, \{\text{hsk}_i\}_{i \in [L]})$: On input the crs and L tuples of the form (pk_i, f_i) with $f_i \in \mathcal{F}_\lambda$, this algorithm outputs a master public key mpk and L helper secret keys $\{\text{hsk}_i\}_{i \in [L]}$. We require **Agg** to be deterministic and assume that hsk_i implicitly contains a description of f_i for $i \in [L]$.

Enc($\text{mpk}, x_{\text{pub}}, x_{\text{pri}}\rangle \rightarrow \text{ct}$: On input the master public key mpk , a public input $x_{\text{pub}} \in \mathcal{X}_{\lambda, \text{pub}}$ and a private input $x_{\text{pri}} \in \mathcal{X}_{\lambda, \text{pri}}$, this algorithm outputs a ciphertext ct . We assume that ct implicitly contains x_{pub} .

Dec($\text{sk}_i, \text{hsk}_i, \text{ct}\rangle \rightarrow y \vee \perp$: On input a secret key sk_i with corresponding helper secret key hsk_i and a ciphertext ct , this algorithm outputs a value $y \in \mathcal{Y}_\lambda$ or a special symbol $\perp$ indicating failure. We require **Dec** to be deterministic.

Completeness. A sRFE scheme is *complete* if for all $\lambda, L \in \mathbb{N}$, $\{\text{param}_i\}_{i \in [L]} \subseteq \text{Param}_\lambda$ and $i^* \in [L]$, it holds that

$$\Pr \left[\text{Ver}(\text{crs}, i^*, \text{pk}_{i^*}) = 1 \mid \begin{array}{l} \text{crs} \leftarrow \text{Setup}(1^\lambda, 1^L, \{\text{param}_i\}_{i \in [L]}) \\ (\text{pk}_{i^*}, \text{sk}_{i^*}) \leftarrow \text{Gen}(\text{crs}, i^*) \end{array} \right] = 1 ,$$

where the probability is taken over the random coins of the algorithms **Setup** and **Gen**.

Correctness. A sRFE scheme is called *correct* if for all $\lambda, L \in \mathbb{N}$, $\{\text{param}_i\}_{i \in [L]} \subseteq \text{Param}_\lambda$ and $i^* \in [L]$, all $\text{crs} \leftarrow \text{Setup}(1^\lambda, 1^L, \{\text{param}_i\}_{i \in [L]})$, $(\text{pk}_{i^*}, \text{sk}_{i^*}) \leftarrow \text{Gen}(\text{crs}, i^*)$, $\{(\text{pk}_i, \text{sk}_i)\}_{i \in [L] \setminus \{i^*\}}$ such that $\text{Ver}(\text{crs}, i, \text{pk}_i) = 1$ for all $i \in [L] \setminus \{i^*\}$, all $(x_{\text{pri}}, x_{\text{pub}}) \in \mathcal{X}_{\lambda, \text{pri}} \times \mathcal{X}_{\lambda, \text{pub}}$ and all $f_1, \dots, f_L \in \mathcal{F}_\lambda$, it holds that

$$\Pr \left[\begin{array}{l} y_{i^*} = y'_{i^*} \\ \text{ct} \leftarrow \text{Enc}(\text{mpk}, x_{\text{pub}}, x_{\text{pri}}) \\ y_{i^*} := \text{Dec}(\text{sk}_{i^*}, \text{hsk}_{i^*}, \text{ct}) \end{array} \mid \begin{array}{l} (\text{mpk}, \{\text{hsk}_i\}_{i \in [L]}) \leftarrow \text{Agg}(\text{crs}, \{(\text{pk}_i, f_i)\}_{i \in [L]}) \end{array} \right] = 1 ,$$

where $y'_{i^*} = f_{i^*}(\text{param}_{i^*}, x_{\text{pub}}, x_{\text{pri}})$ and the probability is taken over the random coins of the algorithms Setup, Gen and Enc.

Compactness. A sRFE scheme for a functionality F is *compact* if for all $\lambda, L \in \mathbb{N}$ and $i \in [L]$, it holds that

$$|\text{mpk}| = \text{poly}(\lambda, \log L) \quad \text{and} \quad |\text{hsk}_i| = \text{poly}(\lambda, \log L) .$$

Security. We define adaptive IND-CPA security and very selective SIM security.

Definition 14 (Adaptive IND-Security of sRFE). A sRFE scheme FE for a functionality $\mathcal{F} = \{\mathcal{F}_\lambda\}_{\lambda \in \mathbb{N}}$ is (adaptively) IND-secure if $\text{Exp}_{\text{FE}, \mathcal{A}}^{\text{srfc-0}}(1^\lambda) \approx_c \text{Exp}_{\text{FE}, \mathcal{A}}^{\text{srfc-1}}(1^\lambda)$ for all PPT adversaries $\mathcal{A}$, where $\text{Exp}_{\text{FE}, \mathcal{A}}^{\text{srfc-}b}$ for $b \in \{0, 1\}$ proceeds as follows.

- *Setup.* Launch $\mathcal{A}(1^\lambda)$ and receive from it $L \in \mathbb{N}$ and a set of parameters $\{\text{param}_i\}_{i \in [L]} \subseteq \text{Param}_\lambda$. Run $\text{crs} \leftarrow \text{Setup}(1^\lambda, 1^L, \{\text{param}_i\}_{i \in [L]})$ and send crs to $\mathcal{A}$. Initialize an empty set $\mathcal{C} := \emptyset$ along with empty dictionaries $\mathcal{D}_1, \dots, \mathcal{D}_L$.
- *Query.* Repeat the following for arbitrarily many rounds determined by $\mathcal{A}$. In each round, $\mathcal{A}$ has two options.
 - QGen(i): upon $\mathcal{A}$ submitting an index $i \in [L]$, run $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(\text{crs}, i)$ and return pk to $\mathcal{A}$. Set $\mathcal{D}_i[\text{pk}] := \text{sk}$.
 - QCor(i, pk): upon $\mathcal{A}$ submitting an index $i \in [L]$ and a public key pk , return $\mathcal{D}_i[\text{pk}]$ to $\mathcal{A}$. Add (i, pk) to $\mathcal{C}$.
- *Challenge.* The adversary submits L tuples $(\text{pk}_i^*, f_i^*)_{i \in [L]}$ with $f_1^*, \dots, f_L^* \in \mathcal{F}_\lambda$ and a pair of messages $(x_{\text{pub}}, x_{\text{pri}}^0), (x_{\text{pub}}, x_{\text{pri}}^1) \in \mathcal{X}_{\lambda, \text{pub}} \times \mathcal{X}_{\lambda, \text{pri}}$. Run $\text{mpk} \leftarrow \text{Agg}(\text{crs}, (\text{pk}_i^*, f_i^*)_{i \in [L]})$ and $\text{ct}^* \leftarrow \text{Enc}(\text{mpk}, x_{\text{pub}}, x_{\text{pri}}^b)$ and return ct^* to $\mathcal{A}$.
- *Guess.* The adversary outputs a bit $b' \in \{0, 1\}$. The outcome of the experiment is b' if $\mathcal{A}$ satisfies the following admissibility condition. Otherwise, the outcome is set to 0. The adversary $\mathcal{A}$ is admissible if
 - $\text{Ver}(\text{crs}, i, \text{pk}_i^*) = 1$ for all $i \in [L]$ such that $\mathcal{D}_i[\text{pk}_i^*] = \perp$, i.e. pk_i^* was not generated by QGen, and
 - $f_i^*(\text{param}_i, x_{\text{pub}}, x_{\text{pri}}^0) = f_i^*(\text{param}_i, x_{\text{pub}}, x_{\text{pri}}^1)$ for all $i \in [L]$ such that $(i, \text{pk}_i^*) \in \mathcal{C}$ or $\mathcal{D}_i[\text{pk}_i^*] = \perp$.

Definition 15 (Very Selective SIM-Security of sRFE). A sRFE scheme FE for a functionality $\mathcal{F} = \{\mathcal{F}_\lambda\}_{\lambda \in \mathbb{N}}$ is very selectively SIM-secure if there exist PPT algorithms $\widetilde{\text{Setup}}$, $\widetilde{\text{Gen}}$ and $\widetilde{\text{Enc}}$ such that $\text{Exp}_{\text{FE}, \mathcal{A}}^{\text{srfc-real}}(1^\lambda) \approx_c \text{Exp}_{\text{FE}, \mathcal{A}}^{\text{srfc-sim}}(1^\lambda)$ for all PPT adversaries $\mathcal{A}$, where $\text{Exp}_{\text{FE}, \mathcal{A}}^{\text{srfc-real}}$ and $\text{Exp}_{\text{FE}, \mathcal{A}}^{\text{srfc-sim}}$ proceed as follows.

- *Setup.* Launch $\mathcal{A}(1^\lambda)$ and receive from it the number of slots $L \in \mathbb{N}$, the set of corrupted slots $\mathcal{I}_{\text{cor}} \subseteq [L]$, the set of malicious slots $\mathcal{I}_{\text{mal}} \subseteq [L] \setminus \mathcal{I}_{\text{cor}}$, the set of parameters $\{\text{param}_i\}_{i \in [L]} \subseteq \text{Param}_\lambda$, the challenge message $(x_{\text{pub}}^*, x_{\text{pri}}^*) \in \mathcal{X}_{\lambda, \text{pub}} \times \mathcal{X}_{\lambda, \text{pri}}$ and the challenge functions $f_1^*, \dots, f_L^* \in \mathcal{F}_\lambda$. Run

$$\begin{aligned} \text{in } \text{Exp}_{\text{FE}, \mathcal{A}}^{\text{srfc-real}}: \quad & \text{crs} \leftarrow \text{Setup}(1^\lambda, 1^L, \{\text{param}_i\}_{i \in [L]}) , \\ \text{in } \text{Exp}_{\text{FE}, \mathcal{A}}^{\text{srfc-sim}}: \quad & (\text{crs}, \text{td}) \leftarrow \widetilde{\text{Setup}}(1^\lambda, 1^L, x_{\text{pub}}^*, \{\text{param}_i, f_i^*\}_{i \in [L]}, \{f_i^*(x_{\text{pub}}^*, x_{\text{pri}}^*)\}_{i \in \mathcal{I}}) , \end{aligned}$$

where $\mathcal{I} = \mathcal{I}_{\text{mal}} \cup \mathcal{I}_{\text{cor}}$, and send crs to $\mathcal{A}$. Initialize an empty set $\mathcal{C} := \emptyset$ and empty dictionaries $\mathcal{D}_1, \dots, \mathcal{D}_L$.

- *Query.* Repeat the following for arbitrarily many rounds determined by $\mathcal{A}$. In each round, $\mathcal{A}$ has two options.

- QGen(i): upon $\mathcal{A}$ submitting an index $i \in [L]$, run

$$\begin{aligned} \text{in } \text{Exp}_{\text{FE}, \mathcal{A}}^{\text{srfc-real}}: \quad & (\text{pk}, \text{sk}) \leftarrow \text{Gen}(\text{crs}, i) , \\ \text{in } \text{Exp}_{\text{FE}, \mathcal{A}}^{\text{srfc-sim}}: \quad & (\text{pk}, \text{sk}) \leftarrow \widetilde{\text{Gen}}(i, \text{td}) , \end{aligned}$$

and return pk to $\mathcal{A}$. Set $\mathcal{D}_i[\text{pk}] := \text{sk}$.

- $\text{QCor}(i, \text{pk})$: upon $\mathcal{A}$ submitting an index $i \in [L]$ and a public key pk , return $\mathcal{D}_i[\text{pk}]$ to $\mathcal{A}$. Add (i, pk) to $\mathcal{C}$.
- *Challenge*. The adversary submits L keys $\{\text{pk}_i^*\}_{i \in [L]}$. Run

$$\begin{aligned} \text{in } \mathbf{Exp}_{\text{FE}, \mathcal{A}}^{\text{srfc-real}}: & \quad \text{mpk} \leftarrow \text{Agg}(\text{crs}, (\text{pk}_i^*, f_i^*)_{i \in [L]}) \\ & \quad \text{ct}^* \leftarrow \text{Enc}(\text{mpk}, x_{\text{pub}}^*, x_{\text{pri}}^*), \\ \text{in } \mathbf{Exp}_{\text{FE}, \mathcal{A}}^{\text{srfc-sim}}: & \quad \text{ct}^* \leftarrow \widetilde{\text{Enc}}(\{\text{pk}_i^*\}_{i \in [L]}, \text{td}), \end{aligned}$$

and return ct^* to $\mathcal{A}$.

- *Guess*. The adversary outputs a guess $\alpha \in \{\text{real}, \text{sim}\}$. The outcome of the experiment is α if $\mathcal{A}$ satisfies the following admissibility condition. Otherwise, the outcome is set to $\perp$. The adversary $\mathcal{A}$ is admissible if
 - $(i, \text{pk}_i^*) \in \mathcal{C}$ for all $i \in \mathcal{I}_{\text{cor}}$,
 - $\mathcal{D}_i[\text{pk}_i^*] = \perp$ and $\text{Ver}(\text{crs}, i, \text{pk}_i^*) = 1$ for all $i \in \mathcal{I}_{\text{mal}}$, and
 - $\mathcal{D}_i[\text{pk}_i^*] \neq \perp$ and $(i, \text{pk}_i^*) \notin \mathcal{C}$ for all $i \in [L] \setminus (\mathcal{I}_{\text{cor}} \cup \mathcal{I}_{\text{mal}})$.

Functionalities. In this work we construct schemes for various variants of the attribute-weighted sums functionality. We give the formal definitions below.

Definition 16 (AB-AWS Functionality). Let $n, n', \hat{n} \in \mathbb{N}$ and $\{p_\lambda\}_{\lambda \in \mathbb{N}}$ be a sequence of prime numbers. For each $\lambda \in \mathbb{N}$, we let $\text{Param}_\lambda = \{\mathbb{T}\}$, $\mathcal{X}_{\lambda, \text{pub}} = \mathbb{Z}_{p_\lambda}^{\hat{n}} \times \bigcup_{N \in \mathbb{N}} (\mathbb{Z}_{p_\lambda}^N)^N$, $\mathcal{X}_{\lambda, \text{pri}} = \bigcup_{N' \in \mathbb{N}} (\mathbb{Z}_{p_\lambda}^{n'})^{N'}$ and $\mathcal{Y}_\lambda = \mathbb{Z}_{p_\lambda}$. The AB-AWS functionality is a family of functions $\{\mathcal{F}_\lambda = \mathcal{F}_{\hat{n}, 1}^{\text{abp}} \times \mathcal{F}_{n, n'}^{\text{abp}}\}_{\lambda \in \mathbb{N}}$, where $f = (g, h) \in \mathcal{F}_\lambda$ represents the function $f: \mathcal{X}_{\lambda, \text{pub}} \times \mathcal{X}_{\lambda, \text{pri}} \rightarrow \mathcal{Y}_\lambda$ ⁶ defined as

$$f((\mathbf{y}, \{\mathbf{x}_j\}_{j \in [N]}), \{\mathbf{z}_j\}_{j \in [N']}) = \begin{cases} \sum_{j \in [N]} h(\mathbf{x}_j)^\top \mathbf{z}_j & \text{if } N = N' \text{ and } g(\mathbf{y}) = 0 \\ \perp & \text{otherwise.} \end{cases}$$

Furthermore, we consider the following special cases of the general AB-AWS functionality:

- AB-1AWS refers to the subset of functions where $N = N' = 1$,
- AWS refers to the subset of functions where $\hat{n} = 0$ and g is the constant zero function, and
- 1AWS refers to the intersection of the previous two cases.

As an intermediate step, we construct sRFE for a functionality called extended AWS which is defined as follows.

Definition 17 (AB-eAWS Functionality). Let $k, n, n', \hat{n} \in \mathbb{N}$ and $\{\mathbb{G}_\lambda\}_{\lambda \in \mathbb{N}}$ be a sequence of pairing groups (parsed as in Section 3.2). For each $\lambda \in \mathbb{N}$, we let $\text{Param}_\lambda = \{\llbracket \mathbf{q} \rrbracket_{1,2} : \mathbf{q} \in \mathbb{Z}_{p_\lambda}^k\}$, $\mathcal{X}_{\lambda, \text{pub}} = \mathbb{Z}_{p_\lambda}^{\hat{n}} \times \mathbb{Z}_{p_\lambda}^n$, $\mathcal{X}_{\lambda, \text{pri}} = \mathbb{Z}_{p_\lambda}^k \times \mathbb{Z}_{p_\lambda}^{n'}$ and $\mathcal{Y}_\lambda = \mathbb{G}_{\lambda, \text{t}}$. The AB-AWS functionality is a family of functions $\{\mathcal{F}_\lambda = \mathcal{F}_{\hat{n}, 1}^{\text{abp}} \times \mathcal{F}_{n, n'}^{\text{abp}}\}_{\lambda \in \mathbb{N}}$, where $f = (g, h) \in \mathcal{F}_\lambda$ represents the function $f: \text{Param}_\lambda \times \mathcal{X}_{\lambda, \text{pub}} \times \mathcal{X}_{\lambda, \text{pri}} \rightarrow \mathcal{Y}_\lambda$ defined as

$$f(\llbracket \mathbf{q} \rrbracket_{1,2}, (\mathbf{y}, \mathbf{x}), (\mathbf{p}, \mathbf{z})) = \begin{cases} \llbracket \mathbf{q}^\top \mathbf{p} + h(\mathbf{x}_j)^\top \mathbf{z}_j \rrbracket_{\text{t}} & \text{if } g(\mathbf{y}) = 0 \\ \perp & \text{otherwise.} \end{cases}$$

Furthermore, we refer by eAWS to the special case of AB-eAWS where $\hat{n} = 0$ and g is the constant zero function. If the output is guaranteed to lie in a polynomial-size set, then we may also consider a scalar variant of the (AB-)eAWS functionality where the output is $\mathbf{q}^\top \mathbf{p} + h(\mathbf{x}_j)^\top \mathbf{z}_j$ instead of $\llbracket \mathbf{q}^\top \mathbf{p} + h(\mathbf{x}_j)^\top \mathbf{z}_j \rrbracket_{\text{t}}$.

Theorem 18 ([FFM⁺23, DPY24]). *If there exists an sRFE scheme for a functionality $\mathcal{F}$, then there exists a (bounded) RFE scheme for the same functionality $\mathcal{F}$.*

⁶Since $\text{Param}_\lambda = \{\mathbb{T}\}$ for all $\lambda \in \mathbb{N}$, we prefer to consider domains of the form $\mathcal{X}_{\lambda, \text{pub}} \times \mathcal{X}_{\lambda, \text{pri}}$ as opposed to $\text{Param}_\lambda \times \mathcal{X}_{\lambda, \text{pub}} \times \mathcal{X}_{\lambda, \text{pri}}$ for notational convenience.

3.6 Quasi-Adaptive Non-Interactive Zero-Knowledge Arguments

We recall the definition of a Quasi-Adaptive Non-Interactive Zero-Knowledge argument (QA-NIZK) for linear spaces over a bilinear group [JR13, KW15]. We follow the presentation of [ZZGQ23, ZLZ⁺24].

Definition 19 (QA-NIZK for Linear Spaces). Let $\mathbb{G} = \{\mathbb{G}_\lambda = (\mathbb{G}_{\lambda,1}, \mathbb{G}_{\lambda,2}, \mathbb{G}_{\lambda,t}, g_{\lambda,1}, g_{\lambda,2}, g_{\lambda,t}, e_\lambda, p_\lambda)\}_{\lambda \in \mathbb{N}}$ be a sequence of pairing groups. A QA-NIZK for linear spaces over $\mathbb{G}$ is a tuple of four efficient algorithms with the following syntax:

$\text{LGen}(1^\lambda, 1^n, 1^m, 1^\ell, [\mathbf{M}]_1) \rightarrow (\text{crs}, \text{td})$: On input the security parameter 1^λ , language parameters $1^n, 1^m, 1^\ell$ and a matrix $[\mathbf{M}]_1 \in \mathbb{G}_{\lambda,1}^{n \times m}$ defining a linear space, this algorithm outputs a common reference string crs and a trapdoor td .

$\text{LPrv}(\text{crs}, [\mathbf{Y}]_1, \mathbf{X}) \rightarrow \pi$: On input the crs , a matrix $[\mathbf{Y}]_1 \in \mathbb{G}_{\lambda,1}^{n \times \ell}$ and a matrix $\mathbf{X} \in \mathbb{Z}_{p_\lambda}^{m \times \ell}$ such that $\mathbf{Y} = \mathbf{M} \times \mathbf{X}$, this algorithm outputs a proof π .

$\text{LVer}(\text{crs}, [\mathbf{Y}]_1, \pi) \rightarrow b$: On input the crs , a matrix $[\mathbf{Y}]_1 \in \mathbb{G}_{\lambda,1}^{n \times \ell}$ and a proof π , this algorithm outputs $b \in \{0, 1\}$ indicating the validity of π .

$\text{LSim}(\text{crs}, \text{td}, [\mathbf{Y}]_1) \rightarrow \tilde{\pi}$: On input the crs with corresponding trapdoor td and a matrix $[\mathbf{Y}]_1 \in \mathbb{G}_{\lambda,1}^{n \times \ell}$, this algorithm outputs a simulated proof $\tilde{\pi}$.

We require the following properties for a QA-NIZK.

Perfect Completeness. A QA-NIZK Π is *perfectly complete* if for all $\lambda, n, m, \ell \in \mathbb{N}$, $\mathbf{M} \in \mathbb{Z}_{p_\lambda}^{n \times m}$, $\mathbf{X} \in \mathbb{Z}_{p_\lambda}^{m \times \ell}$ and $\mathbf{Y} \in \mathbb{Z}_{p_\lambda}^{n \times \ell}$ such that $\mathbf{Y} = \mathbf{M}\mathbf{X}$, it holds that

$$\Pr \left[\text{LVer}(\text{crs}, [\mathbf{Y}]_1, \pi) = 1 \mid \begin{array}{l} (\text{crs}, \top) \leftarrow \text{LGen}(1^\lambda, 1^n, 1^m, 1^\ell, [\mathbf{M}]_1) \\ \pi \leftarrow \text{LPrv}(\text{crs}, [\mathbf{Y}]_1, \mathbf{X}) \end{array} \right] = 1,$$

where the probability is taken over the random coins of the algorithms LGen and LPrv .

Perfect Zero-Knowledge. A QA-NIZK Π satisfies *perfect zero-knowledge* if for all $\lambda, n, m, \ell \in \mathbb{N}$, $\mathbf{M} \in \mathbb{Z}_{p_\lambda}^{n \times m}$, $(\text{crs}, \text{td}) \leftarrow \text{LGen}(1^\lambda, 1^n, 1^m, 1^\ell, [\mathbf{M}]_1)$, $\mathbf{X} \in \mathbb{Z}_{p_\lambda}^{m \times \ell}$ and $\mathbf{Y} \in \mathbb{Z}_{p_\lambda}^{n \times \ell}$ such that $\mathbf{Y} = \mathbf{M}\mathbf{X}$, it holds that

$$\text{LPrv}(\text{crs}, [\mathbf{Y}]_1, \mathbf{X}) \equiv \text{LSim}(\text{crs}, \text{td}, [\mathbf{Y}]_1),$$

where the probability is taken over the random coins of the algorithms LPrv and LSim .

Unbounded Simulation Soundness. A QA-NIZK Π is *unboundedly simulation sound* if for all PPT adversaries $\mathcal{A}$, it holds that $\Pr[\text{Exp}_{\Pi, \mathcal{A}}^{\text{USS}}(1^\lambda) = 1] = \text{negl}(\lambda)$, where $\text{Exp}_{\Pi, \mathcal{A}}^{\text{USS}}$ proceeds as follows.

- *Setup.* Launch $\mathcal{A}(\lambda)$ and receive from it $1^n, 1^m$ and 1^ℓ . Sample a matrix $\mathbf{M} \xleftarrow{\$} \mathbb{Z}_{p_\lambda}^{n \times m}$, run $(\text{crs}, \text{td}) \leftarrow \text{LGen}(1^\lambda, 1^n, 1^m, 1^\ell, [\mathbf{M}]_1)$ and send $(\text{crs}, \mathbf{M})$ to $\mathcal{A}$. Initialize $\mathcal{Q} = \emptyset$.
- *Query.* Repeat the following for arbitrarily many rounds determined by $\mathcal{A}$. In each round, upon $\mathcal{A}$ submitting a matrix $[\mathbf{Y}]_1 \in \mathbb{G}_{\lambda,1}^{n \times \ell}$, generate $\tilde{\pi} \leftarrow \text{LSim}(\text{crs}, \text{td}, [\mathbf{Y}]_1)$ and return $\tilde{\pi}$ to $\mathcal{A}$. Add $([\mathbf{Y}]_1, \tilde{\pi})$ to $\mathcal{Q}$.
- *Challenge.* The adversary submits a pair $([\mathbf{Y}^*]_1, \pi^*)$. The outcome of the experiment is 1 if $([\mathbf{Y}^*]_1, \pi^*) \notin \mathcal{Q}$, $\mathbf{Y}[i]^* \notin \text{span}(\mathbf{M})$ for some $i \in [\ell]$, and $\text{LVer}(\text{crs}, [\mathbf{Y}^*]_1, \pi^*) = 1$. Otherwise, the outcome is set to 0.

4 Adaptively IND-Secure sRFE for eAWS

In this section we present our adaptively IND-secure sRFE scheme for the eAWS functionality which is a generalization of the 1AWS functionality. In Section 5, we show how to extend this construction to AB-eAWS. There are two reason why we consider the construction without access control separately. First, we can prove security in this special case using only MDDH_k whereas the attribute-based version in Section 5 relies on bi-MDDH_k. Second, the secret key size in the non-attribute-based case depends only on the size m of the functions while in the attribute-based case it also depends on the length n of public inputs.

Construction 20 (Adaptively IND-secure sRFE for eAWS). The construction uses the following ingredients:

- A pairing group $\mathbb{G} = (\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_t, g_1, g_2, g_t, e, p)$.
- A QA-NIZK $\Pi = (\text{LGen}, \text{LPrv}, \text{LVer}, \text{LSim})$ for linear spaces over $\mathbb{G}$.

The details of the scheme go as follows:

Setup($1^\lambda, 1^L, \{\mathbf{q}_i\}_{1,2}^{\top} \}_{i \in [L]}$): On input the security parameter 1^λ , the maximum number of slots 1^L and a set of vectors $\{\mathbf{q}_i\}_{1,2}^{\top} \}_{i \in [L]}$ with $\mathbf{q}_i \in \mathbb{Z}_p^k$ for $i \in [L]$, sample vectors $\mathbf{r}_1, \dots, \mathbf{r}_L \xleftarrow{\$} \mathbb{Z}_p^k$ and matrices $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_p^{k \times (2k+1)}$, $\mathbf{B} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times k}$, $\mathbf{V} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)}$, $\mathbf{H}_1, \dots, \mathbf{H}_L \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times k(2k+1)}$, $\mathbf{R}_1, \dots, \mathbf{R}_L \xleftarrow{\$} \mathbb{Z}_p^{(2k+2) \times (2k+1)}$, $\mathbf{W}_1, \dots, \mathbf{W}_L \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)(m+n'-1)}$. Generate

$$(\text{crs}_i, \text{td}_i) \leftarrow \text{LGen}(1^\lambda, \mathbb{G}_1, \{\mathbf{A}_i\}_1),$$

where $\mathbf{A}_i^\top = [\mathbf{A}^\top \parallel \mathbf{R}_i^\top]$, and output the common reference string

$$\text{crs} = \left(\begin{array}{c} \{\mathbf{A}, \mathbf{AV}\}_1, \{\text{crs}_i, [\mathbf{AH}_i, \mathbf{AH}_i(\mathbf{q}_i(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{I}_{2k+1}), \mathbf{AW}_i, \mathbf{R}_i]\}_1\}_{i \in [L]} \\ \{\mathbf{B}\mathbf{r}_j, \mathbf{q}_j(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{B}\mathbf{r}_j\}_2\}_{j \in [L]}, \{\mathbf{H}_i(\mathbf{q}_i(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{B}\mathbf{r}_j), \mathbf{W}_i(\mathbf{I}_{m+n'-1} \otimes \mathbf{B}\mathbf{r}_j)\}_2\}_{\substack{i \in [L], \\ j \in [L] \setminus \{i\} \\ \kappa \in \{i, j\}}} \end{array} \right).$$

Gen(crs, i): On input the crs and an index $i \in [L]$, sample a matrix $\mathbf{U}_i \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)m}$ and generate $\pi_i \leftarrow \text{LPrv}(\text{crs}_i, \{\mathbf{A}_i \mathbf{U}_i\}_1, \mathbf{U}_i)$. Then output $\text{sk}_i = \mathbf{U}_i$ and

$$\text{pk}_i = ([\mathbf{T}_i]_1 := [\mathbf{AU}_i]_1, [\mathbf{Q}_i]_1 := [\mathbf{R}_i \mathbf{U}_i]_1, \{[\mathbf{G}_{i,j}]_2 := [\mathbf{U}_i(\mathbf{I}_m \otimes \mathbf{B}\mathbf{r}_j)]_2\}_{j \in [L] \setminus \{i\}}, \pi_i).$$

Ver($\text{crs}, i, \text{pk}_i$): On input the crs, an index $i \in [L]$ and a public key $\text{pk}_i = ([\mathbf{T}_i]_1, [\mathbf{Q}_i]_1, \{[\mathbf{G}_{i,j}]_2\}_{j \in [L] \setminus \{i\}}, \pi_i)$, define $\mathbf{F}_i^\top = [\mathbf{T}_i^\top \parallel \mathbf{Q}_i^\top]$ and check

$$\text{LVer}(\text{crs}_i, [\mathbf{F}_i]_1, \pi_i) \stackrel{?}{=} 1, \quad \{[\mathbf{T}_i]_1 [\mathbf{I}_m \otimes \mathbf{B}\mathbf{r}_j]_2 \stackrel{?}{=} [\mathbf{A}]_1 [\mathbf{G}_{i,j}]_2\}_{j \in [L] \setminus \{i\}}.$$

Output 1 if all checks pass, and 0 otherwise.

Agg($\text{crs}, (\text{pk}_i, h_i)_{i \in [L]}$): On input the crs and L tuples of the form (pk_i, h_i) with $h_i \in \mathcal{F}_{n,n'}^{\text{abp}}$, parse $\text{pk}_i = ([\mathbf{T}_i, \mathbf{Q}_i]_1, \{[\mathbf{G}_{i,j}]_2\}_{j \in [L] \setminus \{i\}}, \pi_i)$. Obtain the matrices $(\mathbf{L}_{i,0}, \mathbf{L}_{i,1}) \in \mathbb{Z}_p^{(m+n'-1) \times m} \times \mathbb{Z}_p^{(m+n'-1) \times mn}$ by running $(\mathbf{L}_{i,0}, \mathbf{L}_{i,1}) \leftarrow \text{lgen}(h_i)$ and output $(\text{mpk}, \{\text{hsk}_j\}_{j \in [L]})$ as follows:

$$\begin{aligned} \text{mpk} &= \left[\begin{array}{c} \mathbf{A}, \mathbf{AV}, \sum_{i \in [L]} \mathbf{AH}_i, \sum_{i \in [L]} \mathbf{AW}_i, \sum_{i \in [L]} \mathbf{AW}_i(\mathbf{L}_{i,1} \otimes \mathbf{I}_{2k+1}), \\ \sum_{i \in [L]} \mathbf{T}_i + \mathbf{AH}_i(\mathbf{q}_i(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{AW}_i(\mathbf{L}_{i,0} \otimes \mathbf{I}_{2k+1}) \end{array} \right]_1, \\ \text{hsk}_j &= \left[\begin{array}{c} \mathbf{B}\mathbf{r}_j, \mathbf{q}_j(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{B}\mathbf{r}_j, - \sum_{i \in [L] \setminus \{j\}} \mathbf{W}_i(\mathbf{I}_{n'} \otimes \mathbf{B}\mathbf{r}_j), \\ - \sum_{i \in [L] \setminus \{j\}} \mathbf{H}_i(\mathbf{q}_i(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{B}\mathbf{r}_j), \sum_{i \in [L] \setminus \{j\}} \mathbf{W}_i(\mathbf{I}_{m+n'-1} \otimes \mathbf{B}\mathbf{r}_j) \mathbf{L}_{i,1}, \\ \sum_{i \in [L] \setminus \{j\}} \mathbf{G}_{i,j} + \mathbf{H}_i(\mathbf{q}_i(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{B}\mathbf{r}_j) + \mathbf{W}_i(\mathbf{I}_{m+n'-1} \otimes \mathbf{B}\mathbf{r}_j) \mathbf{L}_{i,0} \end{array} \right]_2, \end{aligned}$$

where $\mathbf{W}_i \in \mathbb{Z}_p^{(2k+1) \times (2k+1)n'}$ refers to the matrix composed of the right most $(2k+1)n'$ columns of $\mathbf{W}_i \in \mathbb{Z}_p^{(2k+1) \times (2k+1)(m+n'-1)}$.

Enc(mpk, $\mathbf{x}$, ($\mathbf{p}$, $\mathbf{z}$)): On input the master public key mpk, a public input $\mathbf{x} \in \mathbb{Z}_p^n$ and a private input $(\mathbf{p}, \mathbf{z}) \in \mathbb{Z}_p^k \times \mathbb{Z}_p^{n'}$, sample $\mathbf{s} \xleftarrow{\$} \mathbb{Z}_p^k$ and output

$$\text{ct} = \left\| \begin{array}{l} \mathbf{s}^\top \mathbf{A}, \mathbf{z}^\top \otimes \mathbf{s}^\top \mathbf{A} \mathbf{V} - \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A} \mathbf{W}_i, \mathbf{p}^\top \otimes \mathbf{s}^\top \mathbf{A} \mathbf{V} - \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A} \mathbf{H}_i, \\ \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_i + \mathbf{A} \mathbf{H}_i(\mathbf{q}_i(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{A} \mathbf{W}_i(\mathbf{L}_{i,\mathbf{x}} \otimes \mathbf{I}_{2k+1}), \mathbf{s}^\top \mathbf{A} \mathbf{V} \end{array} \right\|_1,$$

where $\mathbf{L}_{j,\mathbf{x}} := \mathbf{L}_{j,0} + \mathbf{L}_{j,1}(\mathbf{x} \otimes \mathbf{I}_m)$.

Note. Observe that $\text{mpk} = \llbracket \mathbf{K}_0, \mathbf{K}_1, \mathbf{K}_2, \mathbf{K}_3, \mathbf{K}_4, \mathbf{K}_5 \rrbracket_1$ enables the computation of the second last ciphertext component as follows:

$$\begin{aligned} & \mathbf{s}^\top \llbracket \mathbf{K}_5 \rrbracket_1 + \mathbf{s}^\top \llbracket \mathbf{K}_4 \rrbracket_1 (\mathbf{x} \otimes \mathbf{I}_m) \otimes \mathbf{I}_{2k+1} \\ &= \left\| \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_i + \mathbf{A} \mathbf{H}_i(\mathbf{q}_i(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{A} \mathbf{W}_i(\mathbf{L}_{i,0} \otimes \mathbf{I}_{2k+1}) \right. \\ & \quad \left. + \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A} \mathbf{W}_i(\mathbf{L}_{i,1} \otimes \mathbf{I}_{2k+1})(\mathbf{x} \otimes \mathbf{I}_m) \otimes \mathbf{I}_{2k+1} \right\|_1 \\ &= \left\| \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_i + \mathbf{A} \mathbf{H}_i(\mathbf{q}_i(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{A} \mathbf{W}_i(\mathbf{L}_{i,0} \otimes \mathbf{I}_{2k+1} \right. \\ & \quad \left. + (\mathbf{L}_{i,1} \otimes \mathbf{I}_{2k+1})(\mathbf{x} \otimes \mathbf{I}_m) \otimes \mathbf{I}_{2k+1}) \right\|_1 \\ &= \left\| \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_i + \mathbf{A} \mathbf{H}_i(\mathbf{q}_i(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{A} \mathbf{W}_i(\mathbf{L}_{i,0} + \mathbf{L}_{i,1}(\mathbf{x} \otimes \mathbf{I}_m)) \otimes \mathbf{I}_{2k+1} \right\|_1 \\ &= \left\| \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_i + \mathbf{A} \mathbf{H}_i(\mathbf{q}_i(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{A} \mathbf{W}_i \mathbf{L}_{i,\mathbf{x}} \otimes \mathbf{I}_{2k+1} \right\|_1 \end{aligned}$$

Dec($\text{sk}_{i^*}, \text{hsk}_{i^*}, \text{ct}$): Given a secret key sk_{i^*} with corresponding helper secret key hsk_{i^*} and a ciphertext ct, parse $\text{hsk}_{i^*} = (\llbracket \mathbf{d}_0 \rrbracket_2, \llbracket \mathbf{D}_0 \rrbracket_2, \llbracket \mathbf{D}_1 \rrbracket_2, \llbracket \mathbf{D}_2 \rrbracket_2, \llbracket \mathbf{D}_3 \rrbracket_2, \llbracket \mathbf{D}_4 \rrbracket_2)$ and $\text{ct} = (\llbracket \mathbf{c}_0 \rrbracket_1, \llbracket \mathbf{c}_1 \rrbracket_1, \llbracket \mathbf{c}_2 \rrbracket_1, \llbracket \mathbf{c}_3 \rrbracket_1, \llbracket \mathbf{c}_4 \rrbracket_1)$. Compute

$$\begin{aligned} \llbracket \mathbf{c}'_1 \rrbracket_t &= \llbracket \mathbf{c}_1 \rrbracket_1 \llbracket \mathbf{I}_{n'} \otimes \mathbf{d}_0 \rrbracket_2, & \llbracket \mathbf{d}'_1 \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_1 \rrbracket_2, \\ \llbracket \mathbf{c}'_2 \rrbracket_t &= \llbracket \mathbf{c}_2 \rrbracket_1 \llbracket \mathbf{D}_0 \rrbracket_2, & \llbracket \mathbf{d}'_2 \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_2 \rrbracket_2, \\ \llbracket \mathbf{c}'_3 \rrbracket_t &= \llbracket \mathbf{c}_3 \rrbracket_1 \llbracket \mathbf{I}_m \otimes \mathbf{d}_0 \rrbracket_2, & \llbracket \mathbf{d}'_3 \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_3(\mathbf{x} \otimes \mathbf{I}_m) + \mathbf{D}_4 \rrbracket_2. \end{aligned}$$

Using $\text{sk}_{i^*} = \mathbf{U}_{i^*}$, compute the vectors

$$(\llbracket \mathbf{p}_1 \rrbracket_t, \llbracket \mathbf{p}_2 \rrbracket_t) = (\llbracket \mathbf{c}'_1 - \mathbf{d}'_1 \rrbracket_t, \llbracket (\mathbf{c}'_2 - \mathbf{d}'_2) + (\mathbf{c}'_3 - \mathbf{d}'_3) \rrbracket_t - \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{U}_{i^*}(\mathbf{I}_m \otimes \mathbf{d}_0) \rrbracket_2).$$

Run $\mathbf{d}_{h_{i^*}, \mathbf{x}} \leftarrow \text{rec}(h_{i^*}, \mathbf{x})$ and compute $\llbracket \nu' \rrbracket_t = \llbracket (\mathbf{p}_1^\top \parallel \mathbf{p}_2^\top) \cdot \mathbf{d}_{h_{i^*}, \mathbf{x}} \rrbracket_t$. Finally, output the discrete log ν of $\llbracket \nu' \rrbracket_t$ in basis $\llbracket \mathbf{c}_4 \rrbracket_1 \llbracket \mathbf{d}_0 \rrbracket_2$ recovered via brute-force.

Proposition 21 (Completeness, Correctness and Compactness). *The sRFE scheme for the eAWS functionality in Construction 20 is complete, correct and compact.*

Proposition 22 (Security). *Suppose that Π is a QA-NIZK for linear spaces satisfying perfect completeness, perfect zero-knowledge and unbounded simulation soundness. Then the sRFE scheme in Construction 20 is adaptively IND-secure under the MDDH_k assumption on $\mathbb{G}$.*

The proofs of Propositions 21 and 22 can be found in Sections 4.1 and 4.2, respectively. When instantiating Π with the pairing-based QA-NIZK by Kiltz and Wee [KW15] and using the transformation of Theorem 18, we obtain the following theorem, where we note that 1AWS is a special case of eAWS.

Theorem 23. *Assuming MDDH_k on pairings, there exists an adaptively IND-secure (bounded) RFE scheme for the 1AWS functionality.*

4.1 Proof of Completeness, Correctness and Compactness

Proof of Proposition 21. We argue that Construction 20 is complete, correct and compact.

Completeness. For all $\lambda, L \in \mathbb{N}$, $i \in [L]$, $\mathbf{q}_1, \dots, \mathbf{q}_L \in \mathbb{Z}_p^k$, all $\text{crs} \leftarrow \text{Setup}(1^\lambda, 1^L, \{\llbracket \mathbf{q}_i \rrbracket_{1,2}\}_{i \in [L]})$ and $(\text{pk}_i, \text{sk}_i) \leftarrow \text{Gen}(\text{crs}, i)$, we have

$$\text{pk}_i = (\llbracket \mathbf{T}_i \rrbracket_1 := \llbracket \mathbf{A}\mathbf{U}_i \rrbracket_1, \llbracket \mathbf{Q}_i \rrbracket_1 := \llbracket \mathbf{R}_i\mathbf{U}_i \rrbracket_1, \{\llbracket \mathbf{G}_{i,j} \rrbracket_2 := \llbracket \mathbf{U}_i(\mathbf{I}_m \otimes \mathbf{B}\mathbf{r}_j) \rrbracket_2\}_{j \in [L] \setminus \{i\}}, \pi_i)$$

for some $\mathbf{U}_i \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)m}$ and $\pi_i \leftarrow \text{LPrv}(\text{crs}_i, \llbracket \mathbf{A}\mathbf{U}_i \rrbracket_1, \mathbf{U}_i)$, where $(\text{crs}_i, \text{td}_i) \leftarrow \text{LGen}(1^\lambda, \mathbb{G}_1, \llbracket \mathbf{A} \rrbracket_1)$ and $\mathbf{A}_i^\top = [\mathbf{A}^\top \parallel \mathbf{R}_i^\top]$. Let $\mathbf{F}_i^\top = [\mathbf{T}_i^\top \parallel \mathbf{Q}_i^\top]$. We can observe that

- $\text{LVer}(\text{crs}_i, \llbracket \mathbf{F}_i \rrbracket_1, \pi_i) = 1$ from the correctness of Π and the fact that $\mathbf{F}_i = \mathbf{A}_i\mathbf{U}_i$, and
- for each $j \in [L] \setminus \{i\}$, we have $\llbracket \mathbf{T}_i \rrbracket_1 \llbracket \mathbf{I}_m \otimes \mathbf{B}\mathbf{r}_j \rrbracket_2 = \llbracket \mathbf{A} \rrbracket_1 \llbracket \mathbf{G}_{i,j} \rrbracket_2$ which follows from the fact that $\mathbf{A}\mathbf{U}_i \cdot (\mathbf{I}_m \otimes \mathbf{B}\mathbf{r}_j) = \mathbf{A} \cdot \mathbf{U}_i(\mathbf{I}_m \otimes \mathbf{B}\mathbf{r}_j)$.

This implies that $\text{Ver}(\text{crs}, i, \text{pk}_i) = 1$ and concludes the proof of completeness.

Correctness. For all $\lambda, L \in \mathbb{N}$, $i^* \in [L]$, $\mathbf{q}_1, \dots, \mathbf{q}_L \in \mathbb{Z}_p^k$, all $\text{crs} \leftarrow \text{Setup}(1^\lambda, 1^L, \{\llbracket \mathbf{q}_i \rrbracket_{1,2}\}_{i \in [L]})$ and $(\text{pk}_{i^*}, \text{sk}_{i^*}) \leftarrow \text{Gen}(\text{crs}, i^*)$, all $\{\text{pk}_i\}_{i \in [L] \setminus \{i^*\}}$ such that $\text{Ver}(\text{crs}, i, \text{pk}_i) = 1$, all $h_1, \dots, h_L \in \mathcal{F}_{n,n'}^{\text{abp}}$, $\mathbf{x} \in \mathbb{Z}_p^n$ and $(\mathbf{p}, \mathbf{z}) \in \mathbb{Z}_p^k \times \mathbb{Z}_p^{n'}$, all $(\text{mpk}, \{\text{hsk}_j\}_{j \in [L]}) \leftarrow \text{Agg}(\text{crs}, (\text{pk}_i, h_i)_{i \in [L]})$ and $\text{ct} \leftarrow \text{Enc}(\text{mpk}, \mathbf{x}, (\mathbf{p}, \mathbf{z}))$, we have $\text{sk}_{i^*} = \mathbf{U}_{i^*}$ and

$$\begin{aligned} (\mathbf{c}_0, \mathbf{c}_1, \mathbf{c}_2) &= \left(\mathbf{s}^\top \mathbf{A}, \mathbf{z}^\top \otimes \mathbf{s}^\top \mathbf{A}\mathbf{V} - \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A}\mathbf{W}_i, \mathbf{p}^\top \otimes \mathbf{s}^\top \mathbf{A}\mathbf{V} - \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A}\mathbf{H}_i \right) \\ (\mathbf{c}_3, \mathbf{c}_4) &= \left(\mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_i + \mathbf{A}\mathbf{H}_i(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{A}\mathbf{W}_i(\mathbf{L}_{i,\mathbf{x}} \otimes \mathbf{I}_{2k+1}), \mathbf{s}^\top \mathbf{A}\mathbf{V} \right) \\ (\mathbf{d}_0, \mathbf{D}_0, \mathbf{D}_1) &= \left(\mathbf{B}\mathbf{r}_{i^*}, \mathbf{q}_{i^*} \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_{i^*}, - \sum_{i \in [L] \setminus \{i^*\}} \mathbf{W}_i(\mathbf{I}_{n'} \otimes \mathbf{B}\mathbf{r}_{i^*}) \right) \\ (\mathbf{D}_2, \mathbf{D}_3) &= \left(- \sum_{i \in [L] \setminus \{i^*\}} \mathbf{H}_i(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_{i^*}), \sum_{i \in [L] \setminus \{i^*\}} \mathbf{W}_i(\mathbf{I}_{m+n'-1} \otimes \mathbf{B}\mathbf{r}_{i^*}) \mathbf{L}_{i,1} \right) \\ \mathbf{D}_4 &= \sum_{i \in [L] \setminus \{i^*\}} \mathbf{G}_{i,i^*} + \mathbf{H}_i(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_{i^*}) + \mathbf{W}_i(\mathbf{I}_{m+n'-1} \otimes \mathbf{B}\mathbf{r}_{i^*}) \mathbf{L}_{i,0}, \end{aligned}$$

where $\text{ct} = (\llbracket \mathbf{c}_0 \rrbracket_1, \llbracket \mathbf{c}_1 \rrbracket_1, \llbracket \mathbf{c}_2 \rrbracket_1, \llbracket \mathbf{c}_3 \rrbracket_1, \llbracket \mathbf{c}_4 \rrbracket_1)$ and $\text{hsk}_{i^*} = (\llbracket \mathbf{d}_0 \rrbracket_2, \llbracket \mathbf{D}_0 \rrbracket_2, \llbracket \mathbf{D}_1 \rrbracket_2, \llbracket \mathbf{D}_2 \rrbracket_2, \llbracket \mathbf{D}_3 \rrbracket_2, \llbracket \mathbf{D}_4 \rrbracket_2)$. By computing

$$\begin{aligned} \llbracket \mathbf{c}'_1 \rrbracket_t &= \llbracket \mathbf{c}_1 \rrbracket_1 \llbracket \mathbf{I}_{n'} \otimes \mathbf{d}_0 \rrbracket_2, & \llbracket \mathbf{d}'_1 \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_1 \rrbracket_2, \\ \llbracket \mathbf{c}'_2 \rrbracket_t &= \llbracket \mathbf{c}_2 \rrbracket_1 \llbracket \mathbf{D}_0 \rrbracket_2, & \llbracket \mathbf{d}'_2 \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_2 \rrbracket_2, \\ \llbracket \mathbf{c}'_3 \rrbracket_t &= \llbracket \mathbf{c}_3 \rrbracket_1 \llbracket \mathbf{I}_m \otimes \mathbf{d}_0 \rrbracket_2, & \llbracket \mathbf{d}'_3 \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_3(\mathbf{x} \otimes \mathbf{I}_m) + \mathbf{D}_4 \rrbracket_2, \end{aligned}$$

the decryptor obtains

$$\begin{aligned} \mathbf{c}'_1 &= \left(\mathbf{z}^\top \otimes \mathbf{s}^\top \mathbf{A}\mathbf{V} - \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A}\mathbf{W}_i \right) (\mathbf{I}_{n'} \otimes \mathbf{B}\mathbf{r}_{i^*}) \\ &= \mathbf{z}^\top \cdot \mathbf{s}^\top \mathbf{A}\mathbf{V}\mathbf{B}\mathbf{r}_{i^*} - \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A}\mathbf{W}_i(\mathbf{I}_{n'} \otimes \mathbf{B}\mathbf{r}_{i^*}) \\ \mathbf{d}'_1 &= -\mathbf{s}^\top \mathbf{A} \sum_{i \in [L] \setminus \{i^*\}} \mathbf{W}_i(\mathbf{I}_{n'} \otimes \mathbf{B}\mathbf{r}_{i^*}) \\ \mathbf{c}'_2 &= \left(\mathbf{p}^\top \otimes \mathbf{s}^\top \mathbf{A}\mathbf{V} - \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A}\mathbf{H}_i \right) (\mathbf{q}_{i^*} \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_{i^*}) \\ &= \mathbf{p}^\top \mathbf{q}_{i^*} \cdot \mathbf{s}^\top \mathbf{A}\mathbf{V}\mathbf{B}\mathbf{r}_{i^*} \cdot \mathbf{e}_1^\top - \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A}\mathbf{H}_i(\mathbf{q}_{i^*} \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_{i^*}) \\ \mathbf{d}'_2 &= -\mathbf{s}^\top \mathbf{A} \sum_{i \in [L] \setminus \{i^*\}} \mathbf{H}_i(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_{i^*}) \\ \mathbf{c}'_3 &= \mathbf{s}^\top \sum_{i \in [L]} \left(\mathbf{T}_i + \mathbf{A}\mathbf{H}_i(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{A}\mathbf{W}_i(\mathbf{L}_{i,\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \right) (\mathbf{I}_m \otimes \mathbf{B}\mathbf{r}_{i^*}) \end{aligned}$$

$$\begin{aligned}
&= \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_i (\mathbf{I}_m \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{A} \mathbf{H}_i (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{A} \mathbf{W}_i (\mathbf{L}_{i,\mathbf{x}} \otimes \mathbf{B} \mathbf{r}_{i^*}) \\
&= \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_i (\mathbf{I}_m \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{A} \mathbf{H}_i (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{A} \mathbf{W}_i (\mathbf{I}_{m+n'-1} \otimes \mathbf{B} \mathbf{r}_{i^*}) \mathbf{L}_{i,\mathbf{x}} \\
\mathbf{d}'_3 &= \mathbf{s}^\top \mathbf{A} \sum_{i \in [L] \setminus \{i^*\}} \mathbf{W}_i (\mathbf{I}_{m+n'-1} \otimes \mathbf{B} \mathbf{r}_{i^*}) \mathbf{L}_{i,1} (\mathbf{x} \otimes \mathbf{I}_m) \\
&\quad + \mathbf{s}^\top \mathbf{A} \sum_{i \in [L] \setminus \{i^*\}} \mathbf{G}_{i,i^*} + \mathbf{H}_i (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{W}_i (\mathbf{I}_{m+n'-1} \otimes \mathbf{B} \mathbf{r}_{i^*}) \mathbf{L}_{i,0} \\
&= \mathbf{s}^\top \mathbf{A} \sum_{i \in [L] \setminus \{i^*\}} \mathbf{G}_{i,i^*} + \mathbf{H}_i (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{W}_i (\mathbf{I}_{m+n'-1} \otimes \mathbf{B} \mathbf{r}_{i^*}) \mathbf{L}_{i,\mathbf{x}} .
\end{aligned}$$

Since the pair $(\mathbf{pk}_{i^*}, \mathbf{sk}_{i^*})$ was generated honestly, there exists a matrix $\mathbf{U}_{i^*}$ such that

$$\mathbf{pk}_{i^*} = (\llbracket \mathbf{T}_{i^*} \rrbracket_1 := \llbracket \mathbf{A} \mathbf{U}_{i^*} \rrbracket_1, \llbracket \mathbf{Q}_{i^*} \rrbracket_1 := \llbracket \mathbf{R}_{i^*} \mathbf{U}_{i^*} \rrbracket_1, \{\llbracket \mathbf{G}_{i,i^*} \rrbracket_2 := \llbracket \mathbf{U}_{i^*} (\mathbf{I}_m \otimes \mathbf{B} \mathbf{r}_j) \rrbracket_2\}_{i \in [L] \setminus \{i^*\}}, \pi_{i^*})$$

and $\mathbf{sk}_{i^*} = \mathbf{U}_{i^*}$. Furthermore, verification guarantees for all $i \in [L] \setminus \{i^*\}$ that

$$\llbracket \mathbf{T}_i \rrbracket_1 \llbracket \mathbf{I}_{n'+1} \otimes \mathbf{B} \mathbf{r}_i^* \rrbracket_2 \stackrel{?}{=} \llbracket \mathbf{A} \rrbracket_1 \llbracket \mathbf{G}_{i,i^*} \rrbracket_2 .$$

Therefore, the vector $(\llbracket \mathbf{p}_1^\top \rrbracket_t, \llbracket \mathbf{p}_2^\top \rrbracket_t)$ equals

$$\begin{aligned}
\llbracket \mathbf{p}_1^\top \rrbracket_t &= \llbracket \mathbf{c}'_1 - \mathbf{d}'_1 \rrbracket_t \\
&= \mathbf{z}^\top \cdot \mathbf{s}^\top \mathbf{A} \mathbf{V} \mathbf{B} \mathbf{r}_{i^*} - \mathbf{s}^\top \mathbf{A} \mathbf{W}_{i^*} (\mathbf{I}_{n'} \otimes \mathbf{B} \mathbf{r}_{i^*}) \\
\llbracket \mathbf{p}_2^\top \rrbracket_t &= \llbracket (\mathbf{c}'_2 - \mathbf{d}'_2) + (\mathbf{c}'_3 - \mathbf{d}'_3) \rrbracket_t - \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{U}_{i^*} \cdot (\mathbf{I}_m \otimes \mathbf{d}_0) \rrbracket_2 \\
&= \llbracket \mathbf{s}^\top \mathbf{A} \mathbf{W}_{i^*} (\mathbf{I}_{m+n'-1} \otimes \mathbf{B} \mathbf{r}_{i^*}) \mathbf{L}_{i^*,\mathbf{x}} + \mathbf{p}^\top \mathbf{q}_{i^*} \cdot \mathbf{s}^\top \mathbf{A} \mathbf{V} \mathbf{B} \mathbf{r}_{i^*} \cdot \mathbf{e}_1^\top \rrbracket_t ,
\end{aligned}$$

and we observe that

$$(\mathbf{p}_1^\top, \mathbf{p}_2^\top) = \text{pgb}(h_{i^*}, \mathbf{x}, \mathbf{z}^\top \cdot \mathbf{s}^\top \mathbf{A} \mathbf{V} \mathbf{B} \mathbf{r}_{i^*}) + (0, \mathbf{p}^\top \mathbf{q}_{i^*} \cdot \mathbf{s}^\top \mathbf{A} \mathbf{V} \mathbf{B} \mathbf{r}_{i^*} \cdot \mathbf{e}_1^\top) .$$

Then it follows from the correctness of the garbling scheme that

$$v' = (h_{i^*}(\mathbf{x}))^\top \mathbf{z} + \mathbf{p}^\top \mathbf{q}_{i^*} \cdot \mathbf{s}^\top \mathbf{A} \mathbf{V} \mathbf{B} \mathbf{r}_{i^*} ,$$

which implies that the recovered discrete log in basis $\llbracket \mathbf{c}_4 \rrbracket_1 \llbracket \mathbf{d}_0 \rrbracket_2 = \llbracket \mathbf{s}^\top \mathbf{A} \mathbf{V} \mathbf{B} \mathbf{r}_{i^*} \rrbracket_t$ is $v = h_{i^*}(\mathbf{x})^\top \mathbf{z} + \mathbf{p}^\top \mathbf{q}_{i^*}$.

Compactness. Let $j \in [L]$. The scheme has the following properties:

$$\begin{aligned}
|\text{crs}| &= L^2 \cdot (m + n') \cdot \text{poly}(\lambda) & |\text{mpk}| &= (mn + n') \cdot \text{poly}(\lambda) \\
|\text{hsk}_j| &= (mn + n') \cdot \text{poly}(\lambda) & |\text{ct}| &= (m + n') \cdot \text{poly}(\lambda) ,
\end{aligned}$$

where the size of crs follows from the fact that the total size of $\{\text{crs}_i\}_{i \in [L]}$ is $L \cdot \text{poly}(\lambda)$ if we instantiate Π with the QA-NIZK of [KW15], and the fact that the size of the language description is $\text{poly}(\lambda)$. □

4.2 Proof of Security

We will need the following lemma in our proof.

Lemma 24. Let $(\mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3) \stackrel{\$}{\leftarrow} \mathbb{Z}_p^{2k+1 \times k} \times \mathbb{Z}_p^{2k+1} \times \mathbb{Z}_p^{2k+1 \times k}$ be a random basis and let $(\mathbf{B}_1^\parallel, \mathbf{B}_2^\parallel, \mathbf{B}_3^\parallel)$ denote its dual. Furthermore, let $\mathbf{d}^\perp \in \text{span}(\mathbf{B}_3^\parallel)$ such that $\mathbf{B}_1^\top \mathbf{d}^\perp = \mathbf{0}$ and $\mathbf{B}_2^\top \mathbf{d}^\perp = \mathbf{0}$. Then it follows from the MDDH_k assumption in $\mathbb{G}_1$ that the following distributions are indistinguishable:

$$\{C, \llbracket \mathbf{R} \rrbracket_1, \llbracket \mathbf{R} \mathbf{U} \rrbracket_1\} \approx_c \{C, \llbracket \mathbf{R} \rrbracket_1, \llbracket \mathbf{R} \mathbf{U} + \mathbf{P} \otimes (\mathbf{d}^\perp)^\top \rrbracket_1\} ,$$

where $\mathbf{R} \stackrel{\$}{\leftarrow} \mathbb{Z}_p^{(2k+2) \times (2k+1)}$, $\mathbf{U} \stackrel{\$}{\leftarrow} \mathbb{Z}_p^{(2k+1) \times (2k+1)m}$, $\mathbf{P} \stackrel{\$}{\leftarrow} \mathbb{Z}_p^{(2k+2) \times m}$, $\mathbf{A} \stackrel{\$}{\leftarrow} \mathbb{Z}_p^{k \times (2k+1)}$, $\mathbf{c} \stackrel{\$}{\leftarrow} \mathbb{Z}_p^{2k+1}$ and

$$C = \{\mathbf{A}, \mathbf{A} \mathbf{U}, \mathbf{c}, \mathbf{c}^\top \mathbf{U}, \mathbf{B}_1, \mathbf{U} (\mathbf{I}_m \otimes \mathbf{B}_1), \mathbf{B}_2, \mathbf{U} (\mathbf{I}_m \otimes \mathbf{B}_2), \mathbf{d}^\perp\} .$$

Proof. Let $\mathbf{D} \xleftarrow{s} \mathbb{Z}_p^{k \times (2k+1)}$, $\tilde{\mathbf{R}} \xleftarrow{s} \mathbb{Z}_p^{(2k+2) \times k}$ and $\tilde{\mathbf{P}} \xleftarrow{s} \mathbb{Z}_p^{k \times m}$. We consider the following intermediate distributions:

$$\begin{aligned} \{C, \llbracket \mathbf{R} \rrbracket_1, \llbracket \mathbf{R}\mathbf{U} \rrbracket_1\} &\approx_c \{C, \llbracket \tilde{\mathbf{R}}\mathbf{D} \rrbracket_1, \llbracket \tilde{\mathbf{R}}\mathbf{D}\mathbf{U} \rrbracket_1\} \\ &\approx_s \{C, \llbracket \tilde{\mathbf{R}}\mathbf{D} \rrbracket_1, \llbracket \tilde{\mathbf{R}}\mathbf{D}\mathbf{U} + \tilde{\mathbf{R}}\tilde{\mathbf{P}} \otimes (\mathbf{d}^\perp)^\top \rrbracket_1\} \\ &\approx_c \{C, \llbracket \mathbf{R} \rrbracket_1, \llbracket \mathbf{R}\mathbf{U} + \mathbf{P} \otimes (\mathbf{d}^\perp)^\top \rrbracket_1\}. \end{aligned}$$

The first (computational) indistinguishability follows from the MDDH_k assumption w.r.t. $\llbracket \tilde{\mathbf{R}} \rrbracket_1$ which guarantees that $(\llbracket \tilde{\mathbf{R}} \rrbracket_1, \llbracket \mathbf{R} \rrbracket_1) \approx_c (\llbracket \tilde{\mathbf{R}} \rrbracket_1, \llbracket \tilde{\mathbf{R}}\mathbf{D} \rrbracket_1)$. The second (statistical) indistinguishability follows from the change of variables

$$\mathbf{U} \mapsto \mathbf{U} + \mathbf{D}^\perp \tilde{\mathbf{P}} \otimes (\mathbf{d}^\perp)^\top,$$

where $\mathbf{D}^\perp \in \mathbb{Z}_p^{(2k+1) \times k}$ such that $\mathbf{D}\mathbf{D}^\perp = \mathbf{I}_k$, $\mathbf{A}\mathbf{D}^\perp = \mathbf{0}$ and $\mathbf{c}^\top \mathbf{D}^\perp = \mathbf{0}$. Note that such a choice of $\mathbf{D}^\perp$ is possible since $[\mathbf{A}^\top \parallel \mathbf{c} \parallel \mathbf{D}^\top]$ has full rank with high probability. The third (computational) indistinguishability follows again from the MDDH_k assumption w.r.t. $\llbracket \tilde{\mathbf{R}} \rrbracket_1$ which ensures that $(\llbracket \tilde{\mathbf{R}} \rrbracket_1, \llbracket \tilde{\mathbf{R}}(\mathbf{D} \parallel \tilde{\mathbf{P}}) \rrbracket_1) \approx_c (\llbracket \tilde{\mathbf{R}} \rrbracket_1, \llbracket (\mathbf{R} \parallel \mathbf{P}) \rrbracket_1)$. $\square$

Proof of Proposition 22. Let FE denote the sRFE scheme in Construction 20. We consider a sequence of hybrid games $G_0^0, \dots, G_7^0, G_7^1, \dots, G_0^1$ where $G_0^b = \text{Exp}_{\text{FE}, \mathcal{A}}^{\text{srfc-b}}$. We argue $G_7^0 \equiv G_7^1$ and $G_{\ell-1}^b \approx_c G_\ell^b$ for $\ell \in [7]$ and $b \in \{0, 1\}$. Then the proposition follows via a hybrid argument. Modifications between consecutive games are highlighted using gray boxes.

Game G_0^b for $b \in \{0, 1\}$: This is $\text{Exp}_{\text{FE}, \mathcal{A}}^{\text{srfc-b}}$. Specifically, we have:

- The common reference string crs is of the form

$$\text{crs} = \left(\begin{array}{c} \llbracket \mathbf{A}, \mathbf{A}\mathbf{V} \rrbracket_1, \{ \text{crs}_i, \llbracket \mathbf{A}\mathbf{H}_i, \mathbf{A}\mathbf{H}_i(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}), \mathbf{A}\mathbf{W}_i, \mathbf{R}_i \rrbracket_1 \}_{i \in [L]} \\ \{ \llbracket \mathbf{B}\mathbf{r}_j, \mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j \rrbracket_2 \}_{j \in [L]}, \\ \{ \llbracket \mathbf{H}_i(\mathbf{q}_k \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j), \mathbf{W}_i(\mathbf{I}_{m+n'-1} \otimes \mathbf{B}\mathbf{r}_j) \rrbracket_2 \}_{\substack{i \in [L], \\ j \in [L] \setminus \{i\}, \\ \kappa \in \{i, j\}}} \end{array} \right),$$

where $(\text{crs}_i, \text{td}_i) \leftarrow \text{LGen}(1^\lambda, \mathbb{G}_1, \llbracket \mathbf{A}_i \rrbracket_1)$ and $\mathbf{A}_i^\top = [\mathbf{A}^\top \parallel \mathbf{R}_i^\top]$.

- For each $i \in [L]$, each $\text{pk}_i \in \mathcal{D}_i$ is of the form

$$\text{pk}_i = (\llbracket \mathbf{T}_i \rrbracket_1 := \llbracket \mathbf{A}\mathbf{U}_i \rrbracket_1, \llbracket \mathbf{Q}_i \rrbracket_1 := \llbracket \mathbf{R}_i \mathbf{U}_i \rrbracket_1, \{ \llbracket \mathbf{G}_{i,j} \rrbracket_2 := \llbracket \mathbf{U}_i(\mathbf{I}_m \otimes \mathbf{B}\mathbf{r}_j) \rrbracket_2 \}_{j \in [L] \setminus \{i\}}, \pi_i),$$

where $\pi_i \leftarrow \text{LPrv}(\text{crs}_i, \llbracket \mathbf{F}_i \rrbracket_1, \mathbf{U}_i)$, $\mathbf{F}_i^\top = [\mathbf{T}_i^\top \parallel \mathbf{Q}_i^\top]$ and $\text{sk}_i = \mathbf{U}_i$ is the corresponding secret key.

- For each $i \in [L]$, pk_i^* is of the form

$$\text{pk}_i^* = (\llbracket \mathbf{T}_i^* \rrbracket_1, \llbracket \mathbf{Q}_i^* \rrbracket_1, \{ \llbracket \mathbf{G}_{i,j}^* \rrbracket_2 \}_{j \in [L] \setminus \{i\}}, \pi_i^*)$$

such that $\text{Ver}(\text{crs}, i, \text{pk}_i^*) = 1$, i.e. $\text{LVer}(\text{crs}_i, \llbracket \mathbf{F}_i^* \rrbracket_1, \pi_i^*) = 1$ for $\mathbf{F}_i^* = [(\mathbf{T}_i^*)^\top \parallel (\mathbf{Q}_i^*)^\top]^\top$ and $\mathbf{A}\mathbf{G}_{i,j}^* = \mathbf{T}_i^* \mathbf{B}\mathbf{r}_j$ for each $j \in [L] \setminus \{i\}$.

- The challenge ciphertext $\text{ct}^* = (\llbracket \mathbf{c}_0^*, \mathbf{c}_1^*, \mathbf{c}_2^*, \mathbf{c}_3^*, \mathbf{c}_4^* \rrbracket_1)$ for $(\mathbf{x}, (\mathbf{z}^0, \mathbf{p}^0))$ and $(\mathbf{x}, (\mathbf{z}^1, \mathbf{p}^1))$ is of the form

$$\begin{aligned} \mathbf{c}_0^* &= \mathbf{s}^\top \mathbf{A} \\ \mathbf{c}_1^* &= (\mathbf{z}^b)^\top \otimes \mathbf{s}^\top \mathbf{A}\mathbf{V} - \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A}\mathbf{W}_i \\ \mathbf{c}_2^* &= (\mathbf{p}^b)^\top \otimes \mathbf{s}^\top \mathbf{A}\mathbf{V} - \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A}\mathbf{H}_i \\ \mathbf{c}_3^* &= \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_i + \mathbf{A}\mathbf{H}_i(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{A}\mathbf{W}_i(\mathbf{L}_{i,\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \\ \mathbf{c}_4^* &= \mathbf{s}^\top \mathbf{A}\mathbf{V}. \end{aligned}$$

Game G_1^b for $b \in \{0, 1\}$: This is the same as G_0^b , except that for $i \in [L]$ and $\text{pk}_i \in \mathcal{D}_i$, the challenger computes

$$\pi_i \leftarrow \text{LSim}(\text{crs}_i, \text{td}_i, \llbracket \mathbf{F}_i \rrbracket_1) .$$

We have $G_0^b \equiv G_1^b$ which follows from the perfect zero-knowledge of Π .

Game G_2^b for $b \in \{0, 1\}$: This is the same as G_1^b , except that the challenger samples $\mathbf{s} \xleftarrow{\$} \mathbb{Z}_p^k$ and $\tilde{\mathbf{R}}_i \xleftarrow{\$} \mathbb{Z}_p^{(2k+2) \times (2k+2)}$ for $i \in [L]$ along with $\mathbf{A}$, and sets

$$\mathbf{R}_i = \tilde{\mathbf{R}}_i \cdot \begin{bmatrix} \mathbf{s}^\top \mathbf{A} \\ \mathbf{I}_{2k+1} \end{bmatrix} .$$

We observe that the distribution of $\mathbf{R}_i$ does not change from G_1^b to G_2^b (more precisely, $\mathbf{R}_i$ is uniformly random in both games). Hence, we have $G_1^b \equiv G_2^b$.

Game G_3^b for $b \in \{0, 1\}$: This is the same as G_2^b , except that the challenger generates $\mathbf{c}_3^*$ as follows:

$$\mathbf{c}_3^* = \sum_{i \in [L]} (\mathbf{e}_1^{(2k+2)})^\top \tilde{\mathbf{R}}_i^{-1} \mathbf{Q}_i^* + \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A} \mathbf{H}_i (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{A} \mathbf{W}_i (\mathbf{L}_{i,\mathbf{x}} \otimes \mathbf{I}_{2k+1}) .$$

We have $G_2^b \approx_c G_3^b$ which can be seen as follows. First, for $i \in [L]$ such that $\text{pk}_i^* \in \mathcal{D}_i^*$, it is not hard to observe that the distributions of $\mathbf{s}^\top \mathbf{T}_i^*$ and $\mathbf{e}_1^\top \tilde{\mathbf{R}}_i^{-1} \mathbf{Q}_i^*$ are identical. Second, for $i \in [L]$ such that $\text{pk}_i^* \notin \mathcal{D}_i^*$, we rely on the unbounded simulation soundness of Π . In the reduction, the challenger of Π provides $\mathbf{A}_i$ and crs_i . The simulation proceeds as in G_3^b except that, for all $\text{pk}_i \in \mathcal{D}_i$, the simulator submits an oracle query $\llbracket \mathbf{F}_i \rrbracket_1$ to obtain π_i . Finally, upon $\mathcal{A}$ registering pk_i^* , the simulator outputs $(\llbracket \mathbf{F}_i^* \rrbracket_1, \pi_i^*)$. To conclude, we observe that if $\mathbf{e}_1^\top \tilde{\mathbf{R}}_i^{-1} \mathbf{Q}_i^* \neq \mathbf{s}^\top \mathbf{T}_i^*$, then $\mathbf{F}_i^* \notin \text{span}(\mathbf{A}_i)$.

Game G_4^b for $b \in \{0, 1\}$: This is the same as G_3^b , except that the challenger replaces $\mathbf{s}^\top \mathbf{A}$ with a uniformly random vector $\mathbf{c}^\top \xleftarrow{\$} \mathbb{Z}_p^{1 \times (2k+1)}$. Specifically, $\mathbf{R}_i$ and $\text{ct}^* = (\llbracket \mathbf{c}_0^*, \mathbf{c}_1^*, \mathbf{c}_2^*, \mathbf{c}_3^*, \mathbf{c}_4^* \rrbracket_1)$ are computed as follows:

$$\begin{aligned} \mathbf{R}_i &= \tilde{\mathbf{R}}_i \cdot \begin{bmatrix} \mathbf{c}^\top \\ \mathbf{I}_{2k+1} \end{bmatrix} \\ \mathbf{c}_0^* &= \mathbf{c}^\top \\ \mathbf{c}_1^* &= (\mathbf{z}^b)^\top \otimes \mathbf{c}^\top \mathbf{V} - \mathbf{c}^\top \sum_{i \in [L]} \mathbf{W}_i \\ \mathbf{c}_2^* &= (\mathbf{p}^b)^\top \otimes \mathbf{c}^\top \mathbf{V} - \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_i \\ \mathbf{c}_3^* &= \sum_{i \in [L]} \mathbf{e}_1^\top \tilde{\mathbf{R}}_i^{-1} \mathbf{Q}_i^* + \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_i (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_i (\mathbf{L}_{i,\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \\ \mathbf{c}_4^* &= \mathbf{c}^\top \mathbf{V} . \end{aligned}$$

It is not hard to see that we have $G_3^b \approx_c G_4^b$ under the MDDH $_k$ assumption in $\mathbb{G}_1$ which states that $(\llbracket \mathbf{A}^\top \rrbracket_1, \llbracket \mathbf{A}^\top \mathbf{s} \rrbracket_1) \approx_c (\llbracket \mathbf{A}^\top \rrbracket_1, \llbracket \mathbf{c} \rrbracket_1)$ for $\mathbf{c} \xleftarrow{\$} \mathbb{Z}_p^{2k+1}$.

Game G_5^b for $b \in \{0, 1\}$: This is the same as G_4^b , except that the challenger replaces $(\mathbf{A} \mathbf{V}, \mathbf{c}^\top \mathbf{V})$ with $(\tilde{\mathbf{V}}, \mathbf{v}^\top) \xleftarrow{\$}$

$\mathbb{Z}_p^{k \times (2k+1)} \times \mathbb{Z}_p^{1 \times (2k+1)}$. In particular, crs , $\mathbf{c}_1^*$, $\mathbf{c}_2^*$ and $\mathbf{c}_4^*$ are computed as follows:

$$\begin{aligned} \text{crs} &= \left(\begin{aligned} &\llbracket \mathbf{A}, \tilde{\mathbf{V}} \rrbracket_1, \{ \text{crs}_i, \llbracket \mathbf{A} \mathbf{H}_i, \mathbf{A} \mathbf{H}_i (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}), \mathbf{A} \mathbf{W}_i, \mathbf{R}_i \rrbracket_1 \}_{i \in [L]} \\ &\{ \llbracket \mathbf{B} \mathbf{r}_j, \mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_j \rrbracket_2 \}_{j \in [L]}, \\ &\{ \llbracket \mathbf{H}_i (\mathbf{q}_k \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_j), \mathbf{W}_i (\mathbf{I}_{m+n'-1} \otimes \mathbf{B} \mathbf{r}_j) \rrbracket_2 \}_{\substack{i \in [L], \\ j \in [L] \setminus \{i\}, \\ \kappa \in \{i, j\}}} \end{aligned} \right) \\ \mathbf{c}_1^* &= (\mathbf{z}^b)^\top \otimes \mathbf{v}^\top - \mathbf{c}^\top \sum_{i \in [L]} \mathbf{W}_i \\ \mathbf{c}_2^* &= (\mathbf{p}^b)^\top \otimes \mathbf{v}^\top - \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_i \\ \mathbf{c}_4^* &= \mathbf{v}^\top. \end{aligned}$$

We have $G_4^b \approx_s G_5^b$. To see this, we first observe that $\mathbf{V} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)}$ is sampled uniformly at random and not published anywhere else. Second, we note that both $\mathbf{A}$ and $\mathbf{c}^\top$ are full (row) rank with overwhelming probability, in which case $(\mathbf{A} \mathbf{V}, \mathbf{c}^\top \mathbf{V})$ are uniform in $\mathbb{Z}_p^{k \times (2k+1)} \times \mathbb{Z}_p^{1 \times (2k+1)}$.

Game G_6^b for $b \in \{0, 1\}$: This is the same as G_5^b except that, in addition to $\mathbf{B}_1 = \mathbf{B} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times k}$, the challenger also samples $\mathbf{B}_2 \xleftarrow{\$} \mathbb{Z}_p^{2k+1}$ and $\mathbf{B}_3 \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times k}$. Then it computes the dual basis $(\mathbf{B}_1^\parallel, \mathbf{B}_2^\parallel, \mathbf{B}_3^\parallel)$ and generates the ciphertext components $\mathbf{c}_1^*$ and $\mathbf{c}_2^*$ as follows:

$$\begin{aligned} \mathbf{c}_1^* &= (\mathbf{z}^b \otimes \mathbf{v}^{(1,3)})^\top + (\mathbf{z}^0 \otimes \mathbf{v}^{(2)})^\top - \mathbf{c}^\top \sum_{i \in [L]} \mathbf{W}_i \\ \mathbf{c}_2^* &= (\mathbf{p}^b \otimes \mathbf{v}^{(1,3)})^\top + (\mathbf{p}^0 \otimes \mathbf{v}^{(2)})^\top - \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_i. \end{aligned}$$

Below, we prove the following claim.

Claim 25. *We have $G_5^b \equiv G_6^b$.*

Game $G_{6,\ell}^b$ for $\ell \in [0; L]$ and $b \in \{0, 1\}$: This is the same as G_6^b except that, for $j \in [\ell]$, the challenger replaces $\mathbf{B}_1 \mathbf{r}_j$ with $\mathbf{d}_j \xleftarrow{\$} \text{span}(\mathbf{B}_2)$. It is not hard to see that $G_{6,0}^b \equiv G_6^b$. This is because $[0] = \emptyset$, so the two games are identical. Furthermore, we prove the following claim for each $\ell \in [L]$ and $b \in \{0, 1\}$.

Claim 26. *We have $G_{6,\ell-1}^b \approx_c G_{6,\ell}^b$ assuming MDDH $_k$ in G_1 and G_2 .*

Game G_7^b for $b \in \{0, 1\}$: This is the same as $G_{6,L}^b$ except that we generate the ciphertext components $\mathbf{c}_1^*$ and $\mathbf{c}_2^*$ as follows:

$$\begin{aligned} \mathbf{c}_1^* &= (\mathbf{z}^0 \otimes \mathbf{v}^{(1,3)})^\top + (\mathbf{z}^0 \otimes \mathbf{v}^{(2)})^\top - \mathbf{c}^\top \sum_{i \in [L]} \mathbf{W}_i \\ \mathbf{c}_2^* &= (\mathbf{p}^0 \otimes \mathbf{v}^{(1,3)})^\top + (\mathbf{p}^0 \otimes \mathbf{v}^{(2)})^\top - \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_i. \end{aligned}$$

We have $G_{6,L}^b \approx_s G_7^b$ from a statistical argument similar to the proof of Claim 25. Furthermore, we can observe that the challenger's output in G_7^b is independent of the challenge bit b , so it follows $G_7^0 \equiv G_7^1$. This concludes the proof of the proposition. $\square$

We now turn to the claims.

Proof of Claim 25. The claim follows from the fact that the basis $\mathbf{B}_2$ and its dual $\mathbf{B}_2^\parallel$ are not revealed. Therefore, $(\mathbf{c}^\top \mathbf{W}_i)^{(2)}$ and $(\mathbf{c}^\top \mathbf{H}_i)^{(2)}$ are hidden which implies that they can mask $(\mathbf{z}^b \otimes \mathbf{v}^{(2)})^\top$ and $(\mathbf{p}^b \otimes \mathbf{v}^{(2)})^\top$, i.e. the

following distributions are identical:

$$\begin{aligned} \left\{ (\mathbf{z}^b \otimes \mathbf{v}^{(2)})^\top - \sum_{i \in [L]} (\mathbf{c}^\top \mathbf{W}_i)^{(2)} \right\} &\equiv \left\{ - \sum_{i \in [L]} (\mathbf{c}^\top \mathbf{W}_i)^{(2)} \right\} \equiv \left\{ (\mathbf{z}^0 \otimes \mathbf{v}^{(2)})^\top - \sum_{i \in [L]} (\mathbf{c}^\top \mathbf{W}_i)^{(2)} \right\} \\ \left\{ (\mathbf{p}^b \otimes \mathbf{v}^{(2)})^\top - \sum_{i \in [L]} (\mathbf{c}^\top \mathbf{H}_i)^{(2)} \right\} &\equiv \left\{ - \sum_{i \in [L]} (\mathbf{c}^\top \mathbf{H}_i)^{(2)} \right\} \equiv \left\{ (\mathbf{p}^0 \otimes \mathbf{v}^{(2)})^\top - \sum_{i \in [L]} (\mathbf{c}^\top \mathbf{H}_i)^{(2)} \right\}, \end{aligned}$$

and we observe that the left distributions correspond to G_5^b and the right distributions to G_6^b . $\square$

Proof of Claim 26. We consider a sequence of sub-hybrids $\widehat{G}_0, \dots, \widehat{G}_4$ with $\widehat{G}_0 \equiv G_{6,\ell-1}^b$ and $\widehat{G}_4 \equiv G_{6,\ell}^b$.

Game $\widehat{G}_0$: This is the same as $G_{6,\ell-1}^b$ except for the following syntactical changes. Initially, the challenger samples $\mathbf{d}_j \xleftarrow{\$} \text{span}(\mathbf{B}_2)$ for $j \in [\ell-1]$ and sets $\mathbf{B}_2 \mathbf{r}_j := \mathbf{d}_j$. Similarly, it samples $\mathbf{d}_j \xleftarrow{\$} \text{span}(\mathbf{B}_1)$ for $j \in [L] \setminus [\ell-1]$ and sets $\mathbf{B}_1 \mathbf{r}_j := \mathbf{d}_j$. Then it computes crs and $\text{pk}_i \in \mathcal{D}_i$ as follows:

$$\begin{aligned} \text{crs} &= \left(\llbracket \mathbf{A}, \tilde{\mathbf{V}} \rrbracket_1, \{ \text{crs}_i, \llbracket \mathbf{A} \mathbf{H}_i, \mathbf{A} \mathbf{H}_i (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}), \mathbf{A} \mathbf{W}_i, \mathbf{R}_i \rrbracket_1 \}_{i \in [L]}, \{ \llbracket \mathbf{d}_j, \mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{d}_j \rrbracket_2 \}_{j \in [L]} \right) \\ &\quad \{ \llbracket \mathbf{H}_i (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{d}_j), \mathbf{H}_i (\mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{d}_j), \mathbf{W}_i (\mathbf{I}_{m+n'-1} \otimes \mathbf{d}_j) \rrbracket_2 \}_{\substack{i \in [L], \\ j \in [L] \setminus \{i\}}} \\ \text{pk}_i &= (\llbracket \mathbf{T}_i \rrbracket_1 := \llbracket \mathbf{A} \mathbf{U}_i \rrbracket_1, \llbracket \mathbf{Q}_i \rrbracket_1 := \llbracket \mathbf{R}_i \mathbf{U}_i \rrbracket_1, \{ \llbracket \mathbf{G}_{i,j} \rrbracket_2 := \llbracket \mathbf{U}_i (\mathbf{I}_m \otimes \mathbf{d}_j) \rrbracket_2 \}_{j \in [L] \setminus \{i\}}, \pi_i). \end{aligned}$$

Clearly, we have $G_{6,\ell-1}^b \equiv \widehat{G}_0$.

Game $\widehat{G}_1$: This is the same as $\widehat{G}_0$ except that the challenger now samples $\mathbf{d}_\ell \xleftarrow{\$} \text{span}(\mathbf{B}_3)$. We have $\widehat{G}_0 \approx_c \widehat{G}_1$ under the subspace decision assumption $\text{SD}_{\mathbf{B}_1 \rightarrow \mathbf{B}_3}$ which states $\mathbf{d}_\ell \xleftarrow{\$} \text{span}(\mathbf{B}_1)$ (corresponding to $\widehat{G}_0$) and $\mathbf{d}_\ell \xleftarrow{\$} \text{span}(\mathbf{B}_3)$ (corresponding to $\widehat{G}_1$) are indistinguishable given $\llbracket \mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3 \rrbracket_2$, $\text{basis}(\mathbf{B}_1^\parallel, \mathbf{B}_3^\parallel)$ and $\text{basis}(\mathbf{B}_2^\parallel)$.

Game $\widehat{G}_2$: This is the same as $\widehat{G}_1$ except that we generate the ciphertext components $\mathbf{c}_1^*$ and $\mathbf{c}_2^*$ as follows:

$$\begin{aligned} \mathbf{c}_1^* &= (\mathbf{z}^b \otimes \mathbf{v}^{(1)})^\top + (\mathbf{z}^0 \otimes \mathbf{v}^{(2)})^\top + (\mathbf{z}^0 \otimes \mathbf{v}^{(3)})^\top - \mathbf{c}^\top \sum_{i \in [L]} \mathbf{W}_i \\ \mathbf{c}_2^* &= (\mathbf{p}^b \otimes \mathbf{v}^{(1)})^\top + (\mathbf{p}^0 \otimes \mathbf{v}^{(2)})^\top + (\mathbf{p}^0 \otimes \mathbf{v}^{(3)})^\top - \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_i. \end{aligned}$$

We prove the following claim below.

Claim 27. *We have $\widehat{G}_1 \approx_c \widehat{G}_2$ assuming MDDH_k in $\mathbb{G}_1$.*

Game $\widehat{G}_3$: This is the same as $\widehat{G}_2$ except that the challenger now samples $\mathbf{d}_\ell \xleftarrow{\$} \text{span}(\mathbf{B}_2)$ instead of $\mathbf{d}_\ell \xleftarrow{\$} \text{span}(\mathbf{B}_3)$. We have $\widehat{G}_2 \approx_c \widehat{G}_3$ under the subspace decision assumption $\text{SD}_{\mathbf{B}_3 \rightarrow \mathbf{B}_2}$ which states that $\mathbf{d}_\ell \xleftarrow{\$} \text{span}(\mathbf{B}_3)$ (corresponding to $\widehat{G}_2$) and $\mathbf{d}_\ell \xleftarrow{\$} \text{span}(\mathbf{B}_2)$ (corresponding to $\widehat{G}_3$) are indistinguishable given $\llbracket \mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3 \rrbracket_2$, $\text{basis}(\mathbf{B}_2^\parallel, \mathbf{B}_3^\parallel)$ and $\text{basis}(\mathbf{B}_1^\parallel)$.

Game $\widehat{G}_4$: This is the same as $\widehat{G}_3$ except that the challenger computes the ciphertext components $\mathbf{c}_1^*$ and $\mathbf{c}_2^*$ as follows:

$$\begin{aligned} \mathbf{c}_1^* &= (\mathbf{z}^b \otimes \mathbf{v}^{(1)})^\top + (\mathbf{z}^0 \otimes \mathbf{v}^{(2)})^\top + (\mathbf{z}^b \otimes \mathbf{v}^{(3)})^\top - \mathbf{c}^\top \sum_{i \in [L]} \mathbf{W}_i \\ \mathbf{c}_2^* &= (\mathbf{p}^b \otimes \mathbf{v}^{(1)})^\top + (\mathbf{p}^0 \otimes \mathbf{v}^{(2)})^\top + (\mathbf{p}^b \otimes \mathbf{v}^{(3)})^\top - \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_i. \end{aligned}$$

We have $\widehat{G}_3 \approx_c \widehat{G}_4$ under the MDDH_k assumption in $\mathbb{G}_1$. The argument is the same as in the transition $\widehat{G}_1 \rightarrow \widehat{G}_2$ (Claim 27). Furthermore, we observe that $\widehat{G}_4 \equiv G_{6,\ell}^b$ which concludes the proof. $\square$

Proof of Claim 27. We distinguish the cases where pk_ℓ^* is either honest or [malicious or corrupted].

Honest Case. In this case, we have

$$\text{pk}_\ell^* = (\llbracket \mathbf{T}_\ell^* \rrbracket_1, \llbracket \mathbf{Q}_\ell^* \rrbracket_1, \{\llbracket \mathbf{G}_{\ell,j}^* \rrbracket_2\}_{j \in [L] \setminus \{\ell\}}, \pi_\ell^*) \in \mathcal{D}_\ell \setminus \mathcal{C}_\ell$$

and the challenger knows a secret matrix $\mathbf{U}_\ell^*$ such that $\mathbf{T}_\ell^* = \mathbf{A}\mathbf{U}_\ell^*$ and $\mathbf{Q}_\ell^* = \mathbf{R}_\ell^* \mathbf{U}_\ell^*$. In particular, we can write

$$\mathbf{c}_3^* = \left(\mathbf{c}^\top \mathbf{U}_\ell^* + \sum_{i \in [L] \setminus \{\ell\}} \mathbf{e}_1^\top \tilde{\mathbf{R}}_i^{-1} \mathbf{Q}_i^* \right) + \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_i (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_i (\mathbf{L}_{i,\mathbf{x}} \otimes \mathbf{I}_{2k+1}) .$$

Let $\mathbf{c}^\perp \in \mathbb{Z}_p^{2k+1}$ such that $\mathbf{A}\mathbf{c}^\perp = 0$ and $\mathbf{c}^\top \mathbf{c}^\perp = 1$. For $\kappa \in [0; 2]$ and $\beta \in \{b, 0\}$, we define the distribution $D_\kappa^\beta = (C, \hat{D}_\kappa^\beta)$, where

$$C = \left\{ \begin{array}{c} \mathbf{A}, \mathbf{c}^\perp, \llbracket \mathbf{R}_\ell \rrbracket_1, \mathbf{B}_1, \mathbf{B}_2, \mathbf{d}_\ell, \mathbf{A}\mathbf{H}_\ell, \mathbf{A}\mathbf{W}_\ell, \mathbf{A}\mathbf{H}_\ell (\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}), \\ \{\mathbf{H}_\ell (\mathbf{q}_\kappa \mathbf{e}_1^\top \otimes \mathbf{d}_j)\}_{j \in [L] \setminus \{\ell\}}, \{\mathbf{H}_i (\mathbf{q}_\kappa \mathbf{e}_1^\top \otimes \mathbf{d}_\ell)\}_{i \in [L] \setminus \{\ell\}}, \{\mathbf{W}_\ell (\mathbf{L}_{m+n'-1} \otimes \mathbf{B}_d)\}_{d \in \{1,2\}} \end{array} \right\}$$

represents the adversaries view on the crs and $\hat{D}_\kappa^\beta$ the view on the public key and the challenge ciphertext:

$$\begin{aligned} \hat{D}_0^\beta &= \left\{ \begin{array}{c} \text{pk}_\ell^*: \mathbf{A}\mathbf{U}_\ell^*, \llbracket \mathbf{R}_\ell \mathbf{U}_\ell^* \rrbracket_1, \mathbf{U}_\ell^* (\mathbf{I}_m \otimes \mathbf{B}_1), \mathbf{U}_\ell^* (\mathbf{I}_m \otimes \mathbf{B}_2) \\ \mathbf{c}_0^*, \mathbf{c}_1^*, \mathbf{c}_2^*: \mathbf{c}^\top, (\mathbf{z}^\beta \otimes \mathbf{v}^{(3)})^\top - \mathbf{c}^\top \mathbf{W}_\ell, (\mathbf{p}^\beta \otimes \mathbf{v}^{(3)})^\top - \mathbf{c}^\top \mathbf{H}_\ell \\ \mathbf{c}_3^*: \mathbf{c}^\top \mathbf{U}_\ell^* + \mathbf{c}^\top \mathbf{H}_\ell (\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{c}^\top \mathbf{W}_\ell (\mathbf{L}_{\ell,\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \end{array} \right\} \\ \hat{D}_1^\beta &= \left\{ \begin{array}{c} \text{pk}_\ell^*: \mathbf{A}\mathbf{U}_\ell^*, \llbracket \mathbf{R}_\ell \mathbf{U}_\ell^* + \mathbf{P}_\ell \otimes (\mathbf{v}^{(3)})^\top \rrbracket_1, \mathbf{U}_\ell^* (\mathbf{I}_m \otimes \mathbf{B}_1), \mathbf{U}_\ell^* (\mathbf{I}_m \otimes \mathbf{B}_2) \\ \mathbf{c}_0^*, \mathbf{c}_1^*, \mathbf{c}_2^*: \mathbf{c}^\top, (\mathbf{z}^\beta \otimes \mathbf{v}^{(3)})^\top - \mathbf{c}^\top \mathbf{W}_\ell, (\mathbf{p}^\beta \otimes \mathbf{v}^{(3)})^\top - \mathbf{c}^\top \mathbf{H}_\ell \\ \mathbf{c}_3^*: \mathbf{c}^\top \mathbf{U}_\ell^* + \mathbf{c}^\top \mathbf{H}_\ell (\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{c}^\top \mathbf{W}_\ell (\mathbf{L}_{\ell,\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \end{array} \right\} \\ \hat{D}_2^\beta &= \left\{ \begin{array}{c} \text{pk}_\ell^*: \mathbf{A}\mathbf{U}_\ell^*, \llbracket \mathbf{R}_\ell \mathbf{U}_\ell^* + (\mathbf{P}_\ell + \mathbf{R}_\ell \mathbf{c}^\perp (\mathbf{u}_\ell + \mathbf{u}'_\ell)^\top) \otimes (\mathbf{v}^{(3)})^\top \rrbracket_1, \mathbf{U}_\ell^* (\mathbf{I}_m \otimes \mathbf{B}_1), \mathbf{U}_\ell^* (\mathbf{I}_m \otimes \mathbf{B}_2) \\ \mathbf{c}_0^*, \mathbf{c}_1^*, \mathbf{c}_2^*: \mathbf{c}^\top, ((\mathbf{z}^\beta - \mathbf{w}_\ell) \otimes \mathbf{v}^{(3)})^\top - \mathbf{c}^\top \mathbf{W}_\ell, ((\mathbf{p}^\beta - \mathbf{h}_\ell) \otimes \mathbf{v}^{(3)})^\top - \mathbf{c}^\top \mathbf{H}_\ell \\ \mathbf{c}_3^*: \mathbf{c}^\top \mathbf{U}_\ell^* + \mathbf{c}^\top \mathbf{H}_\ell (\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + ((\mathbf{u}_\ell + \mathbf{h}_\ell^\top \mathbf{q}_\ell \cdot \mathbf{e}_1) \otimes \mathbf{v}^{(3)})^\top + \mathbf{c}^\top \mathbf{W}_\ell (\mathbf{L}_{\ell,\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \\ \quad + ((\mathbf{u}'_\ell + \mathbf{L}_{\ell,\mathbf{x}}^\top \mathbf{w}_\ell) \otimes \mathbf{v}^{(3)})^\top \end{array} \right\} \end{aligned}$$

where $\mathbf{P}_\ell \xleftarrow{s} \mathbb{Z}_p^{(2k+2) \times m}$, $\mathbf{u}_\ell, \mathbf{u}'_\ell \xleftarrow{s} \mathbb{Z}_p^m$, $\mathbf{h}_\ell \xleftarrow{s} \mathbb{Z}_p^k$, $\mathbf{w}_\ell \xleftarrow{s} \mathbb{Z}_p^{m+n'-1}$, and $\mathbf{w}_\ell$ denotes the last n' entries of $\mathbf{w}_\ell$. We note that for $\beta = b$ (resp. $\beta = 0$), the distribution D_0^β corresponds to the adversary's view in $\hat{\mathcal{G}}_1$ (resp. $\hat{\mathcal{G}}_2$). Furthermore, we observe that

- $D_0^\beta \approx_c D_1^\beta$ which follows from Lemma 24 when setting $\mathbf{d}^\perp = \mathbf{v}^{(3)}$, $\mathbf{R} = \mathbf{R}_\ell$, $\mathbf{U} = \mathbf{U}_\ell^*$ and $\mathbf{P} = \mathbf{P}_\ell$.
- $D_1^\beta \approx_s D_2^\beta$: this follows from the following change of variables

$$\begin{aligned} \mathbf{U}_\ell^* &\rightarrow \mathbf{U}_\ell^* + \mathbf{c}^\perp ((\mathbf{u}_\ell + \mathbf{u}'_\ell) \otimes \mathbf{v}^{(3)})^\top \\ \mathbf{H}_\ell &\mapsto \mathbf{H}_\ell + \mathbf{c}^\perp (\mathbf{h}_\ell \otimes \mathbf{v}^{(3)})^\top \\ \mathbf{W}_\ell &\mapsto \mathbf{W}_\ell + \mathbf{c}^\perp (\mathbf{w}_\ell \otimes \mathbf{v}^{(3)})^\top . \end{aligned}$$

Here we use the equality $\mathbf{R}_\ell \mathbf{c}^\perp ((\mathbf{u}_\ell + \mathbf{u}'_\ell) \otimes \mathbf{v}^{(3)})^\top = \mathbf{R}_\ell \mathbf{c}^\perp (\mathbf{u}_\ell + \mathbf{u}'_\ell)^\top \otimes (\mathbf{v}^{(3)})^\top$.

- $D_2^\beta \approx_s D_2^1$: this follows from the fact that $\mathbf{P}_\ell$ hides $\mathbf{R}_\ell \mathbf{c}^\perp (\mathbf{u}_\ell + \mathbf{u}'_\ell)^\top$ which implies that $\mathbf{u}_\ell$ hides $\mathbf{h}_\ell^\top \mathbf{q}_\ell \cdot \mathbf{e}_1^\top$ and $\mathbf{h}_\ell$ hides $\mathbf{p}^\beta$. Similarly, $\mathbf{u}'_\ell$ can hide $\mathbf{L}_{\ell,\mathbf{x}}^\top \mathbf{w}_\ell$, so $\mathbf{w}_\ell$ hides $\mathbf{z}^\beta$. In the end, we can observe that the distribution D_2^β is independent of the bit β .

Malicious and Corrupted Case. In this case, we have

$$\text{pk}_\ell^* = (\llbracket \mathbf{T}_\ell^* \rrbracket_1, \llbracket \mathbf{Q}_\ell^* \rrbracket_1, \{\llbracket \mathbf{G}_{\ell,j}^* \rrbracket_2\}_{j \in [L] \setminus \{\ell\}}, \pi_\ell^*) \in \overline{\mathcal{D}_\ell} \cup \mathcal{C}_\ell .$$

Let $\mathbf{c}^\perp \in \mathbb{Z}_p^{2k+1}$ again be such that $\mathbf{A}\mathbf{c}^\perp = 0$ and $\mathbf{c}^\top \mathbf{c}^\perp = 1$. For $\kappa \in [0; 2]$ and $\beta \in \{b, 0\}$, we define the distribution $D_\kappa^\beta = (C, \hat{D}_\kappa^\beta)$, where

$$C = \left\{ \begin{array}{c} \mathbf{A}, \mathbf{c}^\perp, \mathbf{B}_1, \mathbf{B}_2, \mathbf{d}_\ell, \mathbf{A}\mathbf{H}_\ell, \mathbf{A}\mathbf{W}_\ell, \mathbf{A}\mathbf{H}_\ell(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}), \\ \{\mathbf{H}_\ell(\mathbf{q}_\kappa \mathbf{e}_1^\top \otimes \mathbf{d}_j)\}_{j \in [L] \setminus \{\ell\}}, \{\mathbf{H}_i(\mathbf{q}_\kappa \mathbf{e}_1^\top \otimes \mathbf{d}_\ell)\}_{i \in [L] \setminus \{\ell\}}, \{\mathbf{W}_\ell(\mathbf{I}_{m+n'-1} \otimes \mathbf{B}_d)\}_{d \in \{1,2\}} \end{array} \right\}_{\kappa \in \{j, \ell\}}^{\kappa \in \{i, \ell\}}$$

represents the adversaries view on the crs and $\hat{D}_\kappa^\beta$ the view on the challenge ciphertext:

$$\begin{aligned} \hat{D}_0^\beta &= \left\{ \begin{array}{l} \mathbf{c}_0^*, \mathbf{c}_1^*, \mathbf{c}_2^*: \mathbf{c}^\top, (\mathbf{z}^\beta \otimes \mathbf{v}^{(3)})^\top - \mathbf{c}^\top \mathbf{W}_\ell, (\mathbf{p}^\beta \otimes \mathbf{v}^{(3)})^\top - \mathbf{c}^\top \mathbf{H}_\ell \\ \mathbf{c}_3^*: \mathbf{e}_1^\top \tilde{\mathbf{R}}_\ell^{-1} \mathbf{Q}_\ell^* + \mathbf{c}^\top \mathbf{H}_\ell(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{c}^\top \mathbf{W}_\ell(\mathbf{L}_{\ell, \mathbf{x}} \otimes \mathbf{I}_{2k+1}) \end{array} \right\} \\ \hat{D}_1^\beta &= \left\{ \begin{array}{l} \mathbf{c}_0^*, \mathbf{c}_1^*, \mathbf{c}_2^*: \mathbf{c}^\top, (\mathbf{z}^\beta - \mathbf{w}_\ell)^\top \otimes (\mathbf{v}^{(3)})^\top - \mathbf{c}^\top \mathbf{W}_\ell, (\mathbf{p}^\beta \otimes \mathbf{v}^{(3)})^\top - \mathbf{c}^\top \mathbf{H}_\ell \\ \mathbf{c}_3^*: \mathbf{e}_1^\top \tilde{\mathbf{R}}_\ell^{-1} \mathbf{Q}_\ell^* + \mathbf{c}^\top \mathbf{H}_\ell(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{c}^\top \mathbf{W}_\ell(\mathbf{L}_{\ell, \mathbf{x}} \otimes \mathbf{I}_{2k+1}) \\ \quad + (\mathbf{w}_\ell^\top \mathbf{L}_{\ell, \mathbf{x}} + \mathbf{q}_\ell^\top \mathbf{p}^\beta \cdot \mathbf{e}_1^\top) \otimes (\mathbf{v}^{(3)})^\top \end{array} \right\} \\ \hat{D}_2^\beta &= \left\{ \begin{array}{l} \mathbf{c}_0^*, \mathbf{c}_1^*, \mathbf{c}_2^*: \mathbf{c}^\top, (\mathbf{z}^\beta - \mathbf{w}_\ell)^\top \otimes (\mathbf{v}^{(3)})^\top - \mathbf{c}^\top \mathbf{W}_\ell, -\mathbf{c}^\top \mathbf{H}_\ell \\ \mathbf{c}_3^*: \mathbf{e}_1^\top \tilde{\mathbf{R}}_\ell^{-1} \mathbf{Q}_\ell^* + \mathbf{c}^\top \mathbf{H}_\ell(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{c}^\top \mathbf{W}_\ell(\mathbf{L}_{\ell, \mathbf{x}} \otimes \mathbf{I}_{2k+1}) \\ \quad + (\mathbf{w}_\ell^\top \mathbf{L}_{\ell, \mathbf{x}} + (h_\ell^*(\mathbf{x})^\top \mathbf{z}^\beta + \mathbf{q}_\ell^\top \mathbf{p}^\beta) \cdot \mathbf{e}_1^\top) \otimes (\mathbf{v}^{(3)})^\top \end{array} \right\} \end{aligned}$$

where $\mathbf{w}_\ell \xleftarrow{s} \mathbb{Z}_p^{m+n'-1}$ and $\mathbf{w}_\ell$ denotes the last n' entries of $\mathbf{w}_\ell$. We note that for $\beta = b$ (resp. $\beta = 0$), the distribution D_0^β corresponds to the adversary's view in $\hat{G}_1$ (resp. $\hat{G}_2$). Furthermore, we observe that

- $D_0^\beta \approx_s D_1^\beta$: this follows from the changes of variables

$$\mathbf{H}_\ell \mapsto \mathbf{H}_\ell + \mathbf{c}^\perp (\mathbf{p}^\beta \otimes \mathbf{v}^{(3)})^\top, \quad \mathbf{W}_\ell \mapsto \mathbf{W}_\ell + \mathbf{c}^\perp (\mathbf{w}_\ell \otimes \mathbf{v}^{(3)})^\top.$$

- $D_1^\beta \approx_s D_2^\beta$: this follows from the security of pgb. Specifically, we can simulate $\hat{D}_\kappa^\beta$, $\kappa \in \{1, 2\}$, in the form

$$\hat{D}_\kappa^\beta = \left\{ \begin{array}{l} \mathbf{c}_0^*, \mathbf{c}_1^*, \mathbf{c}_2^*: \mathbf{c}^\top, (\mathbf{p}_1 \otimes \mathbf{v}^{(3)})^\top - \mathbf{c}^\top \mathbf{W}_\ell, -\mathbf{c}^\top \mathbf{H}_\ell \\ \mathbf{c}_3^*: \mathbf{e}_1^\top \tilde{\mathbf{R}}_\ell^{-1} \mathbf{Q}_\ell^* + \mathbf{c}^\top \mathbf{H}_\ell(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{c}^\top \mathbf{W}_\ell(\mathbf{L}_{\ell, \mathbf{x}} \otimes \mathbf{I}_{2k+1}) + (\mathbf{p}_2 \otimes \mathbf{v}^{(3)})^\top \end{array} \right\},$$

where the vectors $\mathbf{p}_1 \in \mathbb{Z}_p^{n'}$ and $\mathbf{p}_2 \in \mathbb{Z}_p^m$ are defined as

$$(\mathbf{p}_1^\top, \mathbf{p}_2^\top) \leftarrow \begin{cases} \text{pgb}(h_\ell^*, \mathbf{x}, \mathbf{z}^\beta; \mathbf{w}_\ell) + (\mathbf{0}^\top, \mathbf{q}_\ell^\top \mathbf{p}^\beta \cdot \mathbf{e}_1^\top) & \text{if } \kappa = 1 \\ \text{pgb}^*(h_\ell^*, \mathbf{x}, h_\ell^*(\mathbf{x})^\top \mathbf{z}^\beta + \mathbf{q}_\ell^\top \mathbf{p}^\beta; \mathbf{w}_\ell) & \text{if } \kappa = 2. \end{cases}$$

Then $D_1^\beta \approx_s D_2^\beta$ immediately follows from the privacy of the underlying partial garbling scheme.

- $D_2^0 \equiv D_2^1$: this follows from the adversary's admissibility condition requiring $h_\ell^*(\mathbf{x})^\top \mathbf{z}^0 + \mathbf{q}_\ell^\top \mathbf{p}^0 = h_\ell^*(\mathbf{x})^\top \mathbf{z}^1 + \mathbf{q}_\ell^\top \mathbf{p}^1$.

□

5 Adaptively IND-Secure sRFE for AB-eAWS

Our sRFE scheme for AB-eAWS is an extension of the eAWS scheme presented in Section 4, but the security proof of the attribute-based variant relies on the stronger bi-MDDH_k assumption instead of only MDDH_k.

Construction 28 (Adaptively IND-Secure sRFE for AB-eAWS). The construction uses the following ingredients:

- a pairing group $\mathbb{G} = (\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_t, g_1, g_2, g_t, e, p)$, and

- a QA-NIZK $\Pi = (\text{LGen}, \text{LPrv}, \text{LVer}, \text{LSim})$ for linear spaces over $\mathbb{G}$.

The details of the scheme go as follows:

Setup($1^\lambda, 1^L, \{\llbracket \mathbf{q}_i \rrbracket_{1,2}\}_{i \in [L]}$): On input the security parameter 1^λ , the maximum number of slots 1^L and a set of vectors $\{\llbracket \mathbf{q}_i \rrbracket_{1,2}\}_{i \in [L]}$ with $\mathbf{q}_i \in \mathbb{Z}_p^k$ for $i \in [L]$, sample $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_p^{k \times (2k+1)}$, $\mathbf{B} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times k}$, $\mathbf{V} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)}$ and for $i \in [L]$, $\mathbf{g}_i, \mathbf{r}_i \xleftarrow{\$} \mathbb{Z}_p^k$, $\mathbf{H}_{1,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)(k+n')}$, $\mathbf{H}_{2,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)k}$, $\mathbf{R}_{1,i}, \mathbf{R}_{2,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+2) \times (2k+1)}$ and $\mathbf{W}_i \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)(m+n')}$. Generate

$$(\text{crs}_{1,i}, \text{td}_{1,i}) \leftarrow \text{LGen}(1^\lambda, \mathbb{G}_1, \llbracket \mathbf{A}_{1,i} \rrbracket_1), \quad (\text{crs}_{2,i}, \text{td}_{2,i}) \leftarrow \text{LGen}(1^\lambda, \mathbb{G}_1, \llbracket \mathbf{A}_{2,i} \rrbracket_1)$$

where $\mathbf{A}_{1,i}^\top = [\mathbf{A}^\top \parallel \mathbf{R}_{1,i}^\top]$, $\mathbf{A}_{2,i}^\top = [\mathbf{A}^\top \parallel \mathbf{R}_{2,i}^\top]$, and output the common reference string

$$\text{crs} = \left(\left\{ \text{crs}_{1,i}, \text{crs}_{2,i}, \left\| \begin{array}{c} \mathbf{A}\mathbf{W}_i, \mathbf{R}_{1,i}, \mathbf{R}_{2,i}, \mathbf{A}\mathbf{H}_{1,i}, \mathbf{A}\mathbf{H}_{2,i} \\ \mathbf{A}\mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}), \mathbf{A}\mathbf{H}_{2,i}(\mathbf{q}_i(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{I}_{2k+1}) \end{array} \right\|_1 \right\}_{i \in [L]}, \right. \\ \left. \begin{array}{c} \llbracket \mathbf{A}, \mathbf{A}\mathbf{V} \rrbracket_1, \{\llbracket \mathbf{B}\mathbf{r}_j, \mathbf{I}_{\mathbf{g}_j} \otimes \mathbf{B}\mathbf{r}_j, \mathbf{q}_j(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{B}\mathbf{r}_j \rrbracket_2\}_{j \in [L]}, \\ \{\llbracket \mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_k} \otimes \mathbf{B}\mathbf{r}_j), \mathbf{H}_{2,i}(\mathbf{q}_k(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{B}\mathbf{r}_j), \mathbf{W}_i(\mathbf{I}_{m+n'} \otimes \mathbf{B}\mathbf{r}_j) \rrbracket_2\}_{\substack{i \in [L], \\ j \in [L] \setminus \{i\}, \\ k \in \{i,j\}}} \end{array} \right)$$

where, for $\mathbf{g} \in \mathbb{Z}_p^k$, the matrix $\mathbf{I}_{\mathbf{g}} \in \mathbb{Z}_p^{(k+n') \times (1+n')}$ is defined as

$$\mathbf{I}_{\mathbf{g}} := \begin{bmatrix} \mathbf{g} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{n'} \end{bmatrix}.$$

Gen(crs, i): On input crs and an index $i \in [L]$, sample matrix $\mathbf{U}_{1,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)(n'+1)}$, $\mathbf{U}_{2,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)m}$ and run $\pi_{1,i} \leftarrow \text{LPrv}(\text{crs}_{1,i}, \llbracket \mathbf{A}_{1,i} \mathbf{U}_{1,i} \rrbracket_1, \mathbf{U}_{1,i})$, $\pi_{2,i} \leftarrow \text{LPrv}(\text{crs}_{2,i}, \llbracket \mathbf{A}_{2,i} \mathbf{U}_{2,i} \rrbracket_1, \mathbf{U}_{2,i})$. Output $\text{sk}_i = (\mathbf{U}_{1,i}, \mathbf{U}_{2,i})$ and pk_i as follows:

$$\left(\begin{array}{l} \llbracket \mathbf{T}_{1,i} \rrbracket_1 := \llbracket \mathbf{A}\mathbf{U}_{1,i} \rrbracket_1, \llbracket \mathbf{Q}_{1,i} \rrbracket_1 := \llbracket \mathbf{R}_{1,i} \mathbf{U}_{1,i} \rrbracket_1, \{\llbracket \mathbf{G}_{1,i,j} \rrbracket_2 := \llbracket \mathbf{U}_{1,i}(\mathbf{I}_{n'+1} \otimes \mathbf{B}\mathbf{r}_j) \rrbracket_2\}_{j \in [L] \setminus \{i\}}, \pi_{1,i} \\ \llbracket \mathbf{T}_{2,i} \rrbracket_1 := \llbracket \mathbf{A}\mathbf{U}_{2,i} \rrbracket_1, \llbracket \mathbf{Q}_{2,i} \rrbracket_1 := \llbracket \mathbf{R}_{2,i} \mathbf{U}_{2,i} \rrbracket_1, \{\llbracket \mathbf{G}_{2,i,j} \rrbracket_2 := \llbracket \mathbf{U}_{2,i}(\mathbf{I}_m \otimes \mathbf{B}\mathbf{r}_j) \rrbracket_2\}_{j \in [L] \setminus \{i\}}, \pi_{2,i} \end{array} \right).$$

Ver($\text{crs}, i, \text{pk}_i$): On input crs , an index $i \in [L]$ and a public key $\text{pk}_i = (\llbracket \mathbf{T}_{d,i} \rrbracket_1, \llbracket \mathbf{Q}_{d,i} \rrbracket_1, \{\llbracket \mathbf{G}_{d,i,j} \rrbracket_2\}_{j \in [L] \setminus \{i\}}, \pi_{d,i})_{d \in \{1,2\}}$, define $\mathbf{F}_{d,i}^\top = [\mathbf{T}_{d,i}^\top \parallel \mathbf{Q}_{d,i}^\top]$ and check

$$\begin{array}{l} \text{LVer}(\text{crs}_{1,i}, \llbracket \mathbf{F}_{1,i} \rrbracket_1, \pi_{1,i}) \stackrel{?}{=} 1, \quad \left\{ \begin{array}{l} \llbracket \mathbf{T}_{1,i} \rrbracket_1 \llbracket \mathbf{I}_{n'+1} \otimes \mathbf{B}\mathbf{r}_j \rrbracket_2 \stackrel{?}{=} \llbracket \mathbf{A} \rrbracket_1 \llbracket \mathbf{G}_{1,i,j} \rrbracket_2, \\ \llbracket \mathbf{T}_{2,i} \rrbracket_1 \llbracket \mathbf{I}_m \otimes \mathbf{B}\mathbf{r}_j \rrbracket_2 \stackrel{?}{=} \llbracket \mathbf{A} \rrbracket_1 \llbracket \mathbf{G}_{2,i,j} \rrbracket_2 \end{array} \right\}_{j \in [L] \setminus \{i\}}. \end{array}$$

Output 1 if all checks pass, and 0 otherwise.

Agg($\text{crs}, (\text{pk}_i, f_i)_{i \in [L]}$): On input crs and L tuples of the form (pk_i, f_i) , parse $f_i = (g_i, h_i) \in \mathcal{F}_{\hat{n},1}^{\text{abp}} \times \mathcal{F}_{n,n'}^{\text{abp}}$ and $\text{pk}_i = (\llbracket \mathbf{T}_{d,i} \rrbracket_1, \llbracket \mathbf{Q}_{d,i} \rrbracket_1, \{\llbracket \mathbf{G}_{d,i,j} \rrbracket_2\}_{j \in [L] \setminus \{i\}}, \pi_{d,i})_{d \in \{1,2\}}$. For each $i \in [L]$, define an ABP $\phi_i \in \mathcal{F}_{\hat{n}+n,1+n'}^{\text{abp}}$ as $\phi_i(\mathbf{y}, \mathbf{x})^\top = (g_i(\mathbf{y}), h_i(\mathbf{x})^\top)$ for $\mathbf{y} \in \mathbb{Z}_p^{\hat{n}}$ and $\mathbf{x} \in \mathbb{Z}_p^n$. Then compute the matrices

$$\text{lgen}(\phi_i) \rightarrow (\mathbf{L}_{i,0}, \mathbf{L}_{i,1}) \in \mathbb{Z}_p^{(m+n') \times m} \times \mathbb{Z}_p^{(m+n') \times m(\hat{n}+n)}$$

and output $(\text{mpk}, \{\text{hsk}_j\}_{j \in [L]})$ as follows:

$$\text{mpk} = \left\| \begin{array}{c} \mathbf{A}, \mathbf{A}\mathbf{V}, \sum_{i \in [L]} \mathbf{A}\mathbf{H}_{1,i}, \sum_{i \in [L]} \mathbf{A}\mathbf{H}_{2,i}, \sum_{i \in [L]} \mathbf{T}_{1,i} - \mathbf{A}\mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}) - \mathbf{A}\mathbf{W}_i, \\ \sum_{i \in [L]} \mathbf{A}\mathbf{W}_i(\mathbf{I}_{\mathbf{L}_{i,1}} \otimes \mathbf{I}_{2k+1}), \sum_{i \in [L]} \mathbf{T}_{2,i} - \mathbf{A}\mathbf{H}_{2,i}(\mathbf{q}_i(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{A}\mathbf{W}_i(\mathbf{L}_{i,0} \otimes \mathbf{I}_{2k+1}) \end{array} \right\|_1,$$

$$\text{hsk}_j = \left[\begin{array}{c} \mathbf{Br}_j, \mathbf{I}_{\mathbf{g}_j} \otimes \mathbf{Br}_j, \mathbf{q}_j(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{Br}_j, \sum_{i \in [L] \setminus \{j\}} \mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_j} \otimes \mathbf{Br}_j), \\ \sum_{i \in [L] \setminus \{j\}} \mathbf{G}_{1,i,j} - \mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_j} \otimes \mathbf{Br}_j) - \underline{\mathbf{W}}_i(\mathbf{I}_{n'+1} \otimes \mathbf{Br}_j), \\ \sum_{i \in [L] \setminus \{j\}} \mathbf{H}_{2,i}(\mathbf{q}_j(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{Br}_j), \sum_{i \in [L] \setminus \{j\}} \mathbf{W}_i(\mathbf{I}_{m+n'} \otimes \mathbf{Br}_j) \mathbf{L}_{i,1}, \\ \sum_{i \in [L] \setminus \{j\}} \mathbf{G}_{2,i,j} - \mathbf{H}_{2,i}(\mathbf{q}_j(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{Br}_j) + \mathbf{W}_i(\mathbf{I}_{m+n'} \otimes \mathbf{Br}_j) \mathbf{L}_{i,0} \end{array} \right]_2,$$

where $\underline{\mathbf{W}}_i \in \mathbb{Z}_p^{(2k+1) \times (2k+1)(1+n')}$ refers to the matrix composed of the right most $(2k+1)(1+n')$ columns of $\mathbf{W}_i \in \mathbb{Z}_p^{(2k+1) \times (2k+1)(m+n')}$.

Enc(mpk, (y, x), (p, z)): On input the master public key mpk, a public input $(y, x) \in \mathbb{Z}_p^{\hat{n}} \times \mathbb{Z}_p^n$ and a private input $(p, z) \in \mathbb{Z}_p^k \times \mathbb{Z}_p^{n'}$, sample $\mathbf{f}, \mathbf{s} \xleftarrow{\$} \mathbb{Z}_p^k$ and output

$$\text{ct} = \left[\begin{array}{c} \mathbf{s}^\top \mathbf{A}, (\mathbf{f}^\top \parallel \mathbf{z}^\top) \otimes \mathbf{s}^\top \mathbf{AV} + \mathbf{s}^\top \sum_{i \in [L]} \mathbf{AH}_{1,i}, \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{1,i} - \mathbf{AH}_{1,i}(\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}) - \underline{\mathbf{AW}}_i, \\ \mathbf{p}^\top \otimes \mathbf{s}^\top \mathbf{AV} + \mathbf{s}^\top \sum_{i \in [L]} \mathbf{AH}_{2,i}, \\ \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{2,i} - \mathbf{AH}_{2,i}(\mathbf{q}_i(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{AW}_i(\mathbf{L}_{i,y,x} \otimes \mathbf{I}_{2k+1}), \mathbf{s}^\top \mathbf{AV} \end{array} \right]_1,$$

where $\mathbf{L}_{j,y,x} := \mathbf{L}_{j,0} + \mathbf{L}_{j,1}((\mathbf{y}^\top \parallel \mathbf{x}^\top)^\top \otimes \mathbf{I}_m)$.

Note. Observe that the last two components of mpk denoted as $(\llbracket \mathbf{K}_1 \rrbracket_1, \llbracket \mathbf{K}_0 \rrbracket_1)$ enables the computation of the fifth component of the ciphertext as $\mathbf{s}^\top \llbracket \mathbf{K}_1 \rrbracket_1((\mathbf{y}^\top \parallel \mathbf{x}^\top)^\top \otimes \mathbf{I}_m) \otimes \mathbf{I}_{2k+1} + \mathbf{s}^\top \llbracket \mathbf{K}_0 \rrbracket_1$.

Dec(sk_i^{*}, hsk_i^{*}, ct): Given a secret key sk_i^{*} with corresponding helper secret key hsk_i^{*} = ($\llbracket \mathbf{d}_0 \rrbracket_2, \llbracket \mathbf{D}_{0,1} \rrbracket_2, \llbracket \mathbf{D}_{0,2} \rrbracket_2, \llbracket \mathbf{D}_{1,1} \rrbracket_2, \llbracket \mathbf{D}_{1,2} \rrbracket_2, \llbracket \mathbf{D}_{2,1} \rrbracket_2, \llbracket \mathbf{D}_{2,2} \rrbracket_2, \llbracket \mathbf{D}_{2,3} \rrbracket_2$) and a ciphertext ct = ($\llbracket \mathbf{c}_0 \rrbracket_1, \llbracket \mathbf{c}_{1,1} \rrbracket_1, \llbracket \mathbf{c}_{1,2} \rrbracket_1, \llbracket \mathbf{c}_{2,1} \rrbracket_1, \llbracket \mathbf{c}_{2,2} \rrbracket_1, \llbracket \mathbf{c}_3 \rrbracket_1$), output $\perp$ if $g_{i^*}(\mathbf{y}) \neq 0$. Otherwise, compute

$$\begin{aligned} \llbracket \mathbf{c}'_{1,1} \rrbracket_t &= \llbracket \mathbf{c}_{1,1} \rrbracket_1 \llbracket \mathbf{D}_{0,1} \rrbracket_2, & \llbracket \mathbf{d}'_{1,1} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_{1,1} \rrbracket_2, \\ \llbracket \mathbf{c}'_{1,2} \rrbracket_t &= \llbracket \mathbf{c}_{1,2} \rrbracket_1 \llbracket \mathbf{I}_{n'+1} \otimes \mathbf{d}_0 \rrbracket_2, & \llbracket \mathbf{d}'_{1,2} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_{1,2} \rrbracket_2, \\ \llbracket \mathbf{c}'_{2,1} \rrbracket_t &= \llbracket \mathbf{c}_{2,1} \rrbracket_1 \llbracket \mathbf{D}_{0,2} \rrbracket_2, & \llbracket \mathbf{d}'_{2,1} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_{2,1} \rrbracket_2, \\ \llbracket \mathbf{c}'_{2,2} \rrbracket_t &= \llbracket \mathbf{c}_{2,2} \rrbracket_1 \llbracket \mathbf{I}_m \otimes \mathbf{d}_0 \rrbracket_2, & \llbracket \mathbf{d}'_{2,2} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_{2,2}((\mathbf{y}^\top \parallel \mathbf{x}^\top)^\top \otimes \mathbf{I}_m) + \mathbf{D}_{2,3} \rrbracket_2, \end{aligned}$$

Using sk_i^{*} = ($\mathbf{U}_{1,i^*}, \mathbf{U}_{2,i^*}$), compute the vectors

$$\begin{aligned} \llbracket \mathbf{p}_1^\top \rrbracket_t &= \llbracket (\mathbf{c}'_{1,1} - \mathbf{d}'_{1,1}) + (\mathbf{c}'_{1,2} - \mathbf{d}'_{1,2}) \rrbracket_t - \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{U}_{1,i^*}(\mathbf{I}_{n'+1} \otimes \mathbf{d}_0) \rrbracket_2 \\ \llbracket \mathbf{p}_2^\top \rrbracket_t &= \llbracket (\mathbf{c}'_{2,1} - \mathbf{d}'_{2,1}) + (\mathbf{c}'_{2,2} - \mathbf{d}'_{2,2}) \rrbracket_t - \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{U}_{2,i^*}(\mathbf{I}_m \otimes \mathbf{d}_0) \rrbracket_2. \end{aligned}$$

Run $\mathbf{d}_{\phi_{i^*}, y, x} \leftarrow \text{rec}(\phi_{i^*}, (\mathbf{y}^\top \parallel \mathbf{x}^\top))$ and compute $\llbracket v' \rrbracket_t = \llbracket (\mathbf{p}_1^\top \parallel \mathbf{p}_2^\top) \cdot \mathbf{d}_{\phi_{i^*}, y, x} \rrbracket_t$. Finally, output the discrete log v of $\llbracket v' \rrbracket_t$ in basis $\llbracket \mathbf{c}_3 \rrbracket_1 \llbracket \mathbf{d}_0 \rrbracket_2$ recovered via brute-force.

Proposition 29 (Completeness, Correctness and Compactness). *The sRFE scheme for the AB-eAWS functionality in Construction 28 is complete, correct and compact.*

Proposition 30 (Security). *Suppose that Π is a QA-NIZK for linear spaces satisfying perfect completeness, perfect zero-knowledge and unbounded simulation soundness. Then the sRFE scheme in Construction 28 is adaptively IND-secure under the bi-MDDH_k assumption on $\mathbb{G}$.*

The proofs of Propositions 29 and 30 can be found in Sections 5.1 and 5.2, respectively. When instantiating Π with the pairing-based QA-NIZK by Kiltz and Wee [KW15] and using the transformation of Theorem 18, we obtain the following theorem, where we note that AB-1AWS is a special case of AB-eAWS.

Theorem 31. *Assuming bi-MDDH_k on pairings, there exists an adaptively IND-secure (bounded) RFE scheme for the AB-1AWS functionality.*

5.1 Proof of Completeness, Correctness and Compactness

Proof of Proposition 29. We argue that Construction 28 is complete, correct and compact.

Completeness. Define $t_1 = n' + 1$ and $t_2 = m$. For all $\lambda, L \in \mathbb{N}$, $i \in [L]$, all vectors $\mathbf{q}_1, \dots, \mathbf{q}_L \in \mathbb{Z}_p^k$, all $\text{crs} \leftarrow \text{Setup}(1^\lambda, 1^L, \{\llbracket \mathbf{q}_i \rrbracket_{1,2}\}_{i \in [L]})$ and $(\text{pk}_i, \text{sk}_i) \leftarrow \text{Gen}(\text{crs}, i)$, pk_i is of the following form:

$$(\llbracket \mathbf{T}_{d,i} \rrbracket_1 := \llbracket \mathbf{A} \mathbf{U}_{d,i} \rrbracket_1, \llbracket \mathbf{Q}_{d,i} \rrbracket_1 := \llbracket \mathbf{R}_{d,i} \mathbf{U}_{d,i} \rrbracket_1, \{\llbracket \mathbf{G}_{d,i,j} \rrbracket_2 := \llbracket \mathbf{U}_{d,i} (\mathbf{I}_{n'+1} \otimes \mathbf{B} \mathbf{r}_j) \rrbracket_2\}_{j \in [L] \setminus \{i\}}, \pi_{d,i})_{d \in \{0,1\}},$$

where for $d \in \{0,1\}$, we have $\mathbf{U}_{d,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)t_d}$ and $\pi_{d,i} \leftarrow \text{LPrv}(\text{crs}_{d,i}, \llbracket \mathbf{A}_{d,i} \mathbf{U}_{d,i} \rrbracket_1, \mathbf{U}_{d,i})$ for $(\text{crs}_{d,i}, \text{td}_{d,i}) \leftarrow \text{LGen}(1^\lambda, \mathbb{G}_1, \llbracket \mathbf{A}_{d,i} \rrbracket_1)$ and $\mathbf{A}_{d,i}^\top = [\mathbf{A}^\top \parallel \mathbf{R}_{d,i}^\top]$. Let $\mathbf{F}_{d,i}^\top = [\mathbf{T}_{d,i}^\top \parallel \mathbf{Q}_{d,i}^\top]$. We can observe that

- $\text{LVer}(\text{crs}_{d,i}, \llbracket \mathbf{F}_{d,i} \rrbracket_1, \pi_{d,i}) = 1$ from the correctness of Π and the fact that $\mathbf{F}_{d,i} = \mathbf{A}_{d,i} \mathbf{U}_{d,i}$, and
- for each $j \in [L] \setminus \{i\}$, we have $\llbracket \mathbf{T}_{d,i} \rrbracket_1 \llbracket \mathbf{I}_{t_d} \otimes \mathbf{B} \mathbf{r}_j \rrbracket_2 = \llbracket \mathbf{A} \rrbracket_1 \llbracket \mathbf{G}_{d,i,j} \rrbracket_2$ which follows from the fact that $\mathbf{A} \mathbf{U}_{d,i} \cdot (\mathbf{I}_{t_d} \otimes \mathbf{B} \mathbf{r}_j) = \mathbf{A} \cdot \mathbf{U}_{d,i} (\mathbf{I}_{t_d} \otimes \mathbf{B} \mathbf{r}_j)$.

This implies that $\text{Ver}(\text{crs}, i, \text{pk}_i) = 1$ and concludes the proof of completeness.

Correctness. For all $\lambda, L \in \mathbb{N}$, $i^* \in [L]$, $\mathbf{q}_1, \dots, \mathbf{q}_L \in \mathbb{Z}_p^k$, all $\text{crs} \leftarrow \text{Setup}(1^\lambda, 1^L, \{\llbracket \mathbf{q}_i \rrbracket_{1,2}\}_{i \in [L]})$ and $(\text{pk}_{i^*}, \text{sk}_{i^*}) \leftarrow \text{Gen}(\text{crs}, i^*)$, all $\{\text{pk}_i\}_{i \in [L] \setminus \{i^*\}}$ such that $\text{Ver}(\text{crs}, i, \text{pk}_i) = 1$, all $f_1, \dots, f_L \in \mathcal{F}_{\hat{n},1}^{\text{abp}} \times \mathcal{F}_{n,n'}^{\text{abp}}$, all $(\mathbf{y}, \mathbf{x}) \in \mathbb{Z}_p^{\hat{n}} \times \mathbb{Z}_p^n$ and all $(\mathbf{p}, \mathbf{z}) \in \mathbb{Z}_p^k \times \mathbb{Z}_p^{n'}$, all $(\text{mpk}, \{\text{hsk}_j\}_{j \in [L]}) \leftarrow \text{Agg}(\text{crs}, (\text{pk}_i, f_i)_{i \in [L]})$ and all $\text{ct} \leftarrow \text{Enc}(\text{mpk}, (\mathbf{y}, \mathbf{x}), (\mathbf{p}, \mathbf{z}))$, we have $\text{sk}_{i^*} = (\mathbf{U}_{1,i^*}, \mathbf{U}_{2,i^*})$ and

$$\begin{aligned} (\mathbf{c}_{1,1}, \mathbf{c}_{1,2}) &= \left((\mathbf{f}^\top \parallel \mathbf{z}^\top) \otimes \mathbf{s}^\top \mathbf{A} \mathbf{V} + \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A} \mathbf{H}_{1,i}, \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{1,i} + \mathbf{A} \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}) - \mathbf{A} \mathbf{W}_i \right) \\ (\mathbf{c}_{2,1}, \mathbf{c}_0, \mathbf{c}_3) &= \left(\mathbf{p}^\top \otimes \mathbf{s}^\top \mathbf{A} \mathbf{V} + \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A} \mathbf{H}_{2,i}, \mathbf{s}^\top \mathbf{A}, \mathbf{s}^\top \mathbf{A} \mathbf{V} \right) \\ \mathbf{c}_{2,2} &= \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{2,i} - \mathbf{A} \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{A} \mathbf{W}_i (\mathbf{L}_{i,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \\ (\mathbf{d}_0, \mathbf{D}_{0,1}, \mathbf{D}_{0,2}) &= \left(\mathbf{B} \mathbf{r}_{i^*}, \mathbf{I}_{\mathbf{g}_{i^*}} \otimes \mathbf{B} \mathbf{r}_{i^*}, \mathbf{q}_{i^*} \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_{i^*} \right) \\ (\mathbf{D}_{1,1}, \mathbf{D}_{1,2}) &= \left(\sum_{i \in [L] \setminus \{i^*\}} \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_{i^*}} \otimes \mathbf{B} \mathbf{r}_{i^*}), \sum_{i \in [L] \setminus \{i^*\}} \mathbf{G}_{1,i,i^*} - \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{B} \mathbf{r}_{i^*}) - \mathbf{W}_i (\mathbf{I}_{n'+1} \otimes \mathbf{B} \mathbf{r}_{i^*}) \right) \\ (\mathbf{D}_{2,1}, \mathbf{D}_{2,2}) &= \left(\sum_{i \in [L] \setminus \{i^*\}} \mathbf{H}_{2,i} (\mathbf{q}_{i^*} \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_{i^*}), \sum_{i \in [L] \setminus \{i^*\}} \mathbf{W}_i (\mathbf{I}_{m+n'} \otimes \mathbf{B} \mathbf{r}_{i^*}) \mathbf{L}_{i,1} \right) \\ \mathbf{D}_{2,3} &= \sum_{i \in [L] \setminus \{i^*\}} \mathbf{G}_{2,i,i^*} - \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{W}_i (\mathbf{I}_{m+n'} \otimes \mathbf{B} \mathbf{r}_{i^*}) \mathbf{L}_{i,0}, \end{aligned}$$

where we parse the ciphertext $\text{ct} = (\llbracket \mathbf{c}_0 \rrbracket_1, \llbracket \mathbf{c}_{1,1} \rrbracket_1, \llbracket \mathbf{c}_{1,2} \rrbracket_1, \llbracket \mathbf{c}_{2,1} \rrbracket_1, \llbracket \mathbf{c}_{2,2} \rrbracket_1, \llbracket \mathbf{c}_3 \rrbracket_1)$ and the helper secret key $\text{hsk}_{i^*} = (\llbracket \mathbf{d}_0 \rrbracket_2, \llbracket \mathbf{D}_{0,1} \rrbracket_2, \llbracket \mathbf{D}_{0,2} \rrbracket_2, \llbracket \mathbf{D}_{1,1} \rrbracket_2, \llbracket \mathbf{D}_{1,2} \rrbracket_2, \llbracket \mathbf{D}_{2,1} \rrbracket_2, \llbracket \mathbf{D}_{2,2} \rrbracket_2, \llbracket \mathbf{D}_{2,3} \rrbracket_2)$. By computing

$$\begin{aligned} \llbracket \mathbf{c}'_{1,1} \rrbracket_t &= \llbracket \mathbf{c}_{1,1} \rrbracket_1 \llbracket \mathbf{D}_{0,1} \rrbracket_2, & \llbracket \mathbf{d}'_{1,1} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_{1,1} \rrbracket_2, \\ \llbracket \mathbf{c}'_{1,2} \rrbracket_t &= \llbracket \mathbf{c}_{1,2} \rrbracket_1 \llbracket \mathbf{I}_{n'+1} \otimes \mathbf{d}_0 \rrbracket_2, & \llbracket \mathbf{d}'_{1,2} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_{1,2} \rrbracket_2, \\ \llbracket \mathbf{c}'_{2,1} \rrbracket_t &= \llbracket \mathbf{c}_{2,1} \rrbracket_1 \llbracket \mathbf{D}_{0,2} \rrbracket_2, & \llbracket \mathbf{d}'_{2,1} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_{2,1} \rrbracket_2, \\ \llbracket \mathbf{c}'_{2,2} \rrbracket_t &= \llbracket \mathbf{c}_{2,2} \rrbracket_1 \llbracket \mathbf{I}_m \otimes \mathbf{d}_0 \rrbracket_2, & \llbracket \mathbf{d}'_{2,2} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_{2,2} ((\mathbf{y}^\top \parallel \mathbf{x}^\top)^\top \otimes \mathbf{I}_m) + \mathbf{D}_{2,3} \rrbracket_2, \end{aligned}$$

the decryptor obtains

$$\begin{aligned} \mathbf{c}'_{1,1} &= \left((\mathbf{f}^\top \parallel \mathbf{z}^\top) \otimes \mathbf{s}^\top \mathbf{A} \mathbf{V} + \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A} \mathbf{H}_{1,i} \right) (\mathbf{I}_{\mathbf{g}_{i^*}} \otimes \mathbf{B} \mathbf{r}_{i^*}) \\ &= (\mathbf{f}^\top \mathbf{g}_{i^*} \parallel \mathbf{z}^\top) \cdot \mathbf{s}^\top \mathbf{A} \mathbf{V} \mathbf{B} \mathbf{r}_{i^*} + \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A} \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_{i^*}} \otimes \mathbf{B} \mathbf{r}_{i^*}) \\ \mathbf{d}'_{1,1} &= \mathbf{s}^\top \mathbf{A} \sum_{i \in [L] \setminus \{i^*\}} \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_{i^*}} \otimes \mathbf{B} \mathbf{r}_{i^*}) \end{aligned}$$

$$\begin{aligned}
\mathbf{c}'_{1,2} &= \left(\mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{1,i} - \mathbf{A} \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}) - \mathbf{A} \mathbf{W}_i \right) (\mathbf{I}_{n'+1} \otimes \mathbf{B} \mathbf{r}_{i^*}) \\
&= \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{1,i} (\mathbf{I}_{n'+1} \otimes \mathbf{B} \mathbf{r}_{i^*}) - \mathbf{A} \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{B} \mathbf{r}_{i^*}) - \mathbf{A} \mathbf{W}_i (\mathbf{I}_{n'+1} \otimes \mathbf{B} \mathbf{r}_{i^*}) \\
\mathbf{d}'_{1,2} &= \mathbf{s}^\top \mathbf{A} \sum_{i \in [L] \setminus \{i^*\}} \mathbf{G}_{1,i,i^*} - \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{B} \mathbf{r}_{i^*}) - \mathbf{W}_i (\mathbf{I}_{n'+1} \otimes \mathbf{B} \mathbf{r}_{i^*}) \\
\mathbf{c}'_{2,1} &= \left(\mathbf{p}^\top \otimes \mathbf{s}^\top \mathbf{A} \mathbf{V} + \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A} \mathbf{H}_{2,i} \right) (\mathbf{q}_{i^*} \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_{i^*}) \\
&= \mathbf{p}^\top \mathbf{q}_{i^*} \cdot \mathbf{s}^\top \mathbf{A} \mathbf{V} \mathbf{B} \mathbf{r}_{i^*} \cdot \mathbf{e}_1^\top + \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A} \mathbf{H}_{2,i} (\mathbf{q}_{i^*} \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_{i^*}) \\
\mathbf{d}'_{2,1} &= \mathbf{s}^\top \mathbf{A} \sum_{i \in [L] \setminus \{i^*\}} \mathbf{H}_{2,i} (\mathbf{q}_{i^*} \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_{i^*}) \\
\mathbf{c}'_{2,2} &= \mathbf{s}^\top \sum_{i \in [L]} \left(\mathbf{T}_{2,i} - \mathbf{A} \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{A} \mathbf{W}_i (\mathbf{L}_{i,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \right) (\mathbf{I}_m \otimes \mathbf{B} \mathbf{r}_{i^*}) \\
&= \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{2,i} (\mathbf{I}_m \otimes \mathbf{B} \mathbf{r}_{i^*}) - \mathbf{A} \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{A} \mathbf{W}_i (\mathbf{L}_{i,\mathbf{y},\mathbf{x}} \otimes \mathbf{B} \mathbf{r}_{i^*}) \\
&= \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{2,i} (\mathbf{I}_m \otimes \mathbf{B} \mathbf{r}_{i^*}) - \mathbf{A} \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{A} \mathbf{W}_i (\mathbf{I}_{m+n'} \otimes \mathbf{B} \mathbf{r}_{i^*}) \mathbf{L}_{i,\mathbf{y},\mathbf{x}} \\
\mathbf{d}'_{2,2} &= \mathbf{s}^\top \mathbf{A} \sum_{i \in [L] \setminus \{i^*\}} \mathbf{W}_i (\mathbf{I}_{m+n'} \otimes \mathbf{B} \mathbf{r}_{i^*}) \mathbf{L}_{i,1} ((\mathbf{y}^\top \parallel \mathbf{x}^\top)^\top \otimes \mathbf{I}_m) \\
&\quad + \mathbf{s}^\top \mathbf{A} \sum_{i \in [L] \setminus \{i^*\}} \mathbf{G}_{2,i,i^*} - \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{W}_i (\mathbf{I}_{m+n'} \otimes \mathbf{B} \mathbf{r}_{i^*}) \mathbf{L}_{i,0} \\
&= \mathbf{s}^\top \mathbf{A} \sum_{i \in [L] \setminus \{i^*\}} \mathbf{G}_{2,i,i^*} - \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{W}_i (\mathbf{I}_{m+n'} \otimes \mathbf{B} \mathbf{r}_{i^*}) \mathbf{L}_{i,\mathbf{y},\mathbf{x}} .
\end{aligned}$$

Since the pair $(\text{pk}_{i^*}, \text{sk}_{i^*})$ was generated honestly, there exist two matrices $\mathbf{U}_{1,i^*}, \mathbf{U}_{2,i^*}$ such that pk_{i^*} is of the form

$$\text{pk}_{i^*} = \left(\begin{array}{l} \llbracket \mathbf{T}_{1,i^*} \rrbracket_1 := \llbracket \mathbf{A} \mathbf{U}_{1,i^*} \rrbracket_1, \llbracket \mathbf{Q}_{1,i^*} \rrbracket_1, \{ \llbracket \mathbf{G}_{1,i^*,j} \rrbracket_2 := \llbracket \mathbf{U}_{1,i^*} (\mathbf{I}_{n'+1} \otimes \mathbf{B} \mathbf{r}_j) \rrbracket_2 \}_{j \in [L] \setminus \{i\}}, \pi_{1,i^*} \\ \llbracket \mathbf{T}_{2,i^*} \rrbracket_1 := \llbracket \mathbf{A} \mathbf{U}_{2,i^*} \rrbracket_1, \llbracket \mathbf{Q}_{2,i^*} \rrbracket_1, \{ \llbracket \mathbf{G}_{2,i^*,j} \rrbracket_2 := \llbracket \mathbf{U}_{2,i^*} (\mathbf{I}_m \otimes \mathbf{B} \mathbf{r}_j) \rrbracket_2 \}_{j \in [L] \setminus \{i\}}, \pi_{2,i^*} \end{array} \right),$$

and $\text{sk}_{i^*} = (\mathbf{U}_{1,i^*}, \mathbf{U}_{2,i^*})$. Furthermore, verification guarantees for all $i \in [L] \setminus \{i^*\}$ that

$$\llbracket \mathbf{T}_{1,i} \rrbracket_1 \llbracket \mathbf{I}_{n'+1} \otimes \mathbf{B} \mathbf{r}_{i^*}^* \rrbracket_2 \stackrel{?}{=} \llbracket \mathbf{A} \rrbracket_1 \llbracket \mathbf{G}_{1,i,i^*} \rrbracket_2 \quad \text{and} \quad \llbracket \mathbf{T}_{2,i} \rrbracket_1 \llbracket \mathbf{I}_m \otimes \mathbf{B} \mathbf{r}_{i^*}^* \rrbracket_2 \stackrel{?}{=} \llbracket \mathbf{A} \rrbracket_1 \llbracket \mathbf{G}_{2,i,i^*} \rrbracket_2 .$$

Therefore, the vectors $\llbracket \mathbf{p}_1^\top \rrbracket_t$ and $\llbracket \mathbf{p}_2^\top \rrbracket_t$ equal

$$\begin{aligned}
\llbracket \mathbf{p}_1^\top \rrbracket_t &= \llbracket (\mathbf{c}'_{1,1} - \mathbf{d}'_{1,1}) + (\mathbf{c}'_{1,2} - \mathbf{d}'_{1,2}) \rrbracket_t - \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{U}_{1,i^*} (\mathbf{I}_{n'+1} \otimes \mathbf{d}_0) \rrbracket_2 \\
&= \llbracket (\mathbf{f}^\top \mathbf{g}_{i^*} \parallel \mathbf{z}^\top) \cdot \mathbf{s}^\top \mathbf{A} \mathbf{V} \mathbf{B} \mathbf{r}_{i^*} - \mathbf{s}^\top \mathbf{A} \mathbf{W}_{i^*} (\mathbf{I}_{n'+1} \otimes \mathbf{B} \mathbf{r}_{i^*}) \rrbracket_t \\
\llbracket \mathbf{p}_2^\top \rrbracket_t &= \llbracket (\mathbf{c}'_{2,1} - \mathbf{d}'_{2,1}) + (\mathbf{c}'_{2,2} - \mathbf{d}'_{2,2}) \rrbracket_t - \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{U}_{2,i^*} (\mathbf{I}_m \otimes \mathbf{d}_0) \rrbracket_2 \\
&= \llbracket \mathbf{p}^\top \mathbf{q}_{i^*} \cdot \mathbf{s}^\top \mathbf{A} \mathbf{V} \mathbf{B} \mathbf{r}_{i^*} \cdot \mathbf{e}_1^\top + \mathbf{s}^\top \mathbf{A} \mathbf{W}_{i^*} (\mathbf{I}_{m+n'} \otimes \mathbf{B} \mathbf{r}_{i^*}) \mathbf{L}_{i^*,\mathbf{y},\mathbf{x}} \rrbracket_t ,
\end{aligned}$$

and we observe that

$$(\mathbf{p}_1^\top, \mathbf{p}_2^\top) = \text{pgb}(\phi_{i^*}, (\mathbf{y}, \mathbf{x}), (\mathbf{f}^\top \mathbf{g}_{i^*} \parallel \mathbf{z}^\top) \cdot \mathbf{s}^\top \mathbf{A} \mathbf{V} \mathbf{B} \mathbf{r}_{i^*}) + (\mathbf{0}, \mathbf{p}^\top \mathbf{q}_{i^*} \cdot \mathbf{s}^\top \mathbf{A} \mathbf{V} \mathbf{B} \mathbf{r}_{i^*} \cdot \mathbf{e}_1^\top) .$$

Then it follows from the correctness of the garbling scheme that if $g_{i^*}(\mathbf{y}) = 0$, then

$$v' = (g_{i^*}(\mathbf{y}) \cdot \mathbf{f}^\top \mathbf{g}_{i^*} + h_{i^*}(\mathbf{x})^\top \mathbf{z} + \mathbf{p}^\top \mathbf{q}_{i^*}) \cdot \mathbf{s}^\top \mathbf{A} \mathbf{V} \mathbf{B} \mathbf{r}_{i^*} = (h_{i^*}(\mathbf{x})^\top \mathbf{z} + \mathbf{p}^\top \mathbf{q}_{i^*}) \cdot \mathbf{s}^\top \mathbf{A} \mathbf{V} \mathbf{B} \mathbf{r}_{i^*} ,$$

which implies that the recovered discrete log in basis $\llbracket \mathbf{c}_3 \rrbracket_1 \llbracket \mathbf{d}_0 \rrbracket_2 = \llbracket \mathbf{s}^\top \mathbf{A} \mathbf{V} \mathbf{B} \mathbf{r}_{i^*} \rrbracket_t$ is $v = h_{i^*}(\mathbf{x})^\top \mathbf{z} + \mathbf{p}^\top \mathbf{q}_{i^*}$.

Compactness. Let $j \in [L]$. The scheme has the following properties:

$$\begin{aligned}
|\text{crs}| &= L^2 \cdot (m + n') \cdot \text{poly}(\lambda) & |\text{mpk}| &= (m(\hat{n} + n) + n') \cdot \text{poly}(\lambda) \\
|\text{hsk}_j| &= (m(\hat{n} + n) + n') \cdot \text{poly}(\lambda) & |\text{ct}| &= (m + n') \cdot \text{poly}(\lambda) ,
\end{aligned}$$

where the size of crs follows from the fact that the total size of $\{\text{crs}_i\}_{i \in [L]}$ is $L \cdot \text{poly}(\lambda)$ if we instantiate Π with the QA-NIZK of [KW15], and the fact that the size of the language description is $\text{poly}(\lambda)$. $\square$

5.2 Proof of Security

Proof of Proposition 30. Let FE denote the sRFE scheme in Construction 28. We consider a sequence of hybrid games $G_0^0, \dots, G_7^0, G_7^1, \dots, G_0^1$ where $G_0^b = \text{Exp}_{\text{FE}, \mathcal{A}}^{\text{sRFE}-b}$. We argue $G_7^0 \equiv G_7^1$ and $G_{\ell-1}^b \approx_c G_\ell^b$ for $\ell \in [7]$ and $b \in \{0, 1\}$. Then the proposition follows via a hybrid argument. Modifications between consecutive games are highlighted using gray boxes.

Game G_0^b for $b \in \{0, 1\}$: This is $\text{Exp}_{\text{FE}, \mathcal{A}}^{\text{sRFE}-b}$. Specifically, we have:

- The common reference string crs is of the form

$$\text{crs} = \left(\left\{ \text{crs}_{1,i}, \text{crs}_{2,i}, \left\| \begin{array}{c} \mathbf{A}\mathbf{W}_i, \mathbf{R}_{1,i}, \mathbf{R}_{2,i}, \mathbf{A}\mathbf{H}_{1,i}, \mathbf{A}\mathbf{H}_{2,i} \\ \mathbf{A}\mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}), \mathbf{A}\mathbf{H}_{2,i}(\mathbf{q}_i(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{I}_{2k+1}) \end{array} \right\|_1 \right\}_{i \in [L]}, \right. \\ \left. \begin{array}{c} \llbracket \mathbf{A}, \mathbf{A}\mathbf{V} \rrbracket_1, \{ \llbracket \mathbf{B}\mathbf{r}_j, \mathbf{I}_{\mathbf{g}_j} \otimes \mathbf{B}\mathbf{r}_j, \mathbf{q}_j(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{B}\mathbf{r}_j \rrbracket_2 \}_{j \in [L]}, \\ \{ \llbracket \mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_k} \otimes \mathbf{B}\mathbf{r}_j), \mathbf{H}_{2,i}(\mathbf{q}_k(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{B}\mathbf{r}_j), \mathbf{W}_i(\mathbf{I}_{m+n'} \otimes \mathbf{B}\mathbf{r}_j) \rrbracket_2 \}_{\substack{i \in [L], \\ j \in [L] \setminus \{i\} \\ \kappa \in \{i, j\}}} \end{array} \right),$$

where $(\text{crs}_{d,i}, \text{td}_{d,i}) \leftarrow \text{LGen}(1^\lambda, G_1, \llbracket \mathbf{A}_{d,i} \rrbracket_1)$ and $\mathbf{A}_{d,i}^\top = [\mathbf{A}^\top \parallel \mathbf{R}_{d,i}^\top]$ for $d \in \{1, 2\}$.

- For each $i \in [L]$, each $\text{pk}_i \in \mathcal{D}_i$ is of the form

$$\text{pk}_i = \left(\begin{array}{c} \llbracket \mathbf{T}_{d,i} \rrbracket_1 := \llbracket \mathbf{A}\mathbf{U}_{d,i} \rrbracket_1, \llbracket \mathbf{Q}_{d,i} \rrbracket_1 := \llbracket \mathbf{R}_{d,i}\mathbf{U}_{d,i} \rrbracket_1, \\ \{ \llbracket \mathbf{G}_{d,i,j} \rrbracket_2 := \llbracket \mathbf{U}_{d,i}(\mathbf{I}_{t_d} \otimes \mathbf{B}\mathbf{r}_j) \rrbracket_2 \}_{j \in [L] \setminus \{i\}}, \pi_{d,i} \end{array} \right)_{d \in \{1,2\}},$$

where $t_1 = n' + 1$, $t_2 = m$, $\pi_{d,i} \leftarrow \text{LPrv}(\text{crs}_{d,i}, \llbracket \mathbf{F}_{d,i} \rrbracket_1, \mathbf{U}_{d,i})$, $\mathbf{F}_{d,i}^\top = [\mathbf{T}_{d,i}^\top \parallel \mathbf{Q}_{d,i}^\top]$ and $\text{sk}_i = (\mathbf{U}_{1,i}, \mathbf{U}_{2,i})$ is the corresponding secret key.

- For each $i \in [L]$, pk_i^* is of the form

$$\text{pk}_i^* = (\llbracket \mathbf{T}_{d,i}^* \rrbracket_1, \llbracket \mathbf{Q}_{d,i}^* \rrbracket_1, \{ \llbracket \mathbf{G}_{d,i,j}^* \rrbracket_2 \}_{j \in [L] \setminus \{i\}}, \pi_{d,i}^*)_{d \in \{1,2\}}$$

such that $\text{Ver}(\text{crs}, i, \text{pk}_i^*) = 1$, i.e. $\text{LVer}(\text{crs}_{d,i}, \llbracket \mathbf{F}_{d,i}^* \rrbracket_1, \pi_{d,i}^*) = 1$ for $\mathbf{F}_{d,i}^* = [\mathbf{T}_{d,i}^*]^\top \parallel [\mathbf{Q}_{d,i}^*]^\top$ and $\mathbf{A}\mathbf{G}_{d,i,j}^* = \mathbf{T}_{d,i}^* \mathbf{B}\mathbf{r}_j$ for each $j \in [L] \setminus \{i\}$ and $d \in \{1, 2\}$.

- The challenge ciphertext $\text{ct} = (\llbracket \mathbf{c}_0 \rrbracket_1, \llbracket \mathbf{c}_{1,1} \rrbracket_1, \llbracket \mathbf{c}_{1,2} \rrbracket_1, \llbracket \mathbf{c}_{2,1} \rrbracket_1, \llbracket \mathbf{c}_{2,2} \rrbracket_1, \llbracket \mathbf{c}_3 \rrbracket_1)$ for inputs $((\mathbf{y}, \mathbf{x}), (\mathbf{z}^0, \mathbf{p}^0))$ and $((\mathbf{y}, \mathbf{x}), (\mathbf{z}^1, \mathbf{p}^1))$ is of the form

$$\begin{aligned} \mathbf{c}_0 &= \mathbf{s}^\top \mathbf{A} \\ \mathbf{c}_{1,1} &= (\mathbf{f}^\top \parallel (\mathbf{z}^b)^\top) \otimes \mathbf{s}^\top \mathbf{A}\mathbf{V} + \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A}\mathbf{H}_{1,i} \\ \mathbf{c}_{1,2} &= \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{1,i} - \mathbf{A}\mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}) - \mathbf{A}\mathbf{W}_i \\ \mathbf{c}_{2,1} &= (\mathbf{p}^b)^\top \otimes \mathbf{s}^\top \mathbf{A}\mathbf{V} + \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A}\mathbf{H}_{2,i} \\ \mathbf{c}_{2,2} &= \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{2,i} - \mathbf{A}\mathbf{H}_{2,i}(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{A}\mathbf{W}_i(\mathbf{L}_{i,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \\ \mathbf{c}_3 &= \mathbf{s}^\top \mathbf{A}\mathbf{V}. \end{aligned}$$

Game G_1^b for $b \in \{0, 1\}$: This is the same as G_0^b , except that for $d \in \{1, 2\}$, $i \in [L]$ and $\text{pk}_i \in \mathcal{D}_i$, the challenger computes

$$\pi_{d,i} \leftarrow \text{LSim}(\text{crs}_{d,i}, \text{td}_{d,i}, \llbracket \mathbf{F}_{d,i} \rrbracket_1) .$$

We have $G_0^b \equiv G_1^b$ which follows from the perfect zero-knowledge of Π .

Game G_2^b for $b \in \{0, 1\}$: This is the same as G_1^b except that the challenger picks $\mathbf{s} \xleftarrow{\$} \mathbb{Z}_p^k$ and $\tilde{\mathbf{R}}_{d,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+2) \times (2k+2)}$ for $d \in \{1, 2\}$ and $i \in [L]$ along with $\mathbf{A}$, and sets

$$\mathbf{R}_{d,i} = \tilde{\mathbf{R}}_{d,i} \cdot \begin{bmatrix} \mathbf{s}^\top \mathbf{A} \\ \mathbf{I}_{2k+1} \end{bmatrix} .$$

We observe that the distribution of $\mathbf{R}_{d,i}$ does not change from G_1^b to G_2^b . Hence, we have $G_1^b \equiv G_2^b$.

Game G_3^b for $b \in \{0, 1\}$: This is the same as G_2^b , except that the challenger generates $\mathbf{c}_{1,2}^*$ and $\mathbf{c}_{2,2}^*$ as follows:

$$\begin{aligned} \mathbf{c}_{1,2} &= \sum_{i \in [L]} (\mathbf{e}_1^{(2k+2)})^\top \tilde{\mathbf{R}}_{1,i}^{-1} \mathbf{Q}_{1,i}^* - \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A} \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}) + \underline{\mathbf{A} \mathbf{W}_i} \\ \mathbf{c}_{2,2} &= \sum_{i \in [L]} (\mathbf{e}_1^{(2k+2)})^\top \tilde{\mathbf{R}}_{2,i}^{-1} \mathbf{Q}_{2,i}^* - \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A} \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) - \underline{\mathbf{A} \mathbf{W}_i (\mathbf{L}_{i,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1})} \end{aligned}$$

We have $G_2^b \approx_c G_3^b$ which can be seen as follows. Let $d \in \{1, 2\}$. First, for $i \in [L]$ such that $\text{pk}_i^* \in \mathcal{D}_i^*$, it is not hard to observe that the distributions of $\mathbf{s}^\top \mathbf{T}_{d,i}^*$ and $\mathbf{e}_1^\top \tilde{\mathbf{R}}_{d,i}^{-1} \mathbf{Q}_{d,i}^*$ are identical. Second, for $i \in [L]$ such that $\text{pk}_i^* \notin \mathcal{D}_i^*$, we rely on the unbounded simulation soundness of Π . In the reduction, the challenger of Π provides $\mathbf{A}_{d,i}$ and $\text{crs}_{d,i}$. The simulation proceeds as in G_3^b except that, for all $\text{pk}_i \in \mathcal{D}_i$, the simulator submits an oracle query $\llbracket \mathbf{F}_{d,i} \rrbracket_1$ to obtain $\pi_{d,i}$. Finally, upon $\mathcal{A}$ registering pk_i^* , the simulator outputs $(\llbracket \mathbf{F}_{d,i}^* \rrbracket_1, \pi_{d,i}^*)$. To conclude, we observe that if $\mathbf{e}_1^\top \tilde{\mathbf{R}}_{d,i}^{-1} \mathbf{Q}_{d,i}^* \neq \mathbf{s}^\top \mathbf{T}_{d,i}^*$, then $\mathbf{F}_{d,i}^* \notin \text{span}(\mathbf{A}_{d,i})$.

Game G_4^b for $b \in \{0, 1\}$: This is the same as G_3^b , except that the challenger replaces $\mathbf{s}^\top \mathbf{A}$ with a uniformly random vector $\mathbf{c}^\top \xleftarrow{\$} \mathbb{Z}_p^{1 \times (2k+1)}$. Specifically, $\mathbf{R}_{d,i}$ for $d \in \{1, 2\}$ and $\text{ct} = (\llbracket \mathbf{c}_0 \rrbracket_1, \llbracket \mathbf{c}_{1,1} \rrbracket_1, \llbracket \mathbf{c}_{1,2} \rrbracket_1, \llbracket \mathbf{c}_{2,1} \rrbracket_1, \llbracket \mathbf{c}_{2,2} \rrbracket_1, \llbracket \mathbf{c}_3 \rrbracket_1)$ are computed as follows:

$$\begin{aligned} \mathbf{R}_{d,i} &= \tilde{\mathbf{R}}_{d,i} \cdot \begin{bmatrix} \mathbf{c}^\top \\ \mathbf{I}_{2k+1} \end{bmatrix} \\ \mathbf{c}_0 &= \mathbf{c}^\top \\ \mathbf{c}_{1,1} &= (\mathbf{f}^\top \parallel (\mathbf{z}^b)^\top) \otimes \mathbf{c}^\top \mathbf{V} + \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_{1,i} \\ \mathbf{c}_{1,2} &= \sum_{i \in [L]} \mathbf{e}_1^\top \tilde{\mathbf{R}}_{1,i}^{-1} \mathbf{Q}_{1,i}^* - \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}) + \underline{\mathbf{W}_i} \\ \mathbf{c}_{2,1} &= (\mathbf{p}^b)^\top \otimes \mathbf{c}^\top \mathbf{V} + \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_{2,i} \\ \mathbf{c}_{2,2} &= \sum_{i \in [L]} \mathbf{e}_1^\top \tilde{\mathbf{R}}_{2,i}^{-1} \mathbf{Q}_{2,i}^* - \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) - \underline{\mathbf{W}_i (\mathbf{L}_{i,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1})} \\ \mathbf{c}_3 &= \mathbf{c}^\top \mathbf{V} . \end{aligned}$$

It is not hard to see that we have $G_3^b \approx_c G_4^b$ under the MDDH $_k$ assumption in $\mathbb{G}_1$ which states that $(\llbracket \mathbf{A}^\top \rrbracket_1, \llbracket \mathbf{A}^\top \mathbf{s} \rrbracket_1) \approx_c (\llbracket \mathbf{A}^\top \rrbracket_1, \llbracket \mathbf{c} \rrbracket_1)$ for $\mathbf{c} \xleftarrow{\$} \mathbb{Z}_p^{2k+1}$.

Game G_5^b for $b \in \{0, 1\}$: This is the same as G_4^b , except that the challenger replaces $(\mathbf{A} \mathbf{V}, \mathbf{c}^\top \mathbf{V})$ with $(\tilde{\mathbf{V}}, \mathbf{v}^\top) \xleftarrow{\$}$

$\mathbb{Z}_p^{k \times (2k+1)} \times \mathbb{Z}_p^{1 \times (2k+1)}$. In particular, crs , $\mathbf{c}_{1,1}^*$, $\mathbf{c}_{2,1}^*$ and $\mathbf{c}_3^*$ are computed as follows:

$$\begin{aligned} \text{crs} &= \left(\left\{ \text{crs}_{1,i}, \text{crs}_{2,i}, \left\| \begin{array}{c} \mathbf{A}\mathbf{W}_i, \mathbf{R}_{1,i}, \mathbf{R}_{2,i}, \mathbf{A}\mathbf{H}_{1,i}, \mathbf{A}\mathbf{H}_{2,i} \\ \mathbf{A}\mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}), \mathbf{A}\mathbf{H}_{2,i}(\mathbf{q}_i(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{I}_{2k+1}) \end{array} \right\|_1 \right\}_{i \in [L]} \right. \\ &\quad \left. \left\| \mathbf{A}, \tilde{\mathbf{V}} \right\|_1, \{ \left\| \mathbf{B}\mathbf{r}_j, \mathbf{I}_{\mathbf{g}_j} \otimes \mathbf{B}\mathbf{r}_j, \mathbf{q}_j(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{B}\mathbf{r}_j \right\|_2 \}_{j \in [L]}, \right. \\ &\quad \left. \{ \left\| \mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_\kappa} \otimes \mathbf{B}\mathbf{r}_j), \mathbf{H}_{2,i}(\mathbf{q}_\kappa(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{B}\mathbf{r}_j), \mathbf{W}_i(\mathbf{I}_{m+n'} \otimes \mathbf{B}\mathbf{r}_j) \right\|_2 \}_{\substack{i \in [L], \\ j \in [L] \setminus \{i\} \\ \kappa \in \{i,j\}}} \right) \\ \mathbf{c}_{1,1} &= (\mathbf{f}^\top \parallel (\mathbf{z}^b)^\top) \otimes \mathbf{v}^\top + \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_{1,i} \\ \mathbf{c}_{2,1} &= (\mathbf{p}^b)^\top \otimes \mathbf{v}^\top + \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_{2,i} \\ \mathbf{c}_3^* &= \mathbf{v}^\top. \end{aligned}$$

We have $G_4^b \approx_s G_5^b$. To see this, we first observe that $\mathbf{V} \xleftarrow{s} \mathbb{Z}_p^{(2k+1) \times (2k+1)}$ is sampled uniformly at random and not published anywhere else. Second, we note that both $\mathbf{A}$ and $\mathbf{c}^\top$ are full (row) rank with overwhelming probability, in which case $(\mathbf{A}\mathbf{V}, \mathbf{c}^\top \mathbf{V})$ are uniform in $\mathbb{Z}_p^{k \times (2k+1)} \times \mathbb{Z}_p^{1 \times (2k+1)}$.

Game G_6^b for $b \in \{0, 1\}$: This is the same as G_5^b except that, in addition to $\mathbf{B}_1 = \mathbf{B} \xleftarrow{s} \mathbb{Z}_p^{(2k+1) \times k}$, the challenger also samples $\mathbf{B}_2 \xleftarrow{s} \mathbb{Z}_p^{2k+1}$ and $\mathbf{B}_3 \xleftarrow{s} \mathbb{Z}_p^{(2k+1) \times k}$. Then it computes the dual basis $(\mathbf{B}_1^\parallel, \mathbf{B}_2^\parallel, \mathbf{B}_3^\parallel)$ and generates the ciphertext components $\mathbf{c}_{1,1}^*$ and $\mathbf{c}_{2,1}^*$ as follows:

$$\begin{aligned} \mathbf{c}_{1,1} &= (\mathbf{f}^\top \parallel (\mathbf{z}^b)^\top) \otimes (\mathbf{v}^{(1,3)})^\top + (\mathbf{f}^\top \parallel (\mathbf{z}^0)^\top) \otimes (\mathbf{v}^{(2)})^\top + \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_{1,i} \\ \mathbf{c}_{2,1} &= (\mathbf{p}^b)^\top \otimes (\mathbf{v}^{(1,3)})^\top + (\mathbf{p}^0)^\top \otimes (\mathbf{v}^{(2)})^\top + \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_{2,i}. \end{aligned}$$

Below, we prove the following claim.

Claim 32. *We have $G_5^b \equiv G_6^b$.*

Game $G_{6,\ell}^b$ for $\ell \in [0; L]$ and $b \in \{0, 1\}$: This is the same as G_6^b except that, for $j \in [\ell]$, the challenger replaces $\mathbf{B}_1 \mathbf{r}_j$ with $\mathbf{d}_j \xleftarrow{s} \text{span}(\mathbf{B}_2)$. It is not hard to see that $G_{6,0}^b \equiv G_6^b$. This is because $[0] = \emptyset$, so the two games are identical. Furthermore, we prove the following claim for each $\ell \in [L]$ and $b \in \{0, 1\}$.

Claim 33. *We have $G_{6,\ell-1}^b \approx_c G_{6,\ell}^b$ assuming MDDH $_k$ in $\mathbb{G}_1$ and $\mathbb{G}_2$.*

Game G_7^b for $b \in \{0, 1\}$: This is the same as $G_{6,L}^b$ except that we generate the ciphertext components $\mathbf{c}_1^*$ and $\mathbf{c}_2^*$ as follows:

$$\begin{aligned} \mathbf{c}_{1,1} &= (\mathbf{f}^\top \parallel (\mathbf{z}^0)^\top) \otimes (\mathbf{v}^{(1,3)})^\top + (\mathbf{f}^\top \parallel (\mathbf{z}^b)^\top) \otimes (\mathbf{v}^{(2)})^\top + \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_{1,i} \\ \mathbf{c}_{2,1} &= (\mathbf{p}^0)^\top \otimes (\mathbf{v}^{(1,3)})^\top + (\mathbf{p}^b)^\top \otimes (\mathbf{v}^{(2)})^\top + \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_{2,i}. \end{aligned}$$

We have $G_{6,L}^b \approx_s G_7^b$ from a statistical argument similar to the proof of Claim 32. Furthermore, we can observe that the challenger's output in G_7^b is independent of the challenge bit b , so it follows $G_7^0 \equiv G_7^1$. This concludes the proof of the proposition. $\square$

We now turn to the claims.

Proof of Claim 32. The claim follows from the fact that the basis $\mathbf{B}_2$ and its dual $\mathbf{B}_2^\parallel$ are not revealed. Therefore, $(\mathbf{c}^\top \mathbf{H}_{1,i})^{(2)}$ and $(\mathbf{c}^\top \mathbf{H}_{2,i})^{(2)}$ are hidden which implies that they can mask $(\mathbf{f}^\top \parallel (\mathbf{z}^b)^\top) \otimes (\mathbf{v}^{(2)})^\top$ and $(\mathbf{p}^b)^\top \otimes (\mathbf{v}^{(2)})^\top$, i.e. the following distributions are identical for $\beta \in \{b, 0\}$:

$$\left\{ (\mathbf{f}^\top \parallel (\mathbf{z}^\beta)^\top) \otimes (\mathbf{v}^{(2)})^\top + \sum_{i \in [L]} (\mathbf{c}^\top \mathbf{H}_{1,i})^{(2)} \right\} \equiv \left\{ \sum_{i \in [L]} (\mathbf{c}^\top \mathbf{H}_{1,i})^{(2)} \right\}$$

$$\left\{ (\mathbf{p}^\beta)^\top \otimes (\mathbf{v}^{(2)})^\top + \sum_{i \in [L]} (\mathbf{c}^\top \mathbf{H}_{2,i})^{(2)} \right\} \equiv \left\{ \sum_{i \in [L]} (\mathbf{c}^\top \mathbf{H}_{2,i})^{(2)} \right\},$$

and we observe that the distributions correspond to the adversary's view in G_5^b (resp. G_6^b) if $\beta = b$ (resp. $\beta = 0$). $\square$

Proof of Claim 33. We consider a sequence of sub-hybrids $\widehat{G}_0, \dots, \widehat{G}_4$ with $\widehat{G}_0 \equiv G_{6,\ell-1}^b$ and $\widehat{G}_4 \equiv G_{6,\ell}^b$.

Game $\widehat{G}_0$: This is the same as $G_{6,\ell-1}^b$ except for the following syntactical changes. Initially, the challenger samples $\mathbf{d}_j \xleftarrow{\$} \text{span}(\mathbf{B}_2)$ for $j \in [\ell-1]$ and sets $\mathbf{B}_2 \mathbf{r}_j := \mathbf{d}_j$. Similarly, it samples $\mathbf{d}_j \xleftarrow{\$} \text{span}(\mathbf{B}_1)$ for $j \in [L] \setminus [\ell-1]$ and sets $\mathbf{B}_1 \mathbf{r}_j := \mathbf{d}_j$. Then it computes crs and $\text{pk}_i \in \mathcal{D}_i$ as follows:

$$\text{crs} = \left(\left\{ \text{crs}_{1,i}, \text{crs}_{2,i}, \left\| \begin{array}{c} \mathbf{A} \mathbf{W}_i, \mathbf{R}_{1,i}, \mathbf{R}_{2,i}, \mathbf{A} \mathbf{H}_{1,i}, \mathbf{A} \mathbf{H}_{2,i} \\ \mathbf{A} \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}), \mathbf{A} \mathbf{H}_{2,i} (\mathbf{q}_i (\mathbf{e}_1^{(m)})^\top \otimes \mathbf{I}_{2k+1}) \end{array} \right\|_1 \right\}_{i \in [L]}, \right. \\ \left. \begin{array}{c} [\mathbf{A}, \tilde{\mathbf{V}}]_1, \{ \llbracket \mathbf{d}_j, \mathbf{I}_{\mathbf{g}_j} \otimes \mathbf{d}_j, \mathbf{q}_j (\mathbf{e}_1^{(m)})^\top \otimes \mathbf{d}_j \rrbracket_2 \}_{j \in [L]}, \\ \{ \llbracket \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_k} \otimes \mathbf{d}_j), \mathbf{H}_{2,i} (\mathbf{q}_k (\mathbf{e}_1^{(m)})^\top \otimes \mathbf{d}_j), \mathbf{W}_i (\mathbf{I}_{m+n'} \otimes \mathbf{d}_j) \rrbracket_2 \}_{\substack{i \in [L], \\ j \in [L] \setminus \{i\} \\ \kappa \in \{i,j\}}} \end{array} \right)$$

$$\text{pk}_i = \left(\begin{array}{c} \llbracket \mathbf{T}_{d,i} \rrbracket_1 := \llbracket \mathbf{A} \mathbf{U}_{d,i} \rrbracket_1, \llbracket \mathbf{Q}_{d,i} \rrbracket_1 := \llbracket \mathbf{R}_{d,i} \mathbf{U}_{d,i} \rrbracket_1, \\ \{ \llbracket \mathbf{G}_{d,i,j} \rrbracket_2 := \llbracket \mathbf{U}_{d,i} (\mathbf{I}_{t_d} \otimes \mathbf{d}_j) \rrbracket_2 \}_{j \in [L] \setminus \{i\}}, \pi_{d,i} \end{array} \right)_{d \in \{1,2\}}.$$

Clearly, we have $G_{6,\ell-1}^b \equiv \widehat{G}_0$.

Game $\widehat{G}_1$: This is the same as $\widehat{G}_0$ except that the challenger now samples $\mathbf{d}_\ell \xleftarrow{\$} \text{span}(\mathbf{B}_3)$. We have $\widehat{G}_0 \approx_c \widehat{G}_1$ under the subspace decision assumption $\text{SD}_{\mathbf{B}_1 \rightarrow \mathbf{B}_3}$ which states $\mathbf{d}_\ell \xleftarrow{\$} \text{span}(\mathbf{B}_1)$ (corresponding to $\widehat{G}_0$) and $\mathbf{d}_\ell \xleftarrow{\$} \text{span}(\mathbf{B}_3)$ (corresponding to $\widehat{G}_1$) are indistinguishable given $\llbracket \mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3 \rrbracket_2$, $\text{basis}(\mathbf{B}_1^\parallel, \mathbf{B}_3^\parallel)$ and $\text{basis}(\mathbf{B}_2^\parallel)$.

Game $\widehat{G}_2$: This is the same as $\widehat{G}_1$ except that we generate the ciphertext components $\mathbf{c}_1^*$ and $\mathbf{c}_2^*$ as follows:

$$\mathbf{c}_{1,1} = (\mathbf{f}^\top \parallel (\mathbf{z}^b)^\top) \otimes (\mathbf{v}^{(1)})^\top + (\mathbf{f}^\top \parallel (\mathbf{z}^0)^\top) \otimes (\mathbf{v}^{(2)})^\top + (\mathbf{f}^\top \parallel (\mathbf{z}^0)^\top) \otimes (\mathbf{v}^{(3)})^\top + \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_{1,i}$$

$$\mathbf{c}_{2,1} = (\mathbf{p}^b)^\top \otimes (\mathbf{v}^{(1)})^\top + (\mathbf{p}^0)^\top \otimes (\mathbf{v}^{(2)})^\top + (\mathbf{p}^0)^\top \otimes (\mathbf{v}^{(3)})^\top + \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_{2,i}.$$

We prove the following claim below.

Claim 34. We have $\widehat{G}_1 \approx_c \widehat{G}_2$ assuming bi-MDDH $_k$ in $\mathbb{G}$.

Game $\widehat{G}_3$: This is the same as $\widehat{G}_2$ except that the challenger now samples $\mathbf{d}_\ell \xleftarrow{\$} \text{span}(\mathbf{B}_2)$ instead of $\mathbf{d}_\ell \xleftarrow{\$} \text{span}(\mathbf{B}_3)$. We have $\widehat{G}_2 \approx_c \widehat{G}_3$ under the subspace decision assumption $\text{SD}_{\mathbf{B}_3 \rightarrow \mathbf{B}_2}$ which states that $\mathbf{d}_\ell \xleftarrow{\$} \text{span}(\mathbf{B}_3)$ (corresponding to $\widehat{G}_2$) and $\mathbf{d}_\ell \xleftarrow{\$} \text{span}(\mathbf{B}_2)$ (corresponding to $\widehat{G}_3$) are indistinguishable given $\llbracket \mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3 \rrbracket_2$, $\text{basis}(\mathbf{B}_2^\parallel, \mathbf{B}_3^\parallel)$ and $\text{basis}(\mathbf{B}_1^\parallel)$.

Game $\widehat{G}_4$: This is the same as $\widehat{G}_3$ except that the challenger computes the ciphertext components $\mathbf{c}_1^*$ and $\mathbf{c}_2^*$ as follows:

$$\begin{aligned}\mathbf{c}_{1,1} &= (\mathbf{f}^\top \parallel (\mathbf{z}^b)^\top) \otimes (\mathbf{v}^{(1)})^\top + (\mathbf{f}^\top \parallel (\mathbf{z}^0)^\top) \otimes (\mathbf{v}^{(2)})^\top + (\mathbf{f}^\top \parallel (\mathbf{z}^b)^\top) \otimes (\mathbf{v}^{(3)})^\top + \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_{1,i} \\ \mathbf{c}_{2,1} &= (\mathbf{p}^b)^\top \otimes (\mathbf{v}^{(1)})^\top + (\mathbf{p}^0)^\top \otimes (\mathbf{v}^{(2)})^\top + (\mathbf{p}^b)^\top \otimes (\mathbf{v}^{(3)})^\top + \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_{2,i}.\end{aligned}$$

We have $\widehat{G}_3 \approx_c \widehat{G}_4$ under the bi-MDDH $_k$ assumption in $\mathbb{G}$. The argument is the same as in the transition $\widehat{G}_1 \rightarrow \widehat{G}_2$ (Claim 34). Furthermore, we observe that $\widehat{G}_4 \equiv G_{6,\ell}^b$ which concludes the proof. $\square$

Proof of Claim 34. We distinguish the cases where pk_ℓ^* is either honest or [malicious or corrupted].

Honest Case. In this case, we have

$$\text{pk}_\ell^* = (\llbracket \mathbf{T}_{d,\ell}^* \rrbracket_1, \llbracket \mathbf{Q}_{d,\ell}^* \rrbracket_1, \{\llbracket \mathbf{G}_{d,\ell,j}^* \rrbracket_2\}_{j \in [L] \setminus \{i\}}, \pi_{d,\ell}^*)_{d \in \{1,2\}} \in \mathcal{D}_\ell \setminus \mathcal{C}_\ell$$

and the challenger knows secret matrix $\mathbf{U}_{d,\ell}^*$ such that $\mathbf{T}_{d,\ell}^* = \mathbf{A}\mathbf{U}_{d,\ell}^*$ and $\mathbf{Q}_{d,\ell}^* = \mathbf{R}_{d,\ell}^* \mathbf{U}_{d,\ell}^*$. In particular, we can write

$$\begin{aligned}\mathbf{c}_{1,2} &= \left(\mathbf{c}^\top \mathbf{U}_{1,\ell}^* + \sum_{i \in [L] \setminus \{\ell\}} \mathbf{e}_1^\top \widetilde{\mathbf{R}}_{1,i}^{-1} \mathbf{Q}_{1,i}^* \right) - \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_\ell \\ \mathbf{c}_{2,2} &= \left(\mathbf{c}^\top \mathbf{U}_{2,\ell}^* + \sum_{i \in [L] \setminus \{\ell\}} \mathbf{e}_1^\top \widetilde{\mathbf{R}}_{2,i}^{-1} \mathbf{Q}_{2,i}^* \right) - \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) - \mathbf{W}_\ell (\mathbf{L}_{i,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}).\end{aligned}$$

Let $\mathbf{c}^\perp \in \mathbb{Z}_p^{2k+1}$ such that $\mathbf{A}\mathbf{c}^\perp = 0$ and $\mathbf{c}^\top \mathbf{c}^\perp = 1$. For $\kappa \in [0; 2]$ and $\beta \in \{b, 0\}$, we define the distribution $D_\kappa^\beta = (C, \widehat{D}_\kappa^\beta)$, where

$$C = \left\{ \begin{array}{l} \mathbf{A}, \mathbf{c}^\perp, \llbracket \mathbf{R}_{1,\ell}, \mathbf{R}_{2,\ell} \rrbracket_1, \mathbf{B}_1, \mathbf{B}_2, \mathbf{d}_\ell, \mathbf{A}\mathbf{H}_{1,\ell}, \mathbf{A}\mathbf{H}_{2,\ell}, \mathbf{A}\mathbf{W}_\ell, \\ \mathbf{A}\mathbf{H}_{1,\ell} (\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}), \mathbf{A}\mathbf{H}_{2,\ell} (\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}), \{\mathbf{W}_\ell (\mathbf{I}_{m+n'} \otimes \mathbf{B}_d)\}_{d \in \{1,2\}}, \\ \{\mathbf{H}_{1,\ell} (\mathbf{I}_{\mathbf{g}_\kappa} \otimes \mathbf{d}_j), \mathbf{H}_{2,\ell} (\mathbf{q}_\kappa \mathbf{e}_1^\top \otimes \mathbf{d}_j)\}_{j \in [L] \setminus \{\ell\}}, \{\mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_\kappa} \otimes \mathbf{d}_\ell), \mathbf{H}_{2,i} (\mathbf{q}_\kappa \mathbf{e}_1^\top \otimes \mathbf{d}_\ell)\}_{i \in [L] \setminus \{\ell\}} \end{array} \right\}_{\kappa \in \{j, \ell\}}$$

represents the adversaries view on the crs and $\widehat{D}_\kappa^\beta$ the view on the public key and the challenge ciphertext:

$$\begin{aligned}\widehat{D}_0^\beta &= \left\{ \begin{array}{l} \text{pk}_\ell^*: (\mathbf{A}\mathbf{U}_{d,\ell}^*, \llbracket \mathbf{R}_{d,\ell} \mathbf{U}_{d,\ell}^* \rrbracket_1, \mathbf{U}_{d,\ell}^* (\mathbf{I}_{t_d} \otimes \mathbf{B}_1), \mathbf{U}_{d,\ell}^* (\mathbf{I}_{t_d} \otimes \mathbf{B}_2))_{d \in \{1,2\}} \\ \mathbf{c}_0^*, \mathbf{c}_{1,1}^*, \mathbf{c}_{2,1}^*: \mathbf{c}^\top, (\mathbf{f}^\top \parallel (\mathbf{z}^\beta)^\top) \otimes (\mathbf{v}^{(3)})^\top + \mathbf{c}^\top \mathbf{H}_{1,\ell}, (\mathbf{p}^\beta)^\top \otimes (\mathbf{v}^{(3)})^\top + \mathbf{c}^\top \mathbf{H}_{2,\ell} \\ \mathbf{c}_{1,2}^*: \mathbf{c}^\top \mathbf{U}_{1,\ell}^* - \mathbf{c}^\top \mathbf{H}_{1,\ell} (\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}) - \mathbf{c}^\top \mathbf{W}_\ell \\ \mathbf{c}_{2,2}^*: \mathbf{c}^\top \mathbf{U}_{2,\ell}^* - \mathbf{c}^\top \mathbf{H}_{2,\ell} (\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{c}^\top \mathbf{W}_\ell (\mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \end{array} \right\} \\ \widehat{D}_1^\beta &= \left\{ \begin{array}{l} \text{pk}_\ell^*: (\mathbf{A}\mathbf{U}_{d,\ell}^*, \llbracket \mathbf{R}_{d,\ell} \mathbf{U}_{d,\ell}^* + \mathbf{P}_{d,\ell} \otimes (\mathbf{v}^{(3)})^\top \rrbracket_1, \mathbf{U}_{d,\ell}^* (\mathbf{I}_{t_d} \otimes \mathbf{B}_1), \mathbf{U}_{d,\ell}^* (\mathbf{I}_{t_d} \otimes \mathbf{B}_2))_{d \in \{1,2\}} \\ \mathbf{c}_0^*, \mathbf{c}_{1,1}^*, \mathbf{c}_{2,1}^*: \mathbf{c}^\top, (\mathbf{f}^\top \parallel (\mathbf{z}^\beta)^\top) \otimes (\mathbf{v}^{(3)})^\top + \mathbf{c}^\top \mathbf{H}_{1,\ell}, (\mathbf{p}^\beta)^\top \otimes (\mathbf{v}^{(3)})^\top + \mathbf{c}^\top \mathbf{H}_{2,\ell} \\ \mathbf{c}_{1,2}^*: \mathbf{c}^\top \mathbf{U}_{1,\ell}^* - \mathbf{c}^\top \mathbf{H}_{1,\ell} (\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}) - \mathbf{c}^\top \mathbf{W}_\ell \\ \mathbf{c}_{2,2}^*: \mathbf{c}^\top \mathbf{U}_{2,\ell}^* - \mathbf{c}^\top \mathbf{H}_{2,\ell} (\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{c}^\top \mathbf{W}_\ell (\mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \end{array} \right\} \\ \widehat{D}_2^\beta &= \left\{ \begin{array}{l} \text{pk}_\ell^*: (\mathbf{A}\mathbf{U}_{d,\ell}^*, \llbracket \mathbf{R}_{d,\ell} \mathbf{U}_{d,\ell}^* + (\mathbf{P}_{d,\ell} + \mathbf{R}_{d,\ell} \mathbf{c}^\perp \mathbf{u}_{d,\ell}^\top) \otimes (\mathbf{v}^{(3)})^\top \rrbracket_1, \\ \mathbf{U}_{d,\ell}^* (\mathbf{I}_{t_d} \otimes \mathbf{B}_1), \mathbf{U}_{d,\ell}^* (\mathbf{I}_{t_d} \otimes \mathbf{B}_2))_{d \in \{1,2\}} \\ \mathbf{c}_0^*, \mathbf{c}_{1,1}^*, \mathbf{c}_{2,1}^*: \mathbf{c}^\top, ((\mathbf{f}^\top \parallel (\mathbf{z}^\beta)^\top) + \mathbf{h}_{1,\ell}^\top) \otimes (\mathbf{v}^{(3)})^\top + \mathbf{c}^\top \mathbf{H}_{1,\ell}, (\mathbf{p}^\beta + \mathbf{h}_{2,\ell}^\top)^\top \otimes (\mathbf{v}^{(3)})^\top + \mathbf{c}^\top \mathbf{H}_{2,\ell} \\ \mathbf{c}_{1,2}^*: \mathbf{c}^\top \mathbf{U}_{1,\ell}^* - \mathbf{c}^\top \mathbf{H}_{1,\ell} (\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}) + ((\mathbf{u}_{1,\ell} - \mathbf{h}_{1,\ell}^\top \mathbf{I}_{\mathbf{g}_\ell}) \otimes \mathbf{v}^{(3)})^\top - \mathbf{c}^\top \mathbf{W}_\ell \\ \mathbf{c}_{2,2}^*: \mathbf{c}^\top \mathbf{U}_{2,\ell}^* - \mathbf{c}^\top \mathbf{H}_{2,\ell} (\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + ((\mathbf{u}_{2,\ell} - \mathbf{h}_{2,\ell}^\top \mathbf{q}_\ell \cdot \mathbf{e}_1) \otimes \mathbf{v}^{(3)})^\top \\ + \mathbf{c}^\top \mathbf{W}_\ell (\mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \end{array} \right\}$$

where $t_1 = n' + 1$, $t_2 = m$, $\mathbf{P}_{d,\ell} \xleftarrow{s} \mathbb{Z}_p^{(2k+2) \times t_d}$, $\mathbf{u}_{d,\ell} \xleftarrow{s} \mathbb{Z}_p^{t_d}$, $\mathbf{h}_{1,\ell} \xleftarrow{s} \mathbb{Z}_p^{k+n'}$ and $\mathbf{h}_{2,\ell} \xleftarrow{s} \mathbb{Z}_p^k$. We note that for $\beta = b$ (resp. $\beta = 0$), the distribution D_0^β corresponds to the adversary's view in $\widehat{\mathbf{G}}_1$ (resp. $\widehat{\mathbf{G}}_2$). Furthermore, we observe that

- $D_0^\beta \approx_c D_1^\beta$ which follows from two applications of Lemma 24 when setting $\mathbf{d}^\perp = \mathbf{v}^{(3)}$, $\mathbf{R} = \mathbf{R}_{d,\ell}$, $\mathbf{U} = \mathbf{U}_{d,\ell}^*$ and $\mathbf{P} = \mathbf{P}_{d,\ell}$ for $d \in \{1, 2\}$.
- $D_1^\beta \approx_s D_2^\beta$: this follows from the following changes of variables for $d \in \{1, 2\}$

$$\mathbf{U}_{d,\ell}^* \mapsto \mathbf{U}_{d,\ell}^* + \mathbf{c}^\perp (\mathbf{u}_{d,\ell} \otimes \mathbf{v}^{(3)})^\top, \quad \mathbf{H}_{d,\ell} \mapsto \mathbf{H}_{d,\ell} + \mathbf{c}^\perp (\mathbf{h}_{d,\ell} \otimes \mathbf{v}^{(3)})^\top.$$

Here we use the equality $\mathbf{R}_{d,\ell} \mathbf{c}^\perp (\mathbf{u}_{d,\ell} \otimes \mathbf{v}^{(3)})^\top = \mathbf{R}_{d,\ell} \mathbf{c}^\perp \mathbf{u}_{d,\ell}^\top \otimes (\mathbf{v}^{(3)})^\top$.

- $D_2^\beta \approx_s D_2^1$: this follows from the fact that, for $d \in \{1, 2\}$, $\mathbf{P}_{d,\ell}$ hides $\mathbf{R}_{d,\ell} \mathbf{c}^\perp \mathbf{u}_{d,\ell}^\top$ which implies that $\mathbf{u}_{1,\ell}$ hides $\mathbf{h}_{1,\ell}^\top \mathbf{I}_{\mathbf{g}_\ell}$ and $\mathbf{h}_{1,\ell}$ hides $(\mathbf{f}^\top \parallel (\mathbf{z}^\beta)^\top)$. Similarly, $\mathbf{u}_{2,\ell}$ can hide $\mathbf{h}_{2,\ell}^\top \mathbf{q}_\ell \cdot \mathbf{e}_1^\top$, so $\mathbf{h}_{2,\ell}$ hides $\mathbf{p}^\beta$. In the end, we can observe that the distribution D_2^β is independent of the bit β .

Malicious and Corrupted Case. In this case, we have

$$\text{pk}_\ell^* = (\llbracket \mathbf{T}_{d,\ell}^* \rrbracket_1, \llbracket \mathbf{Q}_{d,\ell}^* \rrbracket_1, \{\llbracket \mathbf{G}_{d,\ell,j}^* \rrbracket_2\}_{j \in [L] \setminus \{i\}}, \pi_{d,\ell}^*)_{d \in \{1,2\}} \in \overline{\mathcal{D}_\ell} \cup \mathcal{C}_\ell.$$

Let $\mathbf{c}^\perp \in \mathbb{Z}_p^{2k+1}$ again be such that $\mathbf{A} \mathbf{c}^\perp = 0$ and $\mathbf{c}^\top \mathbf{c}^\perp = 1$. For $\kappa \in [0; 3]$ and $\beta \in \{b, 0\}$, we define the distribution $D_\kappa^\beta = (C, \widehat{D}_\kappa^\beta)$, where

$$C = \left\{ \begin{array}{l} \mathbf{A}, \mathbf{c}^\perp, \mathbf{B}_1, \mathbf{B}_2, \mathbf{d}_\ell, \llbracket \mathbf{g}_\ell \rrbracket_2, \mathbf{AH}_{1,\ell}, \mathbf{AH}_{2,\ell}, \mathbf{AW}_\ell, \\ \llbracket \mathbf{AH}_{1,\ell} (\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}) \rrbracket_1, \mathbf{AH}_{2,\ell} (\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}), \{\mathbf{W}_\ell (\mathbf{I}_{m+n'} \otimes \mathbf{B}_d)\}_{d \in \{1,2\}}, \\ \{\llbracket \mathbf{H}_{1,\ell} (\mathbf{I}_{\mathbf{g}_\kappa} \otimes \mathbf{d}_j) \rrbracket_1, \mathbf{H}_{2,\ell} (\mathbf{q}_\kappa \mathbf{e}_1^\top \otimes \mathbf{d}_j)\}_{j \in [L] \setminus \{i\}}, \{\llbracket \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_\kappa} \otimes \mathbf{d}_\ell) \rrbracket_1, \mathbf{H}_{2,i} (\mathbf{q}_\kappa \mathbf{e}_1^\top \otimes \mathbf{d}_\ell)\}_{i \in [L] \setminus \{i\}} \\ \kappa \in \{j, \ell\} \end{array} \right\}$$

represents the adversaries view on the crs and $\widehat{D}_\kappa^\beta$ the view on the challenge ciphertext:

$$\begin{aligned} \widehat{D}_0^\beta &= \left\{ \begin{array}{l} \mathbf{c}_0^*, \mathbf{c}_{1,1}^*, \mathbf{c}_{2,1}^*: \mathbf{c}^\top, (\mathbf{f}^\top \parallel (\mathbf{z}^\beta)^\top) \otimes (\mathbf{v}^{(3)})^\top + \mathbf{c}^\top \mathbf{H}_{1,\ell}, (\mathbf{p}^\beta)^\top \otimes (\mathbf{v}^{(3)})^\top + \mathbf{c}^\top \mathbf{H}_{2,\ell} \\ \mathbf{c}_{1,2}^*: \llbracket \mathbf{e}_1^\top \widetilde{\mathbf{R}}_{1,\ell}^{-1} \mathbf{Q}_{1,\ell}^* - \mathbf{c}^\top \mathbf{H}_{1,\ell} (\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}) - \mathbf{c}^\top \mathbf{W}_\ell \rrbracket_1 \\ \mathbf{c}_{2,2}^*: \llbracket \mathbf{e}_1^\top \widetilde{\mathbf{R}}_{2,\ell}^{-1} \mathbf{Q}_{2,\ell}^* - \mathbf{c}^\top \mathbf{H}_{2,\ell} (\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{c}^\top \mathbf{W}_\ell (\mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \rrbracket_1 \end{array} \right\} \\ \widehat{D}_1^\beta &= \left\{ \begin{array}{l} \mathbf{c}_0^*, \mathbf{c}_{1,1}^*, \mathbf{c}_{2,1}^*: \mathbf{c}^\top, (\mathbf{f}^\top \parallel (\mathbf{z}^\beta)^\top) \otimes (\mathbf{v}^{(3)})^\top + \mathbf{c}^\top \mathbf{H}_{1,\ell}, (\mathbf{p}^\beta)^\top \otimes (\mathbf{v}^{(3)})^\top + \mathbf{c}^\top \mathbf{H}_{2,\ell} \\ \mathbf{c}_{1,2}^*: \llbracket \mathbf{e}_1^\top \widetilde{\mathbf{R}}_{1,\ell}^{-1} \mathbf{Q}_{1,\ell}^* - \mathbf{c}^\top \mathbf{H}_{1,\ell} (\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}) - \mathbf{c}^\top \mathbf{W}_\ell \\ \quad + (\mathbf{f}^\top \mathbf{g}_\ell \parallel (\mathbf{z}^\beta)^\top - \mathbf{w}_\ell^\top) \otimes (\mathbf{v}^{(3)})^\top \rrbracket_1 \\ \mathbf{c}_{2,2}^*: \llbracket \mathbf{e}_1^\top \widetilde{\mathbf{R}}_{2,\ell}^{-1} \mathbf{Q}_{2,\ell}^* - \mathbf{c}^\top \mathbf{H}_{2,\ell} (\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{c}^\top \mathbf{W}_\ell (\mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \\ \quad + (\mathbf{w}_\ell^\top \mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} + (\mathbf{p}^\beta)^\top \mathbf{q}_\ell \cdot \mathbf{e}_1^\top) \otimes (\mathbf{v}^{(3)})^\top \rrbracket_1 \end{array} \right\} \\ \widehat{D}_2^\beta &= \left\{ \begin{array}{l} \mathbf{c}_0^*, \mathbf{c}_{1,1}^*, \mathbf{c}_{2,1}^*: \mathbf{c}^\top, \mathbf{c}^\top \mathbf{H}_{1,\ell}, \mathbf{c}^\top \mathbf{H}_{2,\ell} \\ \mathbf{c}_{1,2}^*: \llbracket \mathbf{e}_1^\top \widetilde{\mathbf{R}}_{1,\ell}^{-1} \mathbf{Q}_{1,\ell}^* - \mathbf{c}^\top \mathbf{H}_{1,\ell} (\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}) - \mathbf{c}^\top \mathbf{W}_\ell \rrbracket_1 \\ \quad + (\mathbf{f}^\top \mathbf{g}_\ell \parallel (\mathbf{z}^\beta)^\top - \mathbf{w}_\ell^\top) \otimes (\mathbf{v}^{(3)})^\top \\ \mathbf{c}_{2,2}^*: \llbracket \mathbf{e}_1^\top \widetilde{\mathbf{R}}_{2,\ell}^{-1} \mathbf{Q}_{2,\ell}^* - \mathbf{c}^\top \mathbf{H}_{2,\ell} (\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{c}^\top \mathbf{W}_\ell (\mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \\ \quad + (\mathbf{w}_\ell^\top \mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} + (\mathbf{g}_\ell^*(\mathbf{y}) \cdot \mathbf{f}^\top \mathbf{g}_\ell + h_\ell^*(\mathbf{x})^\top \cdot \mathbf{z}^\beta + (\mathbf{p}^\beta)^\top \mathbf{q}_\ell \cdot \mathbf{e}_1^\top) \otimes (\mathbf{v}^{(3)})^\top \rrbracket_1 \end{array} \right\} \\ \widehat{D}_3^\beta &= \left\{ \begin{array}{l} \mathbf{c}_0^*, \mathbf{c}_{1,1}^*, \mathbf{c}_{2,1}^*: \mathbf{c}^\top, \mathbf{c}^\top \mathbf{H}_{1,\ell}, \mathbf{c}^\top \mathbf{H}_{2,\ell} \\ \mathbf{c}_{1,2}^*: \llbracket \mathbf{e}_1^\top \widetilde{\mathbf{R}}_{1,\ell}^{-1} \mathbf{Q}_{1,\ell}^* - \mathbf{c}^\top \mathbf{H}_{1,\ell} (\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}) - \mathbf{c}^\top \mathbf{W}_\ell - \mathbf{w}_\ell^\top \otimes (\mathbf{v}^{(3)})^\top \rrbracket_1 \\ \mathbf{c}_{2,2}^*: \llbracket \mathbf{e}_1^\top \widetilde{\mathbf{R}}_{2,\ell}^{-1} \mathbf{Q}_{2,\ell}^* - \mathbf{c}^\top \mathbf{H}_{2,\ell} (\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{c}^\top \mathbf{W}_\ell (\mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \\ \quad + (\mathbf{w}_\ell^\top \mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} + (\mathbf{g}_\ell^*(\mathbf{y}) \cdot \alpha_\ell + h_\ell^*(\mathbf{x})^\top \cdot \mathbf{z}^\beta + (\mathbf{p}^\beta)^\top \mathbf{q}_\ell \cdot \mathbf{e}_1^\top) \otimes (\mathbf{v}^{(3)})^\top \rrbracket_1 \end{array} \right\} \end{aligned}$$

where $\mathbf{w}_\ell \stackrel{s}{\leftarrow} \mathbb{Z}_p^{m+n'}$ and $\underline{\mathbf{w}}_\ell$ denotes the last $n' + 1$ entries of $\mathbf{w}_\ell$. We note that for $\beta = b$ (resp. $\beta = 0$), the distribution D_0^β corresponds to the adversary's view in $\hat{G}_1$ (resp. $\hat{G}_2$). Furthermore, we observe that

- $D_0^\beta \approx_s D_1^\beta$: this follows from the changes of variables

$$\begin{aligned} \mathbf{H}_{1,\ell} &\mapsto \mathbf{H}_{1,\ell} + \mathbf{c}^\perp ((\mathbf{f}^\top \parallel \mathbf{z}^\beta)^\top \otimes (\mathbf{v}^{(3)})^\top) , & \mathbf{H}_{2,\ell} &\mapsto \mathbf{H}_{2,\ell} + \mathbf{c}^\perp ((\mathbf{p}^\beta)^\top \otimes (\mathbf{v}^{(3)})^\top) , \\ \mathbf{W}_\ell &\mapsto \mathbf{W}_\ell + \mathbf{c}^\perp (\mathbf{w}_\ell^\top \otimes (\mathbf{v}^{(3)})^\top) . \end{aligned}$$

- $D_1^\beta \approx_s D_2^\beta$: this follows from the security of pgb. Specifically, we can simulate $\hat{D}_\kappa^\beta$, $\kappa \in \{1, 2\}$, as

$$\hat{D}_\kappa^\beta = \left\{ \begin{array}{l} \mathbf{c}_0^*, \mathbf{c}_{1,1}^*, \mathbf{c}_{2,1}^* : \mathbf{c}^\top, \mathbf{c}^\top \mathbf{H}_{1,\ell}, \mathbf{c}^\top \mathbf{H}_{2,\ell} \\ \mathbf{c}_{1,2}^* : \llbracket \mathbf{e}_1^\top \tilde{\mathbf{R}}_{1,\ell}^{-1} \mathbf{Q}_{1,\ell}^* - \mathbf{c}^\top \mathbf{H}_{1,\ell} (\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}) - \mathbf{c}^\top \underline{\mathbf{W}}_\ell + (\mathbf{p}_1 \otimes \mathbf{v}^{(3)})^\top \rrbracket_1 \\ \mathbf{c}_{2,2}^* : \llbracket \mathbf{e}_1^\top \tilde{\mathbf{R}}_{2,\ell}^{-1} \mathbf{Q}_{2,\ell}^* - \mathbf{c}^\top \mathbf{H}_{2,\ell} (\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{c}^\top \mathbf{W}_\ell (\mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) + (\mathbf{p}_2 \otimes \mathbf{v}^{(3)})^\top \rrbracket_1 \end{array} \right\} ,$$

where the vectors $\mathbf{p}_1 \in \mathbb{Z}_p^{n'+1}$ and $\mathbf{p}_2 \in \mathbb{Z}_p^m$ are defined as

$$(\llbracket \mathbf{p}_1^\top \rrbracket_1, \llbracket \mathbf{p}_2^\top \rrbracket_1) \leftarrow \begin{cases} \text{pgb}(\phi_\ell^*, (\mathbf{y}, \mathbf{x}), \llbracket (\mathbf{f}^\top \mathbf{g}_\ell \parallel (\mathbf{z}^\beta)^\top) \rrbracket_1; \llbracket \mathbf{w}_\ell \rrbracket_1) + \llbracket (\mathbf{0}^\top, (\mathbf{p}^\beta)^\top \mathbf{q}_\ell \cdot \mathbf{e}_1^\top) \rrbracket_1 & \text{if } \kappa = 1 \\ \text{pgb}^*(\phi_\ell^*, (\mathbf{y}, \mathbf{x}), \llbracket g_\ell^*(\mathbf{y}) \cdot \mathbf{f}^\top \mathbf{g}_\ell + h_\ell^*(\mathbf{x})^\top \mathbf{z}^\beta + (\mathbf{p}^\beta)^\top \mathbf{q}_\ell \rrbracket_1; \llbracket \mathbf{w}_\ell \rrbracket_1) & \text{if } \kappa = 2 , \end{cases}$$

and we recall that $\phi_\ell^*(\mathbf{y}, \mathbf{x})^\top = (g_\ell^*(\mathbf{y}) \parallel h_\ell^*(\mathbf{x})^\top)$. Then $D_1^\beta \approx_s D_2^\beta$ immediately follows from the privacy of the underlying partial garbling scheme.

- $D_2^\delta \approx_s D_3^\delta$: this follows from the bi-MDDH $_k$ assumption in $\mathbb{G}$ which states that $(\llbracket \mathbf{g}_\ell^\top \rrbracket_{1,2}, \llbracket \mathbf{g}_\ell^\top \mathbf{f} \rrbracket_{1,2}) \approx_c (\llbracket \mathbf{g}_\ell^\top \rrbracket_{1,2}, \llbracket \alpha_\ell \rrbracket_{1,2})$ for $\alpha_\ell \stackrel{s}{\leftarrow} \mathbb{Z}_p$.
- $D_3^0 \equiv D_3^1$: this follows from the adversary's admissibility condition which requires that $h_\ell^*(\mathbf{x})^\top \mathbf{z}^0 + \mathbf{q}_\ell^\top \mathbf{p}^0 = h_\ell^*(\mathbf{x})^\top \mathbf{z}^1 + \mathbf{q}_\ell^\top \mathbf{p}^1$ if $g_\ell^*(\mathbf{y}) = 0$. Otherwise, $g_\ell^*(\mathbf{y}) \cdot \alpha_\ell$ serves as a masking term. In the end, we can observe that the distribution D_3^β is independent of the bit β .

□

6 Very Selectively SIM-Secure sRFE for AB-eAWS

In this section, we present our sRFE scheme for AB-eAWS with very selective SIM-security.

On a technical level, Construction 35 is an extension of Construction 28 in Section 5 which provides the same functionality but achieves “only” adaptive IND-security. Achieving SIM-security is a crucial requirement for the applicability of our compiler from AB-eAWS to AB-AWS presented in Section 7.

Construction 35 (Very Selectively SIM-Secure sRFE for AB-eAWS). The construction uses the following ingredients:

- a pairing group $\mathbb{G} = (\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_t, g_1, g_2, g_t, e, p)$, and
- a QA-NIZK $\Pi = (\text{LGen}, \text{LPrv}, \text{LVer}, \text{LSim})$ for linear spaces over $\mathbb{G}$.

The details of the scheme go as follows:

Setup($1^\lambda, 1^L, \{\llbracket \mathbf{q}_i \rrbracket_{1,2}\}_{i \in [L]}$): On input the security parameter 1^λ , the maximum number of slots 1^L and a set of vectors $\{\llbracket \mathbf{q}_i \rrbracket_{1,2}\}_{i \in [L]}$ with $\mathbf{q}_i \in \mathbb{Z}_p^k$ for $i \in [L]$, sample

$$\begin{aligned} \mathbf{A} &\xleftarrow{\$} \mathbb{Z}_p^{k \times (2k+1)}, \mathbf{B}, \mathbf{D} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times k}, \mathbf{V} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)}, \mathbf{v}, \mathbf{w} \xleftarrow{\$} \mathbb{Z}_p^{2k+1}, \\ \hat{\mathbf{H}}_1 &\xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (k+n')}, \hat{\mathbf{H}}_2 \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times k}, \\ \left\{ \begin{array}{l} \mathbf{E}_i \xleftarrow{\$} \mathbb{Z}_p^{k \times m}, \mathbf{R}_{1,i}, \mathbf{R}_{2,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+2) \times (2k+1)}, \mathbf{g}_i, \mathbf{r}_i \xleftarrow{\$} \mathbb{Z}_p^k, \\ \mathbf{H}_{1,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)(k+n')}, \mathbf{H}_{2,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)k}, \hat{\mathbf{W}}_i \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (m+n')}, \\ \mathbf{W}_{1,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)(m+n')}, \mathbf{W}_{2,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)(2k+1)}, \mathbf{W}_{3,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)} \end{array} \right\}_{i \in [L]}. \end{aligned}$$

Then, generate

$$(\text{crs}_{1,i}, \text{td}_{1,i}) \leftarrow \text{LGen}(1^\lambda, \mathbb{G}_1, \llbracket \mathbf{A}_{1,i} \rrbracket_1), \quad (\text{crs}_{2,i}, \text{td}_{2,i}) \leftarrow \text{LGen}(1^\lambda, \mathbb{G}_1, \llbracket \mathbf{A}_{2,i} \rrbracket_1)$$

where $\mathbf{A}_{1,i}^\top = [\mathbf{A}^\top \parallel \mathbf{R}_{1,i}^\top]$, $\mathbf{A}_{2,i}^\top = [\mathbf{A}^\top \parallel \mathbf{R}_{2,i}^\top]$, and output the common reference string

$$\begin{aligned} \text{crs} = & \left(\llbracket \mathbf{A}, \mathbf{A}\hat{\mathbf{H}}_1, \mathbf{A}\hat{\mathbf{H}}_2, \mathbf{A}\mathbf{V}, \mathbf{A}\mathbf{v} \rrbracket_1, \right. \\ & \left\{ \left\llbracket \begin{array}{l} \mathbf{A}\mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}), \mathbf{A}\mathbf{H}_{2,i}(\mathbf{q}_i(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{I}_{2k+1}), \mathbf{R}_{1,i}, \mathbf{R}_{2,i}, \\ \mathbf{A}\mathbf{W}_{1,i}, \mathbf{A}\mathbf{W}_{2,i}(\mathbf{D}\mathbf{E}_i \otimes \mathbf{I}_{2k+1}) + \mathbf{A}\mathbf{W}_{3,i}(\mathbf{w}^\top \mathbf{D}\mathbf{E}_i \otimes \mathbf{I}_{2k+1}), \text{crs}_{1,i}, \text{crs}_{2,i} \end{array} \right\|_1 \right\}_{i \in [L]}, \\ & \left\{ \left\llbracket \begin{array}{l} \mathbf{B}\mathbf{r}_j, \mathbf{q}_j, \mathbf{g}_j, \mathbf{D}\mathbf{E}_j, \mathbf{w}^\top \mathbf{D}\mathbf{E}_j, \mathbf{W}_{1,j}(\mathbf{I}_{m+n'} \otimes \mathbf{B}\mathbf{r}_j) - \hat{\mathbf{W}}_j, \\ \mathbf{H}_{1,j}(\mathbf{I}_{\mathbf{g}_j} \otimes \mathbf{B}\mathbf{r}_j) + \hat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_j}, \mathbf{H}_{2,j}(\mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j) + \hat{\mathbf{H}}_2 \mathbf{q}_j \mathbf{e}_1^\top \\ + \mathbf{W}_{2,j}(\mathbf{D}\mathbf{E}_j \otimes \mathbf{B}\mathbf{r}_j) + \mathbf{W}_{3,j}(\mathbf{w}^\top \mathbf{D}\mathbf{E}_j \otimes \mathbf{B}\mathbf{r}_j) + \mathbf{V}\mathbf{D}\mathbf{E}_j + \mathbf{v}\mathbf{w}^\top \mathbf{D}\mathbf{E}_j \end{array} \right\|_2 \right\}_{j \in [L]}, \\ & \left. \left\{ \left\llbracket \begin{array}{l} \mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{B}\mathbf{r}_j), \mathbf{W}_{1,i}(\mathbf{I}_{m+n'} \otimes \mathbf{B}\mathbf{r}_j), \\ \mathbf{H}_{2,i}(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j), \mathbf{W}_{2,i}(\mathbf{D}\mathbf{E}_i \otimes \mathbf{B}\mathbf{r}_j) + \mathbf{W}_{3,i}(\mathbf{w}^\top \mathbf{D}\mathbf{E}_i \otimes \mathbf{B}\mathbf{r}_j) \end{array} \right\|_2 \right\}_{\substack{i \in [L], \\ j \in [L] \setminus \{i\}}} \right), \end{aligned}$$

where, for $\mathbf{g} \in \mathbb{Z}_p^k$, the matrix $\mathbf{I}_{\mathbf{g}} \in \mathbb{Z}_p^{(k+n') \times (1+n')}$ is defined as

$$\mathbf{I}_{\mathbf{g}} := \begin{bmatrix} \mathbf{g} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{n'} \end{bmatrix}.$$

Gen(crs, i): On input crs and an index $i \in [L]$, sample $\mathbf{U}_{1,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)(n'+1)}$, $\mathbf{U}_{2,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)m}$ and run $\pi_{1,i} \leftarrow \text{LPrv}(\text{crs}_{1,i}, \llbracket \mathbf{A}_{1,i} \mathbf{U}_{1,i} \rrbracket_1, \mathbf{U}_{1,i})$, $\pi_{2,i} \leftarrow \text{LPrv}(\text{crs}_{2,i}, \llbracket \mathbf{A}_{2,i} \mathbf{U}_{2,i} \rrbracket_1, \mathbf{U}_{2,i})$. Then output $\text{sk}_i = (\mathbf{U}_{1,i}, \mathbf{U}_{2,i})$ and pk_i as follows:

$$\left(\begin{array}{l} \llbracket \mathbf{T}_{1,i} \rrbracket_1 := \llbracket \mathbf{A} \mathbf{U}_{1,i} \rrbracket_1, \llbracket \mathbf{Q}_{1,i} \rrbracket_1 := \llbracket \mathbf{R}_{1,i} \mathbf{U}_{1,i} \rrbracket_1, \{\llbracket \mathbf{G}_{1,i,j} \rrbracket_2 := \llbracket \mathbf{U}_{1,i}(\mathbf{I}_{n'+1} \otimes \mathbf{B}\mathbf{r}_j) \rrbracket_2\}_{j \in [L] \setminus \{i\}}, \pi_{1,i} \\ \llbracket \mathbf{T}_{2,i} \rrbracket_1 := \llbracket \mathbf{A} \mathbf{U}_{2,i} \rrbracket_1, \llbracket \mathbf{Q}_{2,i} \rrbracket_1 := \llbracket \mathbf{R}_{2,i} \mathbf{U}_{2,i} \rrbracket_1, \{\llbracket \mathbf{G}_{2,i,j} \rrbracket_2 := \llbracket \mathbf{U}_{2,i}(\mathbf{I}_m \otimes \mathbf{B}\mathbf{r}_j) \rrbracket_2\}_{j \in [L] \setminus \{i\}}, \pi_{2,i} \end{array} \right).$$

Ver($\text{crs}, i, \text{pk}_i$): On input crs , an index $i \in [L]$ and a public key $\text{pk}_i = (\llbracket \mathbf{T}_{d,i} \rrbracket_1, \llbracket \mathbf{Q}_{d,i} \rrbracket_1, \{\llbracket \mathbf{G}_{d,i,j} \rrbracket_2\}_{j \in [L] \setminus \{i\}}, \pi_{d,i})_{d \in \{1,2\}}$, define $\mathbf{F}_{d,i}^\top = [\mathbf{T}_{d,i}^\top \parallel \mathbf{Q}_{d,i}^\top]$ and check

$$\begin{aligned} \text{LVer}(\text{crs}_{1,i}, \llbracket \mathbf{F}_{1,i} \rrbracket_1, \pi_{1,i}) &\stackrel{?}{=} 1, & \left\{ \begin{array}{l} \llbracket \mathbf{T}_{1,i} \rrbracket_1 \llbracket \mathbf{I}_{n'+1} \otimes \mathbf{B}\mathbf{r}_j \rrbracket_2 \stackrel{?}{=} \llbracket \mathbf{A} \rrbracket_1 \llbracket \mathbf{G}_{1,i,j} \rrbracket_2, \\ \llbracket \mathbf{T}_{2,i} \rrbracket_1 \llbracket \mathbf{I}_m \otimes \mathbf{B}\mathbf{r}_j \rrbracket_2 \stackrel{?}{=} \llbracket \mathbf{A} \rrbracket_1 \llbracket \mathbf{G}_{2,i,j} \rrbracket_2 \end{array} \right\}_{j \in [L] \setminus \{i\}} \\ \text{LVer}(\text{crs}_{2,i}, \llbracket \mathbf{F}_{2,i} \rrbracket_1, \pi_{2,i}) &\stackrel{?}{=} 1, \end{aligned}$$

Output 1 if all checks pass, and 0 otherwise.

$\text{Agg}(\text{crs}, (\text{pk}_i, f_i)_{i \in [L]}):$ On input crs and L tuples of the form (pk_i, f_i) , parse $f_i = (g_i, h_i) \in \mathcal{F}_{\hat{n},1}^{\text{abp}} \times \mathcal{F}_{n,n'}^{\text{abp}}$ and $\text{pk}_i = (\llbracket \mathbf{T}_{d,i} \rrbracket_1, \llbracket \mathbf{Q}_{d,i} \rrbracket_1, \{\llbracket \mathbf{G}_{d,i,j} \rrbracket_2\}_{j \in [L] \setminus \{i\}}, \pi_{d,i})_{d \in \{1,2\}}$. For each $i \in [L]$, define an ABP $\phi_i \in \mathcal{F}_{\hat{n}+n,1+n'}^{\text{abp}}$ as $\phi_i(\mathbf{y}, \mathbf{x})^\top = (g_i(\mathbf{y}), h_i(\mathbf{x})^\top)$ for $\mathbf{y} \in \mathbb{Z}_p^{\hat{n}}$ and $\mathbf{x} \in \mathbb{Z}_p^n$. Then compute the matrices

$$\text{lgen}(\phi_i) \rightarrow (\mathbf{L}_{i,0}, \mathbf{L}_{i,1}) \in \mathbb{Z}_p^{(m+n') \times m} \times \mathbb{Z}_p^{(m+n') \times m(\hat{n}+n)}$$

and output $(\text{mpk}, \{\text{hsk}_j\}_{j \in [L]})$ as follows:

$$\begin{aligned} \text{mpk} &= \left\| \begin{array}{l} \mathbf{A}, \mathbf{A}\hat{\mathbf{H}}_1, \mathbf{A}\hat{\mathbf{H}}_2, \mathbf{A}\mathbf{v}, \mathbf{A}\mathbf{v}, \\ \sum_{i \in [L]} \mathbf{T}_{1,i} + \mathbf{A}\mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}) - \mathbf{A}\mathbf{W}_{1,i}, \\ \mathbf{A} \sum_{i \in [L]} \mathbf{W}_{1,i}(\mathbf{L}_{i,1} \otimes \mathbf{I}_{2k+1}), \\ \sum_{i \in [L]} \mathbf{T}_{2,i} + \mathbf{A}\mathbf{H}_{2,i}(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{A}\mathbf{W}_{1,i}(\mathbf{L}_{i,0} \otimes \mathbf{I}_{2k+1}) \\ + \mathbf{A}\mathbf{W}_{2,i}(\mathbf{D}\mathbf{E}_i \otimes \mathbf{I}_{2k+1}) + \mathbf{A}\mathbf{W}_{3,i}(\mathbf{w}^\top \mathbf{D}\mathbf{E}_i \otimes \mathbf{I}_{2k+1}) \end{array} \right\|_1, \\ \text{hsk}_j &= \left\| \begin{array}{l} \mathbf{B}\mathbf{r}_j, \mathbf{H}_{1,j}(\mathbf{I}_{\mathbf{g}_j} \otimes \mathbf{B}\mathbf{r}_j) - \mathbf{W}_{1,j}(\mathbf{I}_{n'+1} \otimes \mathbf{B}\mathbf{r}_j) + \hat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_j} + \hat{\mathbf{W}}_j, \\ \sum_{i \in [L] \setminus \{j\}} \mathbf{G}_{1,i,j} + \mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{B}\mathbf{r}_j) - \mathbf{W}_{1,i}(\mathbf{I}_{n'+1} \otimes \mathbf{B}\mathbf{r}_j), \\ \mathbf{W}_{1,j}(\mathbf{I}_{m+n'} \otimes \mathbf{B}\mathbf{r}_j) \mathbf{L}_{j,1} - \hat{\mathbf{W}}_j \mathbf{L}_{j,1}, \\ \mathbf{H}_{2,j}(\mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j) + \hat{\mathbf{H}}_2 \mathbf{q}_j \mathbf{e}_1^\top + \mathbf{W}_{1,j}(\mathbf{I}_{m+n'} \otimes \mathbf{B}\mathbf{r}_j) \mathbf{L}_{j,0} - \hat{\mathbf{W}}_j \mathbf{L}_{j,0} \\ + \mathbf{W}_{2,j}(\mathbf{D}\mathbf{E}_j \otimes \mathbf{B}\mathbf{r}_j) + \mathbf{W}_{3,j}(\mathbf{w}^\top \mathbf{D}\mathbf{E}_j \otimes \mathbf{B}\mathbf{r}_j) + \mathbf{V}\mathbf{D}\mathbf{E}_j + \mathbf{v}\mathbf{w}^\top \mathbf{D}\mathbf{E}_j, \\ \sum_{i \in [L] \setminus \{j\}} \mathbf{W}_{1,i}(\mathbf{I}_{m+n'} \otimes \mathbf{B}\mathbf{r}_j) \mathbf{L}_{i,1}, \\ \sum_{i \in [L] \setminus \{j\}} \mathbf{G}_{2,i,j} + \mathbf{H}_{2,i}(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j) + \mathbf{W}_{1,i}(\mathbf{I}_{m+n'} \otimes \mathbf{B}\mathbf{r}_j) \mathbf{L}_{i,0} \\ + \mathbf{W}_{2,i}(\mathbf{D}\mathbf{E}_i \otimes \mathbf{B}\mathbf{r}_j) + \mathbf{W}_{3,i}(\mathbf{w}^\top \mathbf{D}\mathbf{E}_i \otimes \mathbf{B}\mathbf{r}_j), \\ \mathbf{I}_{\mathbf{g}_j}, \mathbf{q}_j \mathbf{e}_1^\top, \mathbf{D}\mathbf{E}_j, \mathbf{w}^\top \mathbf{D}\mathbf{E}_j \end{array} \right\|_2, \end{aligned}$$

where we use the following notations:

- $\mathbf{W}_{1,i} \in \mathbb{Z}_p^{(2k+1) \times (2k+1)(1+n')}$ refers to the matrix composed of the right most $(2k+1)(1+n')$ columns of $\mathbf{W}_{1,i} \in \mathbb{Z}_p^{(2k+1) \times (2k+1)(m+n')}$, and
- $\hat{\mathbf{W}}_j \in \mathbb{Z}_p^{(2k+1) \times (1+n')}$ refers to the matrix composed of the right most $(1+n')$ columns of the matrix $\hat{\mathbf{W}}_j \in \mathbb{Z}_p^{(2k+1) \times (m+n')}$.

$\text{Enc}(\text{mpk}, (\mathbf{y}, \mathbf{x}), (\mathbf{p}, \mathbf{z})):$ On input the master public key mpk , a public input $(\mathbf{y}, \mathbf{x}) \in \mathbb{Z}_p^{\hat{n}} \times \mathbb{Z}_p^n$ and a private input $(\mathbf{p}, \mathbf{z}) \in \mathbb{Z}_p^k \times \mathbb{Z}_p^{n'}$, sample $\mathbf{f}, \mathbf{s} \xleftarrow{\$} \mathbb{Z}_p^k$ and output

$$\text{ct} = \left\| \begin{array}{l} \mathbf{s}^\top \mathbf{A}, (\mathbf{f}^\top \parallel \mathbf{z}^\top) + \mathbf{s}^\top \mathbf{A}\hat{\mathbf{H}}_1, \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{1,i} + \mathbf{A}\mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}) - \mathbf{A}\mathbf{W}_{1,i}, \mathbf{p}^\top + \mathbf{s}^\top \mathbf{A}\hat{\mathbf{H}}_2, \\ \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{2,i} + \mathbf{A}\mathbf{H}_{2,i}(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{A}\mathbf{W}_{1,i}(\mathbf{L}_{i,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \\ + \mathbf{A}\mathbf{W}_{2,i}(\mathbf{D}\mathbf{E}_i \otimes \mathbf{I}_{2k+1}) + \mathbf{A}\mathbf{W}_{3,i}(\mathbf{w}^\top \mathbf{D}\mathbf{E}_i \otimes \mathbf{I}_{2k+1}), \mathbf{s}^\top \mathbf{A}\mathbf{v}, \mathbf{s}^\top \mathbf{A}\mathbf{v} \end{array} \right\|_1,$$

where $\mathbf{L}_{i,\mathbf{y},\mathbf{x}} := \mathbf{L}_{i,0} + \mathbf{L}_{i,1}((\mathbf{y}^\top \parallel \mathbf{x}^\top)^\top \otimes \mathbf{I}_m)$.

Note. Observe that the last two components of mpk denoted as $(\llbracket \mathbf{K}_1 \rrbracket_1, \llbracket \mathbf{K}_0 \rrbracket_1)$ enables the computation of the fifth component of the ciphertext as $\mathbf{s}^\top \llbracket \mathbf{K}_1 \rrbracket_1 ((\mathbf{y}^\top \parallel \mathbf{x}^\top)^\top \otimes \mathbf{I}_m) \otimes \mathbf{I}_{2k+1} + \mathbf{s}^\top \llbracket \mathbf{K}_0 \rrbracket_1$.

$\text{Dec}(\text{sk}_{i^*}, \text{hsk}_{i^*}, \text{ct})$: On input a secret key sk_{i^*} with corresponding helper secret key hsk_{i^*} and a ciphertext ct , parse $\text{hsk}_{i^*} = (\llbracket \mathbf{d}_0 \rrbracket_2, \llbracket \mathbf{D}_{1,1} \rrbracket_2, \llbracket \mathbf{D}_{1,2} \rrbracket_2, \llbracket \mathbf{D}_{2,1,1} \rrbracket_2, \llbracket \mathbf{D}_{2,1,2} \rrbracket_2, \llbracket \mathbf{D}_{2,2,1} \rrbracket_2, \llbracket \mathbf{D}_{2,2,2} \rrbracket_2, \llbracket \mathbf{D}_{0,1} \rrbracket_2, \llbracket \mathbf{D}_{0,2} \rrbracket_2, \llbracket \mathbf{D}_{0,3} \rrbracket_2, \llbracket \mathbf{d}_{0,4} \rrbracket_2)$ and $\text{ct} = (\llbracket \mathbf{c}_0 \rrbracket_1, \llbracket \mathbf{c}_{1,1} \rrbracket_1, \llbracket \mathbf{c}_{1,2} \rrbracket_1, \llbracket \mathbf{c}_{2,1} \rrbracket_1, \llbracket \mathbf{c}_{2,2} \rrbracket_1, \llbracket \mathbf{c}_3 \rrbracket_1, \llbracket \mathbf{c}_4 \rrbracket_1)$. If $g_{i^*}(\mathbf{y}) \neq 0$, output $\perp$. Otherwise, compute

$$\begin{aligned} \llbracket \mathbf{c}'_{1,1} \rrbracket_t &= \llbracket \mathbf{c}_{1,1} \rrbracket_1 \llbracket \mathbf{D}_{0,1} \rrbracket_2, & \llbracket \mathbf{d}'_{1,1} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_{1,1} \rrbracket_2, \\ \llbracket \mathbf{c}'_{1,2} \rrbracket_t &= \llbracket \mathbf{c}_{1,2} \rrbracket_1 \llbracket \mathbf{I}_{n'+1} \otimes \mathbf{d}_0 \rrbracket_2, & \llbracket \mathbf{d}'_{1,2} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_{1,2} \rrbracket_2, \\ \llbracket \mathbf{c}'_{2,1} \rrbracket_t &= \llbracket \mathbf{c}_{2,1} \rrbracket_1 \llbracket \mathbf{D}_{0,2} \rrbracket_2, & \llbracket \mathbf{d}'_{2,1} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_{2,1,1} ((\mathbf{y}^\top \parallel \mathbf{x}^\top)^\top \otimes \mathbf{I}_m) + \mathbf{D}_{2,1,2} \rrbracket_2, \\ \llbracket \mathbf{c}'_{2,2} \rrbracket_t &= \llbracket \mathbf{c}_{2,2} \rrbracket_1 \llbracket \mathbf{I}_m \otimes \mathbf{d}_0 \rrbracket_2, & \llbracket \mathbf{d}'_{2,2} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_{2,2,1} ((\mathbf{y}^\top \parallel \mathbf{x}^\top)^\top \otimes \mathbf{I}_m) + \mathbf{D}_{2,2,2} \rrbracket_2, \\ \llbracket \mathbf{c}'_3 \rrbracket_t &= \llbracket \mathbf{c}_3 \rrbracket_1 \llbracket \mathbf{D}_{0,3} \rrbracket_2, \\ \llbracket \mathbf{c}'_4 \rrbracket_t &= \llbracket \mathbf{c}_4 \rrbracket_1 \llbracket \mathbf{d}_{0,4} \rrbracket_2, \end{aligned}$$

$$\begin{aligned} \llbracket \mathbf{p}_1^\top \rrbracket_t &= \llbracket (\mathbf{c}'_{1,1} - \mathbf{d}'_{1,1}) + (\mathbf{c}'_{1,2} - \mathbf{d}'_{1,2}) \rrbracket_t - \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{U}_{1,i^*} \cdot (\mathbf{I}_{n'+1} \otimes \mathbf{d}_0) \rrbracket_2, \\ \llbracket \mathbf{p}_2^\top \rrbracket_t &= \llbracket (\mathbf{c}'_{2,1} - \mathbf{d}'_{2,1}) + (\mathbf{c}'_{2,2} - \mathbf{d}'_{2,2}) + \mathbf{c}'_3 + \mathbf{c}'_4 \rrbracket_t - \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{U}_{2,i^*} \cdot (\mathbf{I}_m \otimes \mathbf{d}_0) \rrbracket_2. \end{aligned}$$

Run $\mathbf{d}_{\phi_{i^*}, \mathbf{y}, \mathbf{x}} \leftarrow \text{rec}(\phi_{i^*}, (\mathbf{y}, \mathbf{x}))$, then compute and output $\llbracket v \rrbracket_t = \llbracket (\mathbf{p}_1^\top, \mathbf{p}_2^\top) \cdot \mathbf{d}_{\phi_{i^*}, \mathbf{y}, \mathbf{x}} \rrbracket_t$.

Proposition 36 (Compactness, Correctness and Completeness). *The sRFE scheme in Construction 35 is compact, correct and complete.*

Proposition 37 (Security). *Suppose that Π is a quasi-adaptive NIZK for linear spaces satisfying perfect completeness, perfect zero-knowledge and unbounded simulation soundness. Then the sRFE scheme in Construction 35 is very selectively simulation secure under the bilateral MDDH $_k$ assumption on $\mathbb{G}$.*

The proofs of Propositions 36 and 37 can be found in Sections 6.1 and 6.3, respectively. We present the simulator of AB-eAWS in Section 6.2. When instantiating Π with the quasi-adaptive NIZK of [KW15] and using the transformation of Theorem 18, we obtain the following theorem, where we note that AB-1AWS is a special case of AB-eAWS.

Theorem 38. *Assuming bilateral MDDH $_k$ on pairings, there exists a very selectively SIM-secure (bounded) RFE scheme for AB-1AWS functionality.*

6.1 Proof of Completeness, Correctness and Compactness

Proof of Proposition 36. We show that Construction 35 is complete, correct and compact.

Completeness. Public keys have the same form as in Construction 28, so the argument is identical to the proof of Proposition 29.

Correctness. For all $\lambda, L \in \mathbb{N}$, $i^* \in [L]$, $\mathbf{q}_1, \dots, \mathbf{q}_L \in \mathbb{Z}_p^k$, all $\text{crs} \leftarrow \text{Setup}(1^\lambda, 1^L, \{\llbracket \mathbf{q}_i \rrbracket_{1,2}\}_{i \in [L]})$ and $(\text{pk}_{i^*}, \text{sk}_{i^*}) \leftarrow \text{Gen}(\text{crs}, i^*)$, all $\{\text{pk}_i\}_{i \in [L] \setminus \{i^*\}}$ such that $\text{Ver}(\text{crs}, i, \text{pk}_i) = 1$, all $f_1, \dots, f_L \in \mathcal{F}_{n,1}^{\text{abp}} \times \mathcal{F}_{n,n'}^{\text{abp}}$, all $(\mathbf{y}, \mathbf{x}) \in \mathbb{Z}_p^{\hat{n}} \times \mathbb{Z}_p^n$ and all $(\mathbf{p}, \mathbf{z}) \in \mathbb{Z}_p^k \times \mathbb{Z}_p^{n'}$, all $(\text{mpk}, \{\text{hsk}_j\}_{j \in [L]}) \leftarrow \text{Agg}(\text{crs}, \{\text{pk}_i, f_i\}_{i \in [L]})$ and all $\text{ct} \leftarrow \text{Enc}(\text{mpk}, (\mathbf{y}, \mathbf{x}), (\mathbf{p}, \mathbf{z}))$, we have $\text{sk}_{i^*} = (\mathbf{U}_{1,i^*}, \mathbf{U}_{2,i^*})$ and

$$\begin{aligned} (\mathbf{c}_{1,1}, \mathbf{c}_{1,2}) &= \left((\mathbf{f}^\top \parallel \mathbf{z}^\top) + \mathbf{s}^\top \mathbf{A} \hat{\mathbf{H}}_1, \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{1,i} + \mathbf{A} \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}) - \mathbf{A} \mathbf{W}_{1,i} \right) \\ (\mathbf{c}_{2,1}, \mathbf{c}_0, \mathbf{c}_3, \mathbf{c}_4) &= \left(\mathbf{p}^\top + \mathbf{s}^\top \mathbf{A} \hat{\mathbf{H}}_2, \mathbf{s}^\top \mathbf{A}, \mathbf{s}^\top \mathbf{A} \mathbf{V}, \mathbf{s}^\top \mathbf{A} \mathbf{v} \right) \\ \mathbf{c}_{2,2} &= \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{2,i} + \mathbf{A} \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{A} \mathbf{W}_{1,i} (\mathbf{L}_{i,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \end{aligned}$$

$$\begin{aligned}
& + \mathbf{A}\mathbf{W}_{2,i}(\mathbf{D}\mathbf{E}_i \otimes \mathbf{I}_{2k+1}) + \mathbf{A}\mathbf{W}_{3,i}(\mathbf{w}^\top \mathbf{D}\mathbf{E}_i \otimes \mathbf{I}_{2k+1}), \\
(\mathbf{d}_0, \mathbf{D}_{0,1}, \mathbf{D}_{0,2}, \mathbf{D}_{0,3}, \mathbf{d}_{0,4}) = & \left(\mathbf{B}\mathbf{r}_{i^*}, \mathbf{I}_{\mathbf{g}_{i^*}}, \mathbf{q}_{i^*} \mathbf{e}_1^\top, \mathbf{D}\mathbf{E}_{i^*}, \mathbf{w}^\top \mathbf{D}\mathbf{E}_{i^*} \right) \\
\mathbf{D}_{1,1} = & \mathbf{H}_{1,i^*}(\mathbf{I}_{\mathbf{g}_{i^*}} \otimes \mathbf{B}\mathbf{r}_{i^*}) - \underline{\mathbf{W}_{1,i^*}}(\mathbf{I}_{n'+1} \otimes \mathbf{B}\mathbf{r}_{i^*}) + \hat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_{i^*}} + \hat{\mathbf{W}}_{i^*} \\
\mathbf{D}_{1,2} = & \sum_{i \in [L] \setminus \{i^*\}} \mathbf{G}_{1,i,i^*} + \mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{B}\mathbf{r}_{i^*}) - \underline{\mathbf{W}_{1,i}}(\mathbf{I}_{n'+1} \otimes \mathbf{B}\mathbf{r}_{i^*}) \\
\mathbf{D}_{2,1,1} = & \mathbf{W}_{1,i^*}(\mathbf{I}_{m+n'} \otimes \mathbf{B}\mathbf{r}_{i^*}) \mathbf{L}_{i^*,1} - \hat{\mathbf{W}}_{i^*} \mathbf{L}_{i^*,1} \\
\mathbf{D}_{2,1,2} = & \mathbf{H}_{2,i^*}(\mathbf{q}_{i^*} \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_{i^*}) + \hat{\mathbf{H}}_2 \mathbf{q}_{i^*} \mathbf{e}_1^\top + \mathbf{W}_{1,i^*}(\mathbf{I}_{m+n'} \otimes \mathbf{B}\mathbf{r}_{i^*}) \mathbf{L}_{i^*,0} - \hat{\mathbf{W}}_{i^*} \mathbf{L}_{i^*,0} \\
& + \mathbf{W}_{2,i^*}(\mathbf{D}\mathbf{E}_{i^*} \otimes \mathbf{B}\mathbf{r}_{i^*}) + \mathbf{W}_{3,i^*}(\mathbf{w}^\top \mathbf{D}\mathbf{E}_{i^*} \otimes \mathbf{B}\mathbf{r}_{i^*}) + \mathbf{V}\mathbf{D}\mathbf{E}_{i^*} + \mathbf{v}\mathbf{w}^\top \mathbf{D}\mathbf{E}_{i^*} \\
\mathbf{D}_{2,2,1} = & \sum_{i \in [L] \setminus \{i^*\}} \mathbf{W}_{1,i}(\mathbf{I}_{m+n'} \otimes \mathbf{B}\mathbf{r}_{i^*}) \mathbf{L}_{i,1} \\
\mathbf{D}_{2,2,2} = & \sum_{i \in [L] \setminus \{i^*\}} \mathbf{G}_{2,i,i^*} + \mathbf{H}_{2,i}(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_{i^*}) + \mathbf{W}_{1,i}(\mathbf{I}_{m+n'} \otimes \mathbf{B}\mathbf{r}_{i^*}) \mathbf{L}_{i,0} \\
& + \mathbf{W}_{2,i}(\mathbf{D}\mathbf{E}_i \otimes \mathbf{B}\mathbf{r}_{i^*}) + \mathbf{W}_{3,i}(\mathbf{w}^\top \mathbf{D}\mathbf{E}_i \otimes \mathbf{B}\mathbf{r}_{i^*}),
\end{aligned}$$

where we parse the ciphertext $\text{ct} = ([\mathbf{c}_0]_1, [\mathbf{c}_{1,1}]_1, [\mathbf{c}_{1,2}]_1, [\mathbf{c}_{2,1}]_1, [\mathbf{c}_{2,2}]_1, [\mathbf{c}_3]_1, [\mathbf{c}_4]_1)$ and the helper secret key $\text{hsk}_{i^*} = ([\mathbf{d}_0]_2, [\mathbf{D}_{1,1}]_2, [\mathbf{D}_{1,2}]_2, [\mathbf{D}_{2,1,1}]_2, [\mathbf{D}_{2,1,2}]_2, [\mathbf{D}_{2,2,1}]_2, [\mathbf{D}_{2,2,2}]_2, [\mathbf{D}_{0,1}]_2, [\mathbf{D}_{0,2}]_2, [\mathbf{D}_{0,3}]_2, [\mathbf{d}_{0,4}]_2)$. By computing

$$\begin{aligned}
\llbracket \mathbf{c}'_{1,1} \rrbracket_t &= \llbracket \mathbf{c}_{1,1} \rrbracket_1 [\mathbf{D}_{0,1}]_2, & \llbracket \mathbf{d}'_{1,1} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 [\mathbf{D}_{1,1}]_2, \\
\llbracket \mathbf{c}'_{1,2} \rrbracket_t &= \llbracket \mathbf{c}_{1,2} \rrbracket_1 [\mathbf{I}_{n'+1} \otimes \mathbf{d}_0]_2, & \llbracket \mathbf{d}'_{1,2} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 [\mathbf{D}_{1,2}]_2, \\
\llbracket \mathbf{c}'_{2,1} \rrbracket_t &= \llbracket \mathbf{c}_{2,1} \rrbracket_1 [\mathbf{D}_{0,2}]_2, & \llbracket \mathbf{d}'_{2,1} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 [\mathbf{D}_{2,1,1}((\mathbf{y}^\top \parallel \mathbf{x}^\top)^\top \otimes \mathbf{I}_m) + \mathbf{D}_{2,1,2}]_2, \\
\llbracket \mathbf{c}'_{2,2} \rrbracket_t &= \llbracket \mathbf{c}_{2,2} \rrbracket_1 [\mathbf{I}_m \otimes \mathbf{d}_0]_2, & \llbracket \mathbf{d}'_{2,2} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 [\mathbf{D}_{2,2,1}((\mathbf{y}^\top \parallel \mathbf{x}^\top)^\top \otimes \mathbf{I}_m) + \mathbf{D}_{2,2,2}]_2, \\
\llbracket \mathbf{c}'_3 \rrbracket_t &= \llbracket \mathbf{c}_3 \rrbracket_1 [\mathbf{D}_{0,3}]_2, \\
\llbracket \mathbf{c}'_4 \rrbracket_t &= \llbracket \mathbf{c}_4 \rrbracket_1 [\mathbf{d}_{0,4}]_2,
\end{aligned}$$

the decryptor obtains

$$\begin{aligned}
\mathbf{c}'_{1,1} &= \left((\mathbf{f}^\top \parallel \mathbf{z}^\top) + \mathbf{s}^\top \mathbf{A} \hat{\mathbf{H}}_1 \right) \mathbf{I}_{\mathbf{g}_{i^*}} = (\mathbf{f}^\top \mathbf{g}_{i^*} \parallel \mathbf{z}^\top) + \mathbf{s}^\top \mathbf{A} \hat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_{i^*}} \\
\mathbf{d}'_{1,1} &= \mathbf{s}^\top \mathbf{A} \left(\mathbf{H}_{1,i^*}(\mathbf{I}_{\mathbf{g}_{i^*}} \otimes \mathbf{B}\mathbf{r}_{i^*}) - \underline{\mathbf{W}_{1,i^*}}(\mathbf{I}_{n'+1} \otimes \mathbf{B}\mathbf{r}_{i^*}) + \hat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_{i^*}} + \hat{\mathbf{W}}_{i^*} \right) \\
\mathbf{c}'_{1,2} &= \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{1,i}(\mathbf{I}_{n'+1} \otimes \mathbf{B}\mathbf{r}_{i^*}) + \mathbf{A} \mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{B}\mathbf{r}_{i^*}) - \mathbf{A} \underline{\mathbf{W}_{1,i}}(\mathbf{I}_{n'+1} \otimes \mathbf{B}\mathbf{r}_{i^*}) \\
\mathbf{d}'_{1,2} &= \mathbf{s}^\top \mathbf{A} \sum_{i \in [L] \setminus \{i^*\}} \mathbf{G}_{1,i,i^*} + \mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{B}\mathbf{r}_{i^*}) - \underline{\mathbf{W}_{1,i}}(\mathbf{I}_{n'+1} \otimes \mathbf{B}\mathbf{r}_{i^*}) \\
\mathbf{c}'_{2,1} &= \left(\mathbf{p}^\top + \mathbf{s}^\top \mathbf{A} \hat{\mathbf{H}}_2 \right) \mathbf{q}_{i^*} \mathbf{e}_1^\top = (\mathbf{p}^\top \mathbf{q}_{i^*}) \mathbf{e}_1^\top + \mathbf{s}^\top \mathbf{A} \hat{\mathbf{H}}_2 \mathbf{q}_{i^*} \mathbf{e}_1^\top \\
\mathbf{d}'_{2,1} &= \mathbf{s}^\top \mathbf{A} \left(\mathbf{H}_{2,i^*}(\mathbf{q}_{i^*} \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_{i^*}) + \hat{\mathbf{H}}_2 \mathbf{q}_{i^*} \mathbf{e}_1^\top + \mathbf{W}_{1,i^*}(\mathbf{I}_{m+n'} \otimes \mathbf{B}\mathbf{r}_{i^*}) \mathbf{L}_{i^*,\mathbf{y},\mathbf{x}} - \hat{\mathbf{W}}_{i^*} \mathbf{L}_{i^*,\mathbf{y},\mathbf{x}} \right. \\
& \quad \left. + \mathbf{W}_{2,i^*}(\mathbf{D}\mathbf{E}_{i^*} \otimes \mathbf{B}\mathbf{r}_{i^*}) + \mathbf{W}_{3,i^*}(\mathbf{w}^\top \mathbf{D}\mathbf{E}_{i^*} \otimes \mathbf{B}\mathbf{r}_{i^*}) + \mathbf{V}\mathbf{D}\mathbf{E}_{i^*} + \mathbf{v}\mathbf{w}^\top \mathbf{D}\mathbf{E}_{i^*} \right) \\
\mathbf{c}'_{2,2} &= \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{2,i}(\mathbf{I}_m \otimes \mathbf{B}\mathbf{r}_{i^*}) + \mathbf{A} \mathbf{H}_{2,i}(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_{i^*}) + \mathbf{A} \mathbf{W}_{1,i}(\mathbf{L}_{i,\mathbf{y},\mathbf{x}} \otimes \mathbf{B}\mathbf{r}_{i^*}) \\
& \quad + \mathbf{A} \mathbf{W}_{2,i}(\mathbf{D}\mathbf{E}_i \otimes \mathbf{B}\mathbf{r}_{i^*}) + \mathbf{A} \mathbf{W}_{3,i}(\mathbf{w}^\top \mathbf{D}\mathbf{E}_i \otimes \mathbf{B}\mathbf{r}_{i^*}) \\
\mathbf{d}'_{2,2} &= \mathbf{s}^\top \mathbf{A} \sum_{i \in [L] \setminus \{i^*\}} \mathbf{G}_{2,i,i^*} + \mathbf{H}_{2,i}(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_{i^*}) + \mathbf{W}_{1,i}(\mathbf{I}_{m+n'} \otimes \mathbf{B}\mathbf{r}_{i^*}) \mathbf{L}_{i,\mathbf{y},\mathbf{x}} \\
& \quad + \mathbf{W}_{2,i}(\mathbf{D}\mathbf{E}_i \otimes \mathbf{B}\mathbf{r}_{i^*}) + \mathbf{W}_{3,i}(\mathbf{w}^\top \mathbf{D}\mathbf{E}_i \otimes \mathbf{B}\mathbf{r}_{i^*}) \\
\mathbf{c}'_3 &= \mathbf{s}^\top \mathbf{A} \mathbf{V} \cdot \mathbf{D}\mathbf{E}_{i^*} \\
\mathbf{c}'_4 &= \mathbf{s}^\top \mathbf{A} \mathbf{v} \cdot \mathbf{w}^\top \mathbf{D}\mathbf{E}_{i^*}.
\end{aligned}$$

Therefore, the vector $(\llbracket \mathbf{p}_1^\top \rrbracket_t, \llbracket \mathbf{p}_2^\top \rrbracket_t)$ equals

$$\begin{aligned}\llbracket \mathbf{p}_1^\top \rrbracket_t &= \llbracket (\mathbf{c}'_{1,1} - \mathbf{d}'_{1,1}) + (\mathbf{c}'_{1,2} - \mathbf{d}'_{1,2}) \rrbracket_t - \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{U}_{1,i^*} \cdot (\mathbf{I}_{n'+1} \otimes \mathbf{d}_0) \rrbracket_2 \\ &= \llbracket (\mathbf{f}^\top \mathbf{g}_{i^*} \parallel \mathbf{z}^\top) - \mathbf{s}^\top \mathbf{A} \widehat{\mathbf{W}}_{i^*} \rrbracket_t, \\ \llbracket \mathbf{p}_2^\top \rrbracket_t &= \llbracket (\mathbf{c}'_{2,1} - \mathbf{d}'_{2,1}) + (\mathbf{c}'_{2,2} - \mathbf{d}'_{2,2}) + \mathbf{c}'_3 + \mathbf{c}'_4 \rrbracket_t - \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{U}_{2,i^*} \cdot (\mathbf{I}_m \otimes \mathbf{d}_0) \rrbracket_2 \\ &= \llbracket \mathbf{s}^\top \mathbf{A} \widehat{\mathbf{W}}_{i^*} \mathbf{L}_{i^*, \mathbf{y}, \mathbf{x}} + \mathbf{p}^\top \mathbf{q}_{i^*} \cdot \mathbf{e}_1^\top \rrbracket_t,\end{aligned}$$

where the second equation of $\llbracket \mathbf{p}_2^\top \rrbracket_2$ follows from the fact that $(\mathbf{I}_{m+n'} \otimes \mathbf{B} \mathbf{r}_{i^*}) \mathbf{L}_{j, \mathbf{y}, \mathbf{x}} = (\mathbf{L}_{j, \mathbf{y}, \mathbf{x}} \otimes \mathbf{B} \mathbf{r}_{i^*})$. We observe that $(\mathbf{p}_1^\top, \mathbf{p}_2^\top) = \text{pgb}(\phi_{i^*}(\mathbf{y}, \mathbf{x}), (\mathbf{f}^\top \mathbf{g}_{i^*} \parallel \mathbf{z}^\top); \mathbf{s}^\top \mathbf{A} \widehat{\mathbf{W}}_{i^*}) + (\mathbf{0}, \mathbf{p}^\top \mathbf{q}_{i^*} \cdot \mathbf{e}_1^\top)$. Then it follows from the correctness of the garbling scheme that if $g_{i^*}(\mathbf{y}) = 0$, then

$$\llbracket v \rrbracket_t = \llbracket g_{i^*}(\mathbf{y}) \cdot \mathbf{f}^\top \mathbf{g}_{i^*} + h_{i^*}(\mathbf{x})^\top \mathbf{z} + \mathbf{p}^\top \mathbf{q}_{i^*} \rrbracket_t = \llbracket h_{i^*}(\mathbf{x})^\top \mathbf{z} + \mathbf{p}^\top \mathbf{q}_{i^*} \rrbracket_t.$$

Compactness. Let $j \in [L]$. The scheme has the following properties:

$$\begin{aligned}|\text{crs}| &= L^2 \cdot (m + n') \cdot \text{poly}(\lambda) & |\text{mpk}| &= (m(\widehat{n} + n) + n') \cdot \text{poly}(\lambda) \\ |\text{hsk}_j| &= (m(\widehat{n} + n) + n') \cdot \text{poly}(\lambda) & |\text{ct}| &= (m + n') \cdot \text{poly}(\lambda),\end{aligned}$$

where the size of crs follows from the fact that the total size of $\{\text{crs}_i\}_{i \in [L]}$ is $L \cdot \text{poly}(\lambda)$ if we instantiate Π with the QA-NIZK of [KW15], and the fact that the size of the language description is $\text{poly}(\lambda)$. $\square$

6.2 Simulator

Let us denote $\mathcal{I}_{\text{honnauth}} = \{i \in \mathcal{I}_{\text{mal}} \cup \mathcal{I}_{\text{cor}} : g_i^*(\mathbf{y}) = 0\}$, i.e. the set of all corrupt or malicious slots having g_i^* as the challenge predicate satisfying the challenge attribute $\mathbf{y}$.

Setup($1^\lambda, 1^L, (\mathbf{y}, \mathbf{x}), \{\llbracket \mathbf{q}_i \rrbracket_{1,2}, f_i^* = (g_i^*, h_i^*)\}_{i \in [L]}, \{\llbracket \mu_i \rrbracket_{1,2}\}_{i \in \mathcal{I}_{\text{honnauth}}}$): On input the security parameter 1^λ , the maximum number of slots 1^L , a challenge public input $(\mathbf{y}, \mathbf{x})$, a set of parameters $\{\llbracket \mathbf{q}_i \rrbracket_{1,2}\}_{i \in [L]}$ with $\mathbf{q}_i \in \mathbb{Z}_p^k$, a set of functions $\{f_i^* = (g_i^*, h_i^*)\}_{i \in [L]}$ with $f_i^* \in \mathcal{F} = \mathcal{F}_{\widehat{n},1}^{\text{abp}} \times \mathcal{F}_{n,n'}^{\text{abp}}$ for $i \in [L]$, and a set of functional values $\{\llbracket \mu_i \rrbracket_{1,2}\}_{i \in \mathcal{I}_{\text{honnauth}}}$, sample

$$\begin{aligned}\mathbf{A} &\xleftarrow{\$} \mathbb{Z}_p^{k \times (2k+1)}, \mathbf{B}, \mathbf{D} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times k}, \mathbf{V} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)}, \mathbf{v}, \mathbf{w}, \mathbf{c}, \mathbf{c}^\perp \xleftarrow{\$} \mathbb{Z}_p^{2k+1}, \mathbf{f} \xleftarrow{\$} \mathbb{Z}_p^k, \\ \widehat{\mathbf{H}}_1 &\xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (k+n')}, \widehat{\mathbf{H}}_2 \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times k}, \\ \left\{ \begin{array}{l} \mathbf{E}_i \xleftarrow{\$} \mathbb{Z}_p^{k \times m}, \mathbf{R}_{1,i}, \mathbf{R}_{2,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+2) \times (2k+1)}, \widetilde{\mathbf{R}}_{1,i}, \widetilde{\mathbf{R}}_{2,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+2) \times (2k+2)}, \mathbf{g}_i, \mathbf{r}_i \xleftarrow{\$} \mathbb{Z}_p^k, \\ \mathbf{H}_{1,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)(k+n')}, \mathbf{H}_{2,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)k}, \widehat{\mathbf{W}}_i \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (m+n')}, \\ \mathbf{W}_{1,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)(m+n')}, \mathbf{W}_{2,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)(2k+1)}, \mathbf{W}_{3,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)} \end{array} \right\}_{i \in [L]}.\end{aligned}$$

Pick $\mathbf{c} \in \mathbb{Z}_p^{2k+1}$ such that $\mathbf{c}^\top \mathbf{c}^\perp = 1$, $\mathbf{A} \mathbf{c}^\perp = \mathbf{0}$ and define an ABP $\phi_i \in \mathcal{F}_{\widehat{n}+n,1+n'}^{\text{abp}}$ as $\phi_i(\mathbf{y}, \mathbf{x})^\top = (g_i(\mathbf{y}), h_i(\mathbf{x})^\top)$. Then, compute

$$\mathbf{R}_{d,i} = \widetilde{\mathbf{R}}_{d,i} \begin{bmatrix} \mathbf{c}^\top \\ \mathbf{I}_{2k+1} \end{bmatrix}$$

for $d \in \{1, 2\}$ and generate

$$(\text{crs}_{1,i}, \text{td}_{1,i}) \leftarrow \text{LGen}(1^\lambda, \mathbb{G}_1, \llbracket \mathbf{A}_{1,i} \rrbracket_1), \quad (\text{crs}_{2,i}, \text{td}_{2,i}) \leftarrow \text{LGen}(1^\lambda, \mathbb{G}_1, \llbracket \mathbf{A}_{2,i} \rrbracket_1)$$

where $\mathbf{A}_{1,i}^\top = [\mathbf{A}^\top \parallel \mathbf{R}_{1,i}^\top], \mathbf{A}_{2,i}^\top = [\mathbf{A}^\top \parallel \mathbf{R}_{2,i}^\top]$, and output the common reference string

$$\widehat{\text{crs}} = \left(\left[\mathbf{A}, \mathbf{A}\hat{\mathbf{H}}_1, \mathbf{A}\hat{\mathbf{H}}_2, \mathbf{A}\mathbf{V}, \mathbf{A}\mathbf{v} \right]_1, \left\{ \left[\begin{array}{c} \mathbf{A}\mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}), \mathbf{A}\mathbf{H}_{2,i}(\mathbf{q}_i(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{I}_{2k+1}), \mathbf{R}_{1,i}, \mathbf{R}_{2,i}, \\ \mathbf{A}\mathbf{W}_{1,i}, \mathbf{A}\mathbf{W}_{2,i}(\mathbf{D}\mathbf{E}_i \otimes \mathbf{I}_{2k+1}) + \mathbf{A}\mathbf{W}_{3,i}(\mathbf{w}^\top \mathbf{D}\mathbf{E}_i \otimes \mathbf{I}_{2k+1}), \text{crs}_{1,i}, \text{crs}_{2,i} \end{array} \right]_1 \right\}_{i \in [L]}, \left\{ \left[\begin{array}{c} \mathbf{B}\mathbf{r}_j, \mathbf{q}_j, \mathbf{g}_j, \mathbf{D}\mathbf{E}_j, \mathbf{w}^\top \mathbf{D}\mathbf{E}_j + \theta_j \mathbf{e}_1^\top, \mathbf{W}_{1,j}(\mathbf{I}_{m+n'} \otimes \mathbf{B}\mathbf{r}_j) - \hat{\mathbf{W}}_j, \\ \mathbf{H}_{1,j}(\mathbf{I}_{\mathbf{g}_j} \otimes \mathbf{B}\mathbf{r}_j) + \hat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_j}, \mathbf{H}_{2,j}(\mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j) + \hat{\mathbf{H}}_2 \mathbf{q}_j \mathbf{e}_1^\top \\ + \mathbf{W}_{2,j}(\mathbf{D}\mathbf{E}_j \otimes \mathbf{B}\mathbf{r}_j) + \mathbf{W}_{3,j}((\mathbf{w}^\top \mathbf{D}\mathbf{E}_j + \theta_j \mathbf{e}_1^\top) \otimes \mathbf{B}\mathbf{r}_j) \\ + \mathbf{V}\mathbf{D}\mathbf{E}_j + \mathbf{v}(\mathbf{w}^\top \mathbf{D}\mathbf{E}_j + \theta_j \mathbf{e}_1^\top) + \mathbf{1}_{\mathcal{I}_{\text{mal}} \cup \mathcal{I}_{\text{cor}}}(j) \cdot \mathbf{g}_j^*(\mathbf{y}) \cdot \mathbf{g}_j^\top \mathbf{f} \cdot \mathbf{c}^\perp \mathbf{e}_1^\top \end{array} \right]_2 \right\}_{j \in [L]}, \left\{ \left[\begin{array}{c} \mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{B}\mathbf{r}_j), \mathbf{W}_{1,i}(\mathbf{I}_{m+n'} \otimes \mathbf{B}\mathbf{r}_j), \\ \mathbf{H}_{2,i}(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B}\mathbf{r}_j), \mathbf{W}_{2,i}(\mathbf{D}\mathbf{E}_i \otimes \mathbf{B}\mathbf{r}_j) + \mathbf{W}_{3,i}((\mathbf{w}^\top \mathbf{D}\mathbf{E}_i + \theta_i \mathbf{e}_1^\top) \otimes \mathbf{B}\mathbf{r}_j) \end{array} \right]_2 \right\}_{\substack{i \in [L], \\ j \in [L] \setminus \{i\}}} \right),$$

and

$$\text{td} = \left(\left[\begin{array}{c} (\tilde{\mathbf{R}}_{d,i}, \text{td}_{d,i})_{i \in [L], d \in \{1,2\}}, \{\|\mathbf{B}\mathbf{r}_j\|_2\}_{j \in [L]}, \\ \mathbf{c}, \mathbf{A}, \mathbf{c}^\top \hat{\mathbf{H}}_1, \mathbf{c}^\top \hat{\mathbf{H}}_2, \mathbf{c}^\top \mathbf{V} - \mathbf{w}^\top, \mathbf{c}^\top \mathbf{v} + \mathbf{1}, \\ \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}) - \underline{\mathbf{W}}_{1,i}, \\ \mathbf{c}^\top \sum_{i \in [L]} \mathbf{W}_{1,i}(\mathbf{I}_{\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) + \mathbf{H}_{2,i}(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) \\ + \mathbf{W}_{2,i}(\mathbf{D}\mathbf{E}_i \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{3,i}((\mathbf{w}^\top \mathbf{D}\mathbf{E}_i + \theta_i \mathbf{e}_1^\top) \otimes \mathbf{I}_{2k+1}) \end{array} \right]_1 \right).$$

Here, $\mathbf{1}_{\mathcal{I}_{\text{mal}} \cup \mathcal{I}_{\text{cor}}}(j)$ denotes the indicator function that outputs 1 if $j \in \mathcal{I}_{\text{mal}} \cup \mathcal{I}_{\text{cor}}$ and 0 otherwise. Furthermore, we define

$$\theta_j := \begin{cases} \mu_j, & \text{if } j \in \overline{\mathcal{I}_{\text{honor}} \cap \text{auth}} \\ 0, & \text{otherwise} \end{cases},$$

and $\text{lgen}(\phi_i) \rightarrow (\mathbf{L}_{i,0}, \mathbf{L}_{i,1}) \in \mathbb{Z}_p^{(m+n') \times m} \times \mathbb{Z}_p^{(m+n') \times m(\hat{n}+n)}$ with $\mathbf{L}_{i,\mathbf{y},\mathbf{x}} = \mathbf{L}_{i,0} + \mathbf{L}_{i,1}((\mathbf{y}^\top \parallel \mathbf{x}^\top)^\top \otimes \mathbf{I}_m)$.

$\widetilde{\text{Gen}}(\widehat{\text{crs}}, i; \text{td})$: On input the $\widehat{\text{crs}}$, an index $i \in [L]$ and the trapdoor td , sample $\mathbf{U}_{1,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)(n'+1)}$, $\mathbf{U}_{2,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)m}$ and run $\pi_{1,i} \leftarrow \text{LSim}(\text{crs}_{1,i}, \text{td}_{1,i}, \|\mathbf{F}_{1,i}\|_1)$, $\pi_{2,i} \leftarrow \text{LSim}(\text{crs}_{2,i}, \text{td}_{2,i}, \|\mathbf{F}_{2,i}\|_1)$ where $\mathbf{F}_{d,i}^\top = [\mathbf{T}_{d,i}^\top \parallel \mathbf{Q}_{d,i}^\top] = [(\mathbf{A}\mathbf{U}_{d,i})^\top \parallel (\mathbf{R}_{d,i} \mathbf{U}_{d,i})^\top] = (\mathbf{A}_{d,i} \mathbf{U}_{d,i})^\top$. Then output $\text{sk}_i = (\mathbf{U}_{1,i}, \mathbf{U}_{2,i})$ and pk_i as follows:

$$\left(\left[\mathbf{T}_{1,i} \right]_1 := \|\mathbf{A}\mathbf{U}_{1,i}\|_1, \left[\mathbf{Q}_{1,i} \right]_1 := \|\mathbf{R}_{1,i} \mathbf{U}_{1,i}\|_1, \{\|\mathbf{G}_{1,i,j}\|_2 := \|\mathbf{U}_{1,i}(\mathbf{I}_{n'+1} \otimes \mathbf{B}\mathbf{r}_j)\|_2\}_{j \in [L] \setminus \{i\}}, \pi_{1,i} \right) \\ \left(\left[\mathbf{T}_{2,i} \right]_1 := \|\mathbf{A}\mathbf{U}_{2,i}\|_1, \left[\mathbf{Q}_{2,i} \right]_1 := \|\mathbf{R}_{2,i} \mathbf{U}_{2,i}\|_1, \{\|\mathbf{G}_{2,i,j}\|_2 := \|\mathbf{U}_{2,i}(\mathbf{I}_m \otimes \mathbf{B}\mathbf{r}_j)\|_2\}_{j \in [L] \setminus \{i\}}, \pi_{2,i} \right).$$

$\widetilde{\text{Enc}}((\text{pk}_1, \dots, \text{pk}_L), \text{td})$: On input the list of public keys $(\text{pk}_1, \dots, \text{pk}_L)$, a public input $\mathbf{x} \in \mathbb{Z}_p^n$ and a trapdoor td , output

$$\text{ct} = \left[\begin{array}{c} \mathbf{c}^\top, \mathbf{c}^\top \hat{\mathbf{H}}_1, \sum_{i \in [L]} \mathbf{e}_1^\top \tilde{\mathbf{R}}_{1,i}^{-1} \mathbf{Q}_{1,i}^* + \mathbf{c}^\top (\mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}) - \underline{\mathbf{W}}_{1,i}), \mathbf{c}^\top \hat{\mathbf{H}}_2, \\ \sum_{i \in [L]} \mathbf{e}_1^\top \tilde{\mathbf{R}}_{2,i}^{-1} \mathbf{Q}_{2,i}^* + \mathbf{c}^\top (\mathbf{H}_{2,i}(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{1,i}(\mathbf{L}_{i,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \\ + \mathbf{W}_{2,i}(\mathbf{D}\mathbf{E}_i \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{3,i}((\mathbf{w}^\top \mathbf{D}\mathbf{E}_i + \theta_i \mathbf{e}_1^\top) \otimes \mathbf{I}_{2k+1})), \mathbf{c}^\top \mathbf{V} - \mathbf{w}^\top, \mathbf{c}^\top \mathbf{v} + \mathbf{1} \end{array} \right]_1,$$

Sanity Check of the Simulator. Let us consider a simulated ciphertext $ct \leftarrow \widetilde{\text{Enc}}((pk_1, \dots, pk_L), td)$ and some $i^* \in \mathcal{I}_{\text{mal}} \cup \mathcal{I}_{\text{cor}}$. We have $sk_{i^*} = (\mathbf{U}_{1,i^*}, \mathbf{U}_{2,i^*})$ and

$$\begin{aligned}
(\mathbf{c}_{1,1}, \mathbf{c}_{1,2}) &= \left(\mathbf{c}^\top \hat{\mathbf{H}}_1, \sum_{i \in [L]} \mathbf{e}_1^\top \tilde{\mathbf{R}}_{1,i}^{-1} \mathbf{Q}_{1,i}^* + \mathbf{c}^\top (\mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}) - \mathbf{W}_{1,i}) \right) \\
(\mathbf{c}_{2,1}, \mathbf{c}_0, \mathbf{c}_3, c_4) &= \left(\mathbf{c}^\top \hat{\mathbf{H}}_2, \mathbf{c}^\top, \mathbf{c}^\top \mathbf{V} - \mathbf{w}, \mathbf{c}^\top \mathbf{v} + 1 \right) \\
\mathbf{c}_{2,2} &= \sum_{i \in [L]} \mathbf{e}_1^\top \tilde{\mathbf{R}}_{2,i}^{-1} \mathbf{Q}_{2,i}^* + \mathbf{c}^\top \left(\mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{1,i} (\mathbf{L}_{i,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \right. \\
&\quad \left. + \mathbf{W}_{2,i} (\mathbf{DE}_i \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{3,i} ((\mathbf{w}^\top \mathbf{DE}_i + \theta_i \mathbf{e}_1^\top) \otimes \mathbf{I}_{2k+1}) \right) \\
(\mathbf{d}_0, \mathbf{D}_{0,1}, \mathbf{D}_{0,2}, \mathbf{D}_{0,3}, \mathbf{d}_{0,4}) &= \left(\mathbf{Br}_{i^*}, \mathbf{I}_{\mathbf{g}_{i^*}}, \mathbf{q}_{i^*} \mathbf{e}_1^\top, \mathbf{DE}_{i^*}, \mathbf{w}^\top \mathbf{DE}_{i^*} + \theta_{i^*} \mathbf{e}_1^\top \right) \\
\mathbf{D}_{1,1} &= \mathbf{H}_{1,i^*} (\mathbf{I}_{\mathbf{g}_{i^*}} \otimes \mathbf{Br}_{i^*}) - \mathbf{W}_{1,i^*} (\mathbf{I}_{n'+1} \otimes \mathbf{Br}_{i^*}) + \hat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_{i^*}} + \hat{\mathbf{W}}_{i^*} \\
\mathbf{D}_{1,2} &= \sum_{i \in [L] \setminus \{i^*\}} \mathbf{G}_{1,i,i^*} + \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{Br}_{i^*}) - \mathbf{W}_{1,i} (\mathbf{I}_{n'+1} \otimes \mathbf{Br}_{i^*}) \\
\mathbf{D}_{2,1,1} &= \mathbf{W}_{1,i^*} (\mathbf{I}_{m+n'} \otimes \mathbf{Br}_{i^*}) \mathbf{L}_{i^*,1} - \hat{\mathbf{W}}_{i^*} \mathbf{L}_{i^*,1} \\
\mathbf{D}_{2,1,2} &= \mathbf{H}_{2,i^*} (\mathbf{q}_{i^*} \mathbf{e}_1^\top \otimes \mathbf{Br}_{i^*}) + \hat{\mathbf{H}}_2 \mathbf{q}_{i^*} \mathbf{e}_1^\top + \mathbf{W}_{1,i^*} (\mathbf{I}_{m+n'} \otimes \mathbf{Br}_{i^*}) \mathbf{L}_{i^*,0} - \hat{\mathbf{W}}_{i^*} \mathbf{L}_{i^*,0} \\
&\quad + \mathbf{W}_{2,i^*} (\mathbf{DE}_{i^*} \otimes \mathbf{Br}_{i^*}) + \mathbf{W}_{3,i^*} ((\mathbf{w}^\top \mathbf{DE}_{i^*} + \theta_{i^*} \mathbf{e}_1^\top) \otimes \mathbf{Br}_{i^*}) \\
&\quad + \mathbf{VDE}_{i^*} + \mathbf{v} (\mathbf{w}^\top \mathbf{DE}_{i^*} + \theta_{i^*} \mathbf{e}_1^\top) + g_{i^*}^*(\mathbf{y}) \cdot \mathbf{g}_{i^*}^\top \mathbf{f} \cdot \mathbf{c}^\perp \mathbf{e}_1^\top \\
\mathbf{D}_{2,2,1} &= \sum_{i \in [L] \setminus \{i^*\}} \mathbf{W}_{1,i} (\mathbf{I}_{m+n'} \otimes \mathbf{Br}_{i^*}) \mathbf{L}_{i,1} \\
\mathbf{D}_{2,2,2} &= \sum_{i \in [L] \setminus \{i^*\}} \mathbf{G}_{2,i,i^*} + \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{Br}_{i^*}) + \mathbf{W}_{1,i} (\mathbf{I}_{m+n'} \otimes \mathbf{Br}_{i^*}) \mathbf{L}_{i,0} \\
&\quad + \mathbf{W}_{2,i} (\mathbf{DE}_i \otimes \mathbf{Br}_{i^*}) + \mathbf{W}_{3,i} ((\mathbf{w}^\top \mathbf{DE}_i + \theta_i \mathbf{e}_1^\top) \otimes \mathbf{Br}_{i^*}) ,
\end{aligned}$$

where we parse the ciphertext $ct = ([\mathbf{c}_0]_1, [\mathbf{c}_{1,1}]_1, [\mathbf{c}_{1,2}]_1, [\mathbf{c}_{2,1}]_1, [\mathbf{c}_{2,2}]_1, [\mathbf{c}_3]_1, [\mathbf{c}_4]_1)$ and the helper secret key $hsk_{i^*} = ([\mathbf{d}_0]_2, [\mathbf{D}_{1,1}]_2, [\mathbf{D}_{1,2}]_2, [\mathbf{D}_{2,1,1}]_2, [\mathbf{D}_{2,1,2}]_2, [\mathbf{D}_{2,2,1}]_2, [\mathbf{D}_{2,2,2}]_2, [\mathbf{D}_{0,1}]_2, [\mathbf{D}_{0,2}]_2, [\mathbf{D}_{0,3}]_2, [\mathbf{d}_{0,4}]_2)$. By computing

$$\begin{aligned}
[\mathbf{c}'_{1,1}]_t &= [\mathbf{c}_{1,1}]_1 [\mathbf{D}_{0,1}]_2 , & [\mathbf{d}'_{1,1}]_t &= [\mathbf{c}_0]_1 [\mathbf{D}_{1,1}]_2 , \\
[\mathbf{c}'_{1,2}]_t &= [\mathbf{c}_{1,2}]_1 [\mathbf{I}_{n'+1} \otimes \mathbf{d}_0]_2 , & [\mathbf{d}'_{1,2}]_t &= [\mathbf{c}_0]_1 [\mathbf{D}_{1,2}]_2 , \\
[\mathbf{c}'_{2,1}]_t &= [\mathbf{c}_{2,1}]_1 [\mathbf{D}_{0,2}]_2 , & [\mathbf{d}'_{2,1}]_t &= [\mathbf{c}_0]_1 [\mathbf{D}_{2,1,1} ((\mathbf{y}^\top \parallel \mathbf{x}^\top)^\top \otimes \mathbf{I}_m) + \mathbf{D}_{2,1,2}]_2 , \\
[\mathbf{c}'_{2,2}]_t &= [\mathbf{c}_{2,2}]_1 [\mathbf{I}_m \otimes \mathbf{d}_0]_2 , & [\mathbf{d}'_{2,2}]_t &= [\mathbf{c}_0]_1 [\mathbf{D}_{2,2,1} ((\mathbf{y}^\top \parallel \mathbf{x}^\top)^\top \otimes \mathbf{I}_m) + \mathbf{D}_{2,2,2}]_2 , \\
[\mathbf{c}'_3]_t &= [\mathbf{c}_3]_1 [\mathbf{D}_{0,3}]_2 , \\
[\mathbf{c}'_4]_t &= [\mathbf{c}_4]_1 [\mathbf{d}_{0,4}]_2 ,
\end{aligned}$$

the decryptor obtains

$$\begin{aligned}
\mathbf{c}'_{1,1} &= (\mathbf{c}^\top \hat{\mathbf{H}}_1) \mathbf{I}_{\mathbf{g}_{i^*}} = \mathbf{c}^\top \hat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_{i^*}} \\
\mathbf{d}'_{1,1} &= \mathbf{c}^\top \left(\mathbf{H}_{1,i^*} (\mathbf{I}_{\mathbf{g}_{i^*}} \otimes \mathbf{Br}_{i^*}) - \mathbf{W}_{1,i^*} (\mathbf{I}_{n'+1} \otimes \mathbf{Br}_{i^*}) + \hat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_{i^*}} + \hat{\mathbf{W}}_{i^*} \right) \\
\mathbf{c}'_{1,2} &= \sum_{i \in [L]} \mathbf{e}_1^\top \tilde{\mathbf{R}}_{1,i}^{-1} \mathbf{Q}_{1,i}^* (\mathbf{I}_{n'+1} \otimes \mathbf{Br}_{i^*}) + \mathbf{c}^\top \left(\mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{Br}_{i^*}) - \mathbf{W}_{1,i} (\mathbf{I}_{n'+1} \otimes \mathbf{Br}_{i^*}) \right) \\
\mathbf{d}'_{1,2} &= \mathbf{c}^\top \sum_{i \in [L] \setminus \{i^*\}} \mathbf{G}_{1,i,i^*} + \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{Br}_{i^*}) - \mathbf{W}_{1,i} (\mathbf{I}_{n'+1} \otimes \mathbf{Br}_{i^*}) \\
\mathbf{c}'_{2,1} &= (\mathbf{c}^\top \hat{\mathbf{H}}_2) \mathbf{q}_{i^*} \mathbf{e}_1^\top = \mathbf{c}^\top \hat{\mathbf{H}}_2 \mathbf{q}_{i^*} \mathbf{e}_1^\top \\
\mathbf{d}'_{2,1} &= \mathbf{c}^\top \left(\mathbf{H}_{2,i^*} (\mathbf{q}_{i^*} \mathbf{e}_1^\top \otimes \mathbf{Br}_{i^*}) + \hat{\mathbf{H}}_2 \mathbf{q}_{i^*} \mathbf{e}_1^\top + \mathbf{W}_{1,i^*} (\mathbf{I}_{m+n'} \otimes \mathbf{Br}_{i^*}) \mathbf{L}_{i^*,\mathbf{y},\mathbf{x}} - \hat{\mathbf{W}}_{i^*} \mathbf{L}_{i^*,\mathbf{y},\mathbf{x}} \right. \\
&\quad \left. + \mathbf{W}_{2,i^*} (\mathbf{DE}_{i^*} \otimes \mathbf{Br}_{i^*}) + \mathbf{W}_{3,i^*} ((\mathbf{w}^\top \mathbf{DE}_{i^*} + \theta_{i^*} \mathbf{e}_1^\top) \otimes \mathbf{Br}_{i^*}) \right. \\
&\quad \left. + \mathbf{VDE}_{i^*} + \mathbf{v} (\mathbf{w}^\top \mathbf{DE}_{i^*} + \theta_{i^*} \mathbf{e}_1^\top) + g_{i^*}^*(\mathbf{y}) \cdot \mathbf{g}_{i^*}^\top \mathbf{f} \cdot \mathbf{c}^\perp \mathbf{e}_1^\top \right)
\end{aligned}$$

$$\begin{aligned}
\mathbf{c}'_{2,2} &= \sum_{i \in [L]} \mathbf{e}_1^\top \tilde{\mathbf{R}}_{2,i}^{-1} \mathbf{Q}_{2,i}^* (\mathbf{I}_m \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{c}^\top \left(\mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{W}_{1,i} (\mathbf{L}_{i,\mathbf{y},\mathbf{x}} \otimes \mathbf{B} \mathbf{r}_{i^*}) \right. \\
&\quad \left. + \mathbf{W}_{2,i} (\mathbf{D} \mathbf{E}_i \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{W}_{3,i} ((\mathbf{w}^\top \mathbf{D} \mathbf{E}_i + \theta_i \mathbf{e}_1^\top) \otimes \mathbf{B} \mathbf{r}_{i^*}) \right) \\
\mathbf{d}'_{2,2} &= \mathbf{c}^\top \sum_{i \in [L] \setminus \{i^*\}} \mathbf{G}_{2,i,i^*} + \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{W}_{1,i} (\mathbf{I}_{m+n'} \otimes \mathbf{B} \mathbf{r}_{i^*}) \mathbf{L}_{i,\mathbf{y},\mathbf{x}} \\
&\quad + \mathbf{W}_{2,i} (\mathbf{D} \mathbf{E}_i \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{W}_{3,i} ((\mathbf{w}^\top \mathbf{D} \mathbf{E}_i + \theta_i \mathbf{e}_1^\top) \otimes \mathbf{B} \mathbf{r}_{i^*}) \\
\mathbf{c}'_3 &= \mathbf{c}^\top \mathbf{V} \cdot \mathbf{D} \mathbf{E}_{i^*} - \mathbf{w}^\top \mathbf{D} \mathbf{E}_{i^*} \\
\mathbf{c}'_4 &= (\mathbf{c}^\top \mathbf{v} + 1) \cdot (\mathbf{w}^\top \mathbf{D} \mathbf{E}_{i^*} + \theta_{i^*} \mathbf{e}_1^\top) = (\mathbf{c}^\top \mathbf{v}) \cdot (\mathbf{w}^\top \mathbf{D} \mathbf{E}_{i^*} + \theta_{i^*} \mathbf{e}_1^\top) + \mathbf{w}^\top \mathbf{D} \mathbf{E}_{i^*} + \theta_{i^*} \mathbf{e}_1^\top.
\end{aligned}$$

Therefore, the vector $(\llbracket \mathbf{p}_1^\top \rrbracket_t, \llbracket \mathbf{p}_2^\top \rrbracket_t)$ equals

$$\begin{aligned}
\llbracket \mathbf{p}_1^\top \rrbracket_t &= \llbracket (\mathbf{c}'_{1,1} - \mathbf{d}'_{1,1}) + (\mathbf{c}'_{1,2} - \mathbf{d}'_{1,2}) \rrbracket_t - \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{U}_{1,i^*} \cdot (\mathbf{I}_{n'+1} \otimes \mathbf{d}_0) \rrbracket_2 \\
&= \llbracket -\mathbf{c}^\top \widehat{\mathbf{W}}_{i^*} \rrbracket_t, \\
\llbracket \mathbf{p}_2^\top \rrbracket_t &= \llbracket (\mathbf{c}'_{2,1} - \mathbf{d}'_{2,1}) + (\mathbf{c}'_{2,2} - \mathbf{d}'_{2,2}) + \mathbf{c}'_3 + \mathbf{c}'_4 \rrbracket_t - \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{U}_{2,i^*} \cdot (\mathbf{I}_m \otimes \mathbf{d}_0) \rrbracket_2 \\
&= \llbracket \mathbf{c}^\top \widehat{\mathbf{W}}_{i^*} \mathbf{L}_{i^*,\mathbf{y},\mathbf{x}} + (\theta_{i^*} - g_{i^*}^*(\mathbf{y}) \cdot \mathbf{g}_{i^*}^\top \mathbf{f}) \mathbf{e}_1^\top \rrbracket_t.
\end{aligned}$$

We observe that $(\mathbf{p}_1^\top, \mathbf{p}_2^\top) = \text{pgb}^*(\phi_{i^*}, (\mathbf{y}, \mathbf{x}), (\theta_{i^*} - g_{i^*}^*(\mathbf{y}) \cdot \mathbf{g}_{i^*}^\top \mathbf{f} \cdot \mathbf{c}^\perp); \mathbf{c}^\top \widehat{\mathbf{W}}_{i^*})$. Then it follows from the correctness of the garbling scheme that

$$\llbracket \nu \rrbracket_t = \llbracket (\mathbf{p}_1^\top, \mathbf{p}_2^\top) \cdot \mathbf{d}_{\phi_{i^*}, \mathbf{y}, \mathbf{x}} \rrbracket_t = \llbracket \theta_{i^*} - g_{i^*}^*(\mathbf{y}) \cdot \mathbf{g}_{i^*}^\top \mathbf{f} \rrbracket_t.$$

where $\mathbf{d}_{\phi_{i^*}, \mathbf{y}, \mathbf{x}} \leftarrow \text{rec}(\phi_{i^*}, (\mathbf{y}, \mathbf{x}))$. If $i^* \in \mathcal{I}_{\text{hon} \cap \text{auth}}$ then $g_{i^*}^*(\mathbf{y}) = 0, \theta_{i^*} = \mu_{i^*}$ and hence we recover $\llbracket \nu \rrbracket_t = \llbracket \mu_{i^*} \rrbracket_t$.

6.3 Proof of Security

Proof of Proposition 37. Let FE denote the sRFE scheme in Construction 35. We consider a sequence of hybrid games $G_0, \dots, G_6$ and $G_{6,\ell}$ for $\ell \in [L]$ where $G_0 = \text{Exp}_{\text{FE}, \mathcal{A}}^{\text{sRFE-real}}$ and $G_{6,L} = \text{Exp}_{\text{FE}, \mathcal{A}}^{\text{sRFE-ideal}}$. We argue $G_6 \equiv G_{6,0}$ and $G_{6,\ell-1} \approx_c G_{6,\ell}$ for $\ell \in [L]$. Then the proposition follows via a hybrid argument. Modifications between consecutive games are highlighted using **boxes**.

Game G_0 : This is $\text{Exp}_{\text{FE}, \mathcal{A}}^{\text{sRFE-real}}$. Specifically, we have:

- The common reference string crs based on $\{\llbracket \mathbf{q}_i \rrbracket_{1,2}\}_{i \in [L]}$ is of the form

$$\begin{aligned}
\text{crs} &= \left([\mathbf{A}, \mathbf{A} \widehat{\mathbf{H}}_1, \mathbf{A} \widehat{\mathbf{H}}_2, \mathbf{A} \mathbf{V}, \mathbf{A} \mathbf{v}]_1, \right. \\
&\quad \left\{ \left[\begin{array}{l} \mathbf{A} \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}), \mathbf{A} \mathbf{W}_{1,i}, \mathbf{A} \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}), \mathbf{R}_{1,i}, \mathbf{R}_{2,i}, \\ \mathbf{A} \mathbf{W}_{2,i} (\mathbf{D} \mathbf{E}_i \otimes \mathbf{I}_{2k+1}) + \mathbf{A} \mathbf{W}_{3,i} (\mathbf{w}^\top \mathbf{D} \mathbf{E}_i \otimes \mathbf{I}_{2k+1}), \text{crs}_{1,i}, \text{crs}_{2,i} \end{array} \right]_1 \right\}_{i \in [L]}, \\
&\quad \left\{ \left[\begin{array}{l} \mathbf{B} \mathbf{r}_j, \mathbf{q}_j, \mathbf{g}_j, \mathbf{D} \mathbf{E}_j, \mathbf{w}^\top \mathbf{D} \mathbf{E}_j, \mathbf{W}_{1,j} (\mathbf{I}_{m+n'} \otimes \mathbf{B} \mathbf{r}_j) - \widehat{\mathbf{W}}_j, \\ \mathbf{H}_{1,j} (\mathbf{I}_{\mathbf{g}_j} \otimes \mathbf{B} \mathbf{r}_j) + \widehat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_j}, \mathbf{H}_{2,j} (\mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_j) + \widehat{\mathbf{H}}_2 \mathbf{q}_j \mathbf{e}_1^\top \\ + \mathbf{W}_{2,j} (\mathbf{D} \mathbf{E}_j \otimes \mathbf{B} \mathbf{r}_j) + \mathbf{W}_{3,j} (\mathbf{w}^\top \mathbf{D} \mathbf{E}_j \otimes \mathbf{B} \mathbf{r}_j) + \mathbf{V} \mathbf{D} \mathbf{E}_j + \mathbf{v} \mathbf{w}^\top \mathbf{D} \mathbf{E}_j \end{array} \right]_2 \right\}_{j \in [L]}, \\
&\quad \left. \left\{ \left[\begin{array}{l} \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{B} \mathbf{r}_j), \mathbf{W}_{1,i} (\mathbf{I}_{m+n'} \otimes \mathbf{B} \mathbf{r}_j), \\ \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_j), \mathbf{W}_{2,i} (\mathbf{D} \mathbf{E}_i \otimes \mathbf{B} \mathbf{r}_j) + \mathbf{W}_{3,i} (\mathbf{w}^\top \mathbf{D} \mathbf{E}_i \otimes \mathbf{B} \mathbf{r}_j) \end{array} \right]_2 \right\}_{\substack{i \in [L], \\ j \in [L] \setminus \{i\}}} \right),
\end{aligned}$$

where $\mathbf{g}_i \xleftarrow{\$} \mathbb{Z}_p^k$ for $i \in [L]$, $(\text{crs}_{d,i}, \text{td}_{d,i}) \leftarrow \text{LGen}(1^\lambda, \mathbb{G}_1, \llbracket \mathbf{A}_{d,i} \rrbracket_1)$ and $\mathbf{A}_{d,i}^\top = [\mathbf{A}^\top \parallel \mathbf{R}_{d,i}^\top]$ for $d \in \{1, 2\}$.

- For each $i \in [L]$, each $\text{pk}_i \in \mathcal{D}_i$ is of the form

$$\text{pk}_i = \left(\begin{array}{l} \llbracket \mathbf{T}_{d,i} \rrbracket_1 := \llbracket \mathbf{A} \mathbf{U}_{d,i} \rrbracket_1, \llbracket \mathbf{Q}_{d,i} \rrbracket_1 := \llbracket \mathbf{R}_{d,i} \mathbf{U}_{d,i} \rrbracket_1, \\ \llbracket \mathbf{G}_{d,i,j} \rrbracket_2 := \{\llbracket \mathbf{U}_i (\mathbf{I}_{t_d} \otimes \mathbf{B} \mathbf{r}_j) \rrbracket_2\}_{j \in [L] \setminus \{i\}}, \pi_{d,i} \end{array} \right)_{d \in \{1,2\}},$$

where $t_1 = n' + 1, t_2 = m, \pi_{d,i} \leftarrow \text{LPrv}(\text{crs}_i, \llbracket \mathbf{F}_{d,i} \rrbracket_1, \mathbf{U}_{d,i}), \mathbf{F}_{d,i}^\top = [\mathbf{T}_{d,i}^\top \parallel \mathbf{Q}_{d,i}^\top]$ and $\text{sk}_i = (\mathbf{U}_{1,i}, \mathbf{U}_{2,i})$ is the corresponding secret key.

- For each $i \in [L]$, pk_i^* is of the form

$$\text{pk}_i^* = (\llbracket \mathbf{T}_{d,i}^* \rrbracket_1, \llbracket \mathbf{Q}_{d,i}^* \rrbracket_1, \{\llbracket \mathbf{G}_{d,i,j}^* \rrbracket_2\}_{j \in [L] \setminus \{i\}}, \pi_{d,i}^*)_{d \in \{1,2\}}$$

such that $\text{Ver}(\text{crs}, i, \text{pk}_i^*) = 1$, i.e. $\text{LVer}(\text{crs}_i, \llbracket \mathbf{F}_{d,i}^* \rrbracket_1, \pi_{d,i}^*) = 1$ for $\mathbf{F}_{d,i}^* = [(\mathbf{T}_{d,i}^*)^\top \parallel (\mathbf{Q}_{d,i}^*)^\top]^\top$ and $\mathbf{A} \mathbf{G}_{d,i,j}^* = \mathbf{T}_{d,i}^* \mathbf{B} \mathbf{r}_j$ for each $j \in [L] \setminus \{i\}$ and $d \in \{1,2\}$.

- The challenge ciphertext $\text{ct}^* = (\llbracket \mathbf{c}_0^*, \mathbf{c}_{1,1}^*, \mathbf{c}_{1,2}^*, \mathbf{c}_{2,1}^*, \mathbf{c}_{2,2}^*, \mathbf{c}_3^*, \mathbf{c}_4^* \rrbracket_1)$ for $((\mathbf{y}, \mathbf{x}), (\mathbf{z}^*, \mathbf{p}^*))$ is of the form

$$\begin{aligned} \mathbf{c}_0^* &= \mathbf{s}^\top \mathbf{A} \\ \mathbf{c}_{1,1}^* &= (\mathbf{f}^\top \parallel (\mathbf{z}^*)^\top) + \mathbf{s}^\top \mathbf{A} \hat{\mathbf{H}}_1 \\ \mathbf{c}_{1,2}^* &= \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{1,i} + \mathbf{A} \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}) - \mathbf{A} \mathbf{W}_{1,i} \\ \mathbf{c}_{2,1}^* &= (\mathbf{p}^*)^\top + \mathbf{s}^\top \mathbf{A} \hat{\mathbf{H}}_2 \\ \mathbf{c}_{2,2}^* &= \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{2,i} + \mathbf{A} \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{A} \mathbf{W}_{1,i} (\mathbf{L}_{i,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \\ &\quad + \mathbf{A} \mathbf{W}_{2,i} (\mathbf{D} \mathbf{E}_i \otimes \mathbf{I}_{2k+1}) + \mathbf{A} \mathbf{W}_{3,i} (\mathbf{w}^\top \mathbf{D} \mathbf{E}_i \otimes \mathbf{I}_{2k+1}) \\ \mathbf{c}_3^* &= \mathbf{s}^\top \mathbf{A} \mathbf{V} \\ \mathbf{c}_4^* &= \mathbf{s}^\top \mathbf{A} \mathbf{v} \end{aligned}$$

where $\mathbf{L}_{i,\mathbf{y},\mathbf{x}}$'s are obtained from the challenge functions f_i^* 's through the garbling scheme.

Game G_1 : This is the same as G_0 , except that the challenger replaces $\mathbf{w}^\top \mathbf{D} \mathbf{E}_i$ in crs with $\mathbf{w}^\top \mathbf{D} \mathbf{E}_i + \theta_i \mathbf{e}_1^\top$ where

$$\theta_i = \begin{cases} \mu_i & \text{if } [i \in (\mathcal{I}_{\text{cor}} \cup \mathcal{I}_{\text{mal}})] \wedge [g_i^*(\mathbf{y}) = 0] \\ 0 & \text{otherwise.} \end{cases}$$

Note that the challenger receives $\llbracket \mu_i \rrbracket_{1,2}$ for $i \in \overline{\mathcal{I}_{\text{non}} \cap \mathcal{I}_{\text{auth}}}$ (and implicitly we have $\mu_i = h_i^*(\mathbf{x})^\top \mathbf{z}^* + \mathbf{q}_i^\top \mathbf{p}^*$). In particular, the crs is generated as

$$\begin{aligned} \text{crs} = & \left(\llbracket \mathbf{A}, \mathbf{A} \hat{\mathbf{H}}_1, \mathbf{A} \hat{\mathbf{H}}_2, \mathbf{A} \mathbf{V}, \mathbf{A} \mathbf{v} \rrbracket_1, \right. \\ & \left. \left\{ \left\llbracket \begin{array}{l} \mathbf{A} \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}), \mathbf{A} \mathbf{W}_{1,i}, \mathbf{A} \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}), \mathbf{R}_{1,i}, \mathbf{R}_{2,i}, \\ \mathbf{A} \mathbf{W}_{2,i} (\mathbf{D} \mathbf{E}_i \otimes \mathbf{I}_{2k+1}) + \mathbf{A} \mathbf{W}_{3,i} ((\mathbf{w}^\top \mathbf{D} \mathbf{E}_i + \theta_i \mathbf{e}_1^\top) \otimes \mathbf{I}_{2k+1}), \text{crs}_{1,i}, \text{crs}_{2,i} \end{array} \right\|_1 \right\}_{i \in [L]}, \right. \\ & \left. \left\{ \left\llbracket \begin{array}{l} \mathbf{B} \mathbf{r}_j, \mathbf{q}_j, \mathbf{g}_j, \mathbf{D} \mathbf{E}_j, \mathbf{w}^\top \mathbf{D} \mathbf{E}_j + \theta_j \mathbf{e}_1^\top, \mathbf{W}_{1,j} (\mathbf{I}_{m+n'} \otimes \mathbf{B} \mathbf{r}_j) - \hat{\mathbf{W}}_j, \\ \mathbf{H}_{1,j} (\mathbf{I}_{\mathbf{g}_j} \otimes \mathbf{B} \mathbf{r}_j) + \hat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_j}, \mathbf{H}_{2,j} (\mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_j) + \hat{\mathbf{H}}_2 \mathbf{q}_j \mathbf{e}_1^\top + \mathbf{W}_{2,j} (\mathbf{D} \mathbf{E}_j \otimes \mathbf{B} \mathbf{r}_j) \\ + \mathbf{W}_{3,j} ((\mathbf{w}^\top \mathbf{D} \mathbf{E}_j + \theta_j \mathbf{e}_1^\top) \otimes \mathbf{B} \mathbf{r}_j) + \mathbf{V} \mathbf{D} \mathbf{E}_j + \mathbf{v} (\mathbf{w}^\top \mathbf{D} \mathbf{E}_j + \theta_j \mathbf{e}_1^\top) \end{array} \right\|_2 \right\}_{j \in [L]}, \right. \\ & \left. \left\{ \left\llbracket \begin{array}{l} \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{B} \mathbf{r}_j), \mathbf{W}_{1,i} (\mathbf{I}_{m+n'} \otimes \mathbf{B} \mathbf{r}_j), \\ \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_j), \mathbf{W}_{2,i} (\mathbf{D} \mathbf{E}_i \otimes \mathbf{B} \mathbf{r}_j) + \mathbf{W}_{3,i} ((\mathbf{w}^\top \mathbf{D} \mathbf{E}_i + \theta_i \mathbf{e}_1^\top) \otimes \mathbf{B} \mathbf{r}_j) \end{array} \right\|_2 \right\}_{\substack{i \in [L], \\ j \in [L] \setminus \{i\}}} \right) \end{aligned}$$

and the challenge ciphertext component $\mathbf{c}_{2,2}^*$ is computed as

$$\begin{aligned} \mathbf{c}_{2,2}^* = & \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{2,i} + \mathbf{A} \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{A} \mathbf{W}_{1,i} (\mathbf{L}_{i,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) + \mathbf{A} \mathbf{W}_{2,i} (\mathbf{D} \mathbf{E}_i \otimes \mathbf{I}_{2k+1}) \\ & + \mathbf{A} \mathbf{W}_{3,i} ((\mathbf{w}^\top \mathbf{D} \mathbf{E}_i + \theta_i \mathbf{e}_1^\top) \otimes \mathbf{I}_{2k+1}) . \end{aligned}$$

We have $G_0 \approx_c G_1$ under the bi-MDDH $_k$ assumption with respect to $\llbracket \mathbf{D} \rrbracket_{1,2}$. More precisely, it follows from the fact that

$$(\llbracket \mathbf{D} \rrbracket_{1,2}, \llbracket \mathbf{w}^\top \mathbf{D} \mathbf{E}_i \rrbracket_{1,2}) \approx_c (\llbracket \mathbf{D} \rrbracket_{1,2}, \llbracket \mathbf{w}^\top \mathbf{D} \mathbf{E}_i + \theta_i \mathbf{e}_1^\top \rrbracket_{1,2}) ,$$

where $i \in [L]$, $\mathbf{D} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times k}$, $\mathbf{w} \xleftarrow{\$} \mathbb{Z}_p^{2k+1}$, $\mathbf{E}_i \xleftarrow{\$} \mathbb{Z}_p^{k \times m}$.

Game G_2 : This is the same as G_1 , except that for $d \in \{1, 2\}$, $i \in [L]$ and $\text{pk}_i \in \mathcal{D}_i$, the challenger computes

$$\pi_{d,i} \leftarrow \text{LSim}(\text{crs}_{d,i}, \text{td}_{d,i}, \llbracket \mathbf{F}_{d,i} \rrbracket_1) .$$

We have $G_1 \equiv G_2$ which follows from the perfect zero-knowledge of Π .

Game G_3 : This is the same as G_2 except that the challenger samples $\mathbf{s} \xleftarrow{\$} \mathbb{Z}_p^k$ and $\tilde{\mathbf{R}}_{d,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+2) \times (2k+2)}$ for $d \in \{1, 2\}$ and $i \in [L]$ along with $\mathbf{A}$, and sets

$$\mathbf{R}_{d,i} = \tilde{\mathbf{R}}_{d,i} \cdot \begin{bmatrix} \mathbf{s}^\top \mathbf{A} \\ \mathbf{I}_{2k+1} \end{bmatrix} .$$

We observe that the distribution of $\mathbf{R}_{d,i}$ does not change from G_2 to G_3 . Hence, we have $G_2 \equiv G_3$.

Game G_4 : This is the same as G_3 , except that the challenger generates $(\mathbf{c}_{1,2}^*, \mathbf{c}_{2,2}^*)$ as follows:

$$\begin{aligned} \mathbf{c}_{1,2}^* = & \sum_{i \in [L]} (\mathbf{e}_1^{(2k+2)})^\top \tilde{\mathbf{R}}_{1,i}^{-1} \mathbf{Q}_{1,i}^* + \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A} \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}) - \mathbf{A} \mathbf{W}_{1,i} \\ \mathbf{c}_{2,2}^* = & \sum_{i \in [L]} (\mathbf{e}_1^{(2k+2)})^\top \tilde{\mathbf{R}}_{2,i}^{-1} \mathbf{Q}_{2,i}^* + \mathbf{s}^\top \sum_{i \in [L]} \mathbf{A} \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{A} \mathbf{W}_{1,i} (\mathbf{L}_{i,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \\ & + \mathbf{A} \mathbf{W}_{2,i} (\mathbf{D} \mathbf{E}_i \otimes \mathbf{I}_{2k+1}) + \mathbf{A} \mathbf{W}_{3,i} ((\mathbf{w}^\top \mathbf{D} \mathbf{E}_i + \theta_i \mathbf{e}_1^\top) \otimes \mathbf{I}_{2k+1}) . \end{aligned}$$

We have $G_3 \approx_c G_4$ which can be seen as follows. Let $d \in \{1, 2\}$. First, for $i \in [L]$ such that $\text{pk}_i^* \in \mathcal{D}_i^*$, it is not hard to observe that the distributions of $\mathbf{s}^\top \mathbf{T}_{d,i}^*$ and $\mathbf{e}_1^\top \tilde{\mathbf{R}}_{d,i}^{-1} \mathbf{Q}_{d,i}^*$ are identical. Second, for $i \in [L]$ such that $\text{pk}_i^* \notin \mathcal{D}_i^*$, we rely on the unbounded simulation soundness of Π . In the reduction, the challenger of Π provides $\mathbf{A}_{d,i}$ and $\text{crs}_{d,i}$. The simulation proceeds as in G_3^b except that, for all $\text{pk}_i \in \mathcal{D}_i$, the simulator submits an oracle query $\llbracket \mathbf{F}_{d,i} \rrbracket_1$ to obtain $\pi_{d,i}$. Finally, upon $\mathcal{A}$ registering pk_i^* , the simulator outputs $(\llbracket \mathbf{F}_{d,i}^* \rrbracket_1, \pi_{d,i}^*)$. To conclude, we observe that if $\mathbf{e}_1^\top \tilde{\mathbf{R}}_{d,i}^{-1} \mathbf{Q}_{d,i}^* \neq \mathbf{s}^\top \mathbf{T}_{d,i}^*$, then $\mathbf{F}_{d,i}^* \notin \text{span}(\mathbf{A}_{d,i})$.

Game G_5 : This is the same as G_4 , except that the challenger replaces $\mathbf{s}^\top \mathbf{A}$ with a uniformly random vector $\mathbf{c}^\top \xleftarrow{\$} \mathbb{Z}_p^{1 \times (2k+1)}$. Specifically, $\mathbf{R}_{d,i}$ for $d \in \{1, 2\}$ and $\text{ct}^* = (\llbracket \mathbf{c}_0^*, \mathbf{c}_{1,1}^*, \mathbf{c}_{1,2}^*, \mathbf{c}_{2,1}^*, \mathbf{c}_{2,2}^*, \mathbf{c}_3^*, \mathbf{c}_4^* \rrbracket_1)$ are computed

as follows:

$$\begin{aligned}
\mathbf{R}_{d,i} &= \tilde{\mathbf{R}}_{d,i} \cdot \begin{bmatrix} \mathbf{c}^\top \\ \mathbf{I}_{2k+1} \end{bmatrix} \\
\mathbf{c}_0^* &= \mathbf{c}^\top \\
\mathbf{c}_{1,1}^* &= (\mathbf{f}^\top \parallel (\mathbf{z}^*)^\top) + \mathbf{c}^\top \hat{\mathbf{H}}_1 \\
\mathbf{c}_{1,2}^* &= \sum_{i \in [L]} \mathbf{e}_1^\top \tilde{\mathbf{R}}_{1,i}^{-1} \mathbf{Q}_{1,i}^* + \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_{1,i} (\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}) - \underline{\mathbf{W}}_{1,i} \\
\mathbf{c}_{2,1}^* &= (\mathbf{p}^*)^\top + \mathbf{c}^\top \hat{\mathbf{H}}_2 \\
\mathbf{c}_{2,2}^* &= \sum_{i \in [L]} \mathbf{e}_1^\top \tilde{\mathbf{R}}_{2,i}^{-1} \mathbf{Q}_{2,i}^* + \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{1,i} (\mathbf{L}_{i,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \\
&\quad + \mathbf{W}_{2,i} (\mathbf{D}\mathbf{E}_i \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{3,i} ((\mathbf{w}^\top \mathbf{D}\mathbf{E}_i + \theta_i \mathbf{e}_1^\top) \otimes \mathbf{I}_{2k+1}) \\
\mathbf{c}_3^* &= \mathbf{c}^\top \mathbf{V} \\
\mathbf{c}_4^* &= \mathbf{c}^\top \mathbf{v} .
\end{aligned}$$

It is not hard to see that we have $G_4 \approx_c G_5$ under the bi-MDDH $_k$ assumption with respect to $\llbracket \mathbf{A} \rrbracket_{1,2}$. More precisely, it follows from the fact that $(\llbracket \mathbf{A} \rrbracket_{1,2}, \llbracket \mathbf{s}^\top \mathbf{A} \rrbracket_{1,2}) \approx_c (\llbracket \mathbf{A} \rrbracket_{1,2}, \llbracket \mathbf{c}^\top \rrbracket_{1,2})$, where $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_p^{k \times (2k+1)}$, $\mathbf{s} \xleftarrow{\$} \mathbb{Z}_p^k$ and $\mathbf{c} \xleftarrow{\$} \mathbb{Z}_p^{2k+1}$.

For $j \in [L]$, we assign identifiers to the following terms of crs:

$$\begin{aligned}
\text{crs}_{1,j}^* &= \mathbf{H}_{1,j} (\mathbf{I}_{\mathbf{g}_j} \otimes \mathbf{Br}_j) + \hat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_j} \\
\text{crs}_{2,j}^* &= \mathbf{H}_{2,j} (\mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{Br}_j) + \hat{\mathbf{H}}_2 \mathbf{q}_j \mathbf{e}_1^\top + \mathbf{W}_{2,j} (\mathbf{D}\mathbf{E}_j \otimes \mathbf{Br}_j) + \mathbf{W}_{3,j} ((\mathbf{w}^\top \mathbf{D}\mathbf{E}_j + \theta_j \mathbf{e}_1^\top) \otimes \mathbf{Br}_j) \\
&\quad + \mathbf{V}\mathbf{D}\mathbf{E}_j + \mathbf{v}(\mathbf{w}^\top \mathbf{D}\mathbf{E}_j + \theta_j \mathbf{e}_1^\top) \\
\text{crs}_{3,j}^* &= \mathbf{W}_{1,j} (\mathbf{I}_{m+n'} \otimes \mathbf{Br}_j) - \hat{\mathbf{W}}_j .
\end{aligned}$$

Game G_6 : Let $\mathbf{c}^\perp \in \mathbb{Z}_p^{2k+1}$ such that $\mathbf{c}^\top \mathbf{c}^\perp = 1$ and $\mathbf{A}\mathbf{c}^\perp = \mathbf{0}$. This is the same as G_5 except for the following modifications in crs for $j \in [L]$:

$$\begin{aligned}
\text{crs}_{1,j}^* &= \mathbf{H}_{1,j} (\mathbf{I}_{\mathbf{g}_j} \otimes \mathbf{Br}_j) - \mathbf{c}^\perp (\mathbf{f}^\top \mathbf{g}_j \parallel (\mathbf{z}^*)^\top) + \hat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_j} \\
\text{crs}_{2,j}^* &= \mathbf{H}_{2,j} (\mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{Br}_j) + \hat{\mathbf{H}}_2 \mathbf{q}_j \mathbf{e}_1^\top - \mathbf{c}^\perp (\mathbf{p}^*)^\top \mathbf{q}_j \mathbf{e}_1^\top + \mathbf{W}_{2,j} (\mathbf{D}\mathbf{E}_j \otimes \mathbf{Br}_j) \\
&\quad + \mathbf{W}_{3,j} ((\mathbf{w}^\top \mathbf{D}\mathbf{E}_j + \theta_j \mathbf{e}_1^\top) \otimes \mathbf{Br}_j) + \mathbf{V}\mathbf{D}\mathbf{E}_j + \mathbf{v}(\mathbf{w}^\top \mathbf{D}\mathbf{E}_j + \theta_j \mathbf{e}_1^\top) + \theta_j \mathbf{c}^\perp \mathbf{e}_1^\top ,
\end{aligned}$$

and the corresponding changes in the ciphertext:

$$\mathbf{c}_{1,1}^* = (\mathbf{f}^\top \parallel (\mathbf{z}^*)^\top) + \mathbf{c}^\top \hat{\mathbf{H}}_1 , \quad \mathbf{c}_{2,1}^* = (\mathbf{p}^*)^\top + \mathbf{c}^\top \hat{\mathbf{H}}_2 , \quad \mathbf{c}_3^* = \mathbf{c}^\top \mathbf{V} - \mathbf{w}^\top , \quad \mathbf{c}_4^* = \mathbf{c}^\top \mathbf{v} + \mathbf{1} .$$

We have $G_5 \approx_s G_6$ which follows from the following changes of variables:

$$\begin{aligned}
\hat{\mathbf{H}}_1 &\mapsto \hat{\mathbf{H}}_1 - \mathbf{c}^\perp (\mathbf{f}^\top \parallel (\mathbf{z}^*)^\top) & \mathbf{V} &\mapsto \mathbf{V} - \mathbf{c}^\perp \mathbf{w}^\top \\
\hat{\mathbf{H}}_2 &\mapsto \hat{\mathbf{H}}_2 - \mathbf{c}^\perp (\mathbf{p}^*)^\top & \mathbf{v} &\mapsto \mathbf{v} + \mathbf{c}^\perp .
\end{aligned}$$

Game $G_{6,\ell}$ for $\ell \in [0; L]$: This is the same as G_6 except for the following modifications in $(\text{crs}_{1,j}^*, \text{crs}_{2,j}^*)$ for all $j \in [\ell]$:

$$\begin{aligned}
\text{crs}_{1,j}^* &= \mathbf{H}_{1,j} (\mathbf{I}_{\mathbf{g}_j} \otimes \mathbf{Br}_j) - \mathbf{c}^\perp (\mathbf{f}^\top \mathbf{g}_j \parallel (\mathbf{z}^*)^\top) + \hat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_j} \\
\text{crs}_{2,j}^* &= \mathbf{H}_{2,j} (\mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{Br}_j) + \hat{\mathbf{H}}_2 \mathbf{q}_j \mathbf{e}_1^\top - \mathbf{c}^\perp (\mathbf{p}^*)^\top \mathbf{q}_j \mathbf{e}_1^\top + \mathbf{W}_{2,j} (\mathbf{D}\mathbf{E}_j \otimes \mathbf{Br}_j) \\
&\quad + \mathbf{W}_{3,j} ((\mathbf{w}^\top \mathbf{D}\mathbf{E}_j + \theta_j \mathbf{e}_1^\top) \otimes \mathbf{Br}_j) + \mathbf{V}\mathbf{D}\mathbf{E}_j + \mathbf{v}(\mathbf{w}^\top \mathbf{D}\mathbf{E}_j + \theta_j \mathbf{e}_1^\top) + \theta_j \mathbf{c}^\perp \mathbf{e}_1^\top \\
&\quad + \mathbf{g}_\ell^*(\mathbf{y}) \cdot \mathbf{g}_\ell^\top \mathbf{f} \cdot \mathbf{c}^\perp \mathbf{e}_1^\top .
\end{aligned}$$

It is not hard to see that $G_{6,0} \equiv G_6$ as $[0] = \emptyset$, so the two games are identical. Furthermore, we prove the following claim for each $\ell \in [L]$.

Claim 39. *We have $G_{6,\ell-1} \approx_c G_{6,\ell}$ assuming bi-MDDH_k .*

Finally, we observe that $G_{6,L}$ is the same as $\text{Exp}_{\text{FE},\mathcal{A}}^{\text{srfc-ideal}}$. This concludes the proof. $\square$

We now turn to the claim.

Proof of Claim 39. We prove this claim using a sequence of sub-hybrids $\widehat{G}_0, \dots, \widehat{G}_3$ with $\widehat{G}_0 \equiv G_{6,\ell-1}$ and $\widehat{G}_3 \equiv G_{6,\ell}$.

Game $\widehat{G}_0$: This is the same as $G_{6,\ell-1}$ except for the following syntactical change. Initially, the challenger samples $\mathbf{d}_j \xleftarrow{\$} \text{span}(\mathbf{B})$ for all $j \in [L]$ and computes crs and $\text{pk}_i \in \mathcal{D}_i$ as follows:

$$\begin{aligned} \text{crs} = & \left([\mathbf{A}, \mathbf{A}\widehat{\mathbf{H}}_1, \mathbf{A}\widehat{\mathbf{H}}_2, \mathbf{A}\mathbf{V}, \mathbf{A}\mathbf{v}]_1, \right. \\ & \left. \left\{ \left[\begin{array}{l} \mathbf{A}\mathbf{H}_{1,i}(\mathbf{I}_{g_i} \otimes \mathbf{I}_{2k+1}), \mathbf{A}\mathbf{W}_{1,i}, \mathbf{A}\mathbf{H}_{2,i}(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}), \mathbf{R}_{1,i}, \mathbf{R}_{2,i}, \\ \mathbf{A}\mathbf{W}_{2,i}(\mathbf{D}\mathbf{E}_i \otimes \mathbf{I}_{2k+1}) + \mathbf{A}\mathbf{W}_{3,i}((\mathbf{w}^\top \mathbf{D}\mathbf{E}_i + \theta_i \mathbf{e}_1^\top) \otimes \mathbf{I}_{2k+1}), \text{crs}_{1,i}, \text{crs}_{2,i} \end{array} \right]_1 \right\}_{i \in [L]}, \right. \\ & \left. \left\{ \left[\begin{array}{l} \mathbf{d}_j, \mathbf{q}_j, \mathbf{g}_j, \mathbf{D}\mathbf{E}_j, \mathbf{w}^\top \mathbf{D}\mathbf{E}_j + \theta_j \mathbf{e}_1^\top, \\ \text{crs}_{1,j}^* = \begin{cases} \mathbf{H}_{1,j}(\mathbf{I}_{g_j} \otimes \mathbf{d}_j) + \widehat{\mathbf{H}}_1 \mathbf{I}_{g_j} & \text{if } j < \ell \\ \mathbf{H}_{1,j}(\mathbf{I}_{g_j} \otimes \mathbf{d}_j) + \widehat{\mathbf{H}}_1 \mathbf{I}_{g_j} - \mathbf{c}^\perp (\mathbf{f}^\top \mathbf{g}_j \parallel (\mathbf{z}^*)^\top) & \text{if } j \geq \ell \end{cases} \\ \text{crs}_{2,j}^* = \begin{cases} \mathbf{H}_{2,j}(\mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{d}_j) + \widehat{\mathbf{H}}_2 \mathbf{q}_j \mathbf{e}_1^\top + \mathbf{W}_{2,j}(\mathbf{D}\mathbf{E}_j \otimes \mathbf{d}_j) \\ \quad + \mathbf{W}_{3,j}((\mathbf{w}^\top \mathbf{D}\mathbf{E}_j + \theta_j \mathbf{e}_1^\top) \otimes \mathbf{d}_j) + \mathbf{V}\mathbf{D}\mathbf{E}_j & \text{if } j < \ell \\ \quad + \mathbf{v}(\mathbf{w}^\top \mathbf{D}\mathbf{E}_j + \theta_j \mathbf{e}_1^\top) + g_\ell^*(\mathbf{y}) \cdot \mathbf{g}_\ell^\top \mathbf{f} \cdot \mathbf{c}^\perp \mathbf{e}_1^\top & \text{if } j \geq \ell \end{cases} \\ \text{crs}_{3,j}^* = \mathbf{W}_{1,j}(\mathbf{I}_{m+n'} \otimes \mathbf{d}_j) - \widehat{\mathbf{W}}_j \end{array} \right]_2 \right\}_{j \in [L]}, \right. \\ & \left. \left\{ \left[\begin{array}{l} \mathbf{H}_{1,i}(\mathbf{I}_{g_i} \otimes \mathbf{d}_j), \mathbf{W}_{1,i}(\mathbf{I}_{m+n'} \otimes \mathbf{d}_j), \\ \mathbf{H}_{2,i}(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{d}_j), \mathbf{W}_{2,i}(\mathbf{D}\mathbf{E}_i \otimes \mathbf{d}_j) + \mathbf{W}_{3,i}((\mathbf{w}^\top \mathbf{D}\mathbf{E}_i + \theta_i \mathbf{e}_1^\top) \otimes \mathbf{d}_j) \end{array} \right]_2 \right\}_{\substack{i \in [L], \\ j \in [L] \setminus \{i\}}} \right) \\ \text{pk}_i = & \left([\mathbf{T}_{d,i}^*]_1, [\mathbf{Q}_{d,i}^*]_1, \{[\mathbf{G}_{d,i,j}^*]_2 := [\mathbf{U}_{d,i}(\mathbf{I}_{t_d} \otimes \mathbf{d}_j)]_2\}_{j \in [L] \setminus \{i\}}, \pi_{d,i}^* \right)_{d \in \{1,2\}}. \end{aligned}$$

Clearly, we have $G_{5,\ell-1} \equiv \widehat{G}_0$.

Game $\widehat{G}_1$: This is the same as $\widehat{G}_0$ except that the challenger replaces $\mathbf{d}_\ell \xleftarrow{\$} \text{span}(\mathbf{B})$ with $\mathbf{d}_\ell \xleftarrow{\$} \mathbb{Z}_p^{2k+1}$. We have $\widehat{G}_0 \approx_c \widehat{G}_1$ assuming MDDH_k holds in $\mathbb{G}_2$ with respect to $[\mathbf{B}]_2$. More precisely, it follows from the fact that $([\mathbf{B}]_2, [\mathbf{B}\mathbf{r}_\ell]_2) \approx_c ([\mathbf{B}]_2, [\mathbf{d}_\ell]_2)$ where $\mathbf{B} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times k}$, $\mathbf{r}_\ell \xleftarrow{\$} \mathbb{Z}_p^k$ and $\mathbf{d}_\ell \xleftarrow{\$} \mathbb{Z}_p^{2k+1}$.

Game $\widehat{G}_2$: This is the same as $\widehat{G}_1$ except that the challenger modifies the computation of the crs compo-

nents $(\text{crs}_{1,\ell}^*, \text{crs}_{2,\ell}^*)$ as follows:

$$\begin{aligned}\text{crs}_{1,\ell}^* &= \mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{d}_\ell) - \mathbf{c}^\perp (\mathbf{f}^\top \mathbf{g}_\ell \parallel (\mathbf{z}^*)^\top) + \widehat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_\ell} \\ \text{crs}_{2,\ell}^* &= \mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{d}_\ell) + \widehat{\mathbf{H}}_2 \mathbf{q}_\ell \mathbf{e}_1^\top - \mathbf{c}^\perp (\mathbf{p}^*)^\top \mathbf{q}_\ell \mathbf{e}_1^\top + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{d}_\ell) \\ &\quad + \mathbf{W}_{3,\ell}((\mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1) \otimes \mathbf{d}_\ell) + \mathbf{VDE}_\ell + \mathbf{v}(\mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1^\top) + \theta_\ell \mathbf{e}^\perp \mathbf{e}_1^\top \\ &\quad + \mathbf{g}_\ell^*(\mathbf{y}) \cdot \mathbf{g}_\ell^\top \mathbf{f} \cdot \mathbf{c}^\perp \mathbf{e}_1^\top.\end{aligned}$$

We prove $\widehat{\mathbf{G}}_1 \approx_c \widehat{\mathbf{G}}_2$ in the following claim.

Claim 40. *We have $\widehat{\mathbf{G}}_1 \approx_c \widehat{\mathbf{G}}_2$ assuming bi-MDDH_k in $\mathbb{G}$.*

Game $\widehat{\mathbf{G}}_3$: This is the same as $\widehat{\mathbf{G}}_2$ except that the challenger replaces $\mathbf{d}_\ell \xleftarrow{\$} \mathbb{Z}_p^{2k+1}$ with $\mathbf{d}_\ell \xleftarrow{\$} \text{span}(\mathbf{B})$. We have $\widehat{\mathbf{G}}_2 \approx_c \widehat{\mathbf{G}}_3$ assuming MDDH_k holds in $\mathbb{G}_2$ with respect to $\llbracket \mathbf{B} \rrbracket_2$. More precisely, it follows from the fact that $(\llbracket \mathbf{B} \rrbracket_2, \mathbf{d}_\ell) \approx_c (\llbracket \mathbf{B} \rrbracket_2, \llbracket \mathbf{B} \mathbf{r}_\ell \rrbracket_2)$ where $\mathbf{B} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times k}$, $\mathbf{r}_\ell \xleftarrow{\$} \mathbb{Z}_p^k$ and $\mathbf{d}_\ell \xleftarrow{\$} \mathbb{Z}_p^{2k+1}$.

This completes the proof. □

Proof of Claim 40. We distinguish the cases where pk_ℓ^* is either honest or [malicious or corrupted].

Honest Case. In this case, we have $\ell \in [L] \setminus (\mathcal{I}_{\text{cor}} \cup \mathcal{I}_{\text{mal}})$ which implies

$$\theta_\ell = 0, \quad \text{pk}_\ell^* = (\llbracket \mathbf{T}_{d,\ell}^* \rrbracket_1, \llbracket \mathbf{Q}_{d,\ell}^* \rrbracket_1, \{\llbracket \mathbf{G}_{d,\ell,j}^* \rrbracket_2\}_{j \in [L] \setminus \{\ell\}}, \pi_{d,\ell}^*)_{d \in \{1,2\}} \in \mathcal{D}_\ell \setminus \mathcal{C}_\ell,$$

and the challenger knows secret matrices $\mathbf{U}_{d,\ell}^*$ such that $\mathbf{T}_{d,\ell}^* = \mathbf{A} \mathbf{U}_{d,\ell}^*$, $\mathbf{Q}_{d,\ell}^* = \mathbf{R}_{d,\ell}^* \mathbf{U}_{d,\ell}^*$ and $\mathbf{G}_{d,\ell,j} = \mathbf{U}_{d,\ell}^* (\mathbf{I}_{t_d} \otimes \mathbf{d}_\ell)$ for $d \in \{1,2\}$ and $j \in [L] \setminus \{\ell\}$. In particular, we can write

$$\begin{aligned}\mathbf{c}_{1,2}^* &= \left(\mathbf{c}^\top \mathbf{U}_{1,\ell}^* + \sum_{i \in [L] \setminus \ell} \mathbf{e}_1^\top \widetilde{\mathbf{R}}_{1,i}^{-1} \mathbf{Q}_{1,i}^* \right) + \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{I}_{2k+1}) - \underline{\mathbf{W}}_{1,i} \\ \mathbf{c}_{2,2}^* &= \left(\mathbf{c}^\top \mathbf{U}_{2,\ell}^* + \sum_{i \in [L] \setminus \ell} \mathbf{e}_1^\top \widetilde{\mathbf{R}}_{2,i}^{-1} \mathbf{Q}_{2,i}^* \right) + \mathbf{c}^\top \sum_{i \in [L]} \mathbf{H}_{2,i}(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{1,i}(\mathbf{L}_{i,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \\ &\quad + \mathbf{W}_{2,i}(\mathbf{DE}_i \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{3,i}((\mathbf{w}^\top \mathbf{DE}_i + \theta_i \mathbf{e}_1^\top) \otimes \mathbf{I}_{2k+1}).\end{aligned}$$

Recall that we pick $\mathbf{c}^\perp \in \mathbb{Z}_p^{2k+1}$ such that $\mathbf{c}^\top \mathbf{c}^\perp = 1$ and $\mathbf{A} \mathbf{c}^\perp = \mathbf{0}$. For $\kappa \in [0;4]$, we define the distribution $D_\kappa = (C, \widehat{D}_\kappa)$, where

$$C = \left(\begin{array}{l} \mathbf{A}, \mathbf{B}, \mathbf{A}\widehat{\mathbf{H}}_1, \mathbf{A}\widehat{\mathbf{H}}_2, \mathbf{AV}, \mathbf{Av}, \mathbf{c}, \mathbf{c}^\perp, \mathbf{d}_\ell, \mathbf{d}^\perp, \mathbf{DE}_\ell, \mathbf{w}^\top \mathbf{DE}_\ell, \text{crs}_{3,\ell}^*, \\ \mathbf{AH}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}), \mathbf{AH}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}), \llbracket \mathbf{R}_{1,\ell}, \mathbf{R}_{2,\ell} \rrbracket_1, \\ \mathbf{AW}_{1,\ell}, \mathbf{AW}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{I}_{2k+1}) + \mathbf{AW}_{3,\ell}(\mathbf{w}^\top \mathbf{DE}_\ell \otimes \mathbf{I}_{2k+1}), \\ \left\{ \begin{array}{l} \mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{d}_\ell), \mathbf{H}_{2,i}(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{d}_\ell), \\ \mathbf{W}_{1,i}(\mathbf{I}_{m+n'} \otimes \mathbf{d}_\ell), \mathbf{W}_{2,i}(\mathbf{DE}_i \otimes \mathbf{d}_\ell) + \mathbf{W}_{3,i}(\mathbf{w}^\top \mathbf{DE}_i \otimes \mathbf{d}_\ell) \end{array} \right\}_{i \in [L] \setminus \{\ell\}} \end{array} \right),$$

represents the adversaries view on those components of crs that remain the same for all κ , and $\widehat{D}_\kappa$ represents the view on the changing components of crs , the public key and the challenge ciphertext:

$$\widehat{D}_0 = \left\{ \begin{array}{l} \text{crs}_{1,\ell}^*: \mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{d}_\ell) - \mathbf{c}^\perp (\mathbf{f}^\top \mathbf{g}_\ell \parallel (\mathbf{z}^*)^\top) + \widehat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_\ell}, \\ \text{crs}_{2,\ell}^*: \mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{d}_\ell) + \widehat{\mathbf{H}}_2 \mathbf{q}_\ell \mathbf{e}_1^\top - \mathbf{c}^\perp (\mathbf{p}^*)^\top \mathbf{q}_\ell \mathbf{e}_1^\top + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{d}_\ell) \\ \quad + \mathbf{W}_{3,\ell}(\mathbf{w}^\top \mathbf{DE}_\ell \otimes \mathbf{d}_\ell) + \mathbf{VDE}_\ell + \mathbf{vw}^\top \mathbf{DE}_\ell \\ \text{pk}_\ell^*: \mathbf{A} \mathbf{U}_{d,\ell}^*, \llbracket \mathbf{R}_{d,\ell} \mathbf{U}_{d,\ell}^* \rrbracket_1, \mathbf{U}_{d,\ell}^* (\mathbf{I}_m \otimes \mathbf{d}_\ell) \\ \mathbf{c}_{1,2}^*: \mathbf{c}^\top (\mathbf{U}_{1,\ell}^* + \mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}) - \underline{\mathbf{W}}_{1,\ell}) \\ \mathbf{c}_{2,2}^*: \mathbf{c}^\top (\mathbf{U}_{2,\ell}^* + \mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{1,\ell}(\mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{I}_{2k+1}) \\ \quad + \mathbf{W}_{3,\ell}(\mathbf{w}^\top \mathbf{DE}_\ell \otimes \mathbf{I}_{2k+1})) \end{array} \right\}$$

$$\begin{aligned}
\widehat{D}_1 &= \left\{ \begin{aligned} \text{crs}_{1,\ell}^* &: \mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{d}_\ell) - \mathbf{c}^\perp(\mathbf{f}^\top \mathbf{g}_\ell \parallel (\mathbf{z}^*)^\top) + \widehat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_\ell}, \\ \text{crs}_{2,\ell}^* &: \mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{d}_\ell) + \widehat{\mathbf{H}}_2 \mathbf{q}_\ell \mathbf{e}_1^\top - \mathbf{c}^\perp(\mathbf{p}^*)^\top \mathbf{q}_\ell \mathbf{e}_1^\top + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{d}_\ell) \\ &\quad + \mathbf{W}_{3,\ell}(\mathbf{w}^\top \mathbf{DE}_\ell \otimes \mathbf{d}_\ell) + \mathbf{VDE}_\ell + \mathbf{vw}^\top \mathbf{DE}_\ell \\ \text{pk}_\ell^* &: \mathbf{AU}_{d,\ell}^*, \llbracket \mathbf{R}_{d,\ell} \mathbf{U}_{d,\ell}^* + \mathbf{P}_{d,\ell} \otimes (\mathbf{d}^\perp)^\top \rrbracket_1, \mathbf{U}_{d,\ell}^*(\mathbf{I}_m \otimes \mathbf{d}_\ell) \\ \mathbf{c}_{1,2}^* &: \mathbf{c}^\top(\mathbf{U}_{1,\ell}^* + \mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}) - \mathbf{W}_{1,\ell}) \\ \mathbf{c}_{2,2}^* &: \mathbf{c}^\top(\mathbf{U}_{2,\ell}^* + \mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{1,\ell}(\mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{I}_{2k+1}) \\ &\quad + \mathbf{W}_{3,\ell}(\mathbf{w}^\top \mathbf{DE}_\ell \otimes \mathbf{I}_{2k+1})) \end{aligned} \right\} \\
\widehat{D}_2 &= \left\{ \begin{aligned} \text{crs}_{1,\ell}^* &: \mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{d}_\ell) - \mathbf{c}^\perp((\mathbf{f}^\top \mathbf{g}_\ell \parallel (\mathbf{z}^*)^\top) + \mathbf{h}_{1,\ell}^\top \mathbf{I}_{\mathbf{g}_\ell}) + \widehat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_\ell}, \\ \text{crs}_{2,\ell}^* &: \mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{d}_\ell) + \widehat{\mathbf{H}}_2 \mathbf{q}_\ell \mathbf{e}_1^\top + \mathbf{c}^\perp(\mathbf{h}_{2,\ell}^\top \mathbf{q}_\ell - (\mathbf{p}^*)^\top \mathbf{q}_\ell) \mathbf{e}_1^\top + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{d}_\ell) \\ &\quad + \mathbf{W}_{3,\ell}(\mathbf{w}^\top \mathbf{DE}_\ell \otimes \mathbf{d}_\ell) + \mathbf{VDE}_\ell + \mathbf{vw}^\top \mathbf{DE}_\ell \\ \text{pk}_\ell^* &: \mathbf{AU}_{d,\ell}^*, \llbracket \mathbf{R}_{d,\ell} \mathbf{U}_{d,\ell}^* + (\mathbf{P}_{d,\ell} + \mathbf{R}_{d,\ell} \mathbf{c}^\perp \mathbf{u}_{d,\ell}^\top) \otimes (\mathbf{d}^\perp)^\top \rrbracket_1, \mathbf{U}_{d,\ell}^*(\mathbf{I}_m \otimes \mathbf{d}_\ell) \\ \mathbf{c}_{1,2}^* &: \mathbf{c}^\top(\mathbf{U}_{1,\ell}^* + \mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}) - \mathbf{W}_{1,\ell}) + (\mathbf{u}_{1,\ell}^\top + \mathbf{h}_{1,\ell}^\top \mathbf{I}_{\mathbf{g}_\ell}) \otimes (\mathbf{d}^\perp)^\top \\ \mathbf{c}_{2,2}^* &: \mathbf{c}^\top(\mathbf{U}_{2,\ell}^* + \mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{1,\ell}(\mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{I}_{2k+1}) \\ &\quad + \mathbf{W}_{3,\ell}(\mathbf{w}^\top \mathbf{DE}_\ell \otimes \mathbf{I}_{2k+1})) + (\mathbf{u}_{2,\ell}^\top + \mathbf{h}_{2,\ell}^\top \mathbf{q}_\ell \mathbf{e}_1^\top) \otimes (\mathbf{d}^\perp)^\top \end{aligned} \right\} \\
\widehat{D}_3 &= \left\{ \begin{aligned} \text{crs}_{1,\ell}^* &: \mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{d}_\ell) - \mathbf{c}^\perp((\mathbf{f}^\top \mathbf{g}_\ell \parallel (\mathbf{z}^*)^\top) + \mathbf{h}_{1,\ell}^\top \mathbf{I}_{\mathbf{g}_\ell}) + \widehat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_\ell}, \\ \text{crs}_{2,\ell}^* &: \mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{d}_\ell) + \widehat{\mathbf{H}}_2 \mathbf{q}_\ell \mathbf{e}_1^\top + \mathbf{c}^\perp(\mathbf{h}_{2,\ell}^\top \mathbf{q}_\ell - (\mathbf{p}^*)^\top \mathbf{q}_\ell) \mathbf{e}_1^\top \\ &\quad + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{d}_\ell) + \mathbf{W}_{3,\ell}(\mathbf{w}^\top \mathbf{DE}_\ell \otimes \mathbf{d}_\ell) + \mathbf{VDE}_\ell + \mathbf{vw}^\top \mathbf{DE}_\ell \\ \text{pk}_\ell^* &: \mathbf{AU}_{d,\ell}^*, \llbracket \mathbf{R}_{d,\ell} \mathbf{U}_{d,\ell}^* + (\mathbf{P}_{d,\ell} + \mathbf{R}_{d,\ell} \mathbf{c}^\perp \mathbf{u}_{d,\ell}^\top) \otimes (\mathbf{d}^\perp)^\top \rrbracket_1, \mathbf{U}_{d,\ell}^*(\mathbf{I}_m \otimes \mathbf{d}_\ell) \\ \mathbf{c}_{1,2}^* &: \mathbf{c}^\top(\mathbf{U}_{1,\ell}^* + \mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}) - \mathbf{W}_{1,\ell}) + (\mathbf{u}_{1,\ell}^\top + \mathbf{h}_{1,\ell}^\top \mathbf{I}_{\mathbf{g}_\ell}) \otimes (\mathbf{d}^\perp)^\top \\ \mathbf{c}_{2,2}^* &: \mathbf{c}^\top(\mathbf{U}_{2,\ell}^* + \mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{1,\ell}(\mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{I}_{2k+1}) \\ &\quad + \mathbf{W}_{3,\ell}(\mathbf{w}^\top \mathbf{DE}_\ell \otimes \mathbf{I}_{2k+1})) + (\mathbf{u}_{2,\ell}^\top + \mathbf{h}_{2,\ell}^\top \mathbf{q}_\ell \mathbf{e}_1^\top) \otimes (\mathbf{d}^\perp)^\top \end{aligned} \right\} \\
\widehat{D}_4 &= \left\{ \begin{aligned} \text{crs}_{1,\ell}^* &: \mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{d}_\ell) - \mathbf{c}^\perp \mathbf{h}_{1,\ell}^\top \mathbf{I}_{\mathbf{g}_\ell} + \widehat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_\ell}, \\ \text{crs}_{2,\ell}^* &: \mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{d}_\ell) + \widehat{\mathbf{H}}_2 \mathbf{q}_\ell \mathbf{e}_1^\top + \mathbf{c}^\perp \mathbf{h}_{2,\ell}^\top \mathbf{q}_\ell \mathbf{e}_1^\top \\ &\quad + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{d}_\ell) + \mathbf{W}_{3,\ell}(\mathbf{w}^\top \mathbf{DE}_\ell \otimes \mathbf{d}_\ell) + \mathbf{VDE}_\ell + \mathbf{vw}^\top \mathbf{DE}_\ell \\ \text{pk}_\ell^* &: \mathbf{AU}_{d,\ell}^*, \llbracket \mathbf{R}_{d,\ell} \mathbf{U}_{d,\ell}^* + (\mathbf{P}_{d,\ell} + \mathbf{R}_{d,\ell} \mathbf{c}^\perp \mathbf{u}_{d,\ell}^\top) \otimes (\mathbf{d}^\perp)^\top \rrbracket_1, \mathbf{U}_{d,\ell}^*(\mathbf{I}_m \otimes \mathbf{d}_\ell) \\ \mathbf{c}_{1,2}^* &: \mathbf{c}^\top(\mathbf{U}_{1,\ell}^* + \mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}) - \mathbf{W}_{1,\ell}) + (\mathbf{u}_{1,\ell}^\top + \mathbf{h}_{1,\ell}^\top \mathbf{I}_{\mathbf{g}_\ell}) \otimes (\mathbf{d}^\perp)^\top \\ \mathbf{c}_{2,2}^* &: \mathbf{c}^\top(\mathbf{U}_{2,\ell}^* + \mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{1,\ell}(\mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{I}_{2k+1}) \\ &\quad + \mathbf{W}_{3,\ell}(\mathbf{w}^\top \mathbf{DE}_\ell \otimes \mathbf{I}_{2k+1})) + (\mathbf{u}_{2,\ell}^\top + \mathbf{h}_{2,\ell}^\top \mathbf{q}_\ell \mathbf{e}_1^\top) \otimes (\mathbf{d}^\perp)^\top \end{aligned} \right\},
\end{aligned}$$

where $\mathbf{P}_{d,\ell} \stackrel{s}{\leftarrow} \mathbb{Z}_p^{(2k+2) \times t_d}$, $\mathbf{u}_{d,\ell} \stackrel{s}{\leftarrow} \mathbb{Z}_p^{t_d}$, $\mathbf{h}_{1,\ell} \stackrel{s}{\leftarrow} \mathbb{Z}_p^{k+n'}$, $\mathbf{h}_{2,\ell} \stackrel{s}{\leftarrow} \mathbb{Z}_p^k$. We note that the distribution D_0 (resp. D_4) corresponds to the adversary's view in $\widehat{G}_1$ (resp. $\widehat{G}_2$). Furthermore, we observe that

- $D_0 \approx_c D_1$ which follows from two applications of Lemma 24 when setting $\mathbf{R} = \mathbf{R}_{d,\ell}$, $\mathbf{U} = \mathbf{U}_{d,\ell}^*$ and $\mathbf{P} = \mathbf{P}_{d,\ell}$ for $d \in \{1, 2\}$.
- $D_1 \approx_s D_2$: this follows from the following changes of variables for $d \in \{1, 2\}$

$$\mathbf{U}_{d,\ell}^* \mapsto \mathbf{U}_{d,\ell}^* + \mathbf{c}^\perp(\mathbf{u}_{d,\ell} \otimes \mathbf{d}^\perp)^\top, \quad \mathbf{H}_{d,\ell} \mapsto \mathbf{H}_{d,\ell} + \mathbf{c}^\perp(\mathbf{h}_{d,\ell} \otimes \mathbf{d}^\perp)^\top.$$

Here we use the equality $\mathbf{R}_{d,\ell} \mathbf{c}^\perp(\mathbf{u}_{d,\ell} \otimes \mathbf{d}^\perp)^\top = \mathbf{R}_{d,\ell} \mathbf{c}^\perp \mathbf{u}_{d,\ell}^\top \otimes (\mathbf{d}^\perp)^\top$.

- $D_2 \approx_s D_3$: this follows from the fact that for $d \in \{1, 2\}$, $\mathbf{P}_{d,\ell}$ hides $\mathbf{R}_{d,\ell} \mathbf{c}^\perp \mathbf{u}_{d,\ell}^\top$ which implies that $\mathbf{u}_{1,\ell}$ hides $\mathbf{h}_{1,\ell}^\top \mathbf{I}_{\mathbf{g}_\ell}$ and $\mathbf{h}_{1,\ell}^\top \mathbf{I}_{\mathbf{g}_\ell}$ hides $(\mathbf{f}^\top \mathbf{g}_\ell \parallel (\mathbf{z}^*)^\top)$. Similarly, $\mathbf{u}_{2,\ell}$ can hide $\mathbf{h}_{2,\ell}^\top \mathbf{q}_\ell \mathbf{e}_1^\top$, so $\mathbf{h}_{2,\ell}^\top \mathbf{q}_\ell - (\mathbf{p}^*)^\top \mathbf{q}_\ell$ and $\mathbf{h}_{2,\ell}^\top \mathbf{q}_\ell$ are statistically close.
- $D_3 \approx_c D_4$: this follows from the reverse applications of the indistinguishability arguments in $D_1 \approx_s D_2$ and $D_0 \approx_c D_1$.

Corrupted and Malicious Case. In this case, we have $\ell \in (\mathcal{I}_{\text{cor}} \cup \mathcal{I}_{\text{mal}})$ which implies that

$$\text{pk}_\ell^* = (\llbracket \mathbf{T}_{d,\ell}^* \rrbracket_1, \llbracket \mathbf{Q}_{d,\ell}^* \rrbracket_1, \{\llbracket \mathbf{G}_{d,\ell,j}^* \rrbracket_2\}_{j \in [L] \setminus \{\ell\}}, \pi_\ell^*) \in \overline{\mathcal{D}_\ell} \cup \mathcal{C}_\ell.$$

We pick $\mathbf{d}^\perp \in \mathbb{Z}_p^{2k+1}$ such that $(\mathbf{d}^\perp)^\top \mathbf{d}_\ell = 1$ and $(\mathbf{d}^\perp)^\top \mathbf{B} = \mathbf{0}$. For $\kappa \in [0; 3]$, we define the distribution $D_\kappa = (C, \hat{D}_\kappa)$ where, similarly to the previous case,

$$C = \left(\begin{array}{l} \mathbf{A}, \mathbf{B}, \mathbf{A}\hat{\mathbf{H}}_1, \mathbf{A}\hat{\mathbf{H}}_2, \mathbf{AV}, \mathbf{Av}, \mathbf{c}, \mathbf{c}^\perp, \mathbf{d}_\ell, \mathbf{d}^\perp, \mathbf{DE}_\ell, \mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1^\top, \\ \llbracket \mathbf{A}\mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}) \rrbracket_1, \mathbf{AH}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) \\ \mathbf{AW}_{1,\ell}, \mathbf{AW}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{I}_{2k+1}) + \mathbf{AW}_{3,\ell}((\mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1^\top) \otimes \mathbf{I}_{2k+1}), \\ \left\{ \begin{array}{l} \llbracket \mathbf{H}_{1,i}(\mathbf{I}_{\mathbf{g}_i} \otimes \mathbf{d}_j) \rrbracket_2, \mathbf{H}_{2,i}(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{d}_j), \\ \mathbf{W}_{1,i}(\mathbf{I}_{m+n'} \otimes \mathbf{d}_j), \mathbf{W}_{2,i}(\mathbf{DE}_i \otimes \mathbf{d}_j) + \mathbf{W}_{3,i}((\mathbf{w}^\top \mathbf{DE}_i + \theta_i \mathbf{e}_1^\top) \otimes \mathbf{d}_j) \end{array} \right\}_{i \in [L], j \in [L] \setminus \{\ell\}} \end{array} \right),$$

represents the adversaries view on those components of crs that remain the same for all κ , and $\hat{D}_\kappa$ represents the view on the changing components of crs and the challenge ciphertext:

$$\hat{D}_0 = \left\{ \begin{array}{l} \text{crs}_{1,\ell}^*: \llbracket \mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{d}_\ell) - \mathbf{c}^\perp(\mathbf{f}^\top \mathbf{g}_\ell \parallel (\mathbf{z}^*)^\top) + \hat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_\ell} \rrbracket_2, \\ \text{crs}_{2,\ell}^*: \llbracket \mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{d}_\ell) + \hat{\mathbf{H}}_2 \mathbf{q}_\ell \mathbf{e}_1^\top - \mathbf{c}^\perp(\mathbf{p}^*)^\top \mathbf{q}_\ell \mathbf{e}_1^\top + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{d}_\ell) \\ \quad + \mathbf{W}_{3,\ell}((\mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1^\top) \otimes \mathbf{d}_\ell) + \mathbf{VDE}_\ell + \mathbf{v}(\mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1^\top) + \mathbf{c}^\perp \theta_\ell \mathbf{e}_1^\top \rrbracket_2 \\ \text{crs}_{3,\ell}^*: \mathbf{W}_{1,\ell}(\mathbf{I}_{m+n'} \otimes \mathbf{d}_\ell) - \hat{\mathbf{W}}_\ell \\ \mathbf{c}_{1,2}^*: \llbracket \mathbf{e}_1^\top \tilde{\mathbf{R}}_{1,i}^{-1} \mathbf{Q}_{1,i}^* + \mathbf{c}^\top(\mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}) - \mathbf{W}_{1,\ell}) \rrbracket_1 \\ \mathbf{c}_{2,2}^*: \llbracket \mathbf{e}_1^\top \tilde{\mathbf{R}}_{2,i}^{-1} \mathbf{Q}_{2,i}^* + \mathbf{c}^\top(\mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{1,\ell}(\mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{I}_{2k+1}) \\ \quad + \mathbf{W}_{3,\ell}((\mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1^\top) \otimes \mathbf{I}_{2k+1})) \rrbracket_1 \end{array} \right\}$$

$$\hat{D}_1 = \left\{ \begin{array}{l} \text{crs}_{1,\ell}^*: \llbracket \mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{d}_\ell) - \cancel{\mathbf{c}^\perp(\mathbf{f}^\top \mathbf{g}_\ell \parallel (\mathbf{z}^*)^\top)} + \hat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_\ell} \rrbracket_2, \\ \text{crs}_{2,\ell}^*: \llbracket \mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{d}_\ell) + \hat{\mathbf{H}}_2 \mathbf{q}_\ell \mathbf{e}_1^\top - \cancel{\mathbf{c}^\perp(\mathbf{p}^*)^\top \mathbf{q}_\ell \mathbf{e}_1^\top} + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{d}_\ell) \\ \quad + \mathbf{W}_{3,\ell}((\mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1^\top) \otimes \mathbf{d}_\ell) + \mathbf{VDE}_\ell + \mathbf{v}(\mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1^\top) + \cancel{\mathbf{c}^\perp \theta_\ell \mathbf{e}_1^\top} \rrbracket_2 \\ \text{crs}_{3,\ell}^*: \mathbf{W}_{1,\ell}(\mathbf{I}_{m+n'} \otimes \mathbf{d}_\ell) - \cancel{\hat{\mathbf{W}}_\ell} \\ \mathbf{c}_{1,2}^*: \llbracket \mathbf{e}_1^\top \tilde{\mathbf{R}}_{1,i}^{-1} \mathbf{Q}_{1,i}^* + \mathbf{c}^\top(\mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}) - \mathbf{W}_{1,\ell}) + ((\mathbf{f}^\top \mathbf{g}_\ell \parallel (\mathbf{z}^*)^\top) - \mathbf{c}^\top \hat{\mathbf{W}}_\ell) \otimes (\mathbf{d}^\perp)^\top \rrbracket_1 \\ \mathbf{c}_{2,2}^*: \llbracket \mathbf{e}_1^\top \tilde{\mathbf{R}}_{2,i}^{-1} \mathbf{Q}_{2,i}^* + \mathbf{c}^\top(\mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{1,\ell}(\mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{I}_{2k+1}) \\ \quad + \mathbf{W}_{3,\ell}((\mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1^\top) \otimes \mathbf{I}_{2k+1})) + (\mathbf{c}^\top \hat{\mathbf{W}}_\ell \mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} + (\mathbf{p}^*)^\top \mathbf{q}_\ell \mathbf{e}_1^\top - \theta_\ell \mathbf{e}_1^\top) \otimes (\mathbf{d}^\perp)^\top \rrbracket_1 \end{array} \right\}$$

$$\begin{aligned}
\hat{D}_2 &= \left\{ \begin{array}{l} \text{crs}_{1,\ell}^*: \llbracket \mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{d}_\ell) + \hat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_\ell} \rrbracket_2, \\ \text{crs}_{2,\ell}^*: \llbracket \mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{d}_\ell) + \hat{\mathbf{H}}_2 \mathbf{q}_\ell \mathbf{e}_1^\top + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{d}_\ell) + \mathbf{W}_{3,\ell}((\mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1^\top) \otimes \mathbf{d}_\ell) \\ \quad + \mathbf{VDE}_\ell + \mathbf{v}(\mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1^\top) \rrbracket_2 \\ \text{crs}_{3,\ell}^*: \mathbf{W}_{1,\ell}(\mathbf{I}_{m+n'} \otimes \mathbf{d}_\ell) \\ \mathbf{c}_{1,2}^*: \llbracket \mathbf{e}_1^\top \tilde{\mathbf{R}}_{1,i}^{-1} \mathbf{Q}_{1,i}^* + \mathbf{c}^\top (\mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}) - \mathbf{W}_{1,\ell}) + (\mathbf{f}^\top \mathbf{g}_\ell \parallel (\mathbf{z}^*)^\top) - \mathbf{c}^\top \hat{\mathbf{W}}_\ell \otimes (\mathbf{d}^\perp)^\top \rrbracket_1 \\ \mathbf{c}_{2,2}^*: \llbracket \mathbf{e}_1^\top \tilde{\mathbf{R}}_{2,i}^{-1} \mathbf{Q}_{2,i}^* + \mathbf{c}^\top (\mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{1,\ell}(\mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{I}_{2k+1}) \\ \quad + \mathbf{W}_{3,\ell}((\mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1) \otimes \mathbf{I}_{2k+1})) \\ \quad + (\mathbf{c}^\top \hat{\mathbf{W}}_\ell \mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} + (\mathbf{g}_\ell^*(\mathbf{y}) \cdot \mathbf{f}^\top \mathbf{g}_\ell + h_\ell^*(\mathbf{x})^\top \mathbf{z}^* + (\mathbf{p}^*)^\top \mathbf{q}_\ell - \theta_\ell) \cdot \mathbf{e}_1^\top) \otimes (\mathbf{d}^\perp)^\top \rrbracket_1 \end{array} \right\} \\
\hat{D}_3 &= \left\{ \begin{array}{l} \text{crs}_{1,\ell}^*: \llbracket \mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{d}_\ell) + \hat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_\ell} \rrbracket_2, \\ \text{crs}_{2,\ell}^*: \llbracket \mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{d}_\ell) + \hat{\mathbf{H}}_2 \mathbf{q}_\ell \mathbf{e}_1^\top + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{d}_\ell) + \mathbf{W}_{3,\ell}((\mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1^\top) \otimes \mathbf{d}_\ell) \\ \quad + \mathbf{VDE}_\ell + \mathbf{v}(\mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1^\top) - \mathbf{c}^\perp (\mathbf{g}_\ell^*(\mathbf{y}) \cdot \mathbf{f}^\top \mathbf{g}_\ell + h_\ell^*(\mathbf{x})^\top \mathbf{z}^* + (\mathbf{p}^*)^\top \mathbf{q}_\ell - \theta_\ell) \cdot \mathbf{e}_1^\top \rrbracket_2 \\ \text{crs}_{3,\ell}^*: \mathbf{W}_{1,\ell}(\mathbf{I}_{m+n'} \otimes \mathbf{d}_\ell) - \hat{\mathbf{W}}_\ell \\ \mathbf{c}_{1,2}^*: \llbracket \mathbf{e}_1^\top \tilde{\mathbf{R}}_{1,i}^{-1} \mathbf{Q}_{1,i}^* + \mathbf{c}^\top (\mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}) - \mathbf{W}_{1,\ell}) - \mathbf{c}^\top \hat{\mathbf{W}}_\ell \otimes (\mathbf{d}^\perp)^\top \rrbracket_1 \\ \mathbf{c}_{2,2}^*: \llbracket \mathbf{e}_1^\top \tilde{\mathbf{R}}_{2,i}^{-1} \mathbf{Q}_{2,i}^* + \mathbf{c}^\top (\mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{1,\ell}(\mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{I}_{2k+1}) \\ \quad + \mathbf{W}_{3,\ell}((\mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1) \otimes \mathbf{I}_{2k+1})) \\ \quad + (\mathbf{c}^\top \hat{\mathbf{W}}_\ell \mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} + (\mathbf{g}_\ell^*(\mathbf{y}) \cdot \mathbf{f}^\top \mathbf{g}_\ell + h_\ell^*(\mathbf{x})^\top \mathbf{z}^* + (\mathbf{p}^*)^\top \mathbf{q}_\ell - \theta_\ell) \cdot \mathbf{e}_1^\top) \otimes (\mathbf{d}^\perp)^\top \rrbracket_1 \end{array} \right\} \\
\hat{D}_4 &= \left\{ \begin{array}{l} \text{crs}_{1,\ell}^*: \llbracket \mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{d}_\ell) + \hat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_\ell} \rrbracket_2, \\ \text{crs}_{2,\ell}^*: \llbracket \mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{d}_\ell) + \hat{\mathbf{H}}_2 \mathbf{q}_\ell \mathbf{e}_1^\top + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{d}_\ell) + \mathbf{W}_{3,\ell}((\mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1^\top) \otimes \mathbf{d}_\ell) \\ \quad + \mathbf{VDE}_\ell + \mathbf{v}(\mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1^\top) + \mathbf{c}^\perp (\mathbf{g}_\ell^*(\mathbf{y}) \cdot \mathbf{f}^\top \mathbf{g}_\ell + h_\ell^*(\mathbf{x})^\top \mathbf{z}^* + (\mathbf{p}^*)^\top \mathbf{q}_\ell - \theta_\ell) \cdot \mathbf{e}_1^\top \rrbracket_2 \\ \text{crs}_{3,\ell}^*: \mathbf{W}_{1,\ell}(\mathbf{I}_{m+n'} \otimes \mathbf{d}_\ell) - \hat{\mathbf{W}}_\ell \\ \mathbf{c}_{1,2}^*: \llbracket \mathbf{e}_1^\top \tilde{\mathbf{R}}_{1,i}^{-1} \mathbf{Q}_{1,i}^* + \mathbf{c}^\top (\mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}) - \mathbf{W}_{1,\ell}) \rrbracket_1 \\ \mathbf{c}_{2,2}^*: \llbracket \mathbf{e}_1^\top \tilde{\mathbf{R}}_{2,i}^{-1} \mathbf{Q}_{2,i}^* + \mathbf{c}^\top (\mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{1,\ell}(\mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{I}_{2k+1}) \\ \quad + \mathbf{W}_{3,\ell}((\mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1) \otimes \mathbf{I}_{2k+1})) \rrbracket_1 \end{array} \right\}.
\end{aligned}$$

We note that the distribution D_0 (resp. D_4) corresponds to the adversary's view in $\hat{\mathbf{G}}_1$ (resp. $\hat{\mathbf{G}}_2$). Furthermore, we observe that

- $D_0 \approx_s D_1$: this follows from the changes of variables

$$\begin{aligned}
\mathbf{W}_{1,\ell} &\mapsto \mathbf{W}_{1,\ell} + \mathbf{c}^\perp (\mathbf{c}^\top \hat{\mathbf{W}}_\ell \otimes (\mathbf{d}^\perp)^\top), & \mathbf{H}_{1,\ell} &\mapsto \mathbf{H}_{1,\ell} + \mathbf{c}^\perp ((\mathbf{f}^\top \parallel (\mathbf{z}^*)^\top) \otimes (\mathbf{d}^\perp)^\top), \\
\mathbf{W}_{2,\ell} &\mapsto \mathbf{W}_{2,\ell} + \mathbf{c}^\perp (\mathbf{w}^\top \otimes (\mathbf{d}^\perp)^\top), & \mathbf{H}_{2,\ell} &\mapsto \mathbf{H}_{2,\ell} + \mathbf{c}^\perp ((\mathbf{p}^*)^\top \otimes (\mathbf{d}^\perp)^\top), \\
\mathbf{W}_{3,\ell} &\mapsto \mathbf{W}_{3,\ell} - \mathbf{c}^\perp (1 \otimes (\mathbf{d}^\perp)^\top).
\end{aligned}$$

- $D_1 \approx_s D_2$: this follows from the security of pgb. Specifically, we can simulate $\hat{D}_\kappa^\beta$, $\kappa \in \{1, 2\}$, in the form

$$\hat{D}_\kappa = \left\{ \begin{array}{l} \text{crs}_{1,\ell}^*: \llbracket \mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{d}_\ell) + \hat{\mathbf{H}}_1 \mathbf{I}_{\mathbf{g}_\ell} \rrbracket_2, \\ \text{crs}_{2,\ell}^*: \llbracket \mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{d}_\ell) + \hat{\mathbf{H}}_2 \mathbf{q}_\ell \mathbf{e}_1^\top + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{d}_\ell) + \mathbf{W}_{3,\ell}((\mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1^\top) \otimes \mathbf{d}_\ell) \\ \quad + \mathbf{VDE}_\ell + \mathbf{v}(\mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1^\top) \rrbracket_2 \\ \text{crs}_{3,\ell}^*: \mathbf{W}_{1,\ell}(\mathbf{I}_{m+n'} \otimes \mathbf{d}_\ell) \\ \mathbf{c}_{1,2}^*: \llbracket \mathbf{e}_1^\top \tilde{\mathbf{R}}_{1,i}^{-1} \mathbf{Q}_{1,i}^* + \mathbf{c}^\top (\mathbf{H}_{1,\ell}(\mathbf{I}_{\mathbf{g}_\ell} \otimes \mathbf{I}_{2k+1}) - \mathbf{W}_{1,\ell}) + \mathbf{p}_1^\top \otimes (\mathbf{d}^\perp)^\top \rrbracket_1 \\ \mathbf{c}_{2,2}^*: \llbracket \mathbf{e}_1^\top \tilde{\mathbf{R}}_{2,i}^{-1} \mathbf{Q}_{2,i}^* + \mathbf{c}^\top (\mathbf{H}_{2,\ell}(\mathbf{q}_\ell \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{1,\ell}(\mathbf{L}_{\ell,\mathbf{y},\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \\ \quad + \mathbf{W}_{2,\ell}(\mathbf{DE}_\ell \otimes \mathbf{I}_{2k+1}) + \mathbf{W}_{3,\ell}((\mathbf{w}^\top \mathbf{DE}_\ell + \theta_\ell \mathbf{e}_1) \otimes \mathbf{I}_{2k+1})) + (\mathbf{p}_2^\top - \theta_\ell \mathbf{e}_1) \otimes (\mathbf{d}^\perp)^\top \rrbracket_1 \end{array} \right\},$$

where the vectors $\mathbf{p}_1 \in \mathbb{Z}_p^{n'+1}$ and $\mathbf{p}_2 \in \mathbb{Z}_p^m$ are defined as

$$(\llbracket \mathbf{p}_1^\top \rrbracket_1, \llbracket \mathbf{p}_2^\top \rrbracket_1) \leftarrow \begin{cases} \text{pgb}(\phi_\ell^*, (\mathbf{y}, \mathbf{x}), \llbracket (\mathbf{f}^\top \mathbf{g}_\ell \parallel (\mathbf{z}^*)^\top) \rrbracket_1; \llbracket \mathbf{c}^\top \widehat{\mathbf{W}}_\ell \rrbracket_1) + \llbracket (\mathbf{0}^\top \parallel (\mathbf{p}^*)^\top \mathbf{q}_\ell \cdot \mathbf{e}_1^\top) \rrbracket_1 & \text{if } \kappa = 1 \\ \text{pgb}^*(\phi_\ell^*, (\mathbf{y}, \mathbf{x}), \llbracket \mathbf{g}_\ell^*(\mathbf{y}) \cdot \mathbf{f}^\top \mathbf{g}_\ell + h_\ell^*(\mathbf{x})^\top \mathbf{z}^* + (\mathbf{p}^*)^\top \mathbf{q}_\ell \rrbracket_1; \llbracket \mathbf{c}^\top \widehat{\mathbf{W}}_\ell \rrbracket_1) & \text{if } \kappa = 2 \end{cases},$$

and we recall that $\phi_\ell^*(\mathbf{y}, \mathbf{x})^\top = (g_\ell^*(\mathbf{y}) \parallel h_\ell^*(\mathbf{x})^\top)$. Then $D_1^\beta \approx_s D_2^\beta$ immediately follows from the privacy of the underlying partial garbling scheme and the fact that $\mathbf{c}^\top \widehat{\mathbf{W}}_\ell$ is statistically close to uniform as $\widehat{\mathbf{W}}_\ell$ does not appear anywhere else.

- $D_2 \approx_s D_3$: this follows from the changes of variables

$$\mathbf{W}_{1,\ell} \mapsto \mathbf{W}_{1,\ell} - \mathbf{c}^\perp (\mathbf{c}^\top \widehat{\mathbf{W}}_\ell \otimes (\mathbf{d}^\perp)^\top), \quad \mathbf{H}_{2,\ell} \mapsto \mathbf{H}_{2,\ell} - \mathbf{c}^\perp (\gamma_\ell \mathbf{e}_1^\top \otimes (\mathbf{d}^\perp)^\top),$$

where $\gamma_\ell = (g_\ell^*(\mathbf{y}) \cdot \mathbf{f}^\top \mathbf{g}_\ell + h_\ell^*(\mathbf{x})^\top \mathbf{z}^* + (\mathbf{p}^*)^\top \mathbf{q}_\ell - \theta_\ell) / \mathbf{q}_\ell[1]$.

- $D_3 \approx_s D_4$: this follows from the following case distinction. If $\mathbf{g}_\ell^*(\mathbf{y}) = 0$, then we have $\theta_\ell = h_\ell^*(\mathbf{x})^\top \mathbf{z}^* + (\mathbf{p}^*)^\top \mathbf{q}_\ell$ by definition and the cancelation is immediate. If $\mathbf{g}_\ell^*(\mathbf{y}) \neq 0$, then we first rely on the bi-MDDH $_k$ assumption in $\mathbb{G}$ which states that $(\llbracket \mathbf{g}_\ell^\top \rrbracket_{1,2}, \llbracket \mathbf{g}_\ell^\top \mathbf{f} \rrbracket_{1,2}) \approx_c (\llbracket \mathbf{g}_\ell^\top \rrbracket_{1,2}, \llbracket \alpha_\ell \rrbracket_{1,2})$ for $\alpha_\ell \xleftarrow{s} \mathbb{Z}_p$. Subsequently, we observe that α_ℓ and $\alpha_\ell + h_\ell^*(\mathbf{x})^\top \mathbf{z}^* + (\mathbf{p}^*)^\top \mathbf{q}_\ell - \theta_\ell$ are identically distributed, so we can erase $h_\ell^*(\mathbf{x})^\top \mathbf{z}^* + (\mathbf{p}^*)^\top \mathbf{q}_\ell - \theta_\ell$. Finally, we reverse the application of bi-MDDH $_k$ and replace α_ℓ again with $\mathbf{g}_\ell^\top \mathbf{f}$.

This completes the proof. □

6.4 Special Case: Very Selectively SIM-Secure sRFE for eAWS

This section gives a self-contained description of our sRFE for eAWS with very selective SIM-security which is implied by the (more complex) sRFE for AB-eAWS in Section 6 (Construction 35). Using the compiler in Section 7, it can be upgraded to sRFE for the (unbounded-input) AWS functionality.

Construction 41 (Very Selectively SIM-Secure sRFE for eAWS). The construction uses the following ingredients:

- a pairing group $\mathbb{G} = (\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_t, g_1, g_2, g_t, e, p)$, and
- a QA-NIZK $\Pi = (\text{LGen}, \text{LPrv}, \text{LVer}, \text{LSim})$ for linear spaces over $\mathbb{G}$.

The details of the scheme go as follows:

Setup($1^\lambda, 1^L, \{\llbracket \mathbf{q}_i \rrbracket_{1,2}\}_{i \in [L]}$): On input the security parameter 1^λ , the maximum number of slots 1^L and a set of vectors $\{\llbracket \mathbf{q}_i \rrbracket_{1,2}\}_{i \in [L]}$ with $\mathbf{q}_i \in \mathbb{Z}_p^k$ for $i \in [L]$, sample

$$\begin{aligned} \mathbf{A} &\xleftarrow{s} \mathbb{Z}_p^{k \times (2k+1)}, \quad \mathbf{B}, \mathbf{D} \xleftarrow{s} \mathbb{Z}_p^{(2k+1) \times k}, \quad \mathbf{V} \xleftarrow{s} \mathbb{Z}_p^{(2k+1) \times (2k+1)}, \quad \mathbf{v}, \mathbf{w} \xleftarrow{s} \mathbb{Z}_p^{2k+1}, \\ \widehat{\mathbf{H}}_1 &\xleftarrow{s} \mathbb{Z}_p^{(2k+1) \times n'}, \quad \widehat{\mathbf{H}}_2 \xleftarrow{s} \mathbb{Z}_p^{(2k+1) \times k}, \\ \left\{ \begin{array}{l} \mathbf{E}_i \xleftarrow{s} \mathbb{Z}_p^{k \times m}, \quad \mathbf{R}_{1,i}, \mathbf{R}_{2,i} \xleftarrow{s} \mathbb{Z}_p^{(2k+2) \times (2k+1)}, \quad \mathbf{r}_i \xleftarrow{s} \mathbb{Z}_p^k, \\ \mathbf{H}_{2,i} \xleftarrow{s} \mathbb{Z}_p^{(2k+1) \times (2k+1)k}, \quad \widehat{\mathbf{W}}_i \xleftarrow{s} \mathbb{Z}_p^{(2k+1) \times (m+n'-1)}, \\ \mathbf{W}_{1,i} \xleftarrow{s} \mathbb{Z}_p^{(2k+1) \times (2k+1)(m+n'-1)}, \quad \mathbf{W}_{2,i} \xleftarrow{s} \mathbb{Z}_p^{(2k+1) \times (2k+1)(2k+1)}, \quad \mathbf{W}_{3,i} \xleftarrow{s} \mathbb{Z}_p^{(2k+1) \times (2k+1)} \end{array} \right\}_{i \in [L]}. \end{aligned}$$

Then, generate

$$(\text{crs}_{1,i}, \text{td}_{1,i}) \leftarrow \text{LGen}(1^\lambda, \mathbb{G}_1, \llbracket \mathbf{A}_{1,i} \rrbracket_1), \quad (\text{crs}_{2,i}, \text{td}_{2,i}) \leftarrow \text{LGen}(1^\lambda, \mathbb{G}_1, \llbracket \mathbf{A}_{2,i} \rrbracket_1)$$

where $\mathbf{A}_{1,i}^\top = [\mathbf{A}^\top \parallel \mathbf{R}_{1,i}^\top], \mathbf{A}_{2,i}^\top = [\mathbf{A}^\top \parallel \mathbf{R}_{2,i}^\top]$, and output the common reference string

$$\text{crs} = \left(\begin{aligned} & \llbracket \mathbf{A}, \mathbf{A}\hat{\mathbf{H}}_1, \mathbf{A}\hat{\mathbf{H}}_2, \mathbf{AV}, \mathbf{Av} \rrbracket_1, \\ & \left\{ \left\llbracket \begin{aligned} & \mathbf{AH}_{2,i}(\mathbf{q}_i(\mathbf{e}_1^{(m)})^\top \otimes \mathbf{I}_{2k+1}), \mathbf{R}_{1,i}, \mathbf{R}_{2,i}, \\ & \mathbf{AW}_{1,i}, \mathbf{AW}_{2,i}(\mathbf{DE}_i \otimes \mathbf{I}_{2k+1}) + \mathbf{AW}_{3,i}(\mathbf{w}^\top \mathbf{DE}_i \otimes \mathbf{I}_{2k+1}), \text{crs}_{1,i}, \text{crs}_{2,i} \end{aligned} \right\rrbracket_1 \right\}_{i \in [L]}, \\ & \left\{ \left\llbracket \begin{aligned} & \mathbf{Br}_j, \mathbf{q}_j, \mathbf{DE}_j, \mathbf{w}^\top \mathbf{DE}_j, \mathbf{W}_{1,j}(\mathbf{I}_{m+n'-1} \otimes \mathbf{Br}_j) - \hat{\mathbf{W}}_j, \\ & \mathbf{H}_{2,j}(\mathbf{q}_j \mathbf{e}_1^\top \otimes \mathbf{Br}_j) + \hat{\mathbf{H}}_2 \mathbf{q}_j \mathbf{e}_1^\top \\ & + \mathbf{W}_{2,j}(\mathbf{DE}_j \otimes \mathbf{Br}_j) + \mathbf{W}_{3,j}(\mathbf{w}^\top \mathbf{DE}_j \otimes \mathbf{Br}_j) + \mathbf{VDE}_j + \mathbf{vw}^\top \mathbf{DE}_j \end{aligned} \right\rrbracket_2 \right\}_{j \in [L]}, \\ & \left\{ \left\llbracket \begin{aligned} & \mathbf{W}_{1,i}(\mathbf{I}_{m+n'-1} \otimes \mathbf{Br}_j), \\ & \mathbf{H}_{2,i}(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{Br}_j), \mathbf{W}_{2,i}(\mathbf{DE}_i \otimes \mathbf{Br}_j) + \mathbf{W}_{3,i}(\mathbf{w}^\top \mathbf{DE}_i \otimes \mathbf{Br}_j) \end{aligned} \right\rrbracket_2 \right\}_{\substack{i \in [L], \\ j \in [L] \setminus \{i\}}} \end{aligned} \right).$$

Gen(crs, i): On input crs and an index $i \in [L]$, sample matrices $\mathbf{U}_{1,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)n'}$, $\mathbf{U}_{2,i} \xleftarrow{\$} \mathbb{Z}_p^{(2k+1) \times (2k+1)m}$ and run $\pi_{1,i} \leftarrow \text{LPrv}(\text{crs}_{1,i}, \llbracket \mathbf{A}_{1,i} \mathbf{U}_{1,i} \rrbracket_1, \mathbf{U}_{1,i})$, $\pi_{2,i} \leftarrow \text{LPrv}(\text{crs}_{2,i}, \llbracket \mathbf{A}_{2,i} \mathbf{U}_{2,i} \rrbracket_1, \mathbf{U}_{2,i})$. Then output the key pair $\text{sk}_i = (\mathbf{U}_{1,i}, \mathbf{U}_{2,i})$ and pk_i as follows:

$$\left(\begin{aligned} & \llbracket \mathbf{T}_{1,i} \rrbracket_1 := \llbracket \mathbf{AU}_{1,i} \rrbracket_1, \llbracket \mathbf{Q}_{1,i} \rrbracket_1 := \llbracket \mathbf{R}_{1,i} \mathbf{U}_{1,i} \rrbracket_1, \{\llbracket \mathbf{G}_{1,i,j} \rrbracket_2 := \llbracket \mathbf{U}_{1,i}(\mathbf{I}_{n'} \otimes \mathbf{Br}_j) \rrbracket_2\}_{j \in [L] \setminus \{i\}}, \pi_{1,i} \\ & \llbracket \mathbf{T}_{2,i} \rrbracket_1 := \llbracket \mathbf{AU}_{2,i} \rrbracket_1, \llbracket \mathbf{Q}_{2,i} \rrbracket_1 := \llbracket \mathbf{R}_{2,i} \mathbf{U}_{2,i} \rrbracket_1, \{\llbracket \mathbf{G}_{2,i,j} \rrbracket_2 := \llbracket \mathbf{U}_{2,i}(\mathbf{I}_m \otimes \mathbf{Br}_j) \rrbracket_2\}_{j \in [L] \setminus \{i\}}, \pi_{2,i} \end{aligned} \right).$$

Ver(crs, i, pk_i): On input crs, an index $i \in [L]$ and a public key $\text{pk}_i = (\llbracket \mathbf{T}_{d,i} \rrbracket_1, \llbracket \mathbf{Q}_{d,i} \rrbracket_1, \{\llbracket \mathbf{G}_{d,i,j} \rrbracket_2\}_{j \in [L] \setminus \{i\}}, \pi_{d,i})_{d \in \{1,2\}}$, define $\mathbf{F}_{d,i}^\top = [\mathbf{T}_{d,i}^\top \parallel \mathbf{Q}_{d,i}^\top]$ and check

$$\begin{aligned} & \text{LVer}(\text{crs}_{1,i}, \llbracket \mathbf{F}_{1,i} \rrbracket_1, \pi_{1,i}) \stackrel{?}{=} 1, & \left\{ \begin{aligned} & \llbracket \mathbf{T}_{1,i} \rrbracket_1 \llbracket \mathbf{I}_{n'} \otimes \mathbf{Br}_j \rrbracket_2 \stackrel{?}{=} \llbracket \mathbf{A} \rrbracket_1 \llbracket \mathbf{G}_{1,i,j} \rrbracket_2, \\ & \llbracket \mathbf{T}_{2,i} \rrbracket_1 \llbracket \mathbf{I}_m \otimes \mathbf{Br}_j \rrbracket_2 \stackrel{?}{=} \llbracket \mathbf{A} \rrbracket_1 \llbracket \mathbf{G}_{2,i,j} \rrbracket_2 \end{aligned} \right\}_{j \in [L] \setminus \{i\}}. \\ & \text{LVer}(\text{crs}_{2,i}, \llbracket \mathbf{F}_{2,i} \rrbracket_1, \pi_{2,i}) \stackrel{?}{=} 1, \end{aligned}$$

Output 1 if all checks pass, and 0 otherwise.

Agg(crs, (pk_i, f_i)_{i ∈ [L]}): On input crs and L tuples of the form (pk_i, f_i), parse $f_i \in \mathcal{F}_{n,n'}^{\text{abp}}$ and $\text{pk}_i = (\llbracket \mathbf{T}_{d,i} \rrbracket_1, \llbracket \mathbf{Q}_{d,i} \rrbracket_1, \{\llbracket \mathbf{G}_{d,i,j} \rrbracket_2\}_{j \in [L] \setminus \{i\}}, \pi_{d,i})_{d \in \{1,2\}}$. For each $i \in [L]$. Then compute and output (mpk, {hsk_j})_{j ∈ [L]} as follows:

$$\text{mpk} = \left\llbracket \begin{aligned} & \mathbf{A}, \mathbf{A}\hat{\mathbf{H}}_1, \mathbf{A}\hat{\mathbf{H}}_2, \mathbf{AV}, \mathbf{Av}, \\ & \sum_{i \in [L]} \mathbf{T}_{1,i} - \mathbf{AW}_{1,i}, \mathbf{A} \sum_{i \in [L]} \mathbf{W}_{1,i}(\mathbf{I}_{i,1} \otimes \mathbf{I}_{2k+1}), \\ & \sum_{i \in [L]} \mathbf{T}_{2,i} + \mathbf{AH}_{2,i}(\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{AW}_{1,i}(\mathbf{I}_{i,0} \otimes \mathbf{I}_{2k+1}) \\ & + \mathbf{AW}_{2,i}(\mathbf{DE}_i \otimes \mathbf{I}_{2k+1}) + \mathbf{AW}_{3,i}(\mathbf{w}^\top \mathbf{DE}_i \otimes \mathbf{I}_{2k+1}) \end{aligned} \right\rrbracket_1,$$

$$\text{hsk}_j = \left[\begin{array}{c} \mathbf{Br}_j, \quad \underline{\mathbf{W}_{1,j}}(\mathbf{I}_{n'} \otimes \mathbf{Br}_j) + \hat{\mathbf{H}}_1 + \hat{\mathbf{W}}_j, \\ \sum_{i \in [L] \setminus \{j\}} \mathbf{G}_{1,i,j} - \underline{\mathbf{W}_{1,i}}(\mathbf{I}_{n'} \otimes \mathbf{Br}_j), \\ \mathbf{W}_{1,j}(\mathbf{I}_{m+n'-1} \otimes \mathbf{Br}_j)\mathbf{L}_{j,1} - \hat{\mathbf{W}}_j\mathbf{L}_{j,1}, \\ \mathbf{H}_{2,j}(\mathbf{q}_j\mathbf{e}_1^\top \otimes \mathbf{Br}_j) + \hat{\mathbf{H}}_2\mathbf{q}_j\mathbf{e}_1^\top + \mathbf{W}_{1,j}(\mathbf{I}_{m+n'-1} \otimes \mathbf{Br}_j)\mathbf{L}_{j,0} - \hat{\mathbf{W}}_j\mathbf{L}_{j,0} \\ + \mathbf{W}_{2,j}(\mathbf{DE}_j \otimes \mathbf{Br}_j) + \mathbf{W}_{3,j}(\mathbf{w}^\top \mathbf{DE}_j \otimes \mathbf{Br}_j) + \mathbf{VDE}_j + \mathbf{vw}^\top \mathbf{DE}_j, \\ \sum_{i \in [L] \setminus \{j\}} \mathbf{W}_{1,i}(\mathbf{I}_{m+n'-1} \otimes \mathbf{Br}_j)\mathbf{L}_{i,1}, \\ \sum_{i \in [L] \setminus \{j\}} \mathbf{G}_{2,i,j} + \mathbf{H}_{2,i}(\mathbf{q}_i\mathbf{e}_1^\top \otimes \mathbf{Br}_j) + \mathbf{W}_{1,i}(\mathbf{I}_{m+n'-1} \otimes \mathbf{Br}_j)\mathbf{L}_{i,0} \\ + \mathbf{W}_{2,i}(\mathbf{DE}_i \otimes \mathbf{Br}_j) + \mathbf{W}_{3,i}(\mathbf{w}^\top \mathbf{DE}_i \otimes \mathbf{Br}_j), \\ \mathbf{q}_j\mathbf{e}_1^\top, \quad \mathbf{DE}_j, \quad \mathbf{w}^\top \mathbf{DE}_j \end{array} \right]_2,$$

where we use the following notation:

- $\underline{\mathbf{W}_{1,i}} \in \mathbb{Z}_p^{(2k+1) \times (2k+1)n'}$ refers to the matrix composed of the right most $(2k+1)n'$ columns of $\mathbf{W}_{1,i} \in \mathbb{Z}_p^{(2k+1) \times (2k+1)(m+n'-1)}$, and
- $\hat{\mathbf{W}}_j \in \mathbb{Z}_p^{(2k+1) \times n'}$ refers to the right most n' columns of $\hat{\mathbf{W}}_j \in \mathbb{Z}_p^{(2k+1) \times (m+n'-1)}$.

Enc(mpk, $\mathbf{x}$, ($\mathbf{p}$, $\mathbf{z}$)): On input the master public key mpk, a public input $\mathbf{x} \in \mathbb{Z}_p^n$ and a private input $(\mathbf{p}, \mathbf{z}) \in \mathbb{Z}_p^k \times \mathbb{Z}_p^{n'}$, sample $\mathbf{s} \xleftarrow{\$} \mathbb{Z}_p^k$ and output

$$\text{ct} = \left[\begin{array}{c} \mathbf{s}^\top \mathbf{A}, \quad \mathbf{z}^\top + \mathbf{s}^\top \mathbf{A} \hat{\mathbf{H}}_1, \quad \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{1,i} - \underline{\mathbf{AW}_{1,i}}, \quad \mathbf{p}^\top + \mathbf{s}^\top \mathbf{A} \hat{\mathbf{H}}_2, \\ \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{2,i} + \mathbf{AH}_{2,i}(\mathbf{q}_i\mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{AW}_{1,i}(\mathbf{L}_{i,\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \\ + \mathbf{AW}_{2,i}(\mathbf{DE}_i \otimes \mathbf{I}_{2k+1}) + \mathbf{AW}_{3,i}(\mathbf{w}^\top \mathbf{DE}_i \otimes \mathbf{I}_{2k+1}), \quad \mathbf{s}^\top \mathbf{AV}, \quad \mathbf{s}^\top \mathbf{Av} \end{array} \right]_1,$$

where $\mathbf{L}_{i,\mathbf{x}} := \mathbf{L}_{i,0} + \mathbf{L}_{i,1}(\mathbf{x} \otimes \mathbf{I}_m)$.

Note. Observe that the last two components of mpk denoted as $(\llbracket \mathbf{K}_1 \rrbracket_1, \llbracket \mathbf{K}_0 \rrbracket_1)$ enables the computation of the fifth component of the ciphertext as $\mathbf{s}^\top \llbracket \mathbf{K}_1 \rrbracket_1((\mathbf{x} \otimes \mathbf{I}_m) \otimes \mathbf{I}_{2k+1}) + \mathbf{s}^\top \llbracket \mathbf{K}_0 \rrbracket_1$.

Dec($\text{sk}_{i^*}, \text{hsk}_{i^*}, \text{ct}$): On input a secret key sk_{i^*} with corresponding helper secret key $\text{hsk}_{i^*} = (\llbracket \mathbf{d}_0 \rrbracket_2, \llbracket \mathbf{D}_{1,1} \rrbracket_2, \llbracket \mathbf{D}_{1,2} \rrbracket_2, \llbracket \mathbf{D}_{2,1,1} \rrbracket_2, \llbracket \mathbf{D}_{2,1,2} \rrbracket_2, \llbracket \mathbf{D}_{2,2,1} \rrbracket_2, \llbracket \mathbf{D}_{2,2,2} \rrbracket_2, \llbracket \mathbf{D}_{0,2} \rrbracket_2, \llbracket \mathbf{D}_{0,3} \rrbracket_2, \llbracket \mathbf{d}_{0,4} \rrbracket_2)$ and a ciphertext $\text{ct} = (\llbracket \mathbf{c}_0 \rrbracket_1, \llbracket \mathbf{c}_{1,1} \rrbracket_1, \llbracket \mathbf{c}_{1,2} \rrbracket_1, \llbracket \mathbf{c}_{2,1} \rrbracket_1, \llbracket \mathbf{c}_{2,2} \rrbracket_1, \llbracket \mathbf{c}_3 \rrbracket_1, \llbracket \mathbf{c}_4 \rrbracket_1)$, set $\mathbf{D}_{0,1} = \mathbf{I}_{n'}$ and compute

$$\begin{aligned} \llbracket \mathbf{c}'_{1,1} \rrbracket_t &= \llbracket \mathbf{c}_{1,1} \rrbracket_1 \llbracket \mathbf{D}_{0,1} \rrbracket_2, & \llbracket \mathbf{d}'_{1,1} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_{1,1} \rrbracket_2, \\ \llbracket \mathbf{c}'_{1,2} \rrbracket_t &= \llbracket \mathbf{c}_{1,2} \rrbracket_1 \llbracket \mathbf{I}_{n'} \otimes \mathbf{d}_0 \rrbracket_2, & \llbracket \mathbf{d}'_{1,2} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_{1,2} \rrbracket_2, \\ \llbracket \mathbf{c}'_{2,1} \rrbracket_t &= \llbracket \mathbf{c}_{2,1} \rrbracket_1 \llbracket \mathbf{D}_{0,2} \rrbracket_2, & \llbracket \mathbf{d}'_{2,1} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_{2,1,1}(\mathbf{x} \otimes \mathbf{I}_m) + \mathbf{D}_{2,1,2} \rrbracket_2, \\ \llbracket \mathbf{c}'_{2,2} \rrbracket_t &= \llbracket \mathbf{c}_{2,2} \rrbracket_1 \llbracket \mathbf{I}_m \otimes \mathbf{d}_0 \rrbracket_2, & \llbracket \mathbf{d}'_{2,2} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_{2,2,1}(\mathbf{x} \otimes \mathbf{I}_m) + \mathbf{D}_{2,2,2} \rrbracket_2, \\ \llbracket \mathbf{c}'_3 \rrbracket_t &= \llbracket \mathbf{c}_3 \rrbracket_1 \llbracket \mathbf{D}_{0,3} \rrbracket_2, \\ \llbracket \mathbf{c}'_4 \rrbracket_t &= \llbracket \mathbf{c}_4 \rrbracket_1 \llbracket \mathbf{d}_{0,4} \rrbracket_2, \end{aligned}$$

$$\begin{aligned} \llbracket \mathbf{p}_1^\top \rrbracket_t &= \llbracket (\mathbf{c}'_{1,1} - \mathbf{d}'_{1,1}) + (\mathbf{c}'_{1,2} - \mathbf{d}'_{1,2}) \rrbracket_t - \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{U}_{1,i^*} \cdot (\mathbf{I}_{n'} \otimes \mathbf{d}_0) \rrbracket_2, \\ \llbracket \mathbf{p}_2^\top \rrbracket_t &= \llbracket (\mathbf{c}'_{2,1} - \mathbf{d}'_{2,1}) + (\mathbf{c}'_{2,2} - \mathbf{d}'_{2,2}) + \mathbf{c}'_3 + \mathbf{c}'_4 \rrbracket_t - \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{U}_{2,i^*} \cdot (\mathbf{I}_m \otimes \mathbf{d}_0) \rrbracket_2. \end{aligned}$$

Run $\mathbf{d}_{f_{i^*}, \mathbf{x}} \leftarrow \text{rec}(f_{i^*}, \mathbf{x})$, then compute and output $\llbracket v \rrbracket_t = \llbracket \mathbf{p}_1^\top, \mathbf{p}_2^\top \rrbracket \cdot \mathbf{d}_{f_{i^*}, \mathbf{x}} \rrbracket_t$.

Next, we only discuss the proof of correctness and compactness, and note that the proof completeness and the security analysis are the same as in the case of AB-eAWS.

Correctness. For all $\lambda, L \in \mathbb{N}$, $i^* \in [L]$, $\mathbf{q}_1, \dots, \mathbf{q}_L \in \mathbb{Z}_p^k$, all $\text{crs} \leftarrow \text{Setup}(1^\lambda, 1^L, \{\llbracket \mathbf{q}_i \rrbracket_{1,2}\}_{i \in [L]})$ and $(\text{pk}_{i^*}, \text{sk}_{i^*}) \leftarrow \text{Gen}(\text{crs}, i^*)$, all $\{\text{pk}_i\}_{i \in [L] \setminus \{i^*\}}$ such that $\text{Ver}(\text{crs}, i, \text{pk}_i) = 1$, all $f_1, \dots, f_L \in \mathcal{F}_{n,n'}^{\text{abp}}$, all $\mathbf{x} \in \mathbb{Z}_p^n$ and all $(\mathbf{p}, \mathbf{z}) \in \mathbb{Z}_p^k \times \mathbb{Z}_p^{n'}$, all $(\text{mpk}, \{\text{hsk}_j\}_{j \in [L]}) \leftarrow \text{Agg}(\text{crs}, (\text{pk}_i, f_i)_{i \in [L]})$ and all $\text{ct} \leftarrow \text{Enc}(\text{mpk}, \mathbf{x}, (\mathbf{p}, \mathbf{z}))$, we have $\text{sk}_{i^*} = (\mathbf{U}_{1,i^*}, \mathbf{U}_{2,i^*})$ and

$$\begin{aligned}
(\mathbf{c}_{1,1}, \mathbf{c}_{1,2}) &= \left(\mathbf{z}^\top + \mathbf{s}^\top \mathbf{A} \hat{\mathbf{H}}_1, \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{1,i} - \mathbf{A} \mathbf{W}_{1,i} \right) \\
(\mathbf{c}_{2,1}, \mathbf{c}_0, \mathbf{c}_3, \mathbf{c}_4) &= \left(\mathbf{p}^\top + \mathbf{s}^\top \mathbf{A} \hat{\mathbf{H}}_2, \mathbf{s}^\top \mathbf{A}, \mathbf{s}^\top \mathbf{A} \mathbf{V}, \mathbf{s}^\top \mathbf{A} \mathbf{v} \right) \\
\mathbf{c}_{2,2} &= \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{2,i} + \mathbf{A} \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{I}_{2k+1}) + \mathbf{A} \mathbf{W}_{1,i} (\mathbf{I}_{i,\mathbf{x}} \otimes \mathbf{I}_{2k+1}) \\
&\quad + \mathbf{A} \mathbf{W}_{2,i} (\mathbf{D} \mathbf{E}_i \otimes \mathbf{I}_{2k+1}) + \mathbf{A} \mathbf{W}_{3,i} (\mathbf{w}^\top \mathbf{D} \mathbf{E}_i \otimes \mathbf{I}_{2k+1}), \\
(\mathbf{d}_0, \mathbf{D}_{0,2}, \mathbf{D}_{0,3}, \mathbf{d}_{0,4}) &= \left(\mathbf{B} \mathbf{r}_{i^*}, \mathbf{q}_{i^*} \mathbf{e}_1^\top, \mathbf{D} \mathbf{E}_{i^*}, \mathbf{w}^\top \mathbf{D} \mathbf{E}_{i^*} \right) \\
\mathbf{D}_{1,1} &= -\mathbf{W}_{1,i^*} (\mathbf{I}_{n'} \otimes \mathbf{B} \mathbf{r}_{i^*}) + \hat{\mathbf{H}}_1 + \hat{\mathbf{W}}_{i^*} \\
\mathbf{D}_{1,2} &= \sum_{i \in [L] \setminus \{i^*\}} \mathbf{G}_{1,i,i^*} - \mathbf{W}_{1,i} (\mathbf{I}_{n'} \otimes \mathbf{B} \mathbf{r}_{i^*}) \\
\mathbf{D}_{2,1,1} &= \mathbf{W}_{1,i^*} (\mathbf{I}_{m+n'-1} \otimes \mathbf{B} \mathbf{r}_{i^*}) \mathbf{L}_{i^*,1} - \hat{\mathbf{W}}_{i^*} \mathbf{L}_{i^*,1} \\
\mathbf{D}_{2,1,2} &= \mathbf{H}_{2,i^*} (\mathbf{q}_{i^*} \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_{i^*}) + \hat{\mathbf{H}}_2 \mathbf{q}_{i^*} \mathbf{e}_1^\top + \mathbf{W}_{1,i^*} (\mathbf{I}_{m+n'-1} \otimes \mathbf{B} \mathbf{r}_{i^*}) \mathbf{L}_{i^*,0} - \hat{\mathbf{W}}_{i^*} \mathbf{L}_{i^*,0} \\
&\quad + \mathbf{W}_{2,i^*} (\mathbf{D} \mathbf{E}_{i^*} \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{W}_{3,i^*} (\mathbf{w}^\top \mathbf{D} \mathbf{E}_{i^*} \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{V} \mathbf{D} \mathbf{E}_{i^*} + \mathbf{v} \mathbf{w}^\top \mathbf{D} \mathbf{E}_{i^*} \\
\mathbf{D}_{2,2,1} &= \sum_{i \in [L] \setminus \{i^*\}} \mathbf{W}_{1,i} (\mathbf{I}_{m+n'-1} \otimes \mathbf{B} \mathbf{r}_{i^*}) \mathbf{L}_{i,1} \\
\mathbf{D}_{2,2,2} &= \sum_{i \in [L] \setminus \{i^*\}} \mathbf{G}_{2,i,i^*} + \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{W}_{1,i} (\mathbf{I}_{m+n'-1} \otimes \mathbf{B} \mathbf{r}_{i^*}) \mathbf{L}_{i,0} \\
&\quad + \mathbf{W}_{2,i} (\mathbf{D} \mathbf{E}_i \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{W}_{3,i} (\mathbf{w}^\top \mathbf{D} \mathbf{E}_i \otimes \mathbf{B} \mathbf{r}_{i^*}),
\end{aligned}$$

where we parse the ciphertext $\text{ct} = (\llbracket \mathbf{c}_0 \rrbracket_1, \llbracket \mathbf{c}_{1,1} \rrbracket_1, \llbracket \mathbf{c}_{1,2} \rrbracket_1, \llbracket \mathbf{c}_{2,1} \rrbracket_1, \llbracket \mathbf{c}_{2,2} \rrbracket_1, \llbracket \mathbf{c}_3 \rrbracket_1, \llbracket \mathbf{c}_4 \rrbracket_1)$ and the helper secret key $\text{hsk}_{i^*} = (\llbracket \mathbf{d}_0 \rrbracket_2, \llbracket \mathbf{D}_{1,1} \rrbracket_2, \llbracket \mathbf{D}_{1,2} \rrbracket_2, \llbracket \mathbf{D}_{2,1,1} \rrbracket_2, \llbracket \mathbf{D}_{2,1,2} \rrbracket_2, \llbracket \mathbf{D}_{2,2,1} \rrbracket_2, \llbracket \mathbf{D}_{2,2,2} \rrbracket_2, \llbracket \mathbf{D}_{0,2} \rrbracket_2, \llbracket \mathbf{D}_{0,3} \rrbracket_2, \llbracket \mathbf{d}_{0,4} \rrbracket_2)$. By computing

$$\begin{aligned}
\llbracket \mathbf{c}'_{1,1} \rrbracket_t &= \llbracket \mathbf{c}_{1,1} \rrbracket_1 \llbracket \mathbf{I}_{n'} \rrbracket_2, & \llbracket \mathbf{d}'_{1,1} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_{1,1} \rrbracket_2, \\
\llbracket \mathbf{c}'_{1,2} \rrbracket_t &= \llbracket \mathbf{c}_{1,2} \rrbracket_1 \llbracket \mathbf{I}_{n'} \otimes \mathbf{d}_0 \rrbracket_2, & \llbracket \mathbf{d}'_{1,2} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_{1,2} \rrbracket_2, \\
\llbracket \mathbf{c}'_{2,1} \rrbracket_t &= \llbracket \mathbf{c}_{2,1} \rrbracket_1 \llbracket \mathbf{D}_{0,2} \rrbracket_2, & \llbracket \mathbf{d}'_{2,1} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_{2,1,1} (\mathbf{x} \otimes \mathbf{I}_m) + \mathbf{D}_{2,1,2} \rrbracket_2, \\
\llbracket \mathbf{c}'_{2,2} \rrbracket_t &= \llbracket \mathbf{c}_{2,2} \rrbracket_1 \llbracket \mathbf{I}_m \otimes \mathbf{d}_0 \rrbracket_2, & \llbracket \mathbf{d}'_{2,2} \rrbracket_t &= \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{D}_{2,2,1} (\mathbf{x} \otimes \mathbf{I}_m) + \mathbf{D}_{2,2,2} \rrbracket_2, \\
\llbracket \mathbf{c}'_3 \rrbracket_t &= \llbracket \mathbf{c}_3 \rrbracket_1 \llbracket \mathbf{D}_{0,3} \rrbracket_2, \\
\llbracket \mathbf{c}'_4 \rrbracket_t &= \llbracket \mathbf{c}_4 \rrbracket_1 \llbracket \mathbf{d}_{0,4} \rrbracket_2,
\end{aligned}$$

the decryptor obtains

$$\begin{aligned}
\mathbf{c}'_{1,1} &= \left(\mathbf{z}^\top + \mathbf{s}^\top \mathbf{A} \hat{\mathbf{H}}_1 \right) \mathbf{I}_{n'} = \mathbf{z}^\top + \mathbf{s}^\top \mathbf{A} \hat{\mathbf{H}}_1 \\
\mathbf{d}'_{1,1} &= \mathbf{s}^\top \mathbf{A} \left(-\mathbf{W}_{1,i^*} (\mathbf{I}_{n'} \otimes \mathbf{B} \mathbf{r}_{i^*}) + \hat{\mathbf{H}}_1 + \hat{\mathbf{W}}_{i^*} \right) \\
\mathbf{c}'_{1,2} &= \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{1,i} (\mathbf{I}_{n'} \otimes \mathbf{B} \mathbf{r}_{i^*}) - \mathbf{A} \mathbf{W}_{1,i} (\mathbf{I}_{n'} \otimes \mathbf{B} \mathbf{r}_{i^*}) \\
\mathbf{d}'_{1,2} &= \mathbf{s}^\top \mathbf{A} \sum_{i \in [L] \setminus \{i^*\}} \mathbf{G}_{1,i,i^*} - \mathbf{W}_{1,i} (\mathbf{I}_{n'} \otimes \mathbf{B} \mathbf{r}_{i^*}) \\
\mathbf{c}'_{2,1} &= \left(\mathbf{p}^\top + \mathbf{s}^\top \mathbf{A} \hat{\mathbf{H}}_2 \right) \mathbf{q}_{i^*} \mathbf{e}_1^\top = (\mathbf{p}^\top \mathbf{q}_{i^*}) \mathbf{e}_1^\top + \mathbf{s}^\top \mathbf{A} \hat{\mathbf{H}}_2 \mathbf{q}_{i^*} \mathbf{e}_1^\top \\
\mathbf{d}'_{2,1} &= \mathbf{s}^\top \mathbf{A} \left(\mathbf{H}_{2,i^*} (\mathbf{q}_{i^*} \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_{i^*}) + \hat{\mathbf{H}}_2 \mathbf{q}_{i^*} \mathbf{e}_1^\top + \mathbf{W}_{1,i^*} (\mathbf{I}_{m+n'-1} \otimes \mathbf{B} \mathbf{r}_{i^*}) \mathbf{L}_{i^*,\mathbf{x}} - \hat{\mathbf{W}}_{i^*} \mathbf{L}_{i^*,\mathbf{x}} \right. \\
&\quad \left. + \mathbf{W}_{2,i^*} (\mathbf{D} \mathbf{E}_{i^*} \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{W}_{3,i^*} (\mathbf{w}^\top \mathbf{D} \mathbf{E}_{i^*} \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{V} \mathbf{D} \mathbf{E}_{i^*} + \mathbf{v} \mathbf{w}^\top \mathbf{D} \mathbf{E}_{i^*} \right)
\end{aligned}$$

$$\begin{aligned}
\mathbf{c}'_{2,2} &= \mathbf{s}^\top \sum_{i \in [L]} \mathbf{T}_{2,i} (\mathbf{I}_m \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{A} \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{A} \mathbf{W}_{1,i} (\mathbf{L}_{i,\mathbf{x}} \otimes \mathbf{B} \mathbf{r}_{i^*}) \\
&\quad + \mathbf{A} \mathbf{W}_{2,i} (\mathbf{D} \mathbf{E}_i \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{A} \mathbf{W}_{3,i} (\mathbf{w}^\top \mathbf{D} \mathbf{E}_i \otimes \mathbf{B} \mathbf{r}_{i^*}) \\
\mathbf{d}'_{2,2} &= \mathbf{s}^\top \mathbf{A} \sum_{i \in [L] \setminus \{i^*\}} \mathbf{G}_{2,i,i^*} + \mathbf{H}_{2,i} (\mathbf{q}_i \mathbf{e}_1^\top \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{W}_{1,i} (\mathbf{I}_{m+n'-1} \otimes \mathbf{B} \mathbf{r}_{i^*}) \mathbf{L}_{i,\mathbf{x}} \\
&\quad + \mathbf{W}_{2,i} (\mathbf{D} \mathbf{E}_i \otimes \mathbf{B} \mathbf{r}_{i^*}) + \mathbf{W}_{3,i} (\mathbf{w}^\top \mathbf{D} \mathbf{E}_i \otimes \mathbf{B} \mathbf{r}_{i^*}) \\
\mathbf{c}'_3 &= \mathbf{s}^\top \mathbf{A} \mathbf{V} \cdot \mathbf{D} \mathbf{E}_{i^*} \\
\mathbf{c}'_4 &= \mathbf{s}^\top \mathbf{A} \mathbf{v} \cdot \mathbf{w}^\top \mathbf{D} \mathbf{E}_{i^*} .
\end{aligned}$$

Therefore, the vector $(\llbracket \mathbf{p}_1^\top \rrbracket_t, \llbracket \mathbf{p}_2^\top \rrbracket_t)$ equals

$$\begin{aligned}
\llbracket \mathbf{p}_1^\top \rrbracket_t &= \llbracket (\mathbf{c}'_{1,1} - \mathbf{d}'_{1,1}) + (\mathbf{c}'_{1,2} - \mathbf{d}'_{1,2}) \rrbracket_t - \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{U}_{1,i^*} \cdot (\mathbf{I}_{n'} \otimes \mathbf{d}_0) \rrbracket_2 \\
&= \llbracket \mathbf{z}^\top - \mathbf{s}^\top \mathbf{A} \widehat{\mathbf{W}}_{i^*} \rrbracket_t , \\
\llbracket \mathbf{p}_2^\top \rrbracket_t &= \llbracket (\mathbf{c}'_{2,1} - \mathbf{d}'_{2,1}) + (\mathbf{c}'_{2,2} - \mathbf{d}'_{2,2}) + \mathbf{c}'_3 + \mathbf{c}'_4 \rrbracket_t - \llbracket \mathbf{c}_0 \rrbracket_1 \llbracket \mathbf{U}_{2,i^*} \cdot (\mathbf{I}_m \otimes \mathbf{d}_0) \rrbracket_2 \\
&= \llbracket \mathbf{s}^\top \mathbf{A} \widehat{\mathbf{W}}_{i^*} \mathbf{L}_{i^*,\mathbf{x}} + \mathbf{p}^\top \mathbf{q}_{i^*} \cdot \mathbf{e}_1^\top \rrbracket_t ,
\end{aligned}$$

where the second equation of $\llbracket \mathbf{p}_2^\top \rrbracket_2$ follows from the fact that $(\mathbf{I}_{m+n'-1} \otimes \mathbf{B} \mathbf{r}_i) \mathbf{L}_{j,\mathbf{x}} = (\mathbf{L}_{j,\mathbf{x}} \otimes \mathbf{B} \mathbf{r}_i)$. We observe that $(\mathbf{p}_1^\top, \mathbf{p}_2^\top) = \text{pgb}(f_{i^*}, \mathbf{x}, \mathbf{z}; \mathbf{s}^\top \mathbf{A} \widehat{\mathbf{W}}_{i^*}) + (\mathbf{0}, \mathbf{p}^\top \mathbf{q}_{i^*} \cdot \mathbf{e}_1^\top)$. Then it follows from the correctness of the garbling scheme that $\llbracket v \rrbracket_t = \llbracket f_{i^*}(\mathbf{x})^\top \mathbf{z} + \mathbf{p}^\top \mathbf{q}_{i^*} \rrbracket_t$.

Compactness. Let $j \in [L]$. The scheme has the following properties:

$$\begin{aligned}
|\text{crs}| &= L^2 \cdot (m + n' - 1) \cdot \text{poly}(\lambda) & |\text{mpk}| &= (mn + n') \cdot \text{poly}(\lambda) \\
|\text{hsk}_j| &= (mn + n') \cdot \text{poly}(\lambda) & |\text{ct}| &= (m + n' - 1) \cdot \text{poly}(\lambda) ,
\end{aligned}$$

where the size of crs follows from the fact that the total size of $\{\text{crs}_i\}_{i \in [L]}$ is $L \cdot \text{poly}(\lambda)$ if we instantiate Π with the QA-NIZK of [KW15], and the fact that the size of the language description is $\text{poly}(\lambda)$.

Next, we state the following theorem which is a special case of Theorem 38.

Theorem 42. *Assuming bilateral MDDH_k on pairings, there exists a very selectively SIM-secure (bounded) RFE scheme for 1AWS functionality.*

7 Very Selectively SIM-Secure sRFE for AB-AWS

In this section, we present our generic compiler that converts any very selectively SIM-secure sRFE scheme for AB-eAWS into an sRFE scheme for AB-AWS with the same security level.

Construction 43 (Very Selectively SIM-Secure sRFE for AB-AWS). The construction uses two independent instances of a sRFE scheme for AB-eAWS which by denote by $\text{sRFE}_1 = (\text{Setup}_1, \text{Gen}_1, \text{Ver}_1, \text{Agg}_1, \text{Enc}_1, \text{Dec}_1)$ and $\text{sRFE}_2 = (\text{Setup}_2, \text{Gen}_2, \text{Ver}_2, \text{Agg}_2, \text{Enc}_2, \text{Dec}_2)$. The sRFE scheme sRFE for AB-AWS works as follows:

Setup($1^\lambda, 1^L$): On input the security parameter 1^λ and the maximum number of slots 1^L , sample $\mathbf{q}_1, \dots, \mathbf{q}_L \xleftarrow{\$} \mathbb{Z}_p^k$, run

$$\text{crs}_1 \leftarrow \text{Setup}_1(1^\lambda, 1^L, \{\llbracket \mathbf{q}_i \rrbracket_{1,2}\}_{i \in [L]}), \quad \text{crs}_2 \leftarrow \text{Setup}_2(1^\lambda, 1^L, \{\llbracket \mathbf{q}_i \rrbracket_{1,2}\}_{i \in [L]}),$$

and output the common reference string $\text{crs} = (\text{crs}_1, \text{crs}_2)$.

Gen(crs, i): On input crs and an index $i \in [L]$, sample

$$(\text{pk}_{i,1}, \text{sk}_{i,1}) \leftarrow \text{Gen}_1(\text{crs}_1, i), \quad (\text{pk}_{i,2}, \text{sk}_{i,2}) \leftarrow \text{Gen}_2(\text{crs}_2, i),$$

and output $\text{pk}_i = (\text{pk}_{i,1}, \text{pk}_{i,2})$ and $\text{sk}_i = (\text{sk}_{i,1}, \text{sk}_{i,2})$.

$\text{Ver}(\text{crs}, i, \text{pk}_i)$: On input crs , an index $i \in [L]$ and a public key pk_i , parse $\text{crs} = (\text{crs}_1, \text{crs}_2)$, $\text{pk}_i = (\text{pk}_{i,1}, \text{pk}_{i,2})$ and check

$$\text{Ver}_1(\text{crs}_1, i, \text{pk}_{i,1}) \stackrel{?}{=} 1, \quad \text{Ver}_2(\text{crs}_2, i, \text{pk}_{i,2}) \stackrel{?}{=} 1.$$

Output 1 if all checks pass, and 0 otherwise.

$\text{Agg}(\text{crs}, (\text{pk}_i, f_i)_{i \in [L]})$: On input crs and L tuples of the form (pk_i, f_i) , parse $\text{crs} = (\text{crs}_1, \text{crs}_2)$, $\text{pk}_i = (\text{pk}_{i,1}, \text{pk}_{i,2})$ and compute

$$\begin{aligned} (\text{mpk}_1, \{\text{hsk}_{i,1}\}_{i \in [L]}) &\leftarrow \text{Agg}_1(\text{crs}_1, (\text{pk}_{i,1}, f_i)_{i \in [L]}), \\ (\text{mpk}_2, \{\text{hsk}_{i,2}\}_{i \in [L]}) &\leftarrow \text{Agg}_2(\text{crs}_2, (\text{pk}_{i,2}, f_i)_{i \in [L]}). \end{aligned}$$

Output $\text{mpk} = (\text{mpk}_1, \text{mpk}_2)$ and $\{\text{hsk}_i = (\text{hsk}_{i,1}, \text{hsk}_{i,2})\}_{i \in [L]}$.

$\text{Enc}(\text{mpk}, \mathbf{y}, (\mathbf{x}_j, \mathbf{z}_j)_{j \in [N]})$: On input the master public key mpk , an attribute $\mathbf{y} \in \mathbb{Z}_p^{\hat{n}}$ and tuples $(\mathbf{x}_j, \mathbf{z}_j) \in \mathbb{Z}_p^n \times \mathbb{Z}_p^{n'}$ for $j \in [N]$, parse $\text{mpk} = (\text{mpk}_1, \text{mpk}_2)$, sample $\mathbf{p}_2, \dots, \mathbf{p}_L \stackrel{\$}{\leftarrow} \mathbb{Z}_p^k$, compute $\mathbf{p}_1 = -\sum_{j \in [2;N]} \mathbf{p}_j$ and generate

$$\text{ct}_1 \leftarrow \text{Enc}_1(\text{mpk}_1, (\mathbf{y}, \mathbf{x}_1), (\mathbf{z}_1, \mathbf{p}_1)), \quad \{\text{ct}_j \leftarrow \text{Enc}_2(\text{mpk}_2, (\mathbf{y}, \mathbf{x}_j), (\mathbf{z}_j, \mathbf{p}_j))\}_{j \in [2;N]}.$$

Output the ciphertext $\text{ct} = \{\text{ct}_j\}_{j \in [N]}$.

$\text{Dec}(\text{sk}_i, \text{hsk}_i, \text{ct})$: On input a secret key sk_i with corresponding helper secret key hsk_i and a ciphertext ct , parse $\text{hsk}_i = (\text{hsk}_{i,1}, \text{hsk}_{i,2})$ and $\text{ct} = \{\text{ct}_j\}_{j \in [N]}$, run

$$\llbracket v_1 \rrbracket_t = \text{Dec}_1(\text{sk}_{i,1}, \text{hsk}_{i,1}, \text{ct}_1), \quad \{\llbracket v_j \rrbracket_t = \text{Dec}_2(\text{sk}_{i,2}, \text{hsk}_{i,2}, \text{ct}_j)\}_{j \in [2;N]},$$

compute $\llbracket v \rrbracket_t = \sum_{i \in [N]} \llbracket v_i \rrbracket_t$ and output the discrete logarithm v recovered via brute-force.

Proposition 44 (Compactness, Correctness and Completeness). *If sRFE_1 and sRFE_2 are complete, correct and compact, then so is the sRFE scheme sRFE for AB-AWS in Construction 43.*

Proof of Proposition 44. Completeness and compactness readily follow from the completeness and compactness of sRFE_1 and sRFE_2 . For correctness, we consider a slot $i^* \in [L]$ such that $g_{i^*}^*(\mathbf{y}) = 0$. Assuming the correctness of sRFE_1 and sRFE_2 , we have $v_j = h_{i^*}^*(\mathbf{x}_j^*)^\top \mathbf{z}_j^* + \mathbf{p}_j^\top \mathbf{q}_i$ for all $j \in [N]$. Since $\mathbf{p}_1, \dots, \mathbf{p}_N$ are chosen in such a way that $\sum_{j \in [N]} \mathbf{p}_j = \mathbf{0}$, we conclude that $\mathbf{v} = \sum_{j \in [N]} v_j = \sum_{j \in [N]} h_{i^*}^*(\mathbf{x}_j^*)^\top \mathbf{z}_j^*$. $\square$

Proposition 45 (Security). *If sRFE_1 and sRFE_2 are very selectively SIM-secure, then so is the sRFE scheme sRFE for AB-AWS in Construction 43.*

We present the simulator for sRFE in Section 7.1. The proof of Proposition 45 can be found in Sections 7.2. When instantiating sRFE_1 and sRFE_2 with the sRFE schemes for AB-eAWS in Construction 35, and using the transformation of Theorem 18 we obtain the following theorem.

Theorem 46. *Assuming bi-MDDH_k on pairings, there exists a (bounded) RFE scheme for the AB-AWS functionality.*

7.1 Simulator

Let $(\widetilde{\text{Setup}}_1, \widetilde{\text{Gen}}_1, \widetilde{\text{Enc}}_1)$ and $(\widetilde{\text{Setup}}_2, \widetilde{\text{Gen}}_2, \widetilde{\text{Enc}}_2)$ be the simulators of sRFE_1 and sRFE_2 , respectively. The simulator for the sRFE scheme sRFE in Construction 43 works as follows.

$\widetilde{\text{Setup}}(1^\lambda, 1^L, (\mathbf{y}, \{\mathbf{x}_j^*\}_{j \in [N]}), \{f_i^*\}_{i \in [L]}, \{\llbracket \mu_i \rrbracket_{1,2}\}_{i \in \mathcal{I}})$: On input the security parameter 1^λ , the maximum number of slots 1^L , the public inputs $(\mathbf{y}, \{\mathbf{x}_j^*\}_{j \in [N]})$, the challenge functions $\{f_i^*\}_{i \in [L]}$ with $f_i^* = (g_i^*, h_i^*)$, and output

values $\{\|\mu_i\|_{1,2}\}_{i \in \mathcal{I}}$ where $\mathcal{I} = \{i \in \mathcal{I}_{\text{mal}} \cup \mathcal{I}_{\text{cor}} : g_i^*(\mathbf{y}) = 0\}$, sample vectors $\mathbf{p}_2, \dots, \mathbf{p}_N, \mathbf{q}_1, \dots, \mathbf{q}_L \xleftarrow{\$} \mathbb{Z}_p^k$ and set $\mathbf{p}_1 = -\sum_{j \in [2:N]} \mathbf{p}_j$. Then run

$$\begin{aligned} (\text{crs}_1, \text{td}_1) &\leftarrow \widetilde{\text{Setup}}_1(1^\lambda, 1^L, (\mathbf{y}, \mathbf{x}_1^*), (f_i^*, \|\mathbf{q}_i\|_{1,2})_{i \in [L]}, \{\|\mu_i + \mathbf{p}_1^\top \mathbf{q}_i\|_{1,2}\}_{i \in \mathcal{I}}), \\ \text{crs}_2 &\leftarrow \text{Setup}_2(1^\lambda, 1^L, (\|\mathbf{q}_i\|_{1,2})_{i \in [L]}) \end{aligned}$$

and output the pair of common reference string $\text{crs} = (\text{crs}_1, \text{crs}_2)$ and trapdoor $\text{td} = (\text{td}_1, \text{crs}_2, \{\mathbf{p}_j\}_{j \in [2:N]}, \{\mathbf{x}_j^*\}_{j \in [N]}, \{f_i^*\}_{i \in [L]})$.

$\widetilde{\text{Gen}}(i, \text{td})$: On input an index $i \in [L]$ and the trapdoor $\text{td} = (\text{td}_1, \text{crs}_2, \{\mathbf{p}_j\}_{j \in [2:N]}, \{\mathbf{x}_j^*\}_{j \in [N]}, \{f_i^*\}_{i \in [L]})$, run

$$(\text{pk}_{i,1}, \text{sk}_{i,1}) \leftarrow \widetilde{\text{Gen}}_1(i, \text{td}_1), \quad (\text{pk}_{i,2}, \text{sk}_{i,2}) \leftarrow \text{Gen}_2(\text{crs}_2, i),$$

and output $\text{pk}_i = (\text{pk}_{i,1}, \text{pk}_{i,2})$ and $\text{sk}_i = (\text{sk}_{i,1}, \text{sk}_{i,2})$.

$\widetilde{\text{Enc}}(\{\text{pk}_i^*\}_{i \in [L]}, \text{td})$: On input public keys $\{\text{pk}_i^* = (\text{pk}_{i,1}^*, \text{pk}_{i,2}^*)\}_{i \in [L]}$ and the trapdoor $\text{td} = (\text{td}_1, \text{crs}_2, \{\mathbf{p}_j\}_{j \in [2:N]}, \{\mathbf{x}_j^*\}_{j \in [N]}, \{f_i^*\}_{i \in [L]})$, run $\text{mpk}_2 \leftarrow \text{Agg}_2(\text{crs}_2, (\text{pk}_{i,2}^*, f_i^*)_{i \in [L]})$ and

$$\text{ct}_1 \leftarrow \widetilde{\text{Enc}}_1(\{\text{pk}_{i,1}^*\}_{i \in [L]}, \text{td}_1), \quad \{\text{ct}_j \leftarrow \text{Enc}_2(\text{mpk}_2, (\mathbf{y}, \mathbf{x}_j^*), (\mathbf{0}, \mathbf{p}_j))\}_{j \in [2:N]}.$$

Then output the ciphertext $\text{ct} = \{\text{ct}_j\}_{j \in [N]}$.

7.2 Proof of Security

Proof of Proposition 45. We consider a sequence of hybrid games $G_0, \dots, G_N$ where $G_0 = \text{Exp}_{\text{sRFE}, \mathcal{A}}^{\text{srfc-real}}$ and $G_N = \text{Exp}_{\text{sRFE}, \mathcal{A}}^{\text{srfc-sim}}$. We argue $G_{\ell-1}^b \approx_c G_\ell^b$ for $\ell \in [N]$. Then the proposition follows via a hybrid argument. Modifications between consecutive games are highlighted using **boxes**.

Game G_0 : This is $\text{Exp}_{\text{sRFE}, \mathcal{A}}^{\text{srfc-real}}$. Specifically, we have:

- The common reference string is of the form $\text{crs} = (\text{crs}_1, \text{crs}_2)$, where

$$\text{crs}_1 \leftarrow \text{Setup}_2(1^\lambda, 1^L, \{\|\mathbf{q}_i\|_{1,2}\}_{i \in [L]}), \quad \text{crs}_2 \leftarrow \text{Setup}_2(1^\lambda, 1^L, \{\|\mathbf{q}_i\|_{1,2}\}_{i \in [L]}).$$

- For each $i \in [L]$, each public key in $\mathcal{D}_i$ is of the form $\text{pk}_i = (\text{pk}_{i,1}, \text{pk}_{i,2})$, where

$$(\text{pk}_{i,1}, \text{sk}_{i,1}) \leftarrow \text{Gen}_2(\text{crs}_1, i), \quad (\text{pk}_{i,2}, \text{sk}_{i,2}) \leftarrow \text{Gen}_2(\text{crs}_2, i),$$

and the challenger knows the corresponding secret key $\text{sk}_i = (\text{sk}_{i,1}, \text{sk}_{i,2})$.

- For each $i \in [L]$, the challenge public key is of the form $\text{pk}_i^* = (\text{pk}_{i,1}^*, \text{pk}_{i,2}^*)$ such that

$$\text{Ver}_1(\text{crs}_1, i, \text{pk}_{i,1}^*) = 1, \quad \text{Ver}_2(\text{crs}_2, i, \text{pk}_{i,2}^*) = 1.$$

- The challenge ciphertext is of the form $\text{ct}^* = \{\text{ct}_j^*\}_{j \in [N]}$, where vectors $\mathbf{p}_2, \dots, \mathbf{p}_L \xleftarrow{\$} \mathbb{Z}_p^k$ are chosen at random, $\mathbf{p}_1 = -\sum_{j \in [2:N]} \mathbf{p}_j$ and

$$\text{ct}_1 \leftarrow \text{Enc}_1(\text{mpk}_1, (\mathbf{y}, \mathbf{x}_1^*), (\mathbf{z}_1^*, \mathbf{p}_1)), \quad \{\text{ct}_j \leftarrow \text{Enc}_2(\text{mpk}_2, (\mathbf{y}, \mathbf{x}_j^*), (\mathbf{z}_j^*, \mathbf{p}_j))\}_{j \in [2:N]}.$$

Game G_1 : This is the same as G_0 except that we use the simulator for sRFE_1 . Specifically, we apply the following changes:

- The challenger now generates $\text{crs} = (\text{crs}_1, \text{crs}_2)$ as

$$\begin{aligned} (\text{crs}_1, \text{td}_1) &\leftarrow \widetilde{\text{Setup}}_1(1^\lambda, 1^L, (\mathbf{y}, \mathbf{x}_1^*), \{f_i^*, \|\mathbf{q}_i\|_{1,2}\}_{i \in [L]}, \{\|h_i^*(\mathbf{x}_1^*)^\top \mathbf{z}_1^* + \mathbf{p}_1^\top \mathbf{q}_i\|_{1,2}\}_{i \in \mathcal{I}}), \\ \text{crs}_2 &\leftarrow \text{Setup}_2(1^\lambda, 1^L, \{\|\mathbf{q}_i\|_{1,2}\}_{i \in [L]}), \end{aligned}$$

where $f_i^* = (g_i^*, h_i^*)$ for $i \in [L]$, and $\mathcal{I} = \{i \in \mathcal{I}_{\text{mal}} \cup \mathcal{I}_{\text{cor}} : g_i^*(\mathbf{y}) = 0\}$.

- For each $i \in [L]$, each public key in $\mathcal{D}_i$ is of the form $\text{pk}_i = (\text{pk}_{i,1}, \text{pk}_{i,2})$, where

$$(\text{pk}_{i,1}, \text{sk}_{i,1}) \leftarrow \widetilde{\text{Gen}}_1(i, \text{td}_1) , \quad (\text{pk}_{i,2}, \text{sk}_{i,2}) \leftarrow \text{Gen}_2(\text{crs}_2, i) .$$

- The challenge ciphertext is of the form $\text{ct}^* = \{\text{ct}_j^*\}_{j \in [N]}$, where

$$\text{ct}_1 \leftarrow \widetilde{\text{Enc}}_1(\{\text{pk}_{i,1}^*\}_{i \in [L]}, \text{td}_1) , \quad \{\text{ct}_j \leftarrow \text{Enc}_2(\text{mpk}_2, (\mathbf{y}, \mathbf{x}_j^*), (\mathbf{z}_j^*, \mathbf{p}_j))\}_{j \in [2;N]} .$$

We can observe that $G_0 \approx_c G_1$ under the SIM-security of sRFE_2 .

Game G_ℓ for $\ell \in [2; N]$: This is the same as G_1 except for the boxed terms below:

- The challenger now generates $\text{crs} = (\text{crs}_1, \text{crs}_2)$ as

$$(\text{crs}_1, \text{td}_1) \leftarrow \widetilde{\text{Setup}}_1(1^\lambda, 1^L, (\mathbf{y}, \mathbf{x}_1^*), \{f_i^*, \|\mathbf{q}_i\|_{1,2}\}_{i \in [L]}, \{\|\sum_{j \in [\ell]} h_i^*(\mathbf{x}_j^*)^\top \mathbf{z}_j^* + \mathbf{p}_1^\top \mathbf{q}_i\|_{1,2}\}_{i \in \mathcal{I}}) ,$$

$$\text{crs}_2 \leftarrow \text{Setup}_2(1^\lambda, 1^L, \{\|\mathbf{q}_i\|_{1,2}\}_{i \in [L]}) .$$

- The challenge ciphertext is of the form $\text{ct}^* = \{\text{ct}_j^*\}_{j \in [N]}$, where

$$\text{ct}_1 \leftarrow \widetilde{\text{Enc}}_1(\{\text{pk}_{i,1}^*\}_{i \in [L]}, \text{td}_1) , \quad \text{ct}_j \leftarrow \begin{cases} \text{Enc}_2(\text{mpk}_2, (\mathbf{y}, \mathbf{x}_j^*), (\mathbf{0}, \mathbf{p}_j)) & \text{if } j \in [2; \ell] \\ \text{Enc}_2(\text{mpk}_2, (\mathbf{y}, \mathbf{x}_j^*), (\mathbf{z}_j^*, \mathbf{p}_j)) & \text{if } j \in [\ell + 1; N] \end{cases} .$$

Below, we prove the following claim for all $\ell \in [2; N]$.

Claim 47. *If sRFE_2 is very selectively SIM-secure and MDDH_k holds in $\mathbb{G}$, then we have $G_{\ell-1} \approx_c G_\ell$.*

Furthermore, we can observe that $G_N = \mathbf{Exp}_{\text{sRFE}, \mathcal{A}}^{\text{srfc-sim}}$ with respect to the simulator in Section 7.1 on input $\mu_i = \sum_{j \in [N]} h_i^*(\mathbf{x}_j^*)^\top \mathbf{z}_j^*$. This concludes the proof of the proposition. □

We now turn to the claim.

Proof of Claim 47. The proof considers a sequence of sub-hybrids $\widehat{G}_0, \dots, \widehat{G}_3$ with $\widehat{G}_0 \equiv G_{\ell-1}$ and $\widehat{G}_3 \equiv G_\ell$.

Game $\widehat{G}_0$: This is $G_{\ell-1}$.

Game $\widehat{G}_1$: This is the same as $\widehat{G}_0$ except for the boxed terms below:

- The challenger generates $\text{crs} = (\text{crs}_1, \text{crs}_2)$ as

$$(\text{crs}_1, \text{td}_1) \leftarrow \widetilde{\text{Setup}}_1(1^\lambda, 1^L, (\mathbf{y}, \mathbf{x}_1^*), \{f_i^*, \|\mathbf{q}_i\|_{1,2}\}_{i \in [L]}, \{\|\sum_{j \in [\ell-1]} h_i^*(\mathbf{x}_j^*)^\top \mathbf{z}_j^* + \mathbf{p}_1^\top \mathbf{q}_i\|_{1,2}\}_{i \in \mathcal{I}}) ,$$

$$(\text{crs}_2, \text{td}_2) \leftarrow \widetilde{\text{Setup}}_2(1^\lambda, 1^L, (\mathbf{y}, \mathbf{x}_\ell^*), (f_\ell^*, \|\mathbf{q}_\ell\|_{1,2})_{i \in [L]}, \{\|h_i^*(\mathbf{x}_\ell^*)^\top \mathbf{z}_\ell^* + \mathbf{p}_\ell^\top \mathbf{q}_i\|_{1,2}\}_{i \in \mathcal{I}}) .$$

- For each $i \in [L]$, each public key in $\mathcal{D}_i$ is of the form $\text{pk}_i = (\text{pk}_{i,1}, \text{pk}_{i,2})$, where

$$(\text{pk}_{i,1}, \text{sk}_{i,1}) \leftarrow \widetilde{\text{Gen}}_1(i, \text{td}_1) , \quad (\text{pk}_{i,2}, \text{sk}_{i,2}) \leftarrow \widetilde{\text{Gen}}_2(i, \text{td}_2) .$$

- The challenge ciphertext is of the form $\text{ct}^* = \{\text{ct}_j^*\}_{j \in [N]}$, where

$$\text{ct}_1 \leftarrow \widetilde{\text{Enc}}_1(\{\text{pk}_{i,1}^*\}_{i \in [L]}, \text{td}_1) , \quad \text{ct}_j \leftarrow \begin{cases} \text{Enc}_2(\text{mpk}_2, (\mathbf{y}, \mathbf{x}_j^*), (\mathbf{0}, \mathbf{p}_j)) & \text{if } j \in [2; \ell-1] \\ \widetilde{\text{Enc}}_2(\{\text{pk}_{i,2}^*\}_{i \in [L]}, \text{td}_2) & \text{if } j = \ell \\ \text{Enc}_2(\text{mpk}_2, (\mathbf{y}, \mathbf{x}_j^*), (\mathbf{z}_j^*, \mathbf{p}_j)) & \text{if } j \in [\ell+1; N] \end{cases} .$$

We can observe that $\widehat{G}_0 \approx_c \widehat{G}_1$ under the very selective SIM-security of sRFE_2 .

Game $\widehat{G}_2$: This is the same as $\widehat{G}_1$ except for the following modification during the setup procedure. The challenger now generates $\text{crs} = (\text{crs}_1, \text{crs}_2)$ as

$$\begin{aligned} (\text{crs}_1, \text{td}_1) &\leftarrow \widetilde{\text{Setup}}_1(1^\lambda, 1^L, (\mathbf{y}, \mathbf{x}_1^*), \{f_i^*, \llbracket \mathbf{q}_i \rrbracket_{1,2}\}_{i \in [L]}, \{\llbracket \sum_{j \in [\ell]} h_i^*(\mathbf{x}_j^*)^\top \mathbf{z}_j^* + \mathbf{p}_1^\top \mathbf{q}_i \rrbracket_{1,2}\}_{i \in \mathcal{I}}\}, \\ (\text{crs}_2, \text{td}_2) &\leftarrow \widetilde{\text{Setup}}_2(1^\lambda, 1^L, (\mathbf{y}, \mathbf{x}_\ell^*), (f_i^*, \llbracket \mathbf{q}_i \rrbracket_{1,2})_{i \in [L]}, \{\llbracket h_i^*(\mathbf{x}_\ell^*)^\top \mathbf{z}_\ell^* + \mathbf{p}_\ell^\top \mathbf{q}_i \rrbracket_{1,2}\}_{i \in \mathcal{I}}\}. \end{aligned}$$

We have that $\widehat{G}_1 \approx_c \widehat{G}_2$ under the bi-MDDH $_k$ assumption in $\mathbb{G}$ which states that for all $i \in \mathcal{I}$, we have $\{\llbracket \mathbf{q}_i \rrbracket_{1,2}, \llbracket \mathbf{p}_\ell^\top \mathbf{q}_i \rrbracket_{1,2}\}_{\lambda \in \mathbb{N}} \approx_c \{\llbracket \mathbf{q}_i \rrbracket_{1,2}, \llbracket p_{\ell,i} \rrbracket_{1,2}\}_{\lambda \in \mathbb{N}}$ where $p_{\ell,i} \xleftarrow{\$} \mathbb{Z}_p$. Then the indistinguishability follows from the statistical shift $p_{\ell,i} \mapsto p_{\ell,i} - h_i^*(\mathbf{x}_\ell^*)^\top \mathbf{z}_\ell^*$.

Game $\widehat{G}_3$: This game is the same as $\widehat{G}_2$ except that we reverse the modifications from $\widehat{G}_1$.

- The common reference string is of the form $\text{crs} = (\text{crs}_1, \text{crs}_2)$, where

$$\begin{aligned} (\text{crs}_1, \text{td}_1) &\leftarrow \widetilde{\text{Setup}}_1(1^\lambda, 1^L, (\mathbf{y}, \mathbf{x}_1^*), \{f_i^*, \llbracket \mathbf{q}_i \rrbracket_{1,2}\}_{i \in [L]}, \{\llbracket \sum_{j \in [\ell]} h_i^*(\mathbf{x}_j^*)^\top \mathbf{z}_j^* + \mathbf{p}_1^\top \mathbf{q}_i \rrbracket_{1,2}\}_{i \in \mathcal{I}}\}, \\ \text{crs}_2 &\leftarrow \widetilde{\text{Setup}}_2(1^\lambda, 1^L, \{\llbracket \mathbf{q}_i \rrbracket_{1,2}\}_{i \in [L]}). \end{aligned}$$

- For each $i \in [L]$, each public key in $\mathcal{D}_i$ is of the form $\text{pk}_i = (\text{pk}_{i,1}, \text{pk}_{i,2})$, where

$$(\text{pk}_{i,1}, \text{sk}_{i,1}) \leftarrow \widetilde{\text{Gen}}_1(i, \text{td}_1), \quad (\text{pk}_{i,2}, \text{sk}_{i,2}) \leftarrow \widetilde{\text{Gen}}_2(\text{crs}_2, i).$$

- The challenge ciphertext is of the form $\text{ct}^* = \{\text{ct}_j^*\}_{j \in [N]}$, where

$$\text{ct}_1 \leftarrow \widetilde{\text{Enc}}_1(\{\text{pk}_{i,1}^*\}_{i \in [L]}, \text{td}_1), \quad \text{ct}_j \leftarrow \begin{cases} \text{Enc}_2(\text{mpk}_2, (\mathbf{y}, \mathbf{x}_j^*), (\mathbf{0}, \mathbf{p}_j)) & \text{if } j \in [2; \ell - 1] \\ \text{Enc}_2(\text{mpk}_2, (\mathbf{y}, \mathbf{x}_\ell^*), (\mathbf{0}, \mathbf{p}_\ell)) & \text{if } j = \ell \\ \text{Enc}_2(\text{mpk}_2, (\mathbf{y}, \mathbf{x}_j^*), (\mathbf{z}_j^*, \mathbf{p}_j)) & \text{if } j \in [\ell + 1; N] \end{cases}.$$

We have $\widehat{G}_2 \approx_c \widehat{G}_3$ under the very selective SIM-security of sRFE $_2$. Furthermore, we note that $\widehat{G}_3 \equiv G_\ell$. This concludes the proof of the claim. □

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